

Project Data Page:

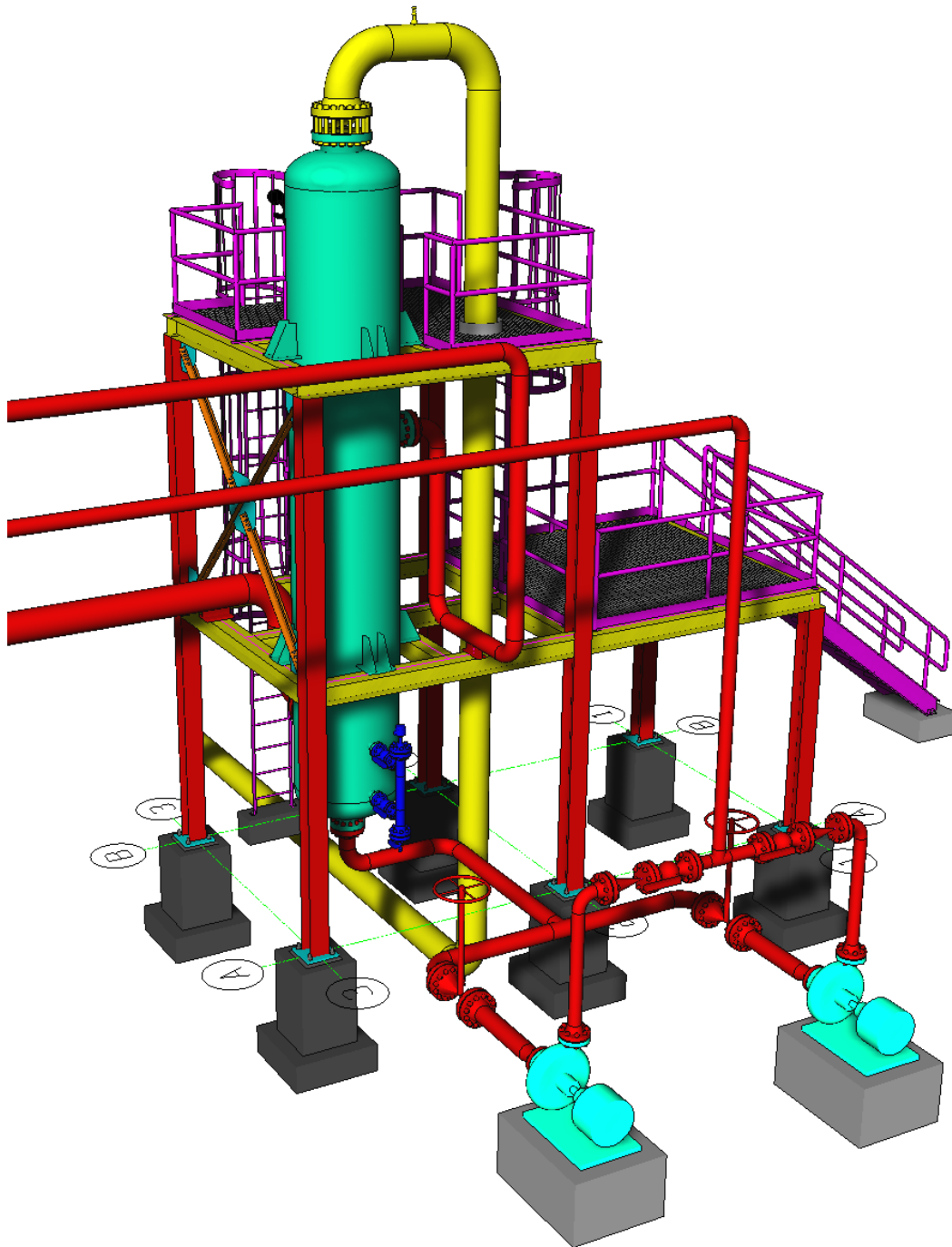




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\122D1LP & HP FLASH COLUMN\\Operating case\\Rev.04 - HLL level.pvdb

Date of Analysis : Apr 14, 2026 5:03pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 6800.000 x 6806.000 x 7200.000 x 7206.000 mm ID
 Vessel Design Length, Tangent to Tangent 41900.00 mm

Distance of Bottom Tangent above Grade 7500.00 mm
 Distance of Base above Grade 0.00 mm
 Specified Datum Line Distance 0.00 mm

Internal Design Temperature 130 °C
 Internal Design Pressure 380.000 KPa.

External Design Temperature 130 °C
 External Design Pressure 51.730 KPa.

Maximum Allowable Working Pressure 483 KPa.
 External Max. Allowable Working Pressure 79 KPa.
 Shop Test Pressure 603.945 KPa.

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C

Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	+IT
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-266 2	...	...	K03506	+IT
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-106 B	...	...	K03006	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-516 70	...	...	K02700	+IT
Basing	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
HP_main skirt-3	...	...	...	1.6000	N/A	No
HP_main skirt-2	...	...	...	1.6000	N/A	No
HP_main skirt-1	...	...	...	1.6000	N/A	No
HP_BTM dish	4207.901	0.00	4722.000	6.0000	N/A	No
HP_shell-1	4172.575	0.00	4274.000	6.0000	N/A	No
HP_shell-2	4141.404	0.00	4306.000	6.0000	N/A	No
HP_shell-3	4110.234	0.00	4337.000	6.0000	N/A	No
HP_shell-4	4104.000	0.00	4301.000	1.0000	N/A	No
HP_shell-5	4104.000	0.00	4211.000	1.0000	N/A	No
HP_shell-6	4104.000	0.00	4211.000	1.0000	N/A	No
HP_shell-7	4104.000	0.00	4211.000	1.0000	N/A	No
HP_TOP dish	4104.000	0.00	4521.000	1.0000	N/A	No
HP_int_skirt-1	...	...	...	1.6000	N/A	No



HP_int_skirt-2	...	...	...	1.6000	N/A	No
LP_int_skirt-2	...	...	...	1.6000	N/A	No
LP_int_skirt-1	...	...	...	1.6000	N/A	No
LP_BTm dish	466.601	51.73	1069.000	6.0000	No	No
LP-shell-1	447.380	51.73	711.000	6.0000	N/A	No
LP-shell-2	416.209	51.73	742.000	6.0000	N/A	No
LP-shell-3	385.039	51.73	687.000	6.0000	N/A	No
LP-shell-4	384.000	51.73	688.000	6.0000	N/A	No
LP-shell-5	384.000	51.73	602.000	6.0000	N/A	No
LP-shell-6	384.000	51.73	622.000	1.0000	N/A	No
LP-shell-7	384.000	51.73	548.000	1.0000	N/A	No
LP-shell-8	384.000	51.73	483.000	1.0000	N/A	No
LP-shell-9	384.000	51.73	548.000	1.0000	N/A	No
LP-TOP dish	384.000	51.73	610.000	1.0000	No	No

Liquid Level: 17950.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.060

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
35650.00	Bar 300.0 x 18.0	Ring R8
38400.00	Bar 252.0 x 18.0	Ring R7
40900.01	Bar 250.0 x 18.0	Ring R1
43399.98	Bar 250.0 x 18.0	Ring R5
45899.98	Bar 250.0 x 18.0	Ring R2
48399.98	Bar 250.0 x 18.0	Ring R4
50899.98	Bar 250.0 x 18.0	Ring R3
53399.98	Bar 250.0 x 18.0	Ring R6
55899.98	Bar 250.0 x 18.0	Ring R9

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	3000.0	3000.0	36.0	36.0	...	...	0.70	0.70
Skirt	6000.0	3000.0	36.0	36.0	...	...	0.70	0.70
Skirt	7500.0	1500.0	38.0	38.0	...	...	0.70	0.55
Sphere	7500.0	0.0	68.0	59.0	52.1	...	1.00	1.00
Cylinder	10500.0	3000.0	102.0	102.0	98.1	...	1.00	1.00
Cylinder	13500.0	3000.0	102.0	102.0	97.4	...	1.00	1.00
Cylinder	15400.0	1900.0	102.0	102.0	96.7	...	1.00	1.00
Cylinder	18400.0	3000.0	96.0	96.0	91.5	...	1.00	1.00
Cylinder	21400.0	3000.0	94.0	94.0	91.5	...	1.00	1.00
Cylinder	23400.0	2000.0	94.0	94.0	91.5	...	1.00	1.00
Cylinder	24900.0	1500.0	94.0	94.0	91.5	...	1.00	1.00
Sphere	24900.0	0.0	58.0	50.6	46.0	...	1.00	1.00
Skirt	26150.0	1250.0	24.0	24.0	...	...	1.00	1.00
Skirt	29150.0	3000.0	24.0	24.0	...	...	1.00	1.00
Skirt	32150.0	3000.0	24.0	24.0	...	...	1.00	1.00
Skirt	33400.0	1250.0	24.0	24.0	...	...	1.00	1.00
Ellipse	33450.0	50.0	34.0	30.0	16.9	...	1.00	1.00
Cylinder	36450.0	3000.0	24.0	24.0	16.3	...	1.00	1.00
Cylinder	39450.0	3000.0	24.0	24.0	15.6	...	1.00	1.00
Cylinder	42450.0	3000.0	22.0	22.0	14.9	...	1.00	1.00
Cylinder	45450.0	3000.0	22.0	22.0	14.9	...	1.00	1.00
Cylinder	48250.0	2800.0	20.0	20.0	14.9	...	1.00	1.00
Cylinder	51250.0	3000.0	18.0	18.0	11.4	...	0.85	0.85
Cylinder	54250.0	3000.0	16.0	16.0	11.4	...	0.85	0.85
Cylinder	56850.0	2600.0	16.0	16.0	12.8	...	0.85	0.85
Cylinder	57850.0	1000.0	16.0	16.0	11.4	...	0.85	0.85
Ellipse	57900.0	50.0	18.0	16.0	10.9	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
------	----	-----------------------	-------------------------	-------------------------------	------------------------



10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	59	No Calc	...	2507.53
50	60	102	No Calc	...	1561.1
60	70	102	No Calc	...	1561.1
70	80	102	No Calc	...	1561.1
80	90	96	No Calc	...	1535.41
90	100	94	No Calc	...	1466.49
100	110	94	No Calc	...	1466.49
110	120	94	No Calc	...	1466.49
120	130	50.6	No Calc	...	2288.54
130	140	...	No Calc	...	No Calc
140	150	...	No Calc	...	No Calc
150	160	...	No Calc	...	No Calc
160	170	...	No Calc	...	No Calc
170	180	30	18.1564	51.73	200.817
180	Ring	24	17.2717	51.73	166.308
Ring	190	24	17.1046	51.73	172.252
190	Ring	24	17.1046	51.73	172.252
Ring	200	24	16.6821	51.73	188.739
200	Ring	22	16.6785	51.73	142.267
Ring	210	22	16.6785	51.73	142.267
210	Ring	22	16.6785	51.73	142.267
Ring	220	22	16.6785	51.73	142.267
220	Ring	20	16.6752	51.73	102.273
Ring	230	20	16.6752	51.73	102.273
230	Ring	18	11.6769	51.73	164.917
Ring	Ring	18	11.6769	51.73	164.909
Ring	240	18	11.6769	51.73	164.917
240	Ring	16	11.6733	51.73	121.482
Ring	250	16	11.6733	51.73	121.482
250	Ring	16	11.6733	51.73	121.482
Ring	260	16	11.8395	51.73	116.675
260	270	16	11.8395	51.73	116.675
270	280	16	13.1654	51.73	78.6246

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	16000	No Calc	No Calc	No Calc
20	30	16000	No Calc	No Calc	No Calc
30	40	16000	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	19667.7	No Calc	No Calc	No Calc
60	70	19667.7	No Calc	No Calc	No Calc
70	80	19667.7	No Calc	No Calc	No Calc
80	90	19667.7	No Calc	No Calc	No Calc
90	100	19667.7	No Calc	No Calc	No Calc
100	110	19667.7	No Calc	No Calc	No Calc
110	120	19667.7	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc
130	140	16000	No Calc	No Calc	No Calc
140	150	16000	No Calc	No Calc	No Calc
150	160	16000	No Calc	No Calc	No Calc
160	170	16000	No Calc	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc
180	Ring	2850	8727.11	No Calc	No Calc
Ring	190	2750	8727.03	12635.8 >>>	11847.8
190	Ring	2750	8727.03	No Calc	No Calc
Ring	200	2500.01	8726.64	12522.2 >>>	7478.23
200	Ring	2500.01	6590.59	No Calc	No Calc
Ring	210	2499.96	6590.47	9310.12 >>>	6902.5
210	Ring	2499.96	6590.47	No Calc	No Calc
Ring	220	2500	6590.57	9310.12 >>>	6902.5
220	Ring	2500	4792.6	No Calc	No Calc



Ring	230	2500	4792.6	6637 >>>	6440.16
230	Ring	2500	7621.01	No Calc	No Calc
Ring	Ring	2500	7620.75	10849.1 >>>	7116.84
Ring	240	2500	7621.02	10849.1 >>>	7116.84
240	Ring	2500	5654.68	No Calc	No Calc
Ring	250	2500	5654.68	7898.49 >>>	6676.47
250	Ring	2500	5654.68	No Calc	No Calc
Ring	260	2600.52	5654.6	7889.62 >>>	6676.47
260	270	2600.52	5654.6	No Calc	No Calc
270	280	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D	WRC-107/537	0.872	n/a	Passed
A	WRC-297	0.972	Shell	Passed
E	WRC-107/537	0.906	n/a	Passed
C	WRC-107/537	0.417	n/a	Passed
B	WRC-297	0.960	Nozzle	Passed
K	WRC-297	0.550	Shell	Passed
H1	WRC-297	0.735	Nozzle	Passed
H2	WRC-297	0.735	Nozzle	Passed
F	WRC-107/537	0.427	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1500	27035.9	200915	831855	7696828
20	30	4500	25926.6	200833	752410	7094194
30	40	6750	24722.7	200591	676435	6492046
40	50	7499.98	23889.2	200260	639976	6191408
50	60	9000	23889.2	196684	639975	6191402
60	70	12000	22745	191658	570022	5608878
70	80	14450	21358	184689	503867	5044346
80	90	16900	20360.4	180702	464233	4697218
90	100	19900	19217.2	175803	404866	4162450
100	110	22400	17824.6	171554	349302	3641403
110	120	24150	16794.7	168217	314682	3301625
120	130	24900	16223.4	162705	289918	3053433
130	140	25525	15940.8	158530	289918	3053429
140	150	27650	15469.3	157947	270286	2855627
150	160	30650	14350.4	151335	225556	2391695
160	170	32775	13205.2	149766	184221	1940034
170	180	33425	12454.2	148469	168184	1753634
180	190	34950	12434.4	136699	167562	1746504
190	200	37950	10990.9	114872	132423	1369141
200	210	40950	9540.75	90691.9	101625	1060790
210	220	43950	8358.23	80304.9	74776.1	804290
220	230	46850	6921.73	71888.7	51855.7	575996
230	240	49750	5558.37	58213.8	34383.2	393848
240	250	52750	4376.17	46836.9	19481.1	236270
250	260	55550	2943.9	42715.8	8500.78	101938
260	270	57350	1661.16	19456.3	2514.08	21112.6
270	280	57875	1267.31	8569.94	1049.82	7099.25

Abs Max of the all of the Stress Ratio's : 0.9442

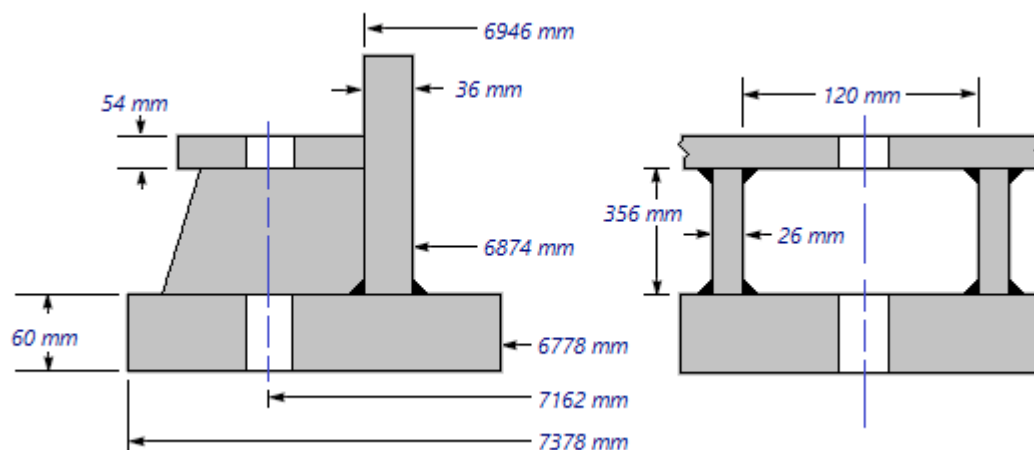
Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	60.0000	mm
Inside Diameter of Basing	6778.0000	mm
Outside Diameter of Basing	7378.0000	mm
Nominal Diameter of Bolts	64.0000	mm



Diameter of Bolt Circle	7162.0000 mm
Number of Bolts	48
Thickness of Gusset Plates	26.0000 mm
Average Width of Gusset Plates	206.0000 mm
Height of Gussets	356.0000 mm
Distance between Gussets	120.0000 mm
Thickness of Top Plate or Ring	54.0000 mm
Radial Width of the Top Plate	216.0000 mm

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	37423. Kg
Total Earthquake Shear on Support	211302. Kg
Wind Moment on Support	2324316. Kg-m.
Earthquake Moment on Support	9189289. Kg-m.
Maximum Initial Bolt Load	70436. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	693041.9 kg.
Shop Test - Fabricated + water (full)	2781996.5 kg.
Shipping - Fab. + rem. intls.+ shipping app.	867207.8 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	1065258.9 kg.
Empty - Fab. + intls. + details + wghts.	1065258.9 kg.
Operating - Empty + operating liquid (no ca)	1796711.1 kg.
Field Test - Empty weight + test medium (full)	2890541.8 kg.



Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
D	4602	...	...	OK	OK	Passed
MH-6	4320	...	...	...	OK	Passed
MH-5	4215	...	...	...	OK	Passed
MH-4	4215	...	...	...	OK	Passed
A	4211	...	...	OK	OK	Passed
E	4525	...	...	OK	OK	Passed
C	718	OK	...	OK	OK	Passed
MH-3	833	OK	...	...	OK	Passed
B	803	OK	...	OK	OK	Passed
K	24014	OK	...	OK	OK	Passed
MH-2	1035	OK	...	...	OK	Passed
MH-1	1035	OK	...	...	OK	Passed
H1	841	OK	...	OK	OK	Passed
H2	841	OK	...	OK	OK	Passed
F	722	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 718 Nozzle : C
 Minimum MAWP Shells/Flanges : 483 Element : LP-shell-8
 Minimum MAPnc Shells/Flanges : 520 Element : LP-shell-8

Computed Vessel M.A.W.P. : 483 KPa.

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

- Error: Nozzle A, was too close to a gross structural discontinuity.

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Inside Radius mm
40	D	0.000	0.000	463.200
60	MH-6	4400.001	205.000	310.800
90	MH-5	12400.001	225.000	304.800
100	MH-4	14900.005	225.000	304.800
110	A	16650.004	180.000	463.200
120	E	0.000	0.000	88.600
170	C	0.000	0.000	597.600
190	MH-3	29550.004	225.000	294.800
200	B	34250.004	180.000	439.200
200	K	34450.004	0.000	71.180
230	MH-2	42850.008	225.000	294.800
240	MH-1	45149.996	225.000	294.800
250	H1	47550.000	180.000	437.200
250	H2	47550.000	0.000	437.200
270	F	0.000	0.000	515.400

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.



No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K	6.000 in	160	WNF	6.625	18.237	322.40	16.00	232.98	150
E	8.000 in	Actual	WNF	8.000	16.000	...	...	313.46	600
MH-6	24.000 in	Actual	WNF	31.874	100.000	...	...	437.40	600
MH-5	24.000 in	Actual	WNF	31.874	100.000	...	...	429.38	600
MH-4	24.000 in	Actual	WNF	31.874	100.000	...	...	429.38	600
MH-3	24.000 in	Actual	WNF	24.000	16.000	1209.60	16.00	348.93	150
MH-2	24.000 in	Actual	WNF	24.000	16.000	1209.60	16.00	348.92	150
MH-1	24.000 in	Actual	WNF	24.000	16.000	1209.60	16.00	348.92	150
D	36.000 in	Actual	WNF	42.299	80.000	...	...	471.86	600
A	36.000 in	Actual	WNF	44.976	114.000	...	...	514.19	600
B	36.000 in	Actual	WNF	36.000	24.000	1824.40	16.00	365.15	150
H1	36.000 in	Actual	WNF	36.000	26.000	1824.40	16.00	365.13	150
H2	36.000 in	Actual	WNF	36.000	26.000	1824.40	16.00	365.13	150
F	42.000 in	Actual	WNF	42.000	24.000	2126.80	16.00	537.93	150
C	48.000 in	Actual	WNF	48.000	18.000	2439.20	16.00	364.65	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K	SA-106 B	16.000	16.000	16.000	16.000	...
E	SA-266 2	14.500	20.000	...	...	...
MH-6	SA-266 2	98.000	10.000	...	...	...
MH-5	SA-266 2	94.000	10.000	...	...	...
MH-4	SA-266 2	94.000	10.000	...	...	...
MH-3	SA-516 70	16.000	16.000	16.000	16.000	...
MH-2	SA-516 70	16.000	16.000	16.000	16.000	...
MH-1	SA-516 70	16.000	16.000	16.000	16.000	...
D	SA-266 2	53.000	10.000	...	...	...
A	SA-266 2	94.000	10.000	...	...	...
B	SA-516 70	16.000	16.000	16.000	16.000	...
H1	SA-516 70	16.000	16.000	16.000	16.000	...
H2	SA-516 70	16.000	16.000	16.000	16.000	...
F	SA-516 70	16.000	16.000	16.000	16.000	...
C	SA-516 70	16.000	16.000	16.000	16.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K	41950.004	0.0	210.00	0.00	LP-shell-3
E	...	0.0	260.00	0.00	HP_TOP dish
MH-6	11900.001	205.0	310.00	0.00	HP_shell-2
MH-5	19900.000	225.0	310.00	0.00	HP_shell-5
MH-4	22400.004	225.0	310.00	0.00	HP_shell-6
MH-3	37050.004	225.0	310.00	0.00	LP-shell-2
MH-2	50350.004	225.0	310.00	0.00	LP-shell-6



MH-1	52650.000	225.0	310.00	0.00	LP-shell-7
D	...	0.0	360.00	0.00	HP_BTM dish
A	24150.004	180.0	360.00	0.00	HP_shell-7
B	41750.004	180.0	310.00	0.00	LP-shell-3
H1	55049.996	180.0	310.00	0.00	LP-shell-8
H2	55049.996	0.0	310.00	0.00	LP-shell-8
F	...	0.0	500.00	0.00	LP-TOP dish
C	...	0.0	310.00	0.00	LP_BTM dish

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 68.0mm THK X 6800.0mm ID X 0.0mm	SA-516 70
1	SPHERICAL HEAD: 58.0mm THK X 6806.0mm ID X 0.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 34.0mm THK X 7200.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 18.0mm THK X 7206.0mm ID X 50.0mm	SA-516 70
14	PLATFORM: 1000mm X 1500mm WIDE	
1	PACKING: 3400.000mm	...
1	INSULATION: 3400mm X 80mm THK	
2	PACKING: 1000.000mm	...
10	INSULATION: 3000mm X 80mm THK	
1	PACKING: 1000.020mm	...
1	INSULATION: 1899mm X 80mm THK	
1	PACKING: 999.976mm	...
4	LINING: 3000.000mm X 3.000mm THK	...
1	INSULATION: 2000mm X 80mm THK	
1	LINING: 2000.000mm X 3.000mm THK	...
1	INSULATION: 1500mm X 80mm THK	
1	LINING: 1500.000mm X 3.000mm THK	...
1	INSULATION: 3403mm X 80mm THK	
1	LINING: 3403.000mm X 3.000mm THK	...
1	INSULATION: 1250mm X 80mm THK	
2	INSULATION: 1000mm X 80mm THK	
1	INSULATION: 1850mm X 80mm THK	
1	BAR RING STIFFENER: 7248.0mm ID X 7848.0mm OD X 18.0mm	SA-516 70
1	BAR RING STIFFENER: 7248.0mm ID X 7752.0mm OD X 18.0mm	SA-516 70
4	BAR RING STIFFENER: 7244.0mm ID X 7744.0mm OD X 18.0mm	SA-516 70
1	PACKING: 1199.950mm	...
1	PACKING: 2800.000mm	...
1	INSULATION: 2800mm X 80mm THK	
3	BAR RING STIFFENER: 7240.0mm ID X 7740.0mm OD X 18.0mm	SA-516 70
1	LADDER: 54800.0mm	...
1	INSULATION: 2600mm X 80mm THK	
1	LINING: 2600.000mm X 3.000mm THK	...
3	TRAYS # 3 SPACING 35.433071 mm	



1	LINING: 1000.000mm X 3.000mm THK	...
1	PLATFORM: 1000mm X 7500mm WIDE	
1	LADDER: 4500.0mm	...
1	INSULATION: 1851mm X 80mm THK	
1	LINING: 1851.500mm X 3.000mm THK	...
3	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
3	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 7378.0mm OD X 6778.0 ID X 60.0mm	SA-516 70
48	BASERING BOLTS: 64.0mm DIA	SA-36
48	BASERING BOLT NUTS: 64.0mm DIA	...
1	BASERING TOP RING: 7378.0mm OD X 6946.0 ID X 54.0mm	...
96	BASERING GUSSET PLATES: 206.0mm X 206.0mm X 356.0mm X 26.0mm	...
8	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 3059.3mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 3119.3mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 1559.6mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 20035.3mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3057.5mm X 1500.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3117.5mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 1558.7mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 20023.7mm X 1500.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 102.0mm THK X 3026.4mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 102.0mm THK X 3085.7mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 102.0mm THK X 18336.1mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 102.0mm THK X 1542.8mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 102.0mm THK X 3026.4mm X 1900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 102.0mm THK X 3085.7mm X 1900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 102.0mm THK X 18276.8mm X 1900.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 102.0mm THK X 1542.8mm X 1900.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 96.0mm THK X 3029.0mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 96.0mm THK X 3088.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 96.0mm THK X 18352.3mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 96.0mm THK X 1544.2mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 94.0mm THK X 3029.0mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 94.0mm THK X 3088.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 94.0mm THK X 19837.1mm X 3000.0mm	SA-516 70



1	CYLINDRICAL SEGMENT 94.0mm THK X 1544.2mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 94.0mm THK X 3029.0mm X 2000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 94.0mm THK X 3088.4mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 94.0mm THK X 19837.1mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 94.0mm THK X 1544.2mm X 2000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 94.0mm THK X 3029.0mm X 1500.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 94.0mm THK X 3088.4mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 94.0mm THK X 1544.2mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 94.0mm THK X 19837.1mm X 1500.0mm	SA-516 70
7	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 2910.8mm X 1250.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 20375.6mm X 1250.0mm	SA-516 70
7	CON SKIRT SEGMENT 24.0mm THK X 2977.9mm X 3000.0mm X 2908.5mm	SA-516 70
1	CON SKIRT SEGMENT 24.0mm THK X 20845.2mm X 3000.0mm X 20359.8mm	SA-516 70
7	CON SKIRT SEGMENT 24.0mm THK X 3049.6mm X 3000.0mm X 2980.2mm	SA-516 70
1	CON SKIRT SEGMENT 24.0mm THK X 21347.5mm X 3000.0mm X 20861.3mm	SA-516 70
7	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 3049.6mm X 1250.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 24.0mm THK X 21347.5mm X 1250.0mm	SA-516 70
13	CYLINDRICAL SEGMENT 24.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 21335.7mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 19571.1mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 1283.4mm X 3000.0mm	SA-516 70
12	CYLINDRICAL SEGMENT 22.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm THK X 19571.1mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm THK X 1283.4mm X 3000.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 20.0mm THK X 3048.0mm X 2800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 19571.1mm X 2800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 1283.4mm X 2800.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 18.0mm THK X 3050.5mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 19587.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1284.5mm X 3000.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 16.0mm THK X 3050.5mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 19587.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 1284.5mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 19587.4mm X 2600.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 16.0mm THK X 3050.5mm X 2600.0mm	SA-516 70



1	CYLINDRICAL SEGMENT 16.0mm THK X 1284.5mm X 2600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 19587.4mm X 1000.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 16.0mm THK X 3050.5mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 1284.5mm X 1000.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
HPBTM dish	[9]	!	-12	Yes			
HPshell-1	[8]	!	-12	Yes			
HPshell-2	[8]	D	-15	Yes	0.850	102.000	1.00
HPshell-3	[8]	D	-15	Yes	0.843	102.000	1.00
HPshell-4	[8]	D	-15	Yes	0.850	96.000	1.00
HPshell-5	[8]	D	-15	Yes	0.869	94.000	1.00
HPshell-6	[8]	!	-12	Yes			
HPshell-7	[8]	D	-15	Yes	0.869	94.000	1.00
HPTOP dish	[9]	D	-26	Yes	0.810	50.600	1.00
LPBTM dish	[10]	D	-18	No	0.498	30.000	1.00
LPBTM dish	[7]	D	-16	No	0.420	34.000	1.00
LP-shell-1	[8]	D	-23	No	0.631	24.000	1.00
LP-shell-2	[8]	D	-23	No	0.595	24.000	1.00
LP-shell-3	[8]	!	-12	No			
LP-shell-4	[8]	D	-24	No	0.628	22.000	1.00
LP-shell-5	[8]	D	-26	No	0.718	20.000	1.00
LP-shell-6	[8]	D	-28	No	0.591	18.000	0.85
LP-shell-7	[8]	D	-30	No	0.670	16.000	0.85
LP-shell-8	[8]	!	-12	No			
LP-shell-9	[8]	D	-30	No	0.670	16.000	0.85
LP-TOP dish	[10]	D	-30	No	0.630	16.000	0.85
LP-TOP dish	[7]	D	-28	No	0.591	18.000	0.85
Ring R8	[23]	D	-28	No	0.947	18.000	1.00
Ring R7	[23]	D	-28	No	0.893	18.000	1.00
Ring R1	[23]	!	-12	No			
Ring R5	[23]	D	-28	No	0.838	18.000	1.00
Ring R2	[23]	D	-28	No	0.838	18.000	1.00
Ring R4	[23]	D	-28	No	0.591	18.000	0.85
Ring R3	[23]	D	-28	No	0.591	18.000	0.85
Ring R6	[23]	D	-30	No	0.670	16.000	0.85
Ring R9	[23]	!	-12	No			
D	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	Yes			
MH-6	[1]	D	-15	Yes	0.847	102.000	1.00
Nozzle Flg	[5]	A	-18	Yes			
MH-5	[1]	D	-15	Yes	0.869	94.000	1.00
Nozzle Flg	[5]	A	-18	Yes			
MH-4	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	Yes			
A	[1]	D	-15	Yes	0.869	94.000	1.00
Nozzle Flg	[5]	A	-18	Yes			
E	[1]	D	-26	Yes	0.809	50.600	1.00
Nozzle Flg	[5]	A	-18	Yes			
C	[1]	D	-30	No	1.000	16.000	1.00
Nozzle Flg	[5]	A	-18	No			
MH-3	[1]	D	-30	No	0.846	16.000	1.00
Nozzle Flg	[5]	A	-18	No			
B	[1]	!	-12	No			
Nozzle Flg	[5]	A	-18	No			
K	[1]	!	-12	No			
Nozzle Flg	[5]	A	-18	No			
MH-2	[1]	D	-30	No	0.528	16.000	1.00
Nozzle Flg	[5]	A	-18	No			
MH-1	[1]	D	-30	No	0.528	16.000	1.00
Nozzle Flg	[5]	A	-18	No			
H1	[1]	!	-12	No			
Nozzle Flg	[5]	A	-18	No			
H2	[1]	!	-12	No			
Nozzle Flg	[5]	A	-18	No			
F	[1]	D	-30	No	0.649	16.000	1.00
Nozzle Flg	[5]	A	-18	No			
Nozl. Bolting	[22]		-48	No	1.000	64.000	1.00



Warmest MDMT: -12 -22

Required Minimum Design Metal Temperature -12.0 °C
Warmest Computed Minimum Design Metal Temperature -12.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Warn. 140 150 Skirt diameter at base is < at the top
Warn. 150 160 Skirt diameter at base is < at the top

Class From To: Check of Additional Element Data

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	380	KPa.
Design Internal Temperature	130.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	Not Applied
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW



Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm
Percent Wind for Hydrotect	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/27
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	HPmain skirt-3
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	6946 mm
Diameter of Skirt at Base	6946 mm
Skirt Thickness	36 mm
Internal Corrosion Allowance	1.6 mm



Nominal Thickness	36	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	175	N./mm ²
Allowable Stress, Operating	161.57	N./mm ²
Allowable Stress, Hydrotest	247	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	242.6	N./mm ²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	
Element From Node	10	
Detail Type	Weight	
Detail ID	HPmanwy wt	
Dist. from "FROM" Node / Offset dist	2200	mm
Miscellaneous Weight	800	Kgf
Offset from Element Centerline	0	mm

Element#	2/27	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	HPmain skirt-2	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	6946	mm
Diameter of Skirt at Base	6946	mm
Skirt Thickness	36	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	36	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	
Element From Node	20	
Detail Type	Platform	
Detail ID	SKRT PLAT-1	
Dist. from "FROM" Node / Offset dist	300	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	15000	cm ²
Platform Weight	1752.9	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm



Element#	3/27
Element From Node	30
Element To Node	40
Element Type	Skirt Support
Description	HPmain skirt-1
Distance "FROM" to "TO"	1500 mm
Skirt Outside Diameter	6946 mm
Diameter of Skirt at Base	6946 mm
Skirt Thickness	38 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.55
Weld is pre-Heated	Yes

Element From Node	30
Detail Type	Platform
Detail ID	PLAT-1
Dist. from "FROM" Node / Offset dist	1300 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	30000 cm ²
Platform Weight	7014.9 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 Kg/mm
Platform Length (top head platform)	0 mm

Element#	4/27
Element From Node	40
Element To Node	50
Element Type	Spherical
Description	HPBTM dish
Distance "FROM" to "TO"	0 mm
Inside Diameter	6800 mm
Element Thickness	59 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4100 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	Yes

Element From Node	40
Detail Type	Packing
Detail ID	HPpack: 11
Dist. from "FROM" Node / Offset dist	-3400 mm
Height of Packed Section	3400 mm
Density	0.000166 kg./cm ³



Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	-3400	mm
Height/Length of Liquid	3400	mm
Liquid Density	0.001058	kg./cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	-3400	mm
Height/Length of Insulation	3400	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	D	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	13357	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	
Element From Node	40	
Detail Type	Weight	
Detail ID	HPVB	
Dist. from "FROM" Node / Offset dist	-3400	mm
Miscellaneous Weight	170	Kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	HPbtm SF dish	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	2020	Kgf
Offset from Element Centerline	0	mm

Element#	5/27	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	HPshell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6800	mm
Element Thickness	102	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	102	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4100	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Packing	



Detail ID	HPPack: 12	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	1000	mm
Density	0.000166	kg./cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001058	kg./cm ³
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	50	
Detail Type	Weight	
Detail ID	HPDPPOVH	
Dist. from "FROM" Node / Offset dist	0.004069	mm
Miscellaneous Weight	3155	Kgf
Offset from Element Centerline	5319	mm

Element#	6/27	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	HPshell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6800	mm
Element Thickness	102	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	102	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4100	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PLAT-2	
Dist. from "FROM" Node / Offset dist	799.97	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	30000	cm ²
Platform Weight	3500.2	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm



Platform Length (top head platform)	0	mm
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001058	kg./cm ³
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	60	
Detail Type	Nozzle	
Detail ID	MH-6	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	205.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1656.5	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element#	7/27	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	HPshell-3	
Distance "FROM" to "TO"	1900	mm
Inside Diameter	6800	mm
Element Thickness	102	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	102	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4100	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	PLAT-3	
Dist. from "FROM" Node / Offset dist	1800	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	30000	cm ²
Platform Weight	3500.2	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm



Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm
Element From Node	70	
Detail Type	Packing	
Detail ID	HPPack: 21	
Dist. from "FROM" Node / Offset dist	899.97	mm
Height of Packed Section	1000	mm
Density	0.000166	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	600	mm
Liquid Density	0.001058	kg./cm ³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1900	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	70	
Detail Type	Weight	
Detail ID	HPB#1btm grd	
Dist. from "FROM" Node / Offset dist	899.97	mm
Miscellaneous Weight	3100	Kgf
Offset from Element Centerline	0	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	HPB#1btm grd	
Dist. from "FROM" Node / Offset dist	899.97	mm
Miscellaneous Weight	300	Kgf
Offset from Element Centerline	0	mm

Element#	8/27	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	HPshell-4	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6806	mm
Element Thickness	96	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	96	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4100	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Packing	
Detail ID	HPPack: 22	
Dist. from "FROM" Node / Offset dist	0.004651	mm
Height of Packed Section	999.98	mm
Density	0.000166	kg./cm ³



Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	80	
Detail Type	Lining	
Detail ID	SS304Lclad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³
Element From Node	80	
Detail Type	Weight	
Detail ID	HPB#1top HDG	
Dist. from "FROM" Node / Offset dist	999.97	mm
Miscellaneous Weight	960	Kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	HPB#1dis. dea	
Dist. from "FROM" Node / Offset dist	1600	mm
Miscellaneous Weight	5900	Kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	HPB#1dis. TA	
Dist. from "FROM" Node / Offset dist	1600	mm
Miscellaneous Weight	300	Kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	HPSHroud	
Dist. from "FROM" Node / Offset dist	2800	mm
Miscellaneous Weight	6000	Kgf
Offset from Element Centerline	0	mm

Element#	9/27	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	HPshell-5	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6806	mm
Element Thickness	94	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	94	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4100	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



Element From Node	90
Detail Type	Platform
Detail ID	PLAT-4
Dist. from "FROM" Node / Offset dist	899.97 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	30000 cm ²
Platform Weight	3496.2 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 Kg/mm
Platform Length (top head platform)	0 mm

Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element From Node	90
Detail Type	Lining
Detail ID	SS304Lclad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³

Element From Node	90
Detail Type	Nozzle
Detail ID	MH-5
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	225.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1648.8 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	90
Detail Type	Weight
Detail ID	HPNozzle wt
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	1400 Kg
Offset from Element Centerline	0 mm

Element#	10/27
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	HPshell-6
Distance "FROM" to "TO"	2000 mm
Inside Diameter	6806 mm
Element Thickness	94 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	94 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4100 KPa.



Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 KPa.
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 100
 Detail Type Platform
 Detail ID PLAT-5
 Dist. from "FROM" Node / Offset dist 1900 mm
 Platform Start Angle (degrees) 270.0
 Platform End Angle (degrees) 90.0
 Platform Wind Area 30000 cm²
 Platform Weight 3496.2 Kg
 Platform Railing Weight 0 Kg/mm
 Platform Grating Weight 0.017 Kgs/cm²
 Platform Width 1500 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 115 mm
 Platform Force Coefficient 1.0
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 0 mm
 Unit Weight of Ladder 0 Kg/mm
 Platform Length (top head platform) 0 mm

Element From Node 100
 Detail Type Insulation
 Detail ID Ins: 70
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 2000 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg./cm³

Element From Node 100
 Detail Type Lining
 Detail ID SS304Lclad
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Lining 2000 mm
 Thickness of Lining 3 mm
 Density 0.008 kg./cm³

Element From Node 100
 Detail Type Nozzle
 Detail ID MH-4
 Dist. from "FROM" Node / Offset dist 1000 mm
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 225.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 1648.8 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2 +IT

Element# 11/27
 Element From Node 110
 Element To Node 120
 Element Type Cylinder
 Description HPshell-7
 Distance "FROM" to "TO" 1500 mm
 Inside Diameter 6806 mm
 Element Thickness 94 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 94 mm



External Corrosion Allowance	0	mm
Design Internal Pressure	4100	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1500	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	110	
Detail Type	Lining	
Detail ID	SS304Lclad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1500	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	A	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2360.5	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	110	
Detail Type	Weight	
Detail ID	HPGallary	
Dist. from "FROM" Node / Offset dist	1100	mm
Miscellaneous Weight	6800	Kgf
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	HPAPPOVH	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	21960	Kgf
Offset from Element Centerline	5319	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	HPEPPOVH	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	2296	Kgf
Offset from Element Centerline	4964	mm

Element#	12/27	
Element From Node	120	
Element To Node	130	
Element Type	Spherical	
Description	HPTOP dish	
Distance "FROM" to "TO"	0	mm
Inside Diameter	6806	mm



Element Thickness	50.6	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	58	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4100	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	120	
Detail Type	Platform	
Detail ID	PLAT-6	
Dist. from "FROM" Node / Offset dist	2400	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	30000	cm ²
Platform Weight	3461.4	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm
Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3403	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	120	
Detail Type	Lining	
Detail ID	SS304Lclad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3403	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³
Element From Node	120	
Detail Type	Nozzle	
Detail ID	E	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	8	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	104.13	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	
Element From Node	120	
Detail Type	Weight	
Detail ID	HPtop SF dish	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1900	Kgf
Offset from Element Centerline	0	mm



Element From Node	120
Detail Type	For./Mom.
Detail ID	HPEFxFz
Dist. from "FROM" Node / Offset dist	3403 mm
Force in X Direction	1457.7 Kgf
Force in Y Direction	-2018.4 Kgf
Force in Z Direction	1457.7 Kgf
Moment about X Axis	897.11 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	897.11 Kg-m.
Force/Moment Combination Method	SRSS

Element#	13/27
Element From Node	130
Element To Node	140
Element Type	Skirt Support
Description	HPintskirt-1
Distance "FROM" to "TO"	1250 mm
Skirt Outside Diameter	6924 mm
Diameter of Skirt at Base	6924 mm
Skirt Thickness	24 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No

Element From Node	130
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1250 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm³

Element#	14/27
Element From Node	140
Element To Node	150
Element Type	Skirt Support
Description	HPintskirt-2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	7087.9 mm
Diameter of Skirt at Base	6924 mm
Skirt Thickness	24 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No

Element From Node	140
Detail Type	Weight
Detail ID	margin
Dist. from "FROM" Node / Offset dist	1650 mm
Miscellaneous Weight	10000 Kgf
Offset from Element Centerline	0 mm



Element From Node	140
Detail Type	Weight
Detail ID	Cleat
Dist. from "FROM" Node / Offset dist	1650 mm
Miscellaneous Weight	17400 Kg
Offset from Element Centerline	0 mm

Element From Node	140
Detail Type	Weight
Detail ID	Lifting wt
Dist. from "FROM" Node / Offset dist	1650 mm
Miscellaneous Weight	20200 Kg
Offset from Element Centerline	0 mm

Element#	15/27
Element From Node	150
Element To Node	160
Element Type	Skirt Support
Description	LPintskirt-2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	7252 mm
Diameter of Skirt at Base	7087.9 mm
Skirt Thickness	24 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No

Element#	16/27
Element From Node	160
Element To Node	170
Element Type	Skirt Support
Description	LPintskirt-1
Distance "FROM" to "TO"	1250 mm
Skirt Outside Diameter	7252 mm
Diameter of Skirt at Base	7252 mm
Skirt Thickness	24 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No

Element From Node	160
Detail Type	Platform
Detail ID	PLAT-7
Dist. from "FROM" Node / Offset dist	899.95 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	30000 cm ²
Platform Weight	3618.9 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm



Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element From Node	160	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element#	17/27	
Element From Node	170	
Element To Node	180	
Element Type	Elliptical	
Description	LPBTM dish	
Distance "FROM" to "TO"	50	mm
Inside Diameter	7200	mm
Element Thickness	30	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	380	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	51.73	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	170	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	-1800	mm
Height/Length of Liquid	1850	mm
Liquid Density	0.001058	kg./cm ³

Element From Node	170	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	-1800	mm
Height/Length of Insulation	1850	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	170	
Detail Type	Nozzle	
Detail ID	C	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	48	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	4427.3	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	

Element From Node	170	
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Detail Type	Weight
Detail ID	LPvB
Dist. from "FROM" Node / Offset dist	-1800 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	LPCPPOVH
Dist. from "FROM" Node / Offset dist	49.95 mm
Miscellaneous Weight	7132 Kg
Offset from Element Centerline	5237 mm

Element#	18/27
Element From Node	180
Element To Node	190
Element Type	Cylinder
Description	LP-shell-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7200 mm
Element Thickness	24 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	380 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	51.73 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	180
Detail Type	Platform
Detail ID	PLAT-8
Dist. from "FROM" Node / Offset dist	1849.9 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	30000 cm ²
Platform Weight	3598 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 Kg/mm
Platform Length (top head platform)	0 mm

Element From Node	180
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	0.001058 kg./cm ³

Element From Node	180
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³



Element From Node	180
Detail Type	Ring
Detail ID	Ring R8
Dist. from "FROM" Node / Offset dist	2200 mm
Inside Diameter of Ring	7248 mm
Thickness of Ring	18 mm
Outside Diameter of Ring	7848 mm
Material Name	SA-516 70 +IT
Height of Section Ring	0 mm
Using Custom Stiffener Section	No

Element#	19/27
Element From Node	190
Element To Node	200
Element Type	Cylinder
Description	LP-shell-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7200 mm
Element Thickness	24 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	380 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	51.73 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	190
Detail Type	Platform
Detail ID	PLAT-9
Dist. from "FROM" Node / Offset dist	2849.9 mm
Platform Start Angle (degrees)	270.0
Platform End Angle (degrees)	90.0
Platform Wind Area	30000 cm ²
Platform Weight	3598 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 Kg/mm
Platform Length (top head platform)	0 mm

Element From Node	190
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	0.001058 kg./cm ³

Element From Node	190
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³



Element From Node 190
 Detail Type Ring
 Detail ID Ring R7
 Dist. from "FROM" Node / Offset dist 1950 mm
 Inside Diameter of Ring 7248 mm
 Thickness of Ring 18 mm
 Outside Diameter of Ring 7752 mm
 Material Name SA-516 70 +IT
 Height of Section Ring 0 mm
 Using Custom Stiffener Section No

Element From Node 190
 Detail Type Nozzle
 Detail ID MH-3
 Dist. from "FROM" Node / Offset dist 600 mm
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 225.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 505.79 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70 +IT

Element# 20/27
 Element From Node 200
 Element To Node 210
 Element Type Cylinder
 Description LP-shell-3
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 7200 mm
 Element Thickness 22 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 22 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 380 KPa.
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 51.73 KPa.
 Design Temperature External Pressure 130 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 200
 Detail Type Liquid
 Detail ID Liquid: 40
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 100 mm
 Liquid Density 0.001058 kg./cm³

Element From Node 200
 Detail Type Insulation
 Detail ID Ins: 70
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 3000 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg./cm³

Element From Node 200
 Detail Type Ring
 Detail ID Ring R1
 Dist. from "FROM" Node / Offset dist 1450 mm
 Inside Diameter of Ring 7244 mm
 Thickness of Ring 18 mm
 Outside Diameter of Ring 7744 mm
 Material Name SA-516 70 +IT



Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	200	
Detail Type	Nozzle	
Detail ID	B	
Dist. from "FROM" Node / Offset dist	2300	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	696.67	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT
Element From Node	200	
Detail Type	Nozzle	
Detail ID	K	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	6	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	42.654	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	+IT
Element From Node	200	
Detail Type	Weight	
Detail ID	LPKPPOVH	
Dist. from "FROM" Node / Offset dist	2500.1	mm
Miscellaneous Weight	7993	Kgf
Offset from Element Centerline	4987	mm
Element From Node	200	
Detail Type	Weight	
Detail ID	LPBPPQVH	
Dist. from "FROM" Node / Offset dist	2299.9	mm
Miscellaneous Weight	30921	Kgf
Offset from Element Centerline	5241	mm

Element#	21/27
Element From Node	210
Element To Node	220
Element Type	Cylinder
Description	LP-shell-4
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7200 mm
Element Thickness	22 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	380 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	51.73 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	210
Detail Type	Platform
Detail ID	PLAT-10
Dist. from "FROM" Node / Offset dist	849.96 mm



Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	30000	cm ²
Platform Weight	3596.4	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element From Node	210	
Detail Type	Packing	
Detail ID	LPPack1	
Dist. from "FROM" Node / Offset dist	1800.1	mm
Height of Packed Section	1199.9	mm
Density	0.000166	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	210	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	210	
Detail Type	Ring	
Detail ID	Ring R5	
Dist. from "FROM" Node / Offset dist	949.97	mm
Inside Diameter of Ring	7244	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	7744	mm
Material Name	SA-516 70 +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	210	
Detail Type	Weight	
Detail ID	LPB#1btm grid	
Dist. from "FROM" Node / Offset dist	1800.1	mm
Miscellaneous Weight	4000	Kgf
Offset from Element Centerline	0	mm

Element From Node	210	
Detail Type	Weight	
Detail ID	LPB#1TA	
Dist. from "FROM" Node / Offset dist	1800.1	mm
Miscellaneous Weight	350	Kgf
Offset from Element Centerline	0	mm

Element#	22/27	
Element From Node	220	
Element To Node	230	
Element Type	Cylinder	
Description	LP-shell-5	
Distance "FROM" to "TO"	2800	mm
Inside Diameter	7200	mm
Element Thickness	20	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	20	mm



External Corrosion Allowance	0	mm
Design Internal Pressure	380	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	51.73	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	220	
Detail Type	Platform	
Detail ID	PLAT-11	
Dist. from "FROM" Node / Offset dist	1849.9	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	30000	cm ²
Platform Weight	3594.8	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element From Node	220	
Detail Type	Packing	
Detail ID	LPPack2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	2800	mm
Density	0.000166	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	220	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2800	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	220	
Detail Type	Ring	
Detail ID	Ring R2	
Dist. from "FROM" Node / Offset dist	449.98	mm
Inside Diameter of Ring	7240	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	7740	mm
Material Name	SA-516 70 +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element#	23/27	
Element From Node	230	
Element To Node	240	
Element Type	Cylinder	
Description	LP-shell-6	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7206	mm
Element Thickness	18	mm
Internal Corrosion Allowance	1	mm



Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	380	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	51.73	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	230	
Detail Type	Packing	
Detail ID	LPPack3	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	1000	mm
Density	0.000166	kg./cm³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	230	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm³
Element From Node	230	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm³
Element From Node	230	
Detail Type	Ring	
Detail ID	Ring R4	
Dist. from "FROM" Node / Offset dist	149.98	mm
Inside Diameter of Ring	7244	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	7744	mm
Material Name	SA-516 70 +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	230	
Detail Type	Ring	
Detail ID	Ring R3	
Dist. from "FROM" Node / Offset dist	2650	mm
Inside Diameter of Ring	7244	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	7744	mm
Material Name	SA-516 70 +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	230	
Detail Type	Nozzle	
Detail ID	MH-2	
Dist. from "FROM" Node / Offset dist	2100	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	225.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	504.41	Kgf
Grade of Attached Flange	GR 1.1	



Nozzle Matl	SA-516 70 +IT
Element From Node	230
Detail Type	Weight
Detail ID	LPB#1HDG
Dist. from "FROM" Node / Offset dist	999.95 mm
Miscellaneous Weight	900 Kg
Offset from Element Centerline	0 mm
Element From Node	230
Detail Type	Weight
Detail ID	LPB#1dist. de
Dist. from "FROM" Node / Offset dist	1600 mm
Miscellaneous Weight	5750 Kg
Offset from Element Centerline	0 mm
Element From Node	230
Detail Type	Weight
Detail ID	LPB#1dist.TA
Dist. from "FROM" Node / Offset dist	1600 mm
Miscellaneous Weight	350 Kg
Offset from Element Centerline	0 mm
Element From Node	230
Detail Type	Weight
Detail ID	LPShroud
Dist. from "FROM" Node / Offset dist	2800.1 mm
Miscellaneous Weight	9800 Kg
Offset from Element Centerline	0 mm
Element From Node	230
Detail Type	Weight
Detail ID	LPNozzle wt
Dist. from "FROM" Node / Offset dist	1100.1 mm
Miscellaneous Weight	850 Kg
Offset from Element Centerline	0 mm

Element#	24/27
Element From Node	240
Element To Node	250
Element Type	Cylinder
Description	LP-shell-7
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7206 mm
Element Thickness	16 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	380 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	51.73 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	240
Detail Type	Platform
Detail ID	PLAT-12
Dist. from "FROM" Node / Offset dist	49.955 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	30000 cm ²
Platform Weight	3594 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm ²



Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element From Node	240	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	240	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element From Node	240	
Detail Type	Ring	
Detail ID	Ring R6	
Dist. from "FROM" Node / Offset dist	2150	mm
Inside Diameter of Ring	7240	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	7740	mm
Material Name	SA-516 70 +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	240	
Detail Type	Nozzle	
Detail ID	MH-1	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	225.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	505.79	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	

Element From Node	240	
Detail Type	Weight	
Detail ID	LPdwncm PP dea	
Dist. from "FROM" Node / Offset dist	2750	mm
Miscellaneous Weight	80	Kgf
Offset from Element Centerline	0	mm

Element#	25/27	
Element From Node	250	
Element To Node	260	
Element Type	Cylinder	
Description	LP-shell-8	
Distance "FROM" to "TO"	2600	mm
Inside Diameter	7206	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm



Design Internal Pressure	380	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	51.73	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	250	
Detail Type	Platform	
Detail ID	PLAT-13	
Dist. from "FROM" Node / Offset dist	1500.1	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	30000	cm ²
Platform Weight	5621.6	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	54800	mm
Unit Weight of Ladder	0.037	Kgf/mm
Platform Length (top head platform)	0	mm
Element From Node	250	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2600	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	250	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	2600	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³
Element From Node	250	
Detail Type	Ring	
Detail ID	Ring R9	
Dist. from "FROM" Node / Offset dist	1650	mm
Inside Diameter of Ring	7240	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	7740	mm
Material Name	SA-516 70	+IT
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	250	
Detail Type	Nozzle	
Detail ID	H1	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	782.24	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT



Element From Node	250	
Detail Type	Nozzle	
Detail ID	H2	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	782.24	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT
Element From Node	250	
Detail Type	Tray	
Detail ID	Tray#1-3	
Dist. from "FROM" Node / Offset dist	650.05	mm
Tray Spacing	900	mm
Number of Trays	3	
Tray Weight per Unit Area	0.0067	Kgs/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg./cm ³
Support Ring and Bolting Bar Weight	210	Kgf
Element From Node	250	
Detail Type	Weight	
Detail ID	LPGallery	
Dist. from "FROM" Node / Offset dist	650.05	mm
Miscellaneous Weight	3100	Kgf
Offset from Element Centerline	0	mm
Element From Node	250	
Detail Type	Weight	
Detail ID	LPFPGdead	
Dist. from "FROM" Node / Offset dist	2450.1	mm
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	0	mm
Element From Node	250	
Detail Type	Weight	
Detail ID	LPFPGTA	
Dist. from "FROM" Node / Offset dist	2450.1	mm
Miscellaneous Weight	20	Kgf
Offset from Element Centerline	0	mm
Element From Node	250	
Detail Type	Weight	
Detail ID	LPH1PPOVH	
Dist. from "FROM" Node / Offset dist	800.05	mm
Miscellaneous Weight	29835	Kgf
Offset from Element Centerline	5241	mm
Element From Node	250	
Detail Type	Weight	
Detail ID	LPH2PPOVH	
Dist. from "FROM" Node / Offset dist	800.05	mm
Miscellaneous Weight	29835	Kgf
Offset from Element Centerline	5241	mm

Element#	26/27	
Element From Node	260	
Element To Node	270	
Element Type	Cylinder	
Description	LP-shell-9	
Distance "FROM" to "TO"	1000	mm
Inside Diameter	7206	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm



Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	380	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	51.73	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	260	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	260	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1000	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element From Node	260	
Detail Type	Weight	
Detail ID	LPDavit	
Dist. from "FROM" Node / Offset dist	0.04651	mm
Miscellaneous Weight	900	Kgf
Offset from Element Centerline	3600	mm

Element From Node	260	
Detail Type	Weight	
Detail ID	LPFPPOVH	
Dist. from "FROM" Node / Offset dist	500.05	mm
Miscellaneous Weight	35704	Kgf
Offset from Element Centerline	5317	mm

Element#	27/27	
Element From Node	270	
Element To Node	280	
Element Type	Elliptical	
Description	LP-TOP dish	
Distance "FROM" to "TO"	50	mm
Inside Diameter	7206	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	380	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	51.73	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	270	
Detail Type	Platform	
Detail ID	TOP PLT	
Dist. from "FROM" Node / Offset dist	2300.1	mm



Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	75000	cm ²
Platform Weight	9733.9	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	7500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	3750	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	315.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4500	mm
Unit Weight of Ladder	0.037	Kgf/mm
Platform Length (top head platform)	7500	mm

Element From Node	270	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1851.5	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	270	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1851.5	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element From Node	270	
Detail Type	Nozzle	
Detail ID	F	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	42	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	837.88	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	

Element From Node	270	
Detail Type	Weight	
Detail ID	LPDemister	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	960	Kgf
Offset from Element Centerline	0	mm

Element From Node	270	
Detail Type	Weight	
Detail ID	PSV	
Dist. from "FROM" Node / Offset dist	1851.5	mm
Miscellaneous Weight	7500	Kgf
Offset from Element Centerline	3750	mm

Element From Node	270	
Detail Type	Weight	
Detail ID	LPDemisterTA	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	0	mm

Element From Node	270	
Detail Type	For./Mom.	
Detail ID	FFxFz	
Dist. from "FROM" Node / Offset dist	1800	mm



Force in X Direction	5887.2	Kgf
Force in Y Direction	-8830.5	Kgf
Force in Z Direction	5887.2	Kgf
Moment about X Axis	17308	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	17308	Kg-m.
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
HP_main skirt-3	...	...	3000	...	3000
HP_main skirt-2	...	...	6000	...	3000
HP_main skirt-1	...	...	7500	...	1500
HP_BTM dish	...	...	7500	...	...
HP_shell-1	...	...	10500	...	3000
HP_shell-2	...	...	13500	...	3000
HP_shell-3	...	...	15400	...	1900
HP_shell-4	...	...	18400	...	3000
HP_shell-5	...	...	21400	...	3000
HP_shell-6	...	...	23400	...	2000
HP_shell-7	...	...	24900	...	1500
HP_TOP dish	...	...	24900	...	...
HP_int_skirt-1	...	...	26150	...	1250
HP_int_skirt-2	...	...	29150	...	3000
LP_int_skirt-2	...	...	32150	...	3000
LP_int_skirt-1	...	...	33400	...	1250
LP_BTM dish	...	...	33450	...	50
LP-shell-1	...	...	36450	...	3000
LP-shell-2	...	...	39450	...	3000
LP-shell-3	...	...	42450	...	3000
LP-shell-4	...	...	45450	...	3000
LP-shell-5	...	...	48250	...	2800
LP-shell-6	...	...	51250	...	3000
LP-shell-7	...	...	54250	...	3000
LP-shell-8	...	...	56850	...	2600
LP-shell-9	...	...	57850	...	1000
LP-TOP dish	...	...	57900	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
HP_main skirt-3	...	...	36	1.6	6874	...	1
HP_main skirt-2	...	...	36	1.6	6874	...	2
HP_main skirt-1	...	...	38	1.6	6870	...	3
HP_BTM dish	4207.9	68	6	6800	156.4	4	
HP_shell-1	4172.6	102	6	6800	156.4	5	
HP_shell-2	4141.4	102	6	6800	156.4	6	
HP_shell-3	4110.2	102	6	6800	156.4	7	
HP_shell-4	4104	96	1	6806	156.4	8	
HP_shell-5	4104	94	1	6806	156.4	9	
HP_shell-6	4104	94	1	6806	156.4	10	
HP_shell-7	4104	94	1	6806	156.4	11	
HP_TOP dish	4104	58	1	6806	156.4	12	
HP_int_skirt-1	...	24	1.6	6876	...	13	
HP_int_skirt-2	...	24	1.6	7039.9	...	14	
LP_int_skirt-2	...	24	1.6	7204	...	15	
LP_int_skirt-1	...	24	1.6	7204	...	16	
LP_BTM dish	466.6	34	6	7200	156.4	17	
LP-shell-1	447.38	24	6	7200	156.4	18	
LP-shell-2	416.21	24	6	7200	156.4	19	
LP-shell-3	385.04	22	6	7200	156.4	20	
LP-shell-4	384	22	6	7200	156.4	21	
LP-shell-5	384	20	6	7200	156.4	22	
LP-shell-6	384	18	1	7206	132.94	23	
LP-shell-7	384	16	1	7206	132.94	24	
LP-shell-8	384	16	1	7206	117.3	25	
LP-shell-9	384	16	1	7206	132.94	26	
LP-TOP dish	384	18	1	7206	132.94	27	

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
HP_main skirt-3	...		No Calc	No Calc	36	No Calc	1
HP_main skirt-2	...		No Calc	No Calc	36	No Calc	2
HP_main skirt-1	...		No Calc	No Calc	38	No Calc	3
HP_BTM dish	4100	4721.74	6021.08	59	52.1311		4
HP_shell-1	4100	4274.4	5172.49	102	98.0966		5
HP_shell-2	4100	4305.57	5172.49	102	97.3995		6
HP_shell-3	4100	4336.74	5172.49	102	96.7023		7
HP_shell-4	4100	4300.81	4868.19	96	91.5099		8
HP_shell-5	4100	4211.41	4768.14	94	91.5099		9
HP_shell-6	4100	4211.41	4768.14	94	91.5099		10
HP_shell-7	4100	4211.41	4768.14	94	91.5099		11
HP_TOP dish	4100	4520.68	5165.61	50.6	45.9581		12
HP_int_skirt-1	...	No Calc	No Calc	No Calc	24	No Calc	13
HP_int_skirt-2	...	No Calc	No Calc	No Calc	24	No Calc	14
LP_int_skirt-2	...	No Calc	No Calc	No Calc	24	No Calc	15
LP_int_skirt-1	...	No Calc	No Calc	No Calc	24	No Calc	16
LP_BTM dish	380	1068.64	1616.53	30	16.8891		17
LP-shell-1	380	711.327	1162.73	24	16.3302		18
LP-shell-2	380	742.497	1162.73	24	15.6097		19
LP-shell-3	380	687.346	1066.11	22	14.8888		20
LP-shell-4	380	688.385	1066.11	22	14.8652		21
LP-shell-5	380	601.998	969.467	20	14.8652		22
LP-shell-6	380	621.566	741.236	18	11.4258		23
LP-shell-7	380	548.11	659.066	16	11.4258		24
LP-shell-8	380	483.156	519.721	16	12.8183		25
LP-shell-9	380	548.11	659.066	16	11.4258		26



LP-TOP dish	380	610.11	733.01	16	10.9329	27
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Minimum 483 520

Warning: The flange MAWP was less than the design pressure!

MAWP: 483 KPa., limited by: LP-shell-8.

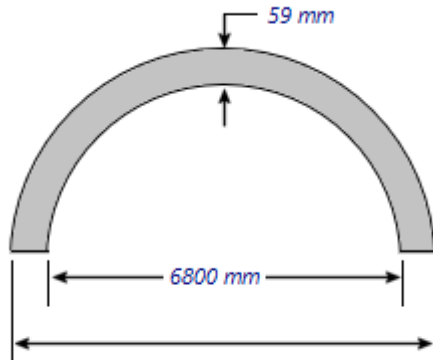
Warning: The computed MAWP is less than the design pressure.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPBTM dish

Spherical Head Design Information:					
Design Pressure	P	4207.901 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6800.000 mm	Actual Stress		134.2 N./mm ²
Spec. Min. Thk t _e , or s _a		59.0000 mm	Spec. Nominal Thk.		68.000 mm
Surface Area		755680.4 cm ²	SG of Contents		1.060
Element Volume		82332864.0 cm ³	Weight of Contents		87219.3 Kgf
Empty Weight		39049.2 Kgf	Operating Weight		126268.5 Kgf
			Comp. All. Ope. Stress		102.8 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 85.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 42.500 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 6812.0/2 \cdot (\exp(0.5 \cdot 4207.901/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 46.1311 + 6.0 + 0.0 = 52.1311 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 156.4 \cdot 1.0 \cdot \ln((2 \cdot 53.0 + 6812.0)/6812.0) - 107.901$
 $= 4829.639 - 107.901 = 4721.7378 \text{ KPa.}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 59.0 + 6800.0)/6800.0)$
 $= 6021.0845 \text{ KPa.}$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 4207.901 \cdot 6812.0^2 / (1.0 (6918.0^2 - 6812.0^2))$$

$$= 134.173 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 4207.901 / (1.0 \cdot \ln((2 \cdot 53.0 + 6812.0) / 6812.0))$$

$$= 136.266 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	1.485 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	36.638 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 755680 / 3.14159)$$

$$= 9808.985 \text{ mm}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

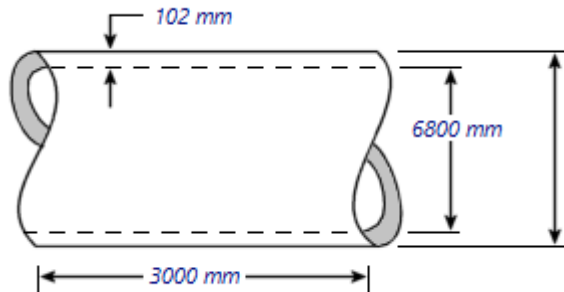
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 85.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 42.500 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPshell-1

Cylindrical Shell Design Information:					
Design Pressure	P	4172.575 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6800.000 mm	Actual Stress	148.0 N./mm ²	
Spec. Min. Thk t _e , or s _a		102.0000 mm	Spec. Nominal Thk.	102.000 mm	
Surface Area		660111.6 cm ²	SG of Contents	1.060	
Element Volume		108969968.0 cm ³	Weight of Contents	115437.3 Kg _f	
Empty Weight		51422.5 Kg _f	Operating Weight	166859.8 Kg _f	
			Comp. All. Ope. Stress	114.2 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6812.0 \cdot (\exp(4172.575/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 92.0966 + 6.0 + 0.0 = 98.0966 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 96.0 + 6812.0) / 6812.0) - 72.575$$

$$= 4346.977 - 72.575 = 4274.4019 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 102.0 + 6800.0) / 6800.0)$$

$$= 5172.4854 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4172.575 \cdot 6812.0 / (1.0 (7004.0 - 6812.0))$$

$$= 148.0482 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4172.575 / (1.0 \cdot \ln(2 \cdot 96.0 + 6812.0) / 6812.0)$$
$$= 150.1252 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.478 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

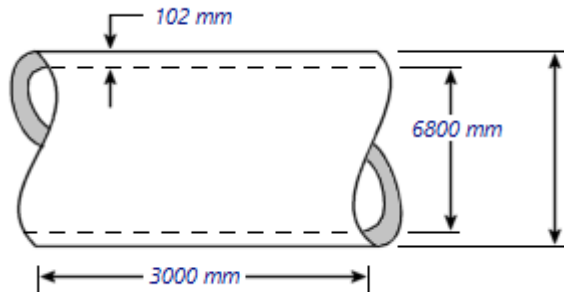
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPshell-2

Cylindrical Shell Design Information:					
Design Pressure	P	4141.404 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6800.000 mm	Actual Stress		146.9 N./mm ²
Spec. Min. Thk t _e , or s _a		102.0000 mm	Spec. Nominal Thk.		102.000 mm
Surface Area		660111.6 cm ²	SG of Contents		1.060
Element Volume		108969968.0 cm ³	Weight of Contents		115437.3 Kg
Empty Weight		51422.5 Kg	Operating Weight		166859.8 Kg
			Comp. All. Ope. Stress		114.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6812.0 \cdot (\exp(4141.404/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 91.3995 + 6.0 + 0.0 = 97.3995 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 96.0 + 6812.0) / 6812.0) - 41.404$$

$$= 4346.977 - 41.404 = 4305.5723 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 102.0 + 6800.0) / 6800.0)$$

$$= 5172.4854 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 4141.404 \cdot 6812.0 / (1.0(7004.0 - 6812.0))$$

$$= 146.9422 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4141.404 / (1.0 \cdot \ln((2 \cdot 96.0 + 6812.0) / 6812.0))$$

$$= 149.0037 \text{ N./mm}^2$$



Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.478 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -22 °C

User entered Minimum Design Temperature : -12 °C

The Curve for this Material is : D

From Figure 3.8M- with PWHT, the MDMT is : -15 °C

Temperature reduction per Graph 3.13 : 8 °C

The computed stress ratio is : 0.850

The computed thickness ratio [sr]:

$= (tr \cdot E^*) / (t_{nom} - ci - c_{ext})$

$= (81.57 \cdot 1.0) / (102.0 - 6.0 - 0.0)$

= 0.850

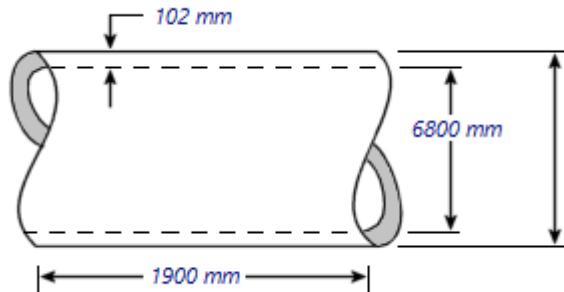
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	4141.404 KPa.
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	102.000 mm	Basic MDMT	-15 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPshell-3

Cylindrical Shell Design Information:					
Design Pressure	P	4110.234 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1900.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6800.000 mm	Actual Stress		145.8 N./mm ²
Spec. Min. Thk t _e , or s _a		102.0000 mm	Spec. Nominal Thk.		102.000 mm
Surface Area		418070.6 cm ²	SG of Contents		1.060
Element Volume		69014312.0 cm ³	Weight of Contents		23087.5 Kg
Empty Weight		32567.6 Kg	Operating Weight		55655.0 Kg
			Comp. All. Ope. Stress		114.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
	100 %		100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6812.0 \cdot (\exp(4110.234/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 90.7023 + 6.0 + 0.0 = 96.7023 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 96.0 + 6812.0) / 6812.0) - 10.234$$

$$= 4346.977 - 10.234 = 4336.7427 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 102.0 + 6800.0) / 6800.0)$$

$$= 5172.4854 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 4110.234 \cdot 6812.0 / (1.0(7004.0 - 6812.0))$$

$$= 145.8362 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4110.234 / (1.0 \cdot \ln((2 \cdot 96.0 + 6812.0) / 6812.0))$$

$$= 147.8822 \text{ N./mm}^2$$



Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.478 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -23 °C

User entered Minimum Design Temperature : -12 °C

The Curve for this Material is : D

From Figure 3.8M- with PWHT, the MDMT is : -15 °C

Temperature reduction per Graph 3.13 : 8 °C

The computed stress ratio is : 0.843

The computed thickness ratio [sr]:

$= (tr \cdot E^*) / (t_{nom} - ci - c_{ext})$

$= (80.948 \cdot 1.0) / (102.0 - 6.0 - 0.0)$

= 0.843

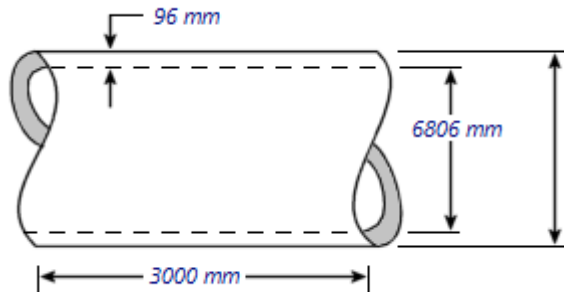
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	4110.234 KPa.
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	102.000 mm	Basic MDMT	-15 °C
Stress/Thk. Ratio	0.843	Temperature Difference	8 °C
		Computed Min. Temp.	-23 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPshell-4

Cylindrical Shell Design Information:					
Design Pressure	P	4104.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6806.000 mm	Actual Stress		147.1 N./mm ²
Spec. Min. Thk t _e , or s _a		96.0000 mm	Spec. Nominal Thk.		96.000 mm
Surface Area		659545.9 cm ²	SG of Contents		0.000
Element Volume		109162336.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		48397.7 Kgf	Operating Weight		48397.7 Kgf
			Comp. All. Ope. Stress		114.0 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6808.0 \cdot (\exp(4104.0/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 90.5099 + 1.0 + 0.0 = 91.5099 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 95.0 + 6808.0) / 6808.0) - 4.0$$

$$= 4304.814 - 4.0 = 4300.8145 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 96.0 + 6806.0) / 6806.0)$$

$$= 4868.1855 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4104.0 \cdot 6808.0 / (1.0 (6998.0 - 6808.0))$$

$$= 147.0613 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4104.0 / (1.0 \cdot \ln((2 \cdot 95.0 + 6808.0) / 6808.0))$$

$$= 149.1041 \text{ N./mm}^2$$



Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.391 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -23 °C

User entered Minimum Design Temperature : -12 °C

The Curve for this Material is : D

From Figure 3.8M- with PWHT, the MDMT is : -15 °C

Temperature reduction per Graph 3.13 : 8 °C

The computed stress ratio is : 0.850

The computed thickness ratio [sr]:

$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$

$= (80.777 \cdot 1.0) / (96.0 - 1.0 - 0.0)$

= 0.850

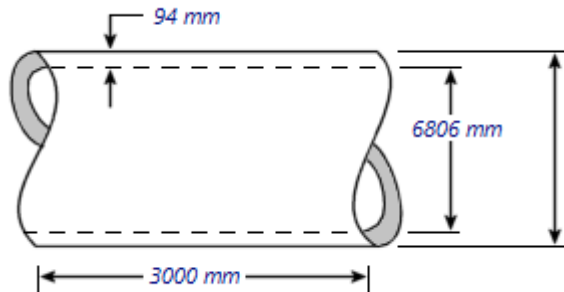
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	4104.000 KPa.
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	96.000 mm	Basic MDMT	-15 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
		Computed Min. Temp.	-23 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPshell-5

Cylindrical Shell Design Information:					
Design Pressure	P	4104.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6806.000 mm	Actual Stress		150.2 N./mm ²
Spec. Min. Thk t _e , or s _a		94.0000 mm	Spec. Nominal Thk.		94.000 mm
Surface Area		659169.1 cm ²	SG of Contents		0.000
Element Volume		109162336.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		47375.8 Kgf	Operating Weight		47375.8 Kgf
			Comp. All. Ope. Stress		113.7 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6808.0 \cdot (\exp(4104.0/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 90.5099 + 1.0 + 0.0 = 91.5099 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 93.0 + 6808.0) / 6808.0) - 4.0$$

$$= 4215.409 - 4.0 = 4211.4087 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 94.0 + 6806.0) / 6806.0)$$

$$= 4768.1362 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4104.0 \cdot 6808.0 / (1.0 (6994.0 - 6808.0))$$

$$= 150.2237 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4104.0 / (1.0 \cdot \ln((2 \cdot 93.0 + 6808.0) / 6808.0))$$

$$= 152.2665 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.362 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -15 °C
 Temperature reduction per Graph 3.13 : 7 °C
 The computed stress ratio is : 0.869

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (80.777 \cdot 1.0) / (94.0 - 1.0 - 0.0)$$

$$= 0.869$$

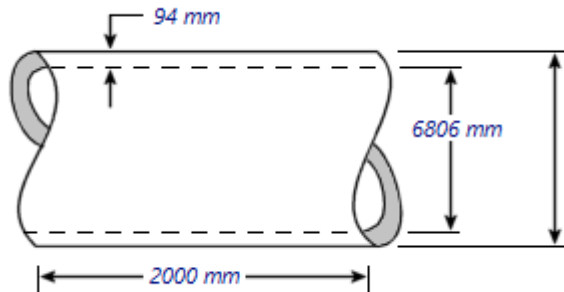
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	4104.000 KPa.
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	94.000 mm	Basic MDMT	-15 °C
Stress/Thk. Ratio	0.869	Temperature Difference	7 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPshell-6

Cylindrical Shell Design Information:					
Design Pressure	P	4104.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		2000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6806.000 mm	Actual Stress	150.2 N./mm ²	
Spec. Min. Thk t _e , or s _a		94.0000 mm	Spec. Nominal Thk.	94.000 mm	
Surface Area		439446.0 cm ²	SG of Contents	0.000	
Element Volume		72774888.0 cm ³	Weight of Contents	0.0 Kg	
Empty Weight		31583.9 Kg	Operating Weight	31583.9 Kg	
			Comp. All. Ope. Stress	113.7 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6808.0 \cdot (\exp(4104.0/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 90.5099 + 1.0 + 0.0 = 91.5099 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 93.0 + 6808.0) / 6808.0) - 4.0$$

$$= 4215.409 - 4.0 = 4211.4087 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 94.0 + 6806.0) / 6806.0)$$

$$= 4768.1362 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4104.0 \cdot 6808.0 / (1.0 (6994.0 - 6808.0))$$

$$= 150.2237 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4104.0 / (1.0 \cdot \ln((2 \cdot 93.0 + 6808.0) / 6808.0))$$
$$= 152.2665 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.362 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

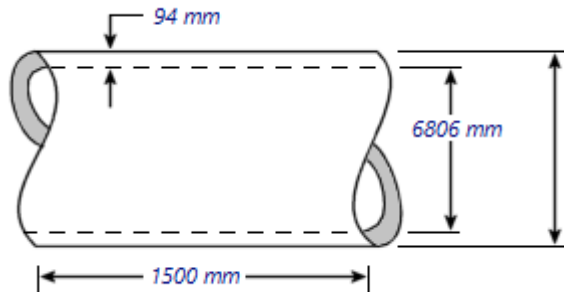
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPshell-7

Cylindrical Shell Design Information:					
Design Pressure	P	4104.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6806.000 mm	Actual Stress		150.2 N./mm ²
Spec. Min. Thk t _e , or s _a		94.0000 mm	Spec. Nominal Thk.		94.000 mm
Surface Area		329584.5 cm ²	SG of Contents		0.000
Element Volume		54581168.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		23687.9 Kgf	Operating Weight		23687.9 Kgf
			Comp. All. Ope. Stress		113.7 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6808.0 \cdot (\exp(4104.0/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 90.5099 + 1.0 + 0.0 = 91.5099 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 93.0 + 6808.0) / 6808.0) - 4.0$$

$$= 4215.409 - 4.0 = 4211.4087 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 94.0 + 6806.0) / 6806.0)$$

$$= 4768.1362 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4104.0 \cdot 6808.0 / (1.0 (6994.0 - 6808.0))$$

$$= 150.2237 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4104.0 / (1.0 \cdot \ln((2 \cdot 93.0 + 6808.0) / 6808.0))$$

$$= 152.2665 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.362 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -22 °C

User entered Minimum Design Temperature : -12 °C

The Curve for this Material is : D

From Figure 3.8M- with PWHT, the MDMT is : -15 °C

Temperature reduction per Graph 3.13 : 7 °C

The computed stress ratio is : 0.869

The computed thickness ratio [sr]:

$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$

$= (80.777 \cdot 1.0) / (94.0 - 1.0 - 0.0)$

= 0.869

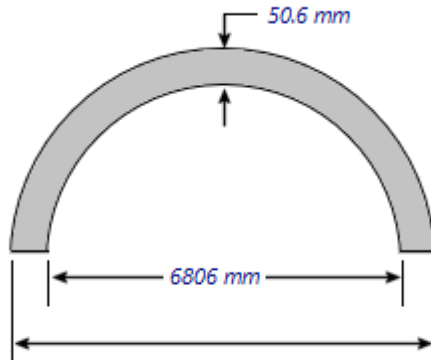
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	4104.000 KPa.
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	94.000 mm	Basic MDMT	-15 °C
Stress/Thk. Ratio	0.869	Temperature Difference	7 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: HPTOP dish

Spherical Head Design Information:					
Design Pressure	P	4104.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	6806.000 mm	Actual Stress		139.8 N./mm ²
Spec. Min. Thk t _e , or sa		50.6000 mm	Spec. Nominal Thk.		58.000 mm
Surface Area		752632.7 cm ²	SG of Contents		0.000
Element Volume		82550984.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		33267.5 Kgf	Operating Weight		33267.5 Kgf
			Comp. All. Ope. Stress		101.3 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 85.075 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 42.538 mm

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2(\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 6808.0/2(\exp(0.5 \cdot 4104.0/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$
 $= 44.9581 + 1.0 + 0.0 = 45.9581 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 156.4 \cdot 1.0 \cdot \ln((2 \cdot 49.6 + 6808.0) / 6808.0) - 4.0$
 $= 4524.681 - 4.0 = 4520.6812 \text{ KPa.}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 50.6 + 6806.0) / 6806.0)$
 $= 5165.6128 \text{ KPa.}$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 4104.0 \cdot 6808.0^2 / (1.0 (6907.2^2 - 6808.0^2))$$

$$= 139.816 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 4104.0 / (1.0 \cdot \ln((2 \cdot 49.6 + 6808.0) / 6808.0))$$

$$= 141.859 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.267 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 36.347 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:
 $= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$
 $= \text{square root}(4 \cdot 752633 / 3.14159)$
 $= 9789.187 \text{ mm}$

MDMT Calculations in the Spherical Portion:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -37 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -26 °C
 Temperature reduction per Graph 3.13 : 10 °C
 The computed stress ratio is : 0.810

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (40.152 \cdot 1.0) / (50.6 - 1.0 - 0.0)$
 $= 0.810$

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	4104.000 KPa.
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	50.600 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.810	Temperature Difference	10 °C
		Computed Min. Temp.	-37 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

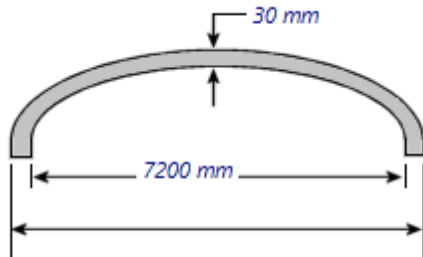
Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 85.075 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 42.538 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LPBTМ dish

Elliptical Head Design Information:					
Design Pressure	P	466.601 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress		63.2 N./mm ²
Spec. Min. Thk t,e, or sa		30.0000 mm	Spec. Nominal Thk.		34.000 mm
Surface Area		584019.1 cm ²	SG of Contents		1.060
Element Volume		50902920.0 cm ³	Weight of Contents		53924.0 Kg _f
Empty Weight		16803.2 Kg _f	Operating Weight		70727.2 Kg _f
			Comp. All. Ope. Stress		79.2 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 90.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 45.000 mm

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 tr = 10.8891 mm

$$\begin{aligned}
 t &= tr + ci + co \\
 &= 10.8891 + 6.0 + 0.0 \\
 &= 16.8891 \text{ mm - see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r.] per 4.3.7.1

$$\begin{aligned}
 h &= D/(2 \cdot \text{Aspect Ratio}) \\
 &= 7200.0/(2 \cdot 2.0) \\
 &= 1800.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 k &= D/(2 \cdot h) \\
 &= 7200.0/(2 \cdot 1800.0) \\
 &= 2.000
 \end{aligned}$$

$$\begin{aligned}
 r &= D(0.5/k - 0.08) + ca \\
 &= 7200.0(0.5/2.0 - 0.08) + 6.0 \\
 &= 1230.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned} L &= D(0.44 \cdot k + 0.02) + ca \\ &= 7200.0(0.44 \cdot 2.0 + 0.02) + 6.0 \\ &= 6486.000 \text{ mm} \end{aligned}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:
 Equations 4.3.8 to 4.3.11 as appropriate
 t (corroded) and P for the calc.: $t = 10.8891 \text{ mm}$, $P = 466.6014 \text{ KPa}$.

Angle [βTh]:

$$\begin{aligned} &= \arccos[(0.5 \cdot D - r)/(L - r)] \\ &= \arccos[(0.5 \cdot 7212.0 - 1230.0)/(6486.0 - 1230.0)] \\ &= 1.1017 \text{ radians - Equation 4.3.8} \end{aligned}$$

$$\begin{aligned} \phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(6486.0 \cdot 10.889) / 1230.0} \\ &= 0.2161 \text{ radians - Equation 4.3.9} \end{aligned}$$

$\phi Th < \beta Th$, Thus:

$$\begin{aligned} Rth &= (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r \\ &= (0.5 \cdot 7212.0 - 1230.0) / \cos(1.102 - 0.216) + 1230.0 \\ &= 4984.897 \text{ mm - Equation 4.3.10} \end{aligned}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$r/D > 0.08$, Thus:

$$\begin{aligned} C1 &= 0.692(r/D) + 0.605 \\ &= 0.692(1230.0/7212.0) + 0.605 \\ &= 0.723 \text{ - Equation 4.3.13} \end{aligned}$$

$$\begin{aligned} C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(1230.0/7212.0) \\ &= 1.0166, \text{ Equation 4.3.15} \end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [P_{eth}]:

$$\begin{aligned} &= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r)) \\ &= 0.723 \cdot 196301.391 \cdot 10.889^2 / (1.017 \cdot 4984.897(0.5 \cdot 4984.897 - 1230.0)) \\ &= 2630.41 \text{ KPa., Equation 4.3.16} \end{aligned}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [P_y]:

$C3 = S_y = 234.4 \text{ N/mm}^2$, Not in the creep range - 4.3.6.1(f)

$$\begin{aligned} P_y &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 234.4 \cdot 10.889 / (1.017 \cdot 4984.897(0.5 \cdot 4984.897 / 1230.0 - 1)) \\ &= 490.705 \text{ KPa., Equation 4.3.17} \end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and P_{ck}]:

$$G = P_{eth}/P_y = 2630.405/490.705 = 5.36$$

$G > 1$, Thus:

Buckling Pressure [P_{ck}]:

$$\begin{aligned} &= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) P_y \\ &= ((0.77508 \cdot 5.36 - 0.20354 \cdot 5.36^2 + 0.019274 \cdot 5.36^3) / \\ &\quad (1 + 0.19014 \cdot 5.36 - 0.089534 \cdot 5.36^2 + 0.0093965 \cdot 5.36^3)) \cdot 490.705 \\ &= 699.9 \text{ KPa., Equation 4.3.19} \end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [P_{ak}]:

$$= P_{ck}/1.5 = 699.9/1.5 = 466.6 \text{ KPa. - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [P_{ac}]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 156.4 \cdot 1.0 / (6486.0/10.889 + 0.5) \\ &= 524.68 \text{ KPa., Equation 4.3.22} \end{aligned}$$

Step 10: Compute final pressure [P_a]:

$$\begin{aligned} &= \min(P_{ak}, P_{ac}) = \min(466.602, 524.676) \\ &= 466.6 \text{ KPa., Equation 4.3.23: (same as original design pressure)} \end{aligned}$$

Allowable pressure in Crown at given thickness [P_{ac}]:

$$= 2 \cdot S \cdot E / (L / t + 0.5)$$

$$= 2 \cdot 156.4 \cdot 1.0 / (6486.0 / 24.0 + 0.5)$$

$$= 1155.242 \text{ KPa.}$$

Computed Maximum Allowable Pressure [MAWP]:
 = MaxPressure - Pliq = 1155.2422 - 86.601 = 1068.6407 KPa.

Computed Maximum Allowable Pressure New and Cold [MAPnc]:
 = 1616.5344 KPa.

Computed Membrane Stress Intensity at Design Pressure [Sm]:
 = 63.2 N./mm²

Required Thickness of Straight Flange = 16.775 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.055 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 18.038 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:
 = square root($4 \cdot \text{Head Surface Area} / 3.14159$)
 = square root($4 \cdot 584019 / 3.14159$)
 = 8623.202 mm

MDMT Calculations in the Knuckle Portion:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -38 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -18 °C
 Temperature reduction per Graph 3.12 : 20 °C
 The computed stress ratio is : 0.498

The computed thickness ratio [sr]:
 = $(tr \cdot E^*) / (t_{nom} - ci - c_{ext})$
 = $(11.951 \cdot 1.0) / (30.0 - 6.0 - 0.0)$
 = 0.498

MDMT Calculations in the Head Straight Flange:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -39 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -16 °C
 Temperature reduction per Graph 3.12 : 23 °C
 The computed stress ratio is : 0.420

The computed thickness ratio [sr]:
 = $(tr \cdot E^*) / (t_{nom} - ci - c_{ext})$
 = $(11.76 \cdot 1.0) / (34.0 - 6.0 - 0.0)$
 = 0.420

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	569.757 KPa.
Governing Thk.	34.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.420	Basic MDMT	-16 °C
		Temperature Difference	23 °C
		Computed Min. Temp.	-39 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 90.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 45.000 mm

EXTERNAL PRESSURE RESULTS for: LPBTМ dish

Elliptical Head From 170 to 180 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 0.19629E+09 (24.0/6484.649)$$

$$= 54.489 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 54.5 / 234.4$$

$$= 0.232$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 54.5/1.9629E+09$$

$$= 0.000278$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.522E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.320E-04
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.124E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.467E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.769E-05
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.120E-04
8	54	-6.189	0.110E-10	0.110E-10	0.290E-08	-0.290E-08	0.195E+09	0.215E-05
9	53	-6.211	0.100E-10	0.100E-10	0.279E-08	-0.279E-08	0.196E+09	0.278E-05
10	54	-6.200	0.105E-10	0.105E-10	0.284E-08	-0.284E-08	0.195E+09	0.314E-06
11	54	-6.194	0.107E-10	0.107E-10	0.287E-08	-0.287E-08	0.195E+09	0.918E-06



12	54	-6.197	0.106E-10	0.106E-10	0.286E-08	-0.286E-08	0.195E+09	0.302E-06
13	54	-6.199	0.105E-10	0.105E-10	0.285E-08	-0.285E-08	0.195E+09	0.605E-08

Factor of Safety [FS]:

Since $Fic [54.26] < 0.55 * 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS
 = $54.263/2.0$
 = 27.131 N./mm^2

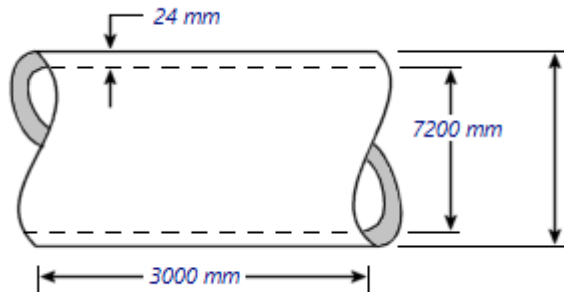
Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha[t/Ro]$
 = $2 \cdot 27.131[24.0/6484.649]$
 = 200.817 KPa.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LP-shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	447.380 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress	89.6 N./mm ²	
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		683107.9 cm ²	SG of Contents	1.060	
Element Volume		122167016.0 cm ³	Weight of Contents	129417.5 Kg	
Empty Weight		12663.9 Kg	Operating Weight	142081.5 Kg	
			Comp. All. Ope. Stress	62.1 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(447.38/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 10.3302 + 6.0 + 0.0 = 16.3302 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 7212.0) / 7212.0) - 67.38$$

$$= 778.707 - 67.38 = 711.3271 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 24.0 + 7200.0) / 7200.0)$$

$$= 1162.7263 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 447.38 \cdot 7212.0 / (1.0 (7248.0 - 7212.0))$$

$$= 89.6300 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 447.38 / (1.0 \cdot \ln((2 \cdot 18.0 + 7212.0) / 7212.0))$$

$$= 89.8543 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.332 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -37 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -23 °C
 Temperature reduction per Graph 3.12 : 15 °C
 The computed stress ratio is : 0.631

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (11.363 \cdot 1.0) / (24.0 - 6.0 - 0.0)$
 $= 0.631$

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	550.535 KPa.
Governing Thk.	24.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.631	Basic MDMT	-23 °C
		Temperature Difference	15 °C
		Computed Min. Temp.	-37 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: LP-shell-1

Cylindrical Shell From 180 to Ring R8 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 2850.0 / \sqrt{3624.0 \cdot 18.0}$
 $= 11.159$

Compute the Factor [Ch]:
 Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$
 $= 0.92 / (11.159 - 0.579)$
 $= 0.0870$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.087 \cdot 0.19629E+09(18.0/7248.0)$
 $= 67.828 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 67.8/234.4$
 $= 0.289$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4/482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 67.8/19629E+09$
 $= 0.000346$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.454E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.359E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.145E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.479E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.498E-05
7	64	-5.969	0.258E-10	0.258E-10	0.414E-08	-0.414E-08	0.194E+09	0.157E-04
8	65	-5.925	0.301E-10	0.301E-10	0.442E-08	-0.442E-08	0.194E+09	0.540E-05
9	66	-5.903	0.325E-10	0.325E-10	0.456E-08	-0.456E-08	0.194E+09	0.224E-06
10	67	-5.892	0.337E-10	0.337E-10	0.463E-08	-0.463E-08	0.194E+09	0.238E-05
11	67	-5.898	0.331E-10	0.331E-10	0.459E-08	-0.459E-08	0.194E+09	0.108E-05
12	67	-5.901	0.328E-10	0.328E-10	0.458E-08	-0.458E-08	0.194E+09	0.425E-06
13	66	-5.902	0.326E-10	0.326E-10	0.457E-08	-0.457E-08	0.194E+09	0.101E-06
14	66	-5.903	0.325E-10	0.325E-10	0.456E-08	-0.456E-08	0.194E+09	0.615E-07
15	66	-5.902	0.326E-10	0.326E-10	0.457E-08	-0.457E-08	0.194E+09	0.196E-07
16	66	-5.902	0.325E-10	0.325E-10	0.456E-08	-0.456E-08	0.194E+09	0.209E-07
17	66	-5.902	0.326E-10	0.326E-10	0.456E-08	-0.456E-08	0.194E+09	0.658E-09

Factor of Safety [FS]:
 Since $Fic [66.97] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 66.97 / 2.0$$

$$= 33.485 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 33483.301 (18.0 / 7248.0)$$

$$= 166.308 \text{ KPa.}$$

Cylindrical Shell From Ring R8 to 190 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2750.0 / \sqrt{(3624.0 \cdot 18.0)}$$

$$= 10.767$$

Compute the Factor [Ch]:

$$\begin{aligned} & \text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579) \\ & = 0.92 / (10.767 - 0.579) \\ & = 0.0903 \end{aligned}$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} & = 1.6 \cdot Ch \cdot E_y(t / D_o) \\ & = 1.6 \cdot 0.09 \cdot 0.19629E+09(18.0/7248.0) \\ & = 70.435 \text{ N./mm}^2 \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} & = Fhe / F_y \\ & = 70.4/234.4 \\ & = 0.300 \end{aligned}$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} & = \sigma_{ys} / \sigma_{uts} \\ & = 234.4/482.6 \\ & = 0.486 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} & = Fhe / E \\ & = 70.4/0.19629E+09 \\ & = 0.000359 \end{aligned}$$

Factor [m1]:

$$\begin{aligned} & = (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys})) \\ & = (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002)) \\ & = 0.157 \end{aligned}$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$\begin{aligned} & = \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1 \\ & = 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301 \\ & = 624.374 \text{ N./mm}^2 \end{aligned}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$\begin{aligned} & = m_2 f(m_2 f_2 - R) \\ & = 0.6(1.0 - 0.486) \\ & = 0.309 \end{aligned}$$

Curve Fitting Constant [A2]:

$$\begin{aligned} & = (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2) \\ & = (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309) \\ & = 944.566 \text{ N./mm}^2 \end{aligned}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$\begin{aligned} & = 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5} \\ & = 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5} \\ & = 0.346 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.441E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.492E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.132E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.347E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.830E-05
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.129E-04
8	69	-5.837	0.405E-10	0.405E-10	0.500E-08	-0.500E-08	0.193E+09	0.221E-05
9	68	-5.859	0.377E-10	0.377E-10	0.485E-08	-0.485E-08	0.193E+09	0.306E-05
10	69	-5.848	0.391E-10	0.391E-10	0.493E-08	-0.493E-08	0.193E+09	0.426E-06
11	69	-5.843	0.398E-10	0.398E-10	0.496E-08	-0.496E-08	0.193E+09	0.893E-06
12	69	-5.846	0.394E-10	0.394E-10	0.494E-08	-0.494E-08	0.193E+09	0.233E-06



13	69	-5.847	0.392E-10	0.392E-10	0.494E-08	-0.494E-08	0.193E+09	0.963E-07
14	69	-5.846	0.393E-10	0.393E-10	0.494E-08	-0.494E-08	0.193E+09	0.685E-07
15	69	-5.847	0.393E-10	0.393E-10	0.494E-08	-0.494E-08	0.193E+09	0.139E-07
16	69	-5.846	0.393E-10	0.393E-10	0.494E-08	-0.494E-08	0.193E+09	0.273E-07
17	69	-5.847	0.393E-10	0.393E-10	0.494E-08	-0.494E-08	0.193E+09	0.673E-08

Factor of Safety [FS]:
 Since $Fic [69.36] < 0.55 * 234.4$, FS = 2

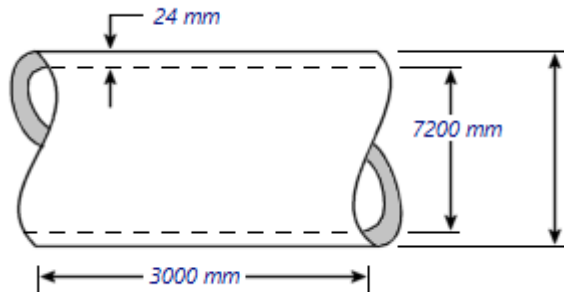
Allowable Circumferential Compressive Stress [Fha]:
 = Fic/FS
 = $69.364/2.0$
 = 34.682 N./mm²

Allowable External Pressure at the given Thickness [Pa]:
 = $2 * Fha(t / Do)$
 = $2 * 34679.984(18.0/7248.0)$
 = 172.252 KPa.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LP-shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	416.209 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress	83.4 N./mm ²	
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		683107.9 cm ²	SG of Contents	1.060	
Element Volume		122167016.0 cm ³	Weight of Contents	129417.5 Kg	
Empty Weight		12663.9 Kg	Operating Weight	142081.5 Kg	
			Comp. All. Ope. Stress	62.1 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(416.209/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 9.6097 + 6.0 + 0.0 = 15.6097 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 7212.0) / 7212.0) - 36.209$$

$$= 778.707 - 36.209 = 742.4975 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 24.0 + 7200.0) / 7200.0)$$

$$= 1162.7263 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 416.209 \cdot 7212.0 / (1.0 (7248.0 - 7212.0))$$

$$= 83.3851 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 416.209 / (1.0 \cdot \ln((2 \cdot 18.0 + 7212.0) / 7212.0))$$

$$= 83.5939 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.332 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -39 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -23 °C
 Temperature reduction per Graph 3.12 : 16 °C
 The computed stress ratio is : 0.595

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (10.718 \cdot 1.0) / (24.0 - 6.0 - 0.0)$
 $= 0.595$

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	519.365 KPa.
Governing Thk.	24.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.595	Basic MDMT	-23 °C
		Temperature Difference	16 °C
		Computed Min. Temp.	-39 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: LP-shell-2

Cylindrical Shell From 190 to Ring R7 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 2750.0 / \sqrt{3624.0 \cdot 18.0}$
 $= 10.767$

Compute the Factor [Ch]:
 Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$
 $= 0.92 / (10.767 - 0.579)$
 $= 0.0903$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.09 \cdot 0.19629E+09(18.0/7248.0)$
 $= 70.435 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 70.4/234.4$
 $= 0.300$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4/482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 70.4/.19629E+09$
 $= 0.000359$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.441E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.492E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.132E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.347E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.830E-05
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.129E-04
8	69	-5.837	0.405E-10	0.405E-10	0.500E-08	-0.500E-08	0.193E+09	0.221E-05
9	68	-5.859	0.377E-10	0.377E-10	0.485E-08	-0.485E-08	0.193E+09	0.306E-05
10	69	-5.848	0.391E-10	0.391E-10	0.493E-08	-0.493E-08	0.193E+09	0.426E-06
11	69	-5.843	0.398E-10	0.398E-10	0.496E-08	-0.496E-08	0.193E+09	0.893E-06
12	69	-5.846	0.394E-10	0.394E-10	0.494E-08	-0.494E-08	0.193E+09	0.233E-06
13	69	-5.847	0.392E-10	0.392E-10	0.494E-08	-0.494E-08	0.193E+09	0.963E-07
14	69	-5.846	0.393E-10	0.393E-10	0.494E-08	-0.494E-08	0.193E+09	0.685E-07
15	69	-5.847	0.393E-10	0.393E-10	0.494E-08	-0.494E-08	0.193E+09	0.139E-07
16	69	-5.846	0.393E-10	0.393E-10	0.494E-08	-0.494E-08	0.193E+09	0.273E-07
17	69	-5.847	0.393E-10	0.393E-10	0.494E-08	-0.494E-08	0.193E+09	0.673E-08

Factor of Safety [FS]:
 Since $Fic [69.36] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 69.364 / 2.0$$

$$= 34.682 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 34679.984 (18.0 / 7248.0)$$

$$= 172.252 \text{ KPa.}$$

Cylindrical Shell From Ring R7 to 200 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2500.01 / \sqrt{(3624.0 \cdot 18.0)}$$

$$= 9.788$$

Compute the Factor [Ch]:

$$\begin{aligned} & \text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579) \\ & = 0.92 / (9.788 - 0.579) \\ & = 0.0999 \end{aligned}$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} & = 1.6 \cdot Ch \cdot E_y (t / D_o) \\ & = 1.6 \cdot 0.1 \cdot 0.19629E+09 (18.0/7248.0) \\ & = 77.921 \text{ N./mm}^2 \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} & = Fhe / F_y \\ & = 77.9/234.4 \\ & = 0.332 \end{aligned}$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} & = \sigma_{ys} / \sigma_{uts} \\ & = 234.4/482.6 \\ & = 0.486 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} & = Fhe / E \\ & = 77.9/1.9629E+09 \\ & = 0.000397 \end{aligned}$$

Factor [m1]:

$$\begin{aligned} & = (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys})) \\ & = (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002)) \\ & = 0.157 \end{aligned}$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$\begin{aligned} & = \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1 \\ & = 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301 \\ & = 624.374 \text{ N./mm}^2 \end{aligned}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$\begin{aligned} & = m_2 f(m_2 f_2 - R) \\ & = 0.6 (1.0 - 0.486) \\ & = 0.309 \end{aligned}$$

Curve Fitting Constant [A2]:

$$\begin{aligned} & = (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2) \\ & = (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309) \\ & = 944.566 \text{ N./mm}^2 \end{aligned}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$\begin{aligned} & = 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5} \\ & = 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5} \\ & = 0.346 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.191E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.403E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.874E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.937E-04
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.347E-05
6	82	-5.530	0.102E-09	0.102E-09	0.737E-08	-0.737E-08	0.189E+09	0.427E-04
7	79	-5.618	0.799E-10	0.799E-10	0.664E-08	-0.664E-08	0.190E+09	0.191E-04
8	77	-5.662	0.702E-10	0.702E-10	0.629E-08	-0.629E-08	0.191E+09	0.771E-05
9	76	-5.684	0.657E-10	0.657E-10	0.612E-08	-0.612E-08	0.191E+09	0.209E-05
10	75	-5.695	0.636E-10	0.636E-10	0.604E-08	-0.604E-08	0.191E+09	0.697E-06
11	76	-5.689	0.647E-10	0.647E-10	0.608E-08	-0.608E-08	0.191E+09	0.696E-06
12	76	-5.692	0.641E-10	0.641E-10	0.606E-08	-0.606E-08	0.191E+09	0.889E-09



Factor of Safety [FS]:

Since $Fic [76.0] < 0.55 * 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= $76.003/2.0$

= 38.002 N./mm^2

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha(t / Do)$

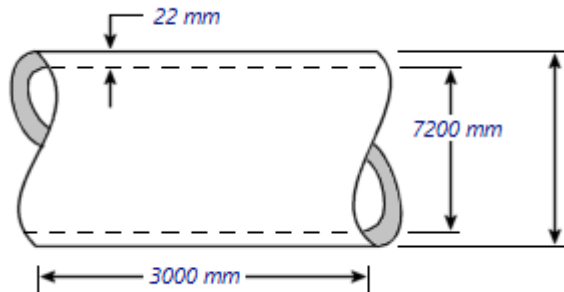
= $2 * 37999.414(18.0/7248.0)$

= 188.739 KPa .

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LP-shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	385.039 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress	86.8 N./mm ²	
Spec. Min. Thk t _e or s _a		22.0000 mm	Spec. Nominal Thk.	22.000 mm	
Surface Area		682730.9 cm ²	SG of Contents	1.060	
Element Volume		122167016.0 cm ³	Weight of Contents	4313.9 Kg	
Empty Weight		11605.2 Kg	Operating Weight	15919.1 Kg	
			Comp. All. Ope. Stress	55.2 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(385.039/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.8888 + 6.0 + 0.0 = 14.8888 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 7212.0) / 7212.0) - 5.039$$

$$= 692.385 - 5.039 = 687.3464 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 22.0 + 7200.0) / 7200.0)$$

$$= 1066.1133 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 385.039 \cdot 7212.0 / (1.0 (7244.0 - 7212.0))$$

$$= 86.7843 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 385.039 / (1.0 \cdot \ln((2 \cdot 16.0 + 7212.0) / 7212.0))$$

$$= 86.9748 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.305 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: LP-shell-3

Cylindrical Shell From 200 to Ring R1 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.01 / \sqrt{(3622.0 \cdot 16.0)}$$

$$= 10.385$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (10.385 - 0.579)$$

$$= 0.0938$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.094 \cdot 0.19629E+09 (16.0 / 7244.0)$$

$$= 65.085 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 65.1 / 234.4$$

$$= 0.278$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 65.1 / 0.19629E+09$$

$$= 0.000332$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

= 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.468E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.220E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.159E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.619E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.190E-04
7	64	-5.969	0.258E-10	0.258E-10	0.414E-08	-0.414E-08	0.194E+09	0.170E-05
8	65	-5.925	0.301E-10	0.301E-10	0.442E-08	-0.442E-08	0.194E+09	0.857E-05
9	65	-5.947	0.278E-10	0.278E-10	0.428E-08	-0.428E-08	0.194E+09	0.342E-05
10	64	-5.958	0.268E-10	0.268E-10	0.421E-08	-0.421E-08	0.194E+09	0.860E-06
11	64	-5.964	0.263E-10	0.263E-10	0.417E-08	-0.417E-08	0.194E+09	0.420E-06
12	64	-5.961	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.219E-06
13	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.101E-06
14	64	-5.962	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.594E-07
15	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.206E-07
16	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.194E-07
17	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.603E-09

Factor of Safety [FS]:

$$\text{Since } F_{ic} [64.41] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 64.415/2.0$$

$$= 32.207 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 32205.605 (16.0/7244.0)$$

$$= 142.267 \text{ KPa.}$$

Cylindrical Shell From Ring R1 to 210 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2499.965 / \sqrt{(3622.0 \cdot 16.0)}$$

$$= 10.385$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.385 - 0.579)$$

$$= 0.0938$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.094 \cdot 0.19629E+09 (16.0/7244.0)$$

$$= 65.086 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 65.1/234.4$$

$$= 0.278$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 65.1/.19629E+09$$

$$= 0.000332$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.468E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.220E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.159E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.619E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.190E-04
7	64	-5.969	0.258E-10	0.258E-10	0.414E-08	-0.414E-08	0.194E+09	0.171E-05
8	65	-5.925	0.301E-10	0.301E-10	0.442E-08	-0.442E-08	0.194E+09	0.857E-05
9	65	-5.947	0.278E-10	0.278E-10	0.428E-08	-0.428E-08	0.194E+09	0.342E-05
10	64	-5.958	0.268E-10	0.268E-10	0.421E-08	-0.421E-08	0.194E+09	0.853E-06
11	64	-5.964	0.263E-10	0.263E-10	0.417E-08	-0.417E-08	0.194E+09	0.427E-06
12	64	-5.961	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.213E-06
13	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.107E-06
14	64	-5.962	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.531E-07
15	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.269E-07
16	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.131E-07
17	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.695E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [64.41] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 64.415/2.0$$

$$= 32.207 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

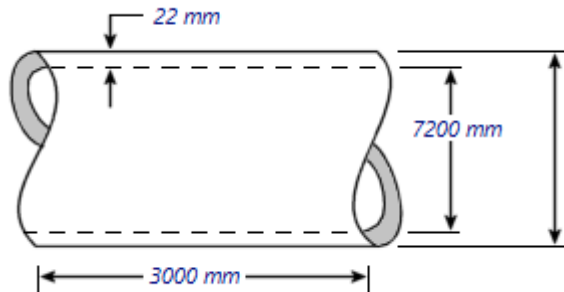
$$= 2 \cdot 32205.605(16.0/7244.0)$$

$$= 142.267 \text{ KPa.}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LP-shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	384.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress		86.6 N./mm ²
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		682730.9 cm ²	SG of Contents		0.000
Element Volume		122167016.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		11605.2 Kgf	Operating Weight		11605.2 Kgf
			Comp. All. Ope. Stress		55.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(384.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.8652 + 6.0 + 0.0 = 14.8652 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 7212.0) / 7212.0) - 4.0$$

$$= 692.385 - 4.0 = 688.3854 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 22.0 + 7200.0) / 7200.0)$$

$$= 1066.1133 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 384.0 \cdot 7212.0 / (1.0 (7244.0 - 7212.0))$$

$$= 86.5501 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 384.0 / (1.0 \cdot \ln((2 \cdot 16.0 + 7212.0) / 7212.0))$$

$$= 86.7401 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.305 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -39 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -24 °C
 Temperature reduction per Graph 3.12 : 15 °C
 The computed stress ratio is : 0.628

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (10.053 \cdot 1.0) / (22.0 - 6.0 - 0.0)$
 $= 0.628$

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	487.156 KPa.
Governing Thk.	22.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.628	Basic MDMT	-24 °C
		Temperature Difference	15 °C
		Computed Min. Temp.	-39 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: LP-shell-4

Cylindrical Shell From 210 to Ring R5 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 2499.965 / \sqrt{3622.0 \cdot 16.0}$
 $= 10.385$

Compute the Factor [Ch]:
 Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$
 $= 0.92 / (10.385 - 0.579)$
 $= 0.0938$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.094 \cdot 0.19629E+09(16.0/7244.0)$
 $= 65.086 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 65.1/234.4$
 $= 0.278$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4/482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 65.1/.19629E+09$
 $= 0.000332$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.468E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.220E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.159E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.619E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.190E-04
7	64	-5.969	0.258E-10	0.258E-10	0.414E-08	-0.414E-08	0.194E+09	0.171E-05
8	65	-5.925	0.301E-10	0.301E-10	0.442E-08	-0.442E-08	0.194E+09	0.857E-05
9	65	-5.947	0.278E-10	0.278E-10	0.428E-08	-0.428E-08	0.194E+09	0.342E-05
10	64	-5.958	0.268E-10	0.268E-10	0.421E-08	-0.421E-08	0.194E+09	0.853E-06
11	64	-5.964	0.263E-10	0.263E-10	0.417E-08	-0.417E-08	0.194E+09	0.427E-06
12	64	-5.961	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.213E-06
13	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.107E-06
14	64	-5.962	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.531E-07
15	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.269E-07
16	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.131E-07
17	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.695E-08

Factor of Safety [FS]:
 Since $Fic [64.41] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 64.415 / 2.0$$

$$= 32.207 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 32205.605 (16.0 / 7244.0)$$

$$= 142.267 \text{ KPa.}$$

Cylindrical Shell From Ring R5 to 220 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2500.002 / \sqrt{(3622.0 \cdot 16.0)}$$

$$= 10.385$$

Compute the Factor [Ch]:

$$\begin{aligned} & \text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579) \\ & = 0.92 / (10.385 - 0.579) \\ & = 0.0938 \end{aligned}$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} & = 1.6 \cdot Ch \cdot E_y(t / D_o) \\ & = 1.6 \cdot 0.094 \cdot 0.19629E+09(16.0/7244.0) \\ & = 65.085 \text{ N./mm}^2 \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} & = Fhe / F_y \\ & = 65.1/234.4 \\ & = 0.278 \end{aligned}$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} & = \sigma_{ys} / \sigma_{uts} \\ & = 234.4/482.6 \\ & = 0.486 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} & = Fhe / E \\ & = 65.1/0.19629E+09 \\ & = 0.000332 \end{aligned}$$

Factor [m1]:

$$\begin{aligned} & = (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys})) \\ & = (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002)) \\ & = 0.157 \end{aligned}$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$\begin{aligned} & = \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1 \\ & = 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301 \\ & = 624.374 \text{ N./mm}^2 \end{aligned}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$\begin{aligned} & = m_2 f(m_2 f_2 - R) \\ & = 0.6(1.0 - 0.486) \\ & = 0.309 \end{aligned}$$

Curve Fitting Constant [A2]:

$$\begin{aligned} & = (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2) \\ & = (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309) \\ & = 944.566 \text{ N./mm}^2 \end{aligned}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$\begin{aligned} & = 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5} \\ & = 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5} \\ & = 0.346 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.468E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.220E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.159E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.619E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.190E-04
7	64	-5.969	0.258E-10	0.258E-10	0.414E-08	-0.414E-08	0.194E+09	0.170E-05
8	65	-5.925	0.301E-10	0.301E-10	0.442E-08	-0.442E-08	0.194E+09	0.857E-05
9	65	-5.947	0.278E-10	0.278E-10	0.428E-08	-0.428E-08	0.194E+09	0.342E-05
10	64	-5.958	0.268E-10	0.268E-10	0.421E-08	-0.421E-08	0.194E+09	0.859E-06
11	64	-5.964	0.263E-10	0.263E-10	0.417E-08	-0.417E-08	0.194E+09	0.422E-06
12	64	-5.961	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.218E-06



13	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.102E-06
14	64	-5.962	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.582E-07
15	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.218E-07
16	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.182E-07
17	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.177E-08

Factor of Safety [FS]:
 Since $Fic [64.41] < 0.55 * 234.4$, $FS = 2$

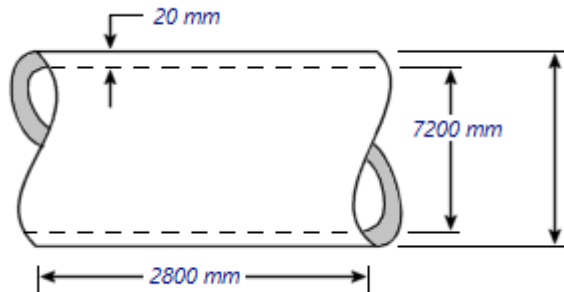
Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 64.415/2.0$
 $= 32.207 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 32205.605(16.0/7244.0)$
 $= 142.267 \text{ KPa.}$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LP-shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	384.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		2800.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress	98.9 N./mm ²	
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.	20.000 mm	
Surface Area		636863.7 cm ²	SG of Contents	0.000	
Element Volume		114022544.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		9844.1 Kgf	Operating Weight	9844.1 Kgf	
			Comp. All. Ope. Stress	48.3 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(384.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.8652 + 6.0 + 0.0 = 14.8652 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 14.0 + 7212.0) / 7212.0) - 4.0$$

$$= 605.998 - 4.0 = 601.9977 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 20.0 + 7200.0) / 7200.0)$$

$$= 969.4675 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 384.0 \cdot 7212.0 / (1.0 (7240.0 - 7212.0))$$

$$= 98.9137 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 384.0 / (1.0 \cdot \ln((2 \cdot 14.0 + 7212.0) / 7212.0))$$

$$= 99.1053 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.277 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -38 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -26 °C
 Temperature reduction per Graph 3.12 : 12 °C
 The computed stress ratio is : 0.718

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (10.053 \cdot 1.0) / (20.0 - 6.0 - 0.0)$
 $= 0.718$

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	487.156 KPa.
Governing Thk.	20.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.718	Basic MDMT	-26 °C
		Temperature Difference	12 °C
		Computed Min. Temp.	-38 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: LP-shell-5

Cylindrical Shell From 220 to Ring R2 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 2500.002 / \sqrt{3620.0 \cdot 14.0}$
 $= 11.105$

Compute the Factor [Ch]:
 Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$
 $= 0.92 / (11.105 - 0.579)$
 $= 0.0874$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.087 \cdot 0.19629E+09(14.0/7240.0)$
 $= 53.083 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 53.1/234.4$
 $= 0.226$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4/482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 53.1/0.19629E+09$
 $= 0.000270$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.529E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.392E-04
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.117E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.395E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.530E-06
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.192E-04
8	54	-6.189	0.110E-10	0.110E-10	0.290E-08	-0.290E-08	0.195E+09	0.932E-05
9	53	-6.211	0.100E-10	0.100E-10	0.279E-08	-0.279E-08	0.196E+09	0.439E-05
10	53	-6.222	0.954E-11	0.954E-11	0.273E-08	-0.273E-08	0.196E+09	0.193E-05
11	53	-6.227	0.932E-11	0.932E-11	0.271E-08	-0.271E-08	0.196E+09	0.699E-06
12	52	-6.230	0.921E-11	0.921E-11	0.269E-08	-0.269E-08	0.196E+09	0.844E-07
13	52	-6.231	0.916E-11	0.916E-11	0.269E-08	-0.269E-08	0.196E+09	0.223E-06
14	52	-6.231	0.918E-11	0.918E-11	0.269E-08	-0.269E-08	0.196E+09	0.691E-07
15	52	-6.230	0.920E-11	0.920E-11	0.269E-08	-0.269E-08	0.196E+09	0.769E-08

Factor of Safety [FS]:
 Since Fic [52.89] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic} / \text{FS}$$

$$= 52.893 / 2.0$$

$$= 26.446 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot \text{Fha} (t / D_o)$$

$$= 2 \cdot 26444.939 (14.0 / 7240.0)$$

$$= 102.273 \text{ KPa.}$$

Cylindrical Shell From Ring R2 to 230 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.0 / \sqrt{(3620.0 \cdot 14.0)}$$

$$= 11.105$$

Compute the Factor [Ch]:

Since $Mx > 1.5$ and $Mx < 13$, $Ch = 0.92 / (Mx - 0.579)$
 $= 0.92 / (11.105 - 0.579)$
 $= 0.0874$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot Ey (t / Do)$
 $= 1.6 \cdot 0.087 \cdot 0.19629E+09 (14.0 / 7240.0)$
 $= 53.083 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / Fy$
 $= 53.1 / 234.4$
 $= 0.226$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4 / 482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 53.1 / 0.19629E+09$
 $= 0.000270$

Factor [m1]:
 $= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(\ln(1 + \epsilon_{psp})) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(\ln(1 + 0.00002)) / \ln(1 + 0.002))$
 $= 0.157$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:
 $= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$
 $= 624.374 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:
 $= m2f(m2f2 - R)$
 $= 0.6(1.0 - 0.486)$
 $= 0.309$

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
 $= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$
 $= 944.566 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$
 $= 0.346$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.529E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.392E-04
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.117E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.395E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.530E-06
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.192E-04
8	54	-6.189	0.110E-10	0.110E-10	0.290E-08	-0.290E-08	0.195E+09	0.932E-05
9	53	-6.211	0.100E-10	0.100E-10	0.279E-08	-0.279E-08	0.196E+09	0.439E-05
10	53	-6.222	0.954E-11	0.954E-11	0.273E-08	-0.273E-08	0.196E+09	0.193E-05
11	53	-6.227	0.932E-11	0.932E-11	0.271E-08	-0.271E-08	0.196E+09	0.698E-06
12	52	-6.230	0.921E-11	0.921E-11	0.269E-08	-0.269E-08	0.196E+09	0.841E-07
13	52	-6.231	0.916E-11	0.916E-11	0.269E-08	-0.269E-08	0.196E+09	0.223E-06
14	52	-6.231	0.918E-11	0.918E-11	0.269E-08	-0.269E-08	0.196E+09	0.694E-07



15	52	-6.230	0.920E-11	0.920E-11	0.269E-08	-0.269E-08	0.196E+09	0.734E-08
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Factor of Safety [FS]:
 Since $Fic [52.89] < 0.55 * 234.4$, FS = 2

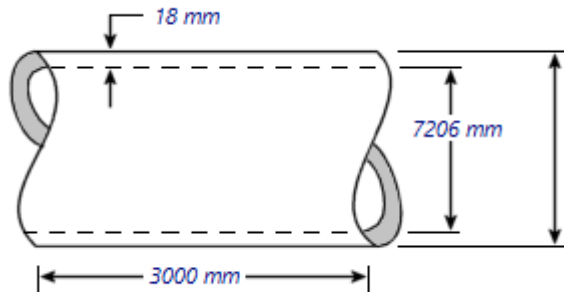
Allowable Circumferential Compressive Stress [Fha]:
 = Fic/FS
 = $52.893/2.0$
 = 26.446 N./mm²

Allowable External Pressure at the given Thickness [Pa]:
 = $2 * Fha(t / Do)$
 = $2 * 26444.939(14.0/7240.0)$
 = 102.273 KPa.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LP-shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	384.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7206.000 mm	Actual Stress	95.8 N./mm ²	
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		682542.5 cm ²	SG of Contents	0.000	
Element Volume		122370720.0 cm ³	Weight of Contents	0.0 Kg	
Empty Weight		9498.0 Kg	Operating Weight	9498.0 Kg	
			Comp. All. Ope. Stress	58.7 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7208.0 \cdot (\exp(384.0/(156.4 \cdot 0.85)) - 1) + 1.0 + 0.0$$

$$= 10.4258 + 1.0 + 0.0 = 11.4258 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 0.85 \cdot \ln((2 \cdot 17.0 + 7208.0) / 7208.0) - 4.0$$

$$= 625.566 - 4.0 = 621.5663 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 0.85 \cdot \ln((2 \cdot 18.0 + 7206.0) / 7206.0)$$

$$= 741.2355 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 384.0 \cdot 7208.0 / (0.85 (7242.0 - 7208.0))$$

$$= 95.7790 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 384.0 / (0.85 \cdot \ln((2 \cdot 17.0 + 7208.0) / 7208.0))$$

$$= 96.0052 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.249 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -45 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -28 °C
 Temperature reduction per Graph 3.12 : 17 °C
 The computed stress ratio is : 0.591

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (11.823 \cdot 0.85) / (18.0 - 1.0 - 0.0)$
 $= 0.591$

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	487.156 KPa.
Governing Thk.	18.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.591	Basic MDMT	-28 °C
		Temperature Difference	17 °C
		Computed Min. Temp.	-45 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: LP-shell-6

Cylindrical Shell From 230 to Ring R4 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 2500.0 / \sqrt{3621.0 \cdot 17.0}$
 $= 10.076$

Compute the Factor [Ch]:
 Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$
 $= 0.92 / (10.076 - 0.579)$
 $= 0.0969$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.097 \cdot 0.19629E+09(17.0/7242.0)$
 $= 71.420 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 71.4/234.4$
 $= 0.305$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4/482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 71.4/19629E+09$
 $= 0.000364$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.436E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.542E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.127E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.296E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.133E-04
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.785E-05
8	69	-5.837	0.405E-10	0.405E-10	0.500E-08	-0.500E-08	0.193E+09	0.280E-05
9	70	-5.815	0.435E-10	0.435E-10	0.515E-08	-0.515E-08	0.193E+09	0.250E-05
10	70	-5.826	0.420E-10	0.420E-10	0.508E-08	-0.508E-08	0.193E+09	0.156E-06
11	70	-5.821	0.428E-10	0.428E-10	0.512E-08	-0.512E-08	0.193E+09	0.117E-05
12	70	-5.824	0.424E-10	0.424E-10	0.510E-08	-0.510E-08	0.193E+09	0.507E-06
13	70	-5.825	0.422E-10	0.422E-10	0.509E-08	-0.509E-08	0.193E+09	0.176E-06
14	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.972E-08

Factor of Safety [FS]:
 Since $Fic [70.26] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 70.259 / 2.0$$

$$= 35.129 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 35127.363 (17.0 / 7242.0)$$

$$= 164.917 \text{ KPa.}$$

Cylindrical Shell From Ring R4 to Ring R3 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2500.004 / \sqrt{(3621.0 \cdot 17.0)}$$

$$= 10.076$$

Compute the Factor [Ch]:
 Since $Mx > 1.5$ and $Mx < 13$, $Ch = 0.92 / (Mx - 0.579)$

$$= 0.92 / (10.076 - 0.579)$$

$$= 0.0969$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.097 \cdot 0.19629E+09(17.0/7242.0)$$

$$= 71.420 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 71.4/234.4$$

$$= 0.305$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 71.4/0.19629E+09$$

$$= 0.000364$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.436E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.542E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.127E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.296E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.133E-04
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.785E-05
8	69	-5.837	0.405E-10	0.405E-10	0.500E-08	-0.500E-08	0.193E+09	0.280E-05
9	70	-5.815	0.435E-10	0.435E-10	0.515E-08	-0.515E-08	0.193E+09	0.250E-05
10	70	-5.826	0.420E-10	0.420E-10	0.508E-08	-0.508E-08	0.193E+09	0.155E-06
11	70	-5.821	0.428E-10	0.428E-10	0.512E-08	-0.512E-08	0.193E+09	0.117E-05
12	70	-5.824	0.424E-10	0.424E-10	0.510E-08	-0.510E-08	0.193E+09	0.508E-06
13	70	-5.825	0.422E-10	0.422E-10	0.509E-08	-0.509E-08	0.193E+09	0.176E-06
14	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.105E-07
15	70	-5.826	0.420E-10	0.420E-10	0.508E-08	-0.508E-08	0.193E+09	0.724E-07

16	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.310E-07
17	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.102E-07

Factor of Safety [FS]:

Since $Fic [70.26] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS
 = $70.255/2.0$
 = 35.128 N./mm^2

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha(t / Do)$
 = $2 \cdot 35125.523(17.0/7242.0)$
 = 164.909 KPa.

Cylindrical Shell From Ring R3 to 240 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

= $L / \sqrt{(Ro \cdot t)}$
 = $2500.0 / \sqrt{(3621.0 \cdot 17.0)}$
 = 10.076

Compute the Factor [Ch]:

Since $Mx > 1.5$ and $Mx < 13$, $Ch = 0.92 / (Mx - 0.579)$
 = $0.92 / (10.076 - 0.579)$
 = 0.0969

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t / Do)$
 = $1.6 \cdot 0.097 \cdot 0.19629E+09(17.0/7242.0)$
 = 71.420 N./mm^2

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy
 = $71.4/234.4$
 = 0.305

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
 = $234.4/482.6$
 = 0.486

Ratio [AE]:

= Fhe / E
 = $71.4/.19629E+09$
 = 0.000364

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 = $(\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
 = 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 = $234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$
 = 624.374 N./mm^2

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f2 - R)$
 = $0.6(1.0 - 0.486)$
 = 0.309

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.436E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.542E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.127E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.296E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.133E-04
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.785E-05
8	69	-5.837	0.405E-10	0.405E-10	0.500E-08	-0.500E-08	0.193E+09	0.280E-05
9	70	-5.815	0.435E-10	0.435E-10	0.515E-08	-0.515E-08	0.193E+09	0.250E-05
10	70	-5.826	0.420E-10	0.420E-10	0.508E-08	-0.508E-08	0.193E+09	0.156E-06
11	70	-5.821	0.428E-10	0.428E-10	0.512E-08	-0.512E-08	0.193E+09	0.117E-05
12	70	-5.824	0.424E-10	0.424E-10	0.510E-08	-0.510E-08	0.193E+09	0.508E-06
13	70	-5.825	0.422E-10	0.422E-10	0.509E-08	-0.509E-08	0.193E+09	0.176E-06
14	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.993E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [70.26] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 70.259 / 2.0$$

$$= 35.129 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

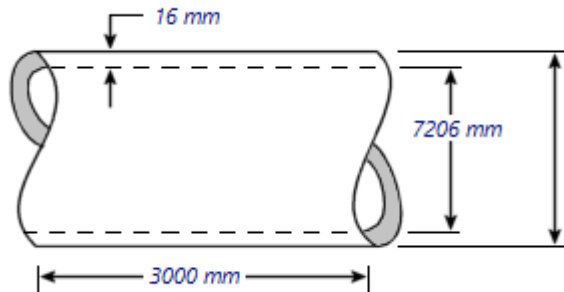
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 35127.363 (17.0 / 7242.0)$$

$$= 164.917 \text{ KPa.}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: LP-shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	384.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7206.000 mm	Actual Stress	108.5 N./mm ²	
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.	16.000 mm	
Surface Area		682165.5 cm ²	SG of Contents	0.000	
Element Volume		122370720.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		8440.4 Kgf	Operating Weight	8440.4 Kgf	
			Comp. All. Ope. Stress	51.8 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7208.0 \cdot (\exp(384.0/(156.4 \cdot 0.85)) - 1) + 1.0 + 0.0$$

$$= 10.4258 + 1.0 + 0.0 = 11.4258 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 0.85 \cdot \ln((2 \cdot 15.0 + 7208.0) / 7208.0) - 4.0$$

$$= 552.11 - 4.0 = 548.1099 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 0.85 \cdot \ln((2 \cdot 16.0 + 7206.0) / 7206.0)$$

$$= 659.0662 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 384.0 \cdot 7208.0 / (0.85 (7238.0 - 7208.0))$$

$$= 108.5485 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 384.0 / (0.85 \cdot \ln((2 \cdot 15.0 + 7208.0) / 7208.0))$$

$$= 108.7783 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.222 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -44 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 13 °C
 The computed stress ratio is : 0.670

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (11.823 \cdot 0.85) / (16.0 - 1.0 - 0.0)$
 $= 0.670$

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	487.156 KPa.
Governing Thk.	16.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.670	Basic MDMT	-30 °C
		Temperature Difference	13 °C
		Computed Min. Temp.	-44 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: LP-shell-7

Cylindrical Shell From 240 to Ring R6 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 2500.0 / \sqrt{3619.0 \cdot 15.0}$
 $= 10.730$

Compute the Factor [Ch]:
 Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$
 $= 0.92 / (10.73 - 0.579)$
 $= 0.0906$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.091 \cdot 0.19629E+09(15.0/7238.0)$
 $= 58.992 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 59.0/234.4$
 $= 0.252$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4/482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 59.0/0.19629E+09$
 $= 0.000301$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.499E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.907E-05
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.147E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.697E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.306E-04
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.109E-04
8	58	-6.101	0.157E-10	0.157E-10	0.336E-08	-0.336E-08	0.195E+09	0.944E-06
9	59	-6.079	0.171E-10	0.171E-10	0.349E-08	-0.349E-08	0.195E+09	0.405E-05
10	58	-6.090	0.164E-10	0.164E-10	0.343E-08	-0.343E-08	0.195E+09	0.155E-05
11	58	-6.096	0.160E-10	0.160E-10	0.340E-08	-0.340E-08	0.195E+09	0.304E-06
12	58	-6.098	0.159E-10	0.159E-10	0.338E-08	-0.338E-08	0.195E+09	0.320E-06
13	58	-6.097	0.160E-10	0.160E-10	0.339E-08	-0.339E-08	0.195E+09	0.830E-08

Factor of Safety [FS]:
 Since $Fic [58.62] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 58.623/2.0$$

$$= 29.311 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 29309.625(15.0/7238.0)$$

$$= 121.482 \text{ KPa.}$$

Cylindrical Shell From Ring R6 to 250 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2500.0 / \sqrt{(3619.0 \cdot 15.0)}$$

$$= 10.730$$

Compute the Factor [Ch]:
 Since $Mx > 1.5$ and $Mx < 13$, $Ch = 0.92 / (Mx - 0.579)$

$$= 0.92 / (10.73 - 0.579)$$

= 0.0906

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.091 \cdot 0.19629E+09(15.0/7238.0)$$

$$= 58.992 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 59.0/234.4$$

$$= 0.252$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.0/.19629E+09$$

$$= 0.000301$$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / (\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.499E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.907E-05
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.147E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.697E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.306E-04
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.109E-04
8	58	-6.101	0.157E-10	0.157E-10	0.336E-08	-0.336E-08	0.195E+09	0.944E-06
9	59	-6.079	0.171E-10	0.171E-10	0.349E-08	-0.349E-08	0.195E+09	0.405E-05
10	58	-6.090	0.164E-10	0.164E-10	0.343E-08	-0.343E-08	0.195E+09	0.155E-05
11	58	-6.096	0.160E-10	0.160E-10	0.340E-08	-0.340E-08	0.195E+09	0.304E-06
12	58	-6.098	0.159E-10	0.159E-10	0.338E-08	-0.338E-08	0.195E+09	0.320E-06
13	58	-6.097	0.160E-10	0.160E-10	0.339E-08	-0.339E-08	0.195E+09	0.830E-08

Factor of Safety [FS]:

$$\text{Since } Fic [58.62] < 0.55 \cdot 234.4, FS = 2$$



Allowable Circumferential Compressive Stress [Fha]:

= F_c/FS
= 58.623/2.0
= 29.311 N./mm²

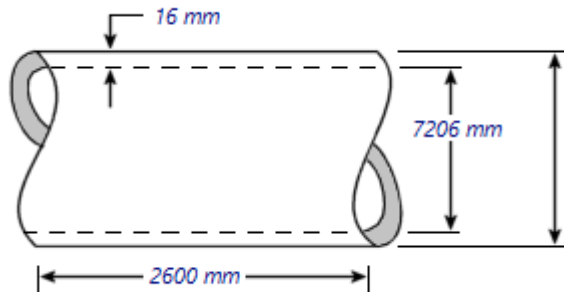
Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot F_{ha}(t / D_o)$
= $2 \cdot 29309.625(15.0/7238.0)$
= 121.482 KPa.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LP-shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	384.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		2600.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	7206.000 mm	Actual Stress		108.5 N./mm ²
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		591210.1 cm ²	SG of Contents		0.000
Element Volume		106054632.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		7315.0 Kgf	Operating Weight		7315.0 Kgf
			Comp. All. Ope. Stress		51.8 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7208.0 \cdot (\exp(384.0/(138.0 \cdot 0.85)) - 1) + 1.0 + 0.0$$

$$= 11.8183 + 1.0 + 0.0 = 12.8183 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 0.85 \cdot \ln((2 \cdot 15.0 + 7208.0) / 7208.0) - 4.0$$

$$= 487.156 - 4.0 = 483.1557 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 138.0 \cdot 0.85 \cdot \ln((2 \cdot 16.0 + 7206.0) / 7206.0)$$

$$= 519.7206 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 384.0 \cdot 7208.0 / (0.85 (7238.0 - 7208.0))$$

$$= 108.5485 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 384.0 / (0.85 \cdot \ln((2 \cdot 15.0 + 7208.0) / 7208.0))$$

$$= 108.7783 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.222 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: LP-shell-8

Cylindrical Shell From 250 to Ring R9 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.0 / \sqrt{(3619.0 \cdot 15.0)}$$

$$= 10.730$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.73 - 0.579)$$

$$= 0.0906$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.091 \cdot 0.19629E+09 (15.0 / 7238.0)$$

$$= 58.992 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 59.0 / 234.4$$

$$= 0.252$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.0 / 0.19629E+09$$

$$= 0.000301$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

= 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.499E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.907E-05
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.147E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.697E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.306E-04
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.109E-04
8	58	-6.101	0.157E-10	0.157E-10	0.336E-08	-0.336E-08	0.195E+09	0.944E-06
9	59	-6.079	0.171E-10	0.171E-10	0.349E-08	-0.349E-08	0.195E+09	0.405E-05
10	58	-6.090	0.164E-10	0.164E-10	0.343E-08	-0.343E-08	0.195E+09	0.155E-05
11	58	-6.096	0.160E-10	0.160E-10	0.340E-08	-0.340E-08	0.195E+09	0.304E-06
12	58	-6.098	0.159E-10	0.159E-10	0.338E-08	-0.338E-08	0.195E+09	0.320E-06
13	58	-6.097	0.160E-10	0.160E-10	0.339E-08	-0.339E-08	0.195E+09	0.830E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [58.62] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 58.623/2.0$$

$$= 29.311 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 29309.625 (15.0/7238.0)$$

$$= 121.482 \text{ KPa.}$$

Cylindrical Shell From Ring R9 to 260 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2600.52 / \sqrt{(3619.0 \cdot 15.0)}$$

$$= 11.161$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (11.161 - 0.579)$$

$$= 0.0869$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.087 \cdot 0.19629E+09 (15.0/7238.0)$$

$$= 56.587 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 56.6/234.4$$

$$= 0.241$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:
 = Fhe / E
 = 56.6/.19629E+09
 = 0.000288

Factor [m1]:
 = (ln(r) + (epsys - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))
 = (ln(0.486) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))
 = 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:
 = $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})$
 = 234.4(1 + 0.002) / ln(1 + 0.002) 0.157301
 = 624.374 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:
 = $m2f(m2f2 - R)$
 = 0.6(1.0 - 0.486)
 = 0.309

Curve Fitting Constant [A2]:
 = ($\sigma_{uts} \cdot \exp(m2)$) / ($m2 m2$)
 = (482.6 • exp(0.309)) / (0.309 0.309)
 = 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:
 = 1.5 • R1.5 - 0.5 • R2.5 - R3.5
 = 1.5 • 0.4861.5 - 0.5 • 0.4862.5 - 0.4863.5
 = 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.511E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.213E-04
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.135E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.574E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.184E-04
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.136E-05
8	54	-6.189	0.110E-10	0.110E-10	0.290E-08	-0.290E-08	0.195E+09	0.854E-05
9	55	-6.167	0.120E-10	0.120E-10	0.301E-08	-0.301E-08	0.195E+09	0.360E-05
10	56	-6.156	0.126E-10	0.126E-10	0.307E-08	-0.307E-08	0.195E+09	0.112E-05
11	56	-6.150	0.129E-10	0.129E-10	0.310E-08	-0.310E-08	0.195E+09	0.117E-06
12	56	-6.153	0.127E-10	0.127E-10	0.308E-08	-0.308E-08	0.195E+09	0.502E-06
13	56	-6.152	0.128E-10	0.128E-10	0.309E-08	-0.309E-08	0.195E+09	0.192E-06
14	56	-6.151	0.128E-10	0.128E-10	0.309E-08	-0.309E-08	0.195E+09	0.375E-07
15	56	-6.151	0.129E-10	0.129E-10	0.310E-08	-0.310E-08	0.195E+09	0.399E-07
16	56	-6.151	0.129E-10	0.129E-10	0.309E-08	-0.309E-08	0.195E+09	0.119E-08

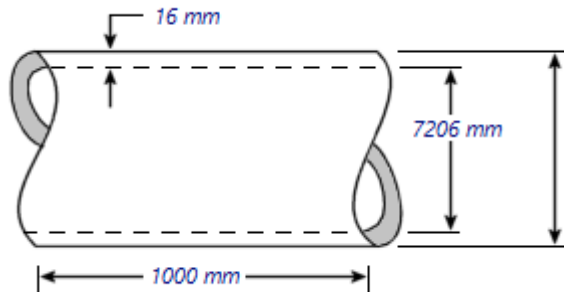
Factor of Safety [FS]:
 Since Fic [56.3] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:
 = Fic/FS
 = 56.303/2.0
 = 28.151 N./mm²

Allowable External Pressure at the given Thickness [Pa]:
 = 2 • Fha(t / Do)
 = 2 • 28149.76(15.0/7238.0)
 = 116.675 KPa.

Internal Pressure Calculation Results:

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Internal Pressure Results for: LP-shell-9

Cylindrical Shell Design Information:					
Design Pressure	P	384.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		1000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7206.000 mm	Actual Stress	108.5 N./mm ²	
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.	16.000 mm	
Surface Area		227388.5 cm ²	SG of Contents	0.000	
Element Volume		40790244.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		2813.4 Kgf	Operating Weight	2813.4 Kgf	
			Comp. All. Ope. Stress	51.8 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7208.0 \cdot (\exp(384.0/(156.4 \cdot 0.85)) - 1) + 1.0 + 0.0$$

$$= 10.4258 + 1.0 + 0.0 = 11.4258 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 0.85 \cdot \ln((2 \cdot 15.0 + 7208.0) / 7208.0) - 4.0$$

$$= 552.11 - 4.0 = 548.1099 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 0.85 \cdot \ln((2 \cdot 16.0 + 7206.0) / 7206.0)$$

$$= 659.0662 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 384.0 \cdot 7208.0 / (0.85 (7238.0 - 7208.0))$$

$$= 108.5485 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 384.0 / (0.85 \cdot \ln((2 \cdot 15.0 + 7208.0) / 7208.0))$$

$$= 108.7783 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.222 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -44 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 13 °C
 The computed stress ratio is : 0.670

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (11.823 \cdot 0.85) / (16.0 - 1.0 - 0.0)$
 $= 0.670$

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	487.156 KPa.
Governing Thk.	16.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.670	Basic MDMT	-30 °C
		Temperature Difference	13 °C
		Computed Min. Temp.	-44 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: LP-shell-9

Cylindrical Shell From 260 to 270 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 2600.52 / \sqrt{3619.0 \cdot 15.0}$
 $= 11.161$

Compute the Factor [Ch]:
 Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$
 $= 0.92 / (11.161 - 0.579)$
 $= 0.0869$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.087 \cdot 0.19629E+09(15.0/7238.0)$
 $= 56.587 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 56.6/234.4$
 $= 0.241$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4/482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 56.6/19629E+09$
 $= 0.000288$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.511E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.213E-04
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.135E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.574E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.184E-04
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.136E-05
8	54	-6.189	0.110E-10	0.110E-10	0.290E-08	-0.290E-08	0.195E+09	0.854E-05
9	55	-6.167	0.120E-10	0.120E-10	0.301E-08	-0.301E-08	0.195E+09	0.360E-05
10	56	-6.156	0.126E-10	0.126E-10	0.307E-08	-0.307E-08	0.195E+09	0.112E-05
11	56	-6.150	0.129E-10	0.129E-10	0.310E-08	-0.310E-08	0.195E+09	0.117E-06
12	56	-6.153	0.127E-10	0.127E-10	0.308E-08	-0.308E-08	0.195E+09	0.502E-06
13	56	-6.152	0.128E-10	0.128E-10	0.309E-08	-0.309E-08	0.195E+09	0.192E-06
14	56	-6.151	0.128E-10	0.128E-10	0.309E-08	-0.309E-08	0.195E+09	0.375E-07
15	56	-6.151	0.129E-10	0.129E-10	0.310E-08	-0.310E-08	0.195E+09	0.399E-07
16	56	-6.151	0.129E-10	0.129E-10	0.309E-08	-0.309E-08	0.195E+09	0.119E-08

Factor of Safety [FS]:
 Since Fic [56.3] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic} / \text{FS}$$

$$= 56.303 / 2.0$$

$$= 28.151 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

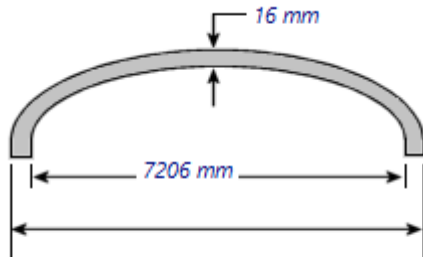
$$= 2 \cdot \text{Fha} (t / D_o)$$

$$= 2 \cdot 28149.76 (15.0 / 7238.0)$$

$$= 116.675 \text{ KPa.}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LP-TOP dish

Elliptical Head Design Information:					
Design Pressure	P	384.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7206.000 mm	Actual Stress		97.8 N./mm ²
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		579888.9 cm ²	SG of Contents		0.000
Element Volume		51028592.0 cm ³	Weight of Contents		0.0 Kg _f
Empty Weight		8871.5 Kg _f	Operating Weight		8871.5 Kg _f
			Comp. All. Ope. Stress		51.8 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 90.075 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 45.038 mm

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

$$t_r = 9.9329 \text{ mm}$$

$$\begin{aligned} t &= t_r + c_i + c_o \\ &= 9.9329 + 1.0 + 0.0 \\ &= 10.9329 \text{ mm} - \text{see below for the derivation} \end{aligned}$$

For Elliptical heads: Compute [h, k, r.] per 4.3.7.1

$$\begin{aligned} h &= D / (2 \cdot \text{Aspect Ratio}) \\ &= 7206.0 / (2 \cdot 2.0) \\ &= 1801.500 \text{ mm} \end{aligned}$$

$$\begin{aligned} k &= D / (2 \cdot h) \\ &= 7206.0 / (2 \cdot 1801.5) \\ &= 2.000 \end{aligned}$$

$$\begin{aligned} r &= D (0.5/k - 0.08) + c_a \\ &= 7206.0 (0.5/2.0 - 0.08) + 1.0 \\ &= 1226.020 \text{ mm} \end{aligned}$$

$$\begin{aligned} L &= D(0.44 \cdot k + 0.02) + ca \\ &= 7206.0(0.44 \cdot 2.0 + 0.02) + 1.0 \\ &= 6486.400 \text{ mm} \end{aligned}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:
 Equations 4.3.8 to 4.3.11 as appropriate
 t (corroded) and P for the calc.: $t = 9.9329 \text{ mm}$, $P = 384.0 \text{ KPa}$.

Angle [βTh]:

$$\begin{aligned} &= \arccos[(0.5 \cdot D - r)/(L - r)] \\ &= \arccos[(0.5 \cdot 7208.0 - 1226.02)/(6486.4 - 1226.02)] \\ &= 1.1017 \text{ radians - Equation 4.3.8} \end{aligned}$$

$$\begin{aligned} \phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(6486.4 \cdot 9.933) / 1226.02} \\ &= 0.207 \text{ radians - Equation 4.3.9} \end{aligned}$$

$\phi Th < \beta Th$, Thus:

$$\begin{aligned} Rth &= (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r \\ &= (0.5 \cdot 7208.0 - 1226.02) / \cos(1.102 - 0.207) + 1226.02 \\ &= 5026.1821 \text{ mm - Equation 4.3.10} \end{aligned}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$r/D > 0.08$, Thus:

$$\begin{aligned} C1 &= 0.692(r/D) + 0.605 \\ &= 0.692(1226.02/7208.0) + 0.605 \\ &= 0.7227 \text{ - Equation 4.3.13} \end{aligned}$$

$$\begin{aligned} C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(1226.02/7208.0) \\ &= 1.0178, \text{ Equation 4.3.15} \end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [P_{eth}]:

$$\begin{aligned} &= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r)) \\ &= 0.723 \cdot 196301.391 \cdot 9.933^2 / (1.018 \cdot 5026.182(0.5 \cdot 5026.182 - 1226.02)) \\ &= 2125.82 \text{ KPa., Equation 4.3.16} \end{aligned}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [P_y]:

$C3 = S_y = 234.4 \text{ N/mm}^2$, Not in the creep range - 4.3.6.1(f)

$$\begin{aligned} P_y &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 234.4 \cdot 9.933 / (1.018 \cdot 5026.182(0.5 \cdot 5026.182 / 1226.02 - 1)) \\ &= 433.532 \text{ KPa., Equation 4.3.17} \end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and P_{ck}]:

$$G = P_{eth}/P_y = 2125.823/433.532 = 4.903$$

$G > 1$, Thus:

Buckling Pressure [P_{ck}]:

$$\begin{aligned} &= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) P_y \\ &= ((0.77508 \cdot 4.903 - 0.20354 \cdot 4.903^2 + 0.019274 \cdot 4.903^3) / \\ &\quad (1 + 0.19014 \cdot 4.903 - 0.089534 \cdot 4.903^2 + 0.0093965 \cdot 4.903^3)) \cdot 433.532 \\ &= 576.0 \text{ KPa., Equation 4.3.19} \end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [P_{ak}]:

$$= P_{ck}/1.5 = 576.0/1.5 = 384.0 \text{ KPa. - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [P_{ac}]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 156.4 \cdot 0.85 / (6486.4/9.933 + 0.5) \\ &= 406.82 \text{ KPa., Equation 4.3.22} \end{aligned}$$

Step 10: Compute final pressure [P_a]:

$$\begin{aligned} &= \min(P_{ak}, P_{ac}) = \min(384.0, 406.82) \\ &= 384.0 \text{ KPa., Equation 4.3.23: (same as original design pressure)} \end{aligned}$$

Allowable pressure in Crown at given thickness [P_{ac}]:



$$\begin{aligned}
 &= 2 \cdot S \cdot E / (L / t + 0.5) \\
 &= 2 \cdot 156.4 \cdot 0.85 / (6486.4 / 15.0 + 0.5) \\
 &= 614.110 \text{ KPa.}
 \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:
 = MaxPressure - Pliq = 614.1099 - 4.0 = 610.1099 KPa.

Computed Maximum Allowable Pressure New and Cold [MAPnc]:
 = 733.0095 KPa.

Computed Membrane Stress Intensity at Design Pressure [Sm]:
 = 97.8 N./mm²

Required Thickness of Straight Flange = 11.426 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.094 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 17.599 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:
 = square root($4 \cdot \text{Head Surface Area} / 3.14159$)
 = square root($4 \cdot 579889 / 3.14159$)
 = 8592.656 mm

MDMT Calculations in the Knuckle Portion:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -45 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 15 °C
 The computed stress ratio is : 0.630

The computed thickness ratio [sr]:
 = $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 = $(11.126 \cdot 0.85) / (16.0 - 1.0 - 0.0)$
 = 0.630

MDMT Calculations in the Head Straight Flange:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -45 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -28 °C
 Temperature reduction per Graph 3.12 : 17 °C
 The computed stress ratio is : 0.591

The computed thickness ratio [sr]:
 = $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 = $(11.823 \cdot 0.85) / (18.0 - 1.0 - 0.0)$
 = 0.591

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	487.156 KPa.
Governing Thk.	16.000 mm	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	0.630	Basic MDMT	-30 °C
		Temperature Difference	15 °C
		Computed Min. Temp.	-45 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 90.075 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 45.038 mm

EXTERNAL PRESSURE RESULTS for: LP-TOP dish

Elliptical Head From 270 to 280 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y(t / R_o)$$

$$= 0.075 \cdot 0.19629E+09(15.0/6489.454)$$

$$= 34.030 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 34.0 / 234.4$$

$$= 0.145$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 34.0/0.19629E+09$$

$$= 0.000173$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.193E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.626E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.136E-03
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.197E-04
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.575E-04
6	37	-6.584	0.150E-11	0.150E-11	0.126E-08	-0.126E-08	0.196E+09	0.188E-04
7	33	-6.672	0.853E-12	0.853E-12	0.995E-09	-0.995E-09	0.196E+09	0.421E-06
8	35	-6.628	0.114E-11	0.114E-11	0.112E-08	-0.112E-08	0.196E+09	0.921E-05
9	34	-6.650	0.988E-12	0.988E-12	0.106E-08	-0.106E-08	0.196E+09	0.439E-05
10	34	-6.661	0.919E-12	0.919E-12	0.103E-08	-0.103E-08	0.196E+09	0.198E-05
11	34	-6.667	0.886E-12	0.886E-12	0.101E-08	-0.101E-08	0.196E+09	0.782E-06



12	34	-6.670	0.869E-12	0.869E-12	0.100E-08	-0.100E-08	0.196E+09	0.180E-06
13	33	-6.671	0.861E-12	0.861E-12	0.999E-09	-0.999E-09	0.196E+09	0.121E-06
14	34	-6.670	0.865E-12	0.865E-12	0.100E-08	-0.100E-08	0.196E+09	0.297E-07
15	34	-6.671	0.863E-12	0.863E-12	0.100E-08	-0.100E-08	0.196E+09	0.455E-07
16	34	-6.670	0.864E-12	0.864E-12	0.100E-08	-0.100E-08	0.196E+09	0.787E-08

Factor of Safety [FS]:
 Since $Fic [34.02] < 0.55 * 234.4$, FS = 2

Allowable Circumferential Compressive Stress [Fha]:
 = Fic/FS
 = $34.017/2.0$
 = 17.009 N./mm²

Allowable External Pressure at the given Thickness [Pa]:
 = $2 * Fha[t/Ro]$
 = $2 * 17.009[15.0/6489.454]$
 = 78.625 KPa.

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Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 483.1557(1.0)$
 $= 603.945 \text{ KPa.}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
HP_BTM dish	19.5	247.0	0.079	670.60
HP_shell-1	22.7	247.0	0.092	670.60
HP_shell-2	22.7	247.0	0.092	670.60
HP_shell-3	22.7	247.0	0.092	670.60
HP_shell-4	24.1	247.0	0.098	670.66
HP_shell-5	24.6	247.0	0.100	670.66
HP_shell-6	24.6	247.0	0.100	670.66
HP_shell-7	24.6	247.0	0.100	670.66
HP_TOP dish	22.7	247.0	0.092	670.66
LP_BTM dish	73.0	247.0	0.296	674.52
LP-shell-1	101.5	247.0	0.411	674.52
LP-shell-2	101.5	247.0	0.411	674.52
LP-shell-3	110.7	247.0	0.448	674.52
LP-shell-4	110.7	247.0	0.448	674.52
LP-shell-5	121.8	247.0	0.493	674.52
LP-shell-6	159.3	247.0	0.645	674.58
LP-shell-7	179.1	247.0	0.725	674.58
LP-shell-8	179.1	234.0	0.765	674.58
LP-shell-9	179.1	247.0	0.725	674.58
LP-TOP dish	161.0	247.0	0.652	674.58

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D	Nozzle	165.00	147.60	1.118
MH-6	Nozzle	165.00	147.60	1.118
MH-5	Nozzle	165.00	147.60	1.118
MH-4	Nozzle	165.00	147.60	1.118
A	Nozzle	165.00	147.60	1.118
E	Nozzle	138.00	138.00	1.000
C	Nozzle	175.00	156.40	1.119
C	Pad	175.00	156.40	1.119
MH-3	Nozzle	175.00	156.40	1.119
MH-3	Pad	175.00	156.40	1.119
B	Nozzle	175.00	156.40	1.119
B	Pad	175.00	156.40	1.119
K	Nozzle	161.00	143.60	1.121
K	Pad	175.00	156.40	1.119
MH-2	Nozzle	175.00	156.40	1.119
MH-2	Pad	175.00	156.40	1.119
MH-1	Nozzle	175.00	156.40	1.119
MH-1	Pad	175.00	156.40	1.119
H1	Nozzle	175.00	156.40	1.119
H1	Pad	175.00	156.40	1.119
H2	Nozzle	175.00	156.40	1.119
H2	Pad	175.00	156.40	1.119
F	Nozzle	175.00	156.40	1.119
F	Pad	175.00	156.40	1.119

Minimum

1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R8	175.00	156.40	1.119
Ring R7	175.00	156.40	1.119
Ring R1	175.00	156.40	1.119
Ring R5	175.00	156.40	1.119
Ring R2	175.00	156.40	1.119
Ring R4	175.00	156.40	1.119
Ring R3	175.00	156.40	1.119
Ring R6	175.00	156.40	1.119
Ring R9	175.00	156.40	1.119

Minimum 1.119

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
HP_BTM dish	175.00	156.40	1.119
HP_shell-1	175.00	156.40	1.119
HP_shell-2	175.00	156.40	1.119
HP_shell-3	175.00	156.40	1.119
HP_shell-4	175.00	156.40	1.119
HP_shell-5	175.00	156.40	1.119
HP_shell-6	175.00	156.40	1.119
HP_shell-7	175.00	156.40	1.119
HP_TOP dish	175.00	156.40	1.119
LP_BTM dish	175.00	156.40	1.119
LP-shell-1	175.00	156.40	1.119
LP-shell-2	175.00	156.40	1.119
LP-shell-3	175.00	156.40	1.119
LP-shell-4	175.00	156.40	1.119
LP-shell-5	175.00	156.40	1.119
LP-shell-6	175.00	156.40	1.119
LP-shell-7	175.00	156.40	1.119
LP-shell-8	138.00	138.00	1.000
LP-shell-9	175.00	156.40	1.119
LP-TOP dish	175.00	156.40	1.119

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D	4.16	237.50	0.018
MH-6	2.36	237.50	0.010
MH-5	2.36	237.50	0.010
MH-4	2.36	237.50	0.010
A	3.01	237.50	0.013
E	3.91	225.00	0.017
C	22.51	247.00	0.091
MH-3	12.51	247.00	0.051
B	12.51	247.00	0.051
K	3.21	228.00	0.014
MH-2	12.51	247.00	0.051
MH-1	12.51	247.00	0.051
H1	11.52	247.00	0.047
H2	11.52	247.00	0.047
F	14.65	247.00	0.059

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

MDMT Calculations for: Ring R8, Shell to Ring Junction, Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -31 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -28 °C
 Temperature reduction per Graph 3.12 : 2 °C
 The computed stress ratio is : 0.947

The computed thickness ratio [sr]:

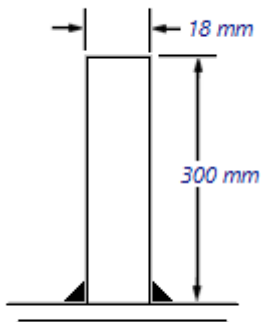
$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (11.363 \cdot 1.0) / (18.0 - 6.0 - 0.0)$$

$$= 0.947$$

Stiffening Ring Calculations for: Ring R8, SA-516 70 , Bar Ring: 300.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$300.0/18.0 \leq 0.375 \cdot \sqrt{0.19629 \times 10^9 / 234}$$

$$16.667 \leq 10.852 \quad \text{Failed Geometry Check}$$

Effective Length of Shell: 397 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	71.517	9.0000	643.655
Ring :	54.000	168.0000	9071.995
Total:	125.517		9715.650

Centroid of Ring plus Shell: 77 mm

	Inertia	Distance	A*Dist ²
Shell:	19.310	68.4049	334645.750
Ring :	4049.992	-90.5950	443202.656
Total:	4069.302		777848.375

Available Moment of Inertia, Ring + Shell: 0.118E+05 cm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2800.0 / \sqrt{(3624.0 \cdot 18.0)}$$

$$= 10.963$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (10.963 - 0.579)$$

$$= 0.0886$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey (t / Do)$$

$$= 1.6 \cdot 0.089 \cdot 0.19629E+09 (18.0 / 7248.0)$$

$$= 69.107 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 69.1 / 234.4$$

$$= 0.295$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 69.1 / 0.19629E+09$$

$$= 0.000352$$

Factor [m1]:

$$= (\ln(r) + (\text{epsys} - \text{epsp})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.448E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.425E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.139E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.414E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.153E-05
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.196E-04
8	69	-5.837	0.405E-10	0.405E-10	0.500E-08	-0.500E-08	0.193E+09	0.898E-05
9	68	-5.859	0.377E-10	0.377E-10	0.485E-08	-0.485E-08	0.193E+09	0.371E-05



10	68	-5.870	0.363E-10	0.363E-10	0.478E-08	-0.478E-08	0.194E+09	0.108E-05
11	68	-5.876	0.356E-10	0.356E-10	0.474E-08	-0.474E-08	0.194E+09	0.224E-06
12	68	-5.873	0.360E-10	0.360E-10	0.476E-08	-0.476E-08	0.194E+09	0.430E-06
13	68	-5.874	0.358E-10	0.358E-10	0.475E-08	-0.475E-08	0.194E+09	0.103E-06
14	68	-5.875	0.357E-10	0.357E-10	0.474E-08	-0.474E-08	0.194E+09	0.610E-07
15	68	-5.875	0.358E-10	0.358E-10	0.475E-08	-0.475E-08	0.194E+09	0.208E-07
16	68	-5.875	0.357E-10	0.357E-10	0.475E-08	-0.475E-08	0.194E+09	0.201E-07
17	68	-5.875	0.357E-10	0.357E-10	0.475E-08	-0.475E-08	0.194E+09	0.321E-09

Factor of Safety [FS]:

Since $Fic [68.15] < 0.55 * 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

 $= Fic/FS$ $= 68.149/2.0$ $= 34.074 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:

 $= 2 * Fha(t / Do)$ $= 2 * 34072.441(18.0/7248.0)$ $= 169.234 \text{ KPa.}$

Determine the value of [n]:

 $= \sqrt[3]{(2 * Do^{1.5} / (3 * LB * (t)))}$ $= \sqrt[3]{(2 * 7248.01.5 / (3 * 25700.5 * (24.0)))}$ $= 2.000$ Note: Rounded to Nearest Integer ValueBecause $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

 $= 1.5 * Fhe * Ls * Rc^2 * t / (E * (n^2 - 1))$ [eq 6-1] or [4.4.30] $= 1.5 * 69 * 2800.0 * 3774.0^2 * 24.0 / (0.19629E+09 * (2.0^2 - 1))$ $= 12635.812 \text{ cm}^4$ Failed, Inertia less than required !

Factor of Safety [FS]:

Since $Fic [118.62] < 0.55 * 234.4$, $FS = 2$

Engineering Yield to Tensile Stress ratio [R]:

 $= \sigma_{ys} / \sigma_{uts}$ $= 234.4/482.6$ $= 0.486$

Ratio [AE]:

 $= Fhe / E$ $= 178.3/1.9629E+09$ $= 0.000908$

Factor [m1]:

 $= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$ $= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$ $= 0.157$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

 $= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$ $= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$ $= 624.374 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

 $= m2f(m2f2 - R)$ $= 0.6(1.0 - 0.486)$ $= 0.309$

Curve Fitting Constant [A2]:

 $= (\sigma_{uts} * \exp(m2)) / (m2^{m2})$ $= (482.6 * \exp(0.309)) / (0.309^{0.309})$ $= 944.566 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:

 $= 1.5 * R1.5 - 0.5 * R2.5 - R3.5$ $= 1.5 * 0.4861.5 - 0.5 * 0.4862.5 - 0.4863.5$

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.186E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.109E-03
3	180	-3.245	0.670E-08	0.665E-08	0.423E-07	-0.419E-07	0.529E+08	0.251E-02
4	150	-3.948	0.252E-08	0.252E-08	0.281E-07	-0.281E-07	0.980E+08	0.630E-03
5	135	-4.299	0.143E-08	0.143E-08	0.222E-07	-0.222E-07	0.125E+09	0.176E-03
6	128	-4.475	0.106E-08	0.106E-08	0.196E-07	-0.195E-07	0.139E+09	0.172E-04
7	124	-4.563	0.900E-09	0.899E-09	0.183E-07	-0.183E-07	0.145E+09	0.493E-04
8	126	-4.519	0.975E-09	0.975E-09	0.189E-07	-0.189E-07	0.142E+09	0.170E-04
9	127	-4.497	0.101E-08	0.101E-08	0.192E-07	-0.192E-07	0.140E+09	0.143E-06
10	127	-4.486	0.103E-08	0.103E-08	0.194E-07	-0.194E-07	0.139E+09	0.846E-05
11	127	-4.492	0.102E-08	0.102E-08	0.193E-07	-0.193E-07	0.140E+09	0.414E-05
12	127	-4.494	0.102E-08	0.102E-08	0.193E-07	-0.193E-07	0.140E+09	0.200E-05
13	127	-4.496	0.102E-08	0.102E-08	0.193E-07	-0.192E-07	0.140E+09	0.926E-06
14	127	-4.496	0.102E-08	0.102E-08	0.192E-07	-0.192E-07	0.140E+09	0.391E-06
15	127	-4.497	0.102E-08	0.101E-08	0.192E-07	-0.192E-07	0.140E+09	0.124E-06
16	127	-4.497	0.101E-08	0.101E-08	0.192E-07	-0.192E-07	0.140E+09	0.923E-08

Factor of Safety [FS]:

Since Fic [127.27] < 0.55 * 234.4, FS = 2

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

= $0.184 \cdot C_v \cdot M_s 0.8 \cdot t^3 \cdot L_s$ [eq 6-5] or [eq 4.4.43]

= $0.184 \cdot 0.4572 \cdot 10.9630.8 \cdot 18.0^3 \cdot 2800.0$

= 932.901 cm⁴

MDMT Calculations for: Ring R7, Shell to Ring Junction, Curve: D

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -33 °C

User entered Minimum Design Temperature : -12 °C

The Curve for this Material is : D

From Figure 3.7M- Non PWHT, the MDMT is : -28 °C

Temperature reduction per Graph 3.12 : 5 °C

The computed stress ratio is : 0.893

The computed thickness ratio [sr]:

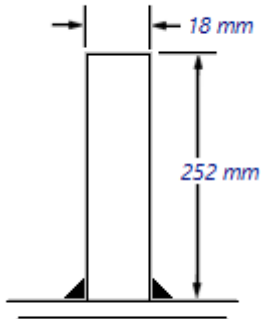
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$

= $(10.718 \cdot 1.0) / (18.0 - 6.0 - 0.0)$

= 0.893

Stiffening Ring Calculations for: Ring R7, SA-516 70 , Bar Ring: 252.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$252.0/18.0 \leq 0.375 \cdot \sqrt{0.19629\text{E}+09/234}$$

$$14.0 \leq 10.852 \quad \text{Failed Geometry Check}$$

Effective Length of Shell: 397 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	71.517	9.0000	643.655
Ring :	45.360	143.9999	6531.828
Total:	116.877		7175.482

Centroid of Ring plus Shell: 61 mm

	Inertia	Distance	A*Dist ²
Shell:	19.310	52.3934	196319.359
Ring :	2400.441	-82.6065	309528.844
Total:	2419.751		505848.188

Available Moment of Inertia, Ring + Shell: 0.748E+04 cm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 2625.005 / \sqrt{3624.0 \cdot 18.0}$$

$$= 10.278$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.278 - 0.579)$$

$$= 0.0949$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.095 \cdot 0.19629\text{E}+09 (18.0/7248.0)$$

$$= 73.989 \text{ N/mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 74.0/234.4$$

$$= 0.316$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

= 0.486

Ratio [AE]:

= Fhe / E

= 74.0/19629E+09

= 0.000377

Factor [m1]:

= (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))

= (ln(0.486) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys}) m1$

= 234.4 (1 + 0.002) / ln(1 + 0.002) 0.157301

= 624.374 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= m2f(m2f2 - R)

= 0.6(1.0 - 0.486)

= 0.309

Curve Fitting Constant [A2]:

= ($\sigma_{uts} \cdot \exp(m2)$) / (m2 m2)

= (482.6 • exp(0.309)) / (0.309 0.309)

= 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= 1.5 • R1.5 - 0.5 • R2.5 - R3.5

= 1.5 • 0.4861.5 - 0.5 • 0.4862.5 - 0.4863.5

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.191E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.423E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.673E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.114E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.166E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.264E-04
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.524E-05
8	73	-5.749	0.537E-10	0.537E-10	0.563E-08	-0.563E-08	0.192E+09	0.557E-05
9	72	-5.771	0.501E-10	0.501E-10	0.547E-08	-0.547E-08	0.192E+09	0.143E-06
10	72	-5.782	0.484E-10	0.484E-10	0.539E-08	-0.539E-08	0.193E+09	0.255E-05
11	72	-5.777	0.493E-10	0.493E-10	0.543E-08	-0.543E-08	0.193E+09	0.121E-05
12	72	-5.774	0.497E-10	0.497E-10	0.545E-08	-0.545E-08	0.193E+09	0.532E-06
13	72	-5.773	0.499E-10	0.499E-10	0.546E-08	-0.546E-08	0.193E+09	0.194E-06
14	72	-5.772	0.500E-10	0.500E-10	0.546E-08	-0.546E-08	0.193E+09	0.256E-07
15	72	-5.772	0.501E-10	0.501E-10	0.546E-08	-0.546E-08	0.192E+09	0.589E-07
16	72	-5.772	0.500E-10	0.500E-10	0.546E-08	-0.546E-08	0.193E+09	0.166E-07
17	72	-5.772	0.500E-10	0.500E-10	0.546E-08	-0.546E-08	0.193E+09	0.446E-08

Factor of Safety [FS]:

Since Fic [72.56] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 72.56/2.0

= 36.280 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 36278.023(18.0/7248.0)

= 180.189 KPa.

Determine the value of [n]:

= $\sqrt[3]{ 2 \cdot Do^{1.5} / (3 \cdot LB \cdot (t)) }$

$$= \sqrt{(2 \cdot 7248.015 / (3 \cdot 25700.5 \cdot (24.0)))}$$

= 2.000 Note: Rounded to Nearest Integer Value

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 74 \cdot 2625.005 \cdot 3750.0^2 \cdot 24.0 / (0.19629E+09 \cdot (2.0^2 - 1))$$

= 12522.160 cm⁴ Failed, Inertia less than required !

Factor of Safety [FS]:

Since $F_{ic} [119.27] < 0.55 \cdot 234.4$, FS = 2

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

= 0.486

Ratio [AE]:

$$= F_{he} / E$$

$$= 184.8 / 1.9629E+09$$

= 0.000941

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

= 624.374 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

= 0.309

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

= 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.186E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.141E-03
3	180	-3.245	0.670E-08	0.665E-08	0.423E-07	-0.419E-07	0.529E+08	0.248E-02
4	150	-3.948	0.252E-08	0.252E-08	0.281E-07	-0.281E-07	0.980E+08	0.597E-03
5	135	-4.299	0.143E-08	0.143E-08	0.222E-07	-0.222E-07	0.125E+09	0.143E-03
6	128	-4.475	0.106E-08	0.106E-08	0.196E-07	-0.195E-07	0.139E+09	0.157E-04
7	131	-4.387	0.123E-08	0.123E-08	0.209E-07	-0.208E-07	0.132E+09	0.589E-04
8	130	-4.431	0.114E-08	0.114E-08	0.202E-07	-0.202E-07	0.135E+09	0.205E-04
9	129	-4.453	0.110E-08	0.110E-08	0.199E-07	-0.199E-07	0.137E+09	0.216E-05
10	128	-4.464	0.108E-08	0.108E-08	0.197E-07	-0.197E-07	0.138E+09	0.683E-05
11	128	-4.459	0.109E-08	0.109E-08	0.198E-07	-0.198E-07	0.137E+09	0.235E-05
12	129	-4.456	0.109E-08	0.109E-08	0.198E-07	-0.198E-07	0.137E+09	0.101E-06
13	129	-4.455	0.110E-08	0.109E-08	0.199E-07	-0.198E-07	0.137E+09	0.103E-05
14	129	-4.455	0.109E-08	0.109E-08	0.198E-07	-0.198E-07	0.137E+09	0.463E-06
15	129	-4.456	0.109E-08	0.109E-08	0.198E-07	-0.198E-07	0.137E+09	0.181E-06
16	129	-4.456	0.109E-08	0.109E-08	0.198E-07	-0.198E-07	0.137E+09	0.399E-07
17	129	-4.456	0.109E-08	0.109E-08	0.198E-07	-0.198E-07	0.137E+09	0.305E-07

Factor of Safety [FS]:
 Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 129.03/234.4$
 $= 1.999$

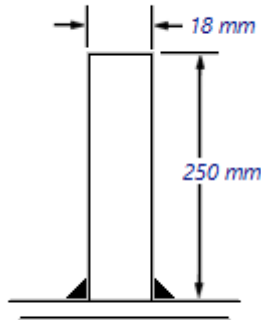
Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:
 $= 0.184 \cdot C_v \cdot M_s 0.8 \cdot t^3 \cdot L_s$ [eq 6-5] or [eq 4.4.43]
 $= 0.184 \cdot 0.4737 \cdot 10.2780.8 \cdot 18.0^3 \cdot 2625.0049$
 $= 860.642 \text{ cm}^4$

MDMT Calculations for: Ring R1, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R1, SA-516 70 , Bar Ring: 250.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:
 $h/t \leq 0.375 \cdot \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]
 $250.0/18.0 \leq 0.375 \cdot \sqrt{0.19629E+09/234}$
 $13.889 \leq 10.852$ *Failed Geometry Check*

Effective Length of Shell: 374 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	59.919	8.0000	479.349
Ring :	45.000	141.0001	6345.012
Total:	104.919		6824.361

Centroid of Ring plus Shell: 65 mm

	Inertia	Distance	A*Dist ²
Shell:	12.783	57.0443	194978.328
Ring :	2343.755	-75.9558	259618.203
Total:	2356.537		454596.531

Available Moment of Inertia, Ring + Shell: 0.690E+04 cm⁴

Determine Parameter [Mx]:
 $= L / \sqrt{(R_o \cdot t)}$
 $= 2499.988 / \sqrt{(3622.0 \cdot 16.0)}$
 $= 10.385$

Compute the Factor [Ch]:

Since $Mx > 1.5$ and $Mx < 13$, $Ch = 0.92 / (Mx - 0.579)$
 $= 0.92 / (10.385 - 0.579)$
 $= 0.0938$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot Ey (t / Do)$
 $= 1.6 \cdot 0.094 \cdot 0.19629E+09 (16.0 / 7244.0)$
 $= 65.085 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / Fy$
 $= 65.1 / 234.4$
 $= 0.278$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4 / 482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 65.1 / 0.19629E+09$
 $= 0.000332$

Factor [m1]:
 $= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(\ln(1 + \epsilon_{psp})) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(\ln(1 + 0.00002)) / \ln(1 + 0.002))$
 $= 0.157$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:
 $= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$
 $= 624.374 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:
 $= m2f(m2f2 - R)$
 $= 0.6 (1.0 - 0.486)$
 $= 0.309$

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
 $= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$
 $= 944.566 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$
 $= 0.346$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.468E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.220E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.159E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.619E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.190E-04
7	64	-5.969	0.258E-10	0.258E-10	0.414E-08	-0.414E-08	0.194E+09	0.170E-05
8	65	-5.925	0.301E-10	0.301E-10	0.442E-08	-0.442E-08	0.194E+09	0.857E-05
9	65	-5.947	0.278E-10	0.278E-10	0.428E-08	-0.428E-08	0.194E+09	0.342E-05
10	64	-5.958	0.268E-10	0.268E-10	0.421E-08	-0.421E-08	0.194E+09	0.857E-06
11	64	-5.964	0.263E-10	0.263E-10	0.417E-08	-0.417E-08	0.194E+09	0.424E-06
12	64	-5.961	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.216E-06
13	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.104E-06
14	64	-5.962	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.562E-07

15	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.238E-07
16	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.162E-07
17	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.378E-08

Factor of Safety [FS]:

Since $F_{ic} [64.41] < 0.55 * 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$\begin{aligned} &= \text{Fic}/\text{FS} \\ &= 64.415/2.0 \\ &= 32.207 \text{ N./mm}^2 \end{aligned}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / Do)$$

$$= 2 \cdot 32205.605(16.0/7244.0)$$

$$= 142.267 \text{ KPa.}$$

Determine the value of $[n]$:

$$= \sqrt{(2 \cdot \text{Do}1.5 / (3 \cdot \text{LB} \cdot (t)))}$$

$$= \sqrt{(2 \cdot 7244.015 / (3 \cdot 25700.5 \cdot (22.0)))}$$

$$= 2.000 \text{ Note: Rounded to Nearest Integer Value}$$

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot \text{Fhe} \cdot \text{Ls} \cdot \text{Rc}^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 65 \cdot 2499.988 \cdot 3747.0^2 \cdot 22.0 / (0.19629 \text{E} + 09 \cdot (2.0^2 - 1))$$

$$= 9310.120 \text{ cm}^4 \text{ Failed, Inertia less than required !}$$

Factor of Safety [FS]:

Since $F_{ic} [117.13] < 0.55 * 234.4$, $FS = 2$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= 163.4 / 1.9629 \times 10^9$$

Factor [m1]:

$$= (\ln(r) + (\text{epsps} - \text{epsys})) / \ln(\ln(1 + \text{epsps}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= 0.6(1.0 - 0.486) = 0.309$$

Curve Fitting Constant [A2]:

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309) = 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.48615 - 0.5 \cdot 0.48625 - 0.48635$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Modulus	Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.187E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.325E-04
3	180	-3.245	0.670E-08	0.665E-08	0.423E-07	-0.419E-07	0.529E+08	0.259E-02
4	150	-3.948	0.252E-08	0.252E-08	0.281E-07	-0.281E-07	0.980E+08	0.706E-03
5	135	-4.299	0.143E-08	0.143E-08	0.222E-07	-0.222E-07	0.125E+09	0.252E-03
6	128	-4.475	0.106E-08	0.106E-08	0.196E-07	-0.195E-07	0.139E+09	0.932E-04
7	124	-4.563	0.900E-09	0.899E-09	0.183E-07	-0.183E-07	0.145E+09	0.268E-04
8	122	-4.607	0.829E-09	0.829E-09	0.177E-07	-0.177E-07	0.148E+09	0.369E-05
9	123	-4.585	0.864E-09	0.863E-09	0.180E-07	-0.180E-07	0.146E+09	0.113E-04
10	123	-4.596	0.846E-09	0.846E-09	0.178E-07	-0.178E-07	0.147E+09	0.377E-05
11	122	-4.601	0.837E-09	0.837E-09	0.178E-07	-0.177E-07	0.148E+09	0.266E-07
12	122	-4.604	0.833E-09	0.833E-09	0.177E-07	-0.177E-07	0.148E+09	0.184E-05
13	122	-4.603	0.835E-09	0.835E-09	0.177E-07	-0.177E-07	0.148E+09	0.905E-06
14	122	-4.602	0.836E-09	0.836E-09	0.177E-07	-0.177E-07	0.148E+09	0.440E-06
15	122	-4.602	0.837E-09	0.837E-09	0.177E-07	-0.177E-07	0.148E+09	0.207E-06
16	122	-4.602	0.837E-09	0.837E-09	0.177E-07	-0.177E-07	0.148E+09	0.900E-07
17	122	-4.602	0.837E-09	0.837E-09	0.177E-07	-0.177E-07	0.148E+09	0.317E-07

Factor of Safety [FS]:

Since $Fic [122.78] < 0.55 * 234.4$, $FS = 2$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls \quad [\text{eq 6-5}] \text{ or } [\text{eq 4.4.43}]$$

$$= 0.184 \cdot 0.471 \cdot 10.3850.8 \cdot 16.0^3 \cdot 2499.9875$$

$$= 577.134 \text{ cm}^4$$

MDMT Calculations for: Ring R5, Shell to Ring Junction, Curve: D

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -35 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -28 °C
 Temperature reduction per Graph 3.12 : 7 °C
 The computed stress ratio is : 0.838

The computed thickness ratio [sr]:

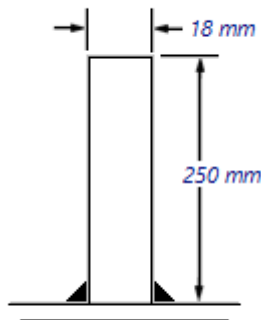
$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (10.053 \cdot 1.0) / (18.0 - 6.0 - 0.0)$$

$$= 0.838$$

Stiffening Ring Calculations for: Ring R5, SA-516 70 , Bar Ring: 250.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:
 $h/t \leq 0.375 \cdot \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]

$250.0/18.0 \leq 0.375 * \sqrt{0.19629E+09/234}$
 $13.889 \leq 10.852$ Failed Geometry Check

Effective Length of Shell: 374 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	59.919	8.0000	479.349
Ring :	45.000	141.0001	6345.012
Total:	104.919		6824.361

Centroid of Ring plus Shell: 65 mm

	Inertia	Distance	A*Dist ²
Shell:	12.783	57.0443	194978.328
Ring :	2343.755	-75.9558	259618.203
Total:	2356.537		454596.531

Available Moment of Inertia, Ring + Shell: 0.690E+04 cm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2499.983 / \sqrt{(3622.0 \cdot 16.0)}$$

$$= 10.385$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.385 - 0.579)$$

$$= 0.0938$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.094 \cdot 0.19629E+09 (16.0/7244.0)$$

$$= 65.086 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 65.1/234.4$$

$$= 0.278$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 65.1/0.19629E+09$$

$$= 0.000332$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2')$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.468E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.220E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.159E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.619E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.190E-04
7	64	-5.969	0.258E-10	0.258E-10	0.414E-08	-0.414E-08	0.194E+09	0.170E-05
8	65	-5.925	0.301E-10	0.301E-10	0.442E-08	-0.442E-08	0.194E+09	0.857E-05
9	65	-5.947	0.278E-10	0.278E-10	0.428E-08	-0.428E-08	0.194E+09	0.342E-05
10	64	-5.958	0.268E-10	0.268E-10	0.421E-08	-0.421E-08	0.194E+09	0.856E-06
11	64	-5.964	0.263E-10	0.263E-10	0.417E-08	-0.417E-08	0.194E+09	0.424E-06
12	64	-5.961	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.216E-06
13	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.104E-06
14	64	-5.962	0.265E-10	0.265E-10	0.419E-08	-0.419E-08	0.194E+09	0.556E-07
15	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.244E-07
16	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.156E-07
17	64	-5.962	0.264E-10	0.264E-10	0.418E-08	-0.418E-08	0.194E+09	0.436E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [64.41] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 64.415/2.0$$

$$= 32.207 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 32205.605 (16.0/7244.0)$$

$$= 142.267 \text{ KPa.}$$

Determine the value of [n]:

$$= \sqrt[3]{(2 \cdot D_o^{1.5} / (3 \cdot L_B \cdot (t)))}$$

$$= \sqrt[3]{(2 \cdot 7244.01.5 / (3 \cdot 25700.5 \cdot (22.0)))}$$

$$= 2.000 \text{ Note: Rounded to Nearest Integer Value}$$

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 65 \cdot 2499.983 \cdot 3747.0^2 \cdot 22.0 / (0.19629E+09 \cdot (2.0^2 - 1))$$

$$= 9310.123 \text{ cm}^4 \text{ Failed, Inertia less than required !}$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [117.13] < 0.55 \cdot 234.4, FS = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 163.4/1.9629E+09$$

$$= 0.000832$$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ N/mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N/mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.187E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.325E-04
3	180	-3.245	0.670E-08	0.665E-08	0.423E-07	-0.419E-07	0.529E+08	0.259E-02
4	150	-3.948	0.252E-08	0.252E-08	0.281E-07	-0.281E-07	0.980E+08	0.706E-03
5	135	-4.299	0.143E-08	0.143E-08	0.222E-07	-0.222E-07	0.125E+09	0.252E-03
6	128	-4.475	0.106E-08	0.106E-08	0.196E-07	-0.195E-07	0.139E+09	0.932E-04
7	124	-4.563	0.900E-09	0.899E-09	0.183E-07	-0.183E-07	0.145E+09	0.268E-04
8	122	-4.607	0.829E-09	0.829E-09	0.177E-07	-0.177E-07	0.148E+09	0.369E-05
9	123	-4.585	0.864E-09	0.863E-09	0.180E-07	-0.180E-07	0.146E+09	0.113E-04
10	123	-4.596	0.846E-09	0.846E-09	0.178E-07	-0.178E-07	0.147E+09	0.377E-05
11	122	-4.601	0.837E-09	0.837E-09	0.178E-07	-0.177E-07	0.148E+09	0.259E-07
12	122	-4.604	0.833E-09	0.833E-09	0.177E-07	-0.177E-07	0.148E+09	0.184E-05
13	122	-4.603	0.835E-09	0.835E-09	0.177E-07	-0.177E-07	0.148E+09	0.906E-06
14	122	-4.602	0.836E-09	0.836E-09	0.177E-07	-0.177E-07	0.148E+09	0.440E-06
15	122	-4.602	0.837E-09	0.837E-09	0.177E-07	-0.177E-07	0.148E+09	0.207E-06
16	122	-4.602	0.837E-09	0.837E-09	0.177E-07	-0.177E-07	0.148E+09	0.907E-07
17	122	-4.602	0.837E-09	0.837E-09	0.177E-07	-0.177E-07	0.148E+09	0.324E-07

Factor of Safety [FS]:
 Since $Fic [122.78] < 0.55 \cdot 234.4$, FS = 2

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot C_v \cdot M_s 0.8 \cdot t^3 \cdot L_s \text{ [eq 6-5] or [eq 4.4.43]}$$

$$= 0.184 \cdot 0.471 \cdot 10.3850.8 \cdot 16.0^3 \cdot 2499.9834$$

$$= 577.133 \text{ cm}^4$$

MDMT Calculations for: Ring R2, Shell to Ring Junction, Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -35 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -28 °C
 Temperature reduction per Graph 3.12 : 7 °C
 The computed stress ratio is : 0.838

The computed thickness ratio [sr]:

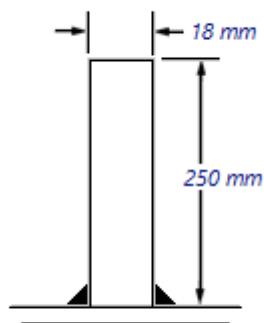
$$= (tr \cdot E^*) / (t_{nom} - ci - c_{ext})$$

$$= (10.053 \cdot 1.0) / (18.0 - 6.0 - 0.0)$$

$$= 0.838$$

Stiffening Ring Calculations for: Ring R2, SA-516 70 , Bar Ring: 250.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$250.0/18.0 \leq 0.375 \cdot \sqrt{0.19629\text{E}+09/234}$$

$$13.889 \leq 10.852 \quad \text{Failed Geometry Check}$$

Effective Length of Shell: 350 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	49.029	7.0000	343.204
Ring :	45.000	139.0001	6255.012
Total:	94.029		6598.215

Centroid of Ring plus Shell: 70 mm

	Inertia	Distance	A*Dist ²
Shell:	8.008	63.1720	195660.656
Ring :	2343.755	-68.8281	213179.078
Total:	2351.763		408839.750

Available Moment of Inertia, Ring + Shell: 0.644E+04 cm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 2500.0 / \sqrt{3620.0 \cdot 14.0}$$

$$= 11.105$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, \text{ Ch} = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (11.105 - 0.579)$$

$$= 0.0874$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.087 \cdot 0.19629\text{E}+09 (14.0/7240.0)$$

$$= 53.083 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 53.1/234.4$$

$$= 0.226$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 53.1/19629E+09$$

$$= 0.000270$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.529E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.392E-04
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.117E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.395E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.530E-06
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.192E-04
8	54	-6.189	0.110E-10	0.110E-10	0.290E-08	-0.290E-08	0.195E+09	0.932E-05
9	53	-6.211	0.100E-10	0.100E-10	0.279E-08	-0.279E-08	0.196E+09	0.439E-05
10	53	-6.222	0.954E-11	0.954E-11	0.273E-08	-0.273E-08	0.196E+09	0.193E-05
11	53	-6.227	0.932E-11	0.932E-11	0.271E-08	-0.271E-08	0.196E+09	0.698E-06
12	52	-6.230	0.921E-11	0.921E-11	0.269E-08	-0.269E-08	0.196E+09	0.843E-07
13	52	-6.231	0.916E-11	0.916E-11	0.269E-08	-0.269E-08	0.196E+09	0.223E-06
14	52	-6.231	0.918E-11	0.918E-11	0.269E-08	-0.269E-08	0.196E+09	0.692E-07
15	52	-6.230	0.920E-11	0.920E-11	0.269E-08	-0.269E-08	0.196E+09	0.751E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [52.89] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 52.893/2.0$$

$$= 26.446 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

$$= 2 \cdot 26444.939(14.0/7240.0)$$

$$= 102.273 \text{ KPa.}$$

Determine the value of [n]:

$$= \sqrt[3]{2 \cdot Do^{1.5} / (3 \cdot LB \cdot (t))}$$

$$= \sqrt[3]{2 \cdot 7240.015 / (3 \cdot 25700.5 \cdot (20.0))}$$

= 2.000 Note: Rounded to Nearest Integer Value

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot Fhe \cdot Ls \cdot Rc^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 53 \cdot 2500.0 \cdot 3745.0^2 \cdot 20.0 / (0.19629E+09 \cdot (2.0^2 - 1))$$

= 6637.001 cm⁴ Failed, Inertia less than required !

Factor of Safety [FS]:

$$\text{Since } Fic [114.53] < 0.55 \cdot 234.4, FS = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

= 0.486

Ratio [AE]:

$$= Fhe / E$$

$$= 137.4 / 0.19629E+09$$

= 0.000700

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

= 624.374 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

= 0.309

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

= 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.188E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.997E-04
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.390E-03
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.209E-03
5	105	-5.002	0.373E-09	0.373E-09	0.127E-07	-0.127E-07	0.171E+09	0.831E-04
6	113	-4.827	0.540E-09	0.540E-09	0.148E-07	-0.148E-07	0.162E+09	0.111E-05
7	116	-4.739	0.644E-09	0.643E-09	0.159E-07	-0.159E-07	0.157E+09	0.466E-04
8	115	-4.783	0.590E-09	0.590E-09	0.153E-07	-0.153E-07	0.159E+09	0.221E-04
9	114	-4.805	0.564E-09	0.564E-09	0.150E-07	-0.150E-07	0.161E+09	0.103E-04
10	113	-4.816	0.552E-09	0.552E-09	0.149E-07	-0.149E-07	0.161E+09	0.458E-05
11	113	-4.821	0.546E-09	0.546E-09	0.148E-07	-0.148E-07	0.162E+09	0.172E-05
12	113	-4.824	0.543E-09	0.543E-09	0.148E-07	-0.148E-07	0.162E+09	0.303E-06
13	113	-4.825	0.541E-09	0.541E-09	0.148E-07	-0.148E-07	0.162E+09	0.406E-06
14	113	-4.825	0.542E-09	0.542E-09	0.148E-07	-0.148E-07	0.162E+09	0.517E-07



15	113	-4.824	0.543E-09	0.542E-09	0.148E-07	-0.148E-07	0.162E+09	0.126E-06
16	113	-4.824	0.542E-09	0.542E-09	0.148E-07	-0.148E-07	0.162E+09	0.370E-07
17	113	-4.825	0.542E-09	0.542E-09	0.148E-07	-0.148E-07	0.162E+09	0.735E-08

Factor of Safety [FS]:

Since $Fic [113.21] < 0.55 * 234.4$, $FS = 2$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls$ [eq 6-5] or [eq 4.4.43]

$= 0.184 \cdot 0.454 \cdot 11.1050.8 \cdot 14.0^3 \cdot 2500.0002$

$= 393.196 \text{ cm}^4$

MDMT Calculations for: Ring R4, Shell to Ring Junction, Curve: D

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -45 °C

User entered Minimum Design Temperature : -12 °C

The Curve for this Material is : D

From Figure 3.7M- Non PWHT, the MDMT is : -28 °C

Temperature reduction per Graph 3.12 : 17 °C

The computed stress ratio is : 0.591

The computed thickness ratio [sr]:

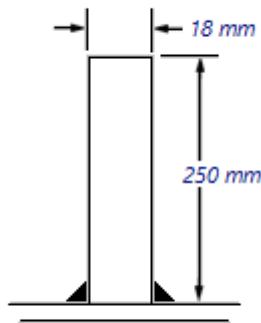
$= (tr \cdot E^*) / (tnom - ci - cext)$

$= (11.823 \cdot 0.85) / (18.0 - 1.0 - 0.0)$

$= 0.591$

Stiffening Ring Calculations for: Ring R4, SA-516 70 , Bar Ring: 250.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$h/t \leq 0.375 \cdot \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]

$250.0/18.0 \leq 0.375 \cdot \sqrt{0.19629E+09/234}$

$13.889 \leq 10.852$ Failed Geometry Check

Effective Length of Shell:

386 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	65.614	8.5000	557.718
Ring :	45.000	141.9999	6389.993
Total:	110.614		6947.711

Centroid of Ring plus Shell:

63 mm

Inertia	Distance	A*Dist ²
---------	----------	---------------------

Shell:	15.802	54.3105	193536.719
Ring :	2343.744	-79.1894	282193.125
Total:	2359.546		475729.844

Available Moment of Inertia, Ring + Shell: 0.712E+04 cm⁴

Determine Parameter [Mx]:
 $= L / \sqrt{(Ro \cdot t)}$
 $= 2500.001 / \sqrt{(3621.0 \cdot 17.0)}$
 $= 10.076$

Compute the Factor [Ch]:
 Since $Mx > 1.5$ and $Mx < 13$, $Ch = 0.92 / (Mx - 0.579)$
 $= 0.92 / (10.076 - 0.579)$
 $= 0.0969$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot Ey(t / Do)$
 $= 1.6 \cdot 0.097 \cdot 0.19629E+09(17.0/7242.0)$
 $= 71.420 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / Fy$
 $= 71.4/234.4$
 $= 0.305$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 234.4/482.6$
 $= 0.486$

Ratio [AE]:
 $= Fhe / E$
 $= 71.4/0.19629E+09$
 $= 0.000364$

Factor [m1]:
 $= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
 $= 0.157$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:
 $= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$
 $= 624.374 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:
 $= m2f / (m2f^2 - R)$
 $= 0.6 / (1.0 - 0.486)$
 $= 0.309$

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
 $= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$
 $= 944.566 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$
 $= 0.346$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.436E-03



3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.542E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.127E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.296E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.133E-04
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.785E-05
8	69	-5.837	0.405E-10	0.405E-10	0.500E-08	-0.500E-08	0.193E+09	0.280E-05
9	70	-5.815	0.435E-10	0.435E-10	0.515E-08	-0.515E-08	0.193E+09	0.250E-05
10	70	-5.826	0.420E-10	0.420E-10	0.508E-08	-0.508E-08	0.193E+09	0.156E-06
11	70	-5.821	0.428E-10	0.428E-10	0.512E-08	-0.512E-08	0.193E+09	0.117E-05
12	70	-5.824	0.424E-10	0.424E-10	0.510E-08	-0.510E-08	0.193E+09	0.508E-06
13	70	-5.825	0.422E-10	0.422E-10	0.509E-08	-0.509E-08	0.193E+09	0.176E-06
14	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.100E-07
15	70	-5.826	0.420E-10	0.420E-10	0.508E-08	-0.508E-08	0.193E+09	0.729E-07
16	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.315E-07
17	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.107E-07

Factor of Safety [FS]:

Since $Fic [70.26] < 0.55 * 234.4$, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS = $70.255/2.0$ = 35.128 N./mm^2

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha(t / Do)$ = $2 * 35125.523(17.0/7242.0)$ = 164.909 KPa .

Determine the value of [n]:

= $\sqrt[3]{(2 * Do^{1.5} / (3 * LB * (t)))}$ = $\sqrt[3]{(2 * 7242.015 / (3 * 25700.5 * (18.0)))}$

= 2.000 Note: Rounded to Nearest Integer Value

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

= $1.5 * Fhe * Ls * Rc^2 * t / (E * (n^2 - 1))$ [eq 6-1] or [4.4.30]= $1.5 * 71 * 2500.001 * 3746.0^2 * 18.0 / (0.19629E+09 * (2.0^2 - 1))$ = 10849.061 cm^4 Failed, Inertia less than required !

Factor of Safety [FS]:

Since $Fic [118.45] < 0.55 * 234.4$, FS = 2

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$ = $234.4/482.6$

= 0.486

Ratio [AE]:

= Fhe / E = $176.6/1.9629E+09$

= 0.000900

Factor [m1]:

= $(\ln(r) + (epsys - epsys)) / \ln(\ln(1 + epsp) / \ln(1 + epsys))$ = $(\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + epsys) / \ln(1 + epsys) m1$ = $234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301$ = 624.374 N./mm^2

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f2 - R)$ = $0.6(1.0 - 0.486)$

= 0.309

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2')$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.566 \text{ N/mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.186E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.998E-04
3	180	-3.245	0.670E-08	0.665E-08	0.423E-07	-0.419E-07	0.529E+08	0.252E-02
4	150	-3.948	0.252E-08	0.252E-08	0.281E-07	-0.281E-07	0.980E+08	0.639E-03
5	135	-4.299	0.143E-08	0.143E-08	0.222E-07	-0.222E-07	0.125E+09	0.184E-03
6	128	-4.475	0.106E-08	0.106E-08	0.196E-07	-0.195E-07	0.139E+09	0.260E-04
7	124	-4.563	0.900E-09	0.899E-09	0.183E-07	-0.183E-07	0.145E+09	0.404E-04
8	126	-4.519	0.975E-09	0.975E-09	0.189E-07	-0.189E-07	0.142E+09	0.817E-05
9	127	-4.497	0.101E-08	0.101E-08	0.192E-07	-0.192E-07	0.140E+09	0.867E-05
10	126	-4.508	0.995E-09	0.994E-09	0.191E-07	-0.191E-07	0.141E+09	0.192E-06
11	126	-4.514	0.985E-09	0.984E-09	0.190E-07	-0.190E-07	0.141E+09	0.400E-05
12	126	-4.511	0.990E-09	0.989E-09	0.190E-07	-0.190E-07	0.141E+09	0.191E-05
13	126	-4.509	0.992E-09	0.992E-09	0.191E-07	-0.190E-07	0.141E+09	0.859E-06
14	126	-4.509	0.993E-09	0.993E-09	0.191E-07	-0.191E-07	0.141E+09	0.334E-06
15	126	-4.508	0.994E-09	0.994E-09	0.191E-07	-0.191E-07	0.141E+09	0.708E-07
16	126	-4.508	0.994E-09	0.994E-09	0.191E-07	-0.191E-07	0.141E+09	0.607E-07
17	126	-4.508	0.994E-09	0.994E-09	0.191E-07	-0.191E-07	0.141E+09	0.503E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [126.78] < 0.55 \cdot 234.4, \text{ FS} = 2$$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot C_v \cdot M_s 0.8 \cdot t^3 \cdot L_s \quad [\text{eq 6-5}] \text{ or } [\text{eq 4.4.43}]$$

$$= 0.184 \cdot 0.479 \cdot 10.0760.8 \cdot 17.0^3 \cdot 2500.0015$$

$$= 687.187 \text{ cm}^4$$

MDMT Calculations for: Ring R3, Shell to Ring Junction, Curve: D

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -45 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -28 °C
 Temperature reduction per Graph 3.12 : 17 °C
 The computed stress ratio is : 0.591

The computed thickness ratio [sr]:

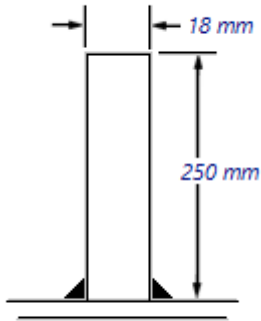
$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (11.823 \cdot 0.85) / (18.0 - 1.0 - 0.0)$$

$$= 0.591$$

Stiffening Ring Calculations for: Ring R3, SA-516 70 , Bar Ring: 250.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$250.0/18.0 \leq 0.375 \cdot \sqrt{0.19629\text{E}+09/234}$$

$$13.889 \leq 10.852 \quad \text{Failed Geometry Check}$$

Effective Length of Shell: 386 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	65.614	8.5000	557.718
Ring :	45.000	141.9999	6389.993
Total:	110.614		6947.711

Centroid of Ring plus Shell: 63 mm

	Inertia	Distance	A*Dist ²
Shell:	15.802	54.3105	193536.719
Ring :	2343.744	-79.1894	282193.125
Total:	2359.546		475729.844

Available Moment of Inertia, Ring + Shell: 0.712E+04 cm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 2500.002 / \sqrt{3621.0 \cdot 17.0}$$

$$= 10.076$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, \text{ Ch} = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.076 - 0.579)$$

$$= 0.0969$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.097 \cdot 0.19629\text{E}+09 (17.0/7242.0)$$

$$= 71.420 \text{ N/mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 71.4/234.4$$

$$= 0.305$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

= 0.486

Ratio [AE]:

= Fhe / E

= 71.4/19629E+09

= 0.000364

Factor [m1]:

= (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))

= (ln(0.486) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$

= 234.4(1 + 0.002) / ln(1 + 0.002) 0.157301

= 624.374 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= m2f(m2f² - R)

= 0.6(1.0 - 0.486)

= 0.309

Curve Fitting Constant [A2]:

= ($\sigma_{uts} \cdot \exp(m2)$) / (m2 m2)

= (482.6 • exp(0.309)) / (0.309 0.309)

= 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= 1.5 • R1.5 - 0.5 • R2.5 - R3.5

= 1.5 • 0.4861.5 - 0.5 • 0.4862.5 - 0.4863.5

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.436E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.542E-04
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.127E-03
5	75	-5.706	0.615E-10	0.615E-10	0.596E-08	-0.596E-08	0.192E+09	0.296E-04
6	67	-5.881	0.350E-10	0.350E-10	0.470E-08	-0.470E-08	0.194E+09	0.133E-04
7	71	-5.793	0.467E-10	0.467E-10	0.531E-08	-0.531E-08	0.193E+09	0.785E-05
8	69	-5.837	0.405E-10	0.405E-10	0.500E-08	-0.500E-08	0.193E+09	0.280E-05
9	70	-5.815	0.435E-10	0.435E-10	0.515E-08	-0.515E-08	0.193E+09	0.250E-05
10	70	-5.826	0.420E-10	0.420E-10	0.508E-08	-0.508E-08	0.193E+09	0.156E-06
11	70	-5.821	0.428E-10	0.428E-10	0.512E-08	-0.512E-08	0.193E+09	0.117E-05
12	70	-5.824	0.424E-10	0.424E-10	0.510E-08	-0.510E-08	0.193E+09	0.508E-06
13	70	-5.825	0.422E-10	0.422E-10	0.509E-08	-0.509E-08	0.193E+09	0.176E-06
14	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.102E-07
15	70	-5.826	0.420E-10	0.420E-10	0.508E-08	-0.508E-08	0.193E+09	0.727E-07
16	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.313E-07
17	70	-5.826	0.421E-10	0.421E-10	0.508E-08	-0.508E-08	0.193E+09	0.105E-07

Factor of Safety [FS]:

Since Fic [70.26] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 70.255/2.0

= 35.128 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 35125.523(17.0/7242.0)

= 164.909 KPa.

Determine the value of [n]:

= $\sqrt[3]{2 \cdot Do^{1.5} / (3 \cdot LB \cdot (t))}$

$$= \sqrt{(2 \cdot 7242.015 / (3 \cdot 25700.5 \cdot (18.0)))}$$

= 2.000 Note: Rounded to Nearest Integer Value

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 71 \cdot 2500.002 \cdot 3746.0^2 \cdot 18.0 / (0.19629E+09 \cdot (2.0^2 - 1))$$

= 10849.059 cm⁴ Failed, Inertia less than required !

Factor of Safety [FS]:

Since $F_{ic} [118.45] < 0.55 \cdot 234.4$, FS = 2

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

= 0.486

Ratio [AE]:

$$= F_{he} / E$$

$$= 176.6 / 0.19629E+09$$

= 0.000900

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

= 624.374 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

= 0.309

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

= 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.186E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.998E-04
3	180	-3.245	0.670E-08	0.665E-08	0.423E-07	-0.419E-07	0.529E+08	0.252E-02
4	150	-3.948	0.252E-08	0.252E-08	0.281E-07	-0.281E-07	0.980E+08	0.639E-03
5	135	-4.299	0.143E-08	0.143E-08	0.222E-07	-0.222E-07	0.125E+09	0.184E-03
6	128	-4.475	0.106E-08	0.106E-08	0.196E-07	-0.195E-07	0.139E+09	0.260E-04
7	124	-4.563	0.900E-09	0.899E-09	0.183E-07	-0.183E-07	0.145E+09	0.404E-04
8	126	-4.519	0.975E-09	0.975E-09	0.189E-07	-0.189E-07	0.142E+09	0.817E-05
9	127	-4.497	0.101E-08	0.101E-08	0.192E-07	-0.192E-07	0.140E+09	0.867E-05
10	126	-4.508	0.995E-09	0.994E-09	0.191E-07	-0.191E-07	0.141E+09	0.192E-06
11	126	-4.514	0.985E-09	0.984E-09	0.190E-07	-0.190E-07	0.141E+09	0.400E-05
12	126	-4.511	0.990E-09	0.989E-09	0.190E-07	-0.190E-07	0.141E+09	0.191E-05
13	126	-4.509	0.992E-09	0.992E-09	0.191E-07	-0.190E-07	0.141E+09	0.859E-06
14	126	-4.509	0.993E-09	0.993E-09	0.191E-07	-0.191E-07	0.141E+09	0.334E-06
15	126	-4.508	0.994E-09	0.994E-09	0.191E-07	-0.191E-07	0.141E+09	0.707E-07
16	126	-4.508	0.994E-09	0.994E-09	0.191E-07	-0.191E-07	0.141E+09	0.608E-07
17	126	-4.508	0.994E-09	0.994E-09	0.191E-07	-0.191E-07	0.141E+09	0.497E-08

Factor of Safety [FS]:

Since $Fic [126.78] < 0.55 * 234.4$, $FS = 2$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls$ [eq 6-5] or [eq 4.4.43]

$= 0.184 \cdot 0.479 \cdot 10.0760.8 \cdot 17.0^3 \cdot 2500.002$

$= 687.187 \text{ cm}^4$

MDMT Calculations for: Ring R6, Shell to Ring Junction, Curve: D

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -44 °C

User entered Minimum Design Temperature : -12 °C

The Curve for this Material is : D

From Figure 3.7M- Non PWHT, the MDMT is : -30 °C

Temperature reduction per Graph 3.12 : 13 °C

The computed stress ratio is : 0.670

The computed thickness ratio [sr]:

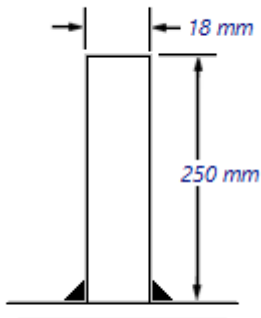
$= (tr \cdot E^*) / (tnom - ci - cext)$

$= (11.823 \cdot 0.85) / (16.0 - 1.0 - 0.0)$

$= 0.670$

Stiffening Ring Calculations for: Ring R6, SA-516 70 , Bar Ring: 250.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$h/t \leq 0.375 \cdot \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]

$250.0/18.0 \leq 0.375 \cdot \sqrt{0.19629E+09/234}$

$13.889 \leq 10.852$ **Failed Geometry Check**

Effective Length of Shell:

362 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	54.367	7.5000	407.756
Ring :	45.000	139.9999	6299.993
Total:	99.367		6707.749

Centroid of Ring plus Shell:

68 mm

	Inertia	Distance	A*Dist ²
Shell:	10.194	60.0045	195752.234
Ring :	2343.744	-72.4954	236501.188
Total:	2353.938		432253.438

Available Moment of Inertia, Ring + Shell: 0.668E+04 cm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.0 / \sqrt{(3619.0 \cdot 15.0)}$$

$$= 10.730$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (10.73 - 0.579)$$

$$= 0.0906$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.091 \cdot 0.19629E+09 (15.0 / 7238.0)$$

$$= 58.992 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 59.0 / 234.4$$

$$= 0.252$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.0 / 0.19629E+09$$

$$= 0.000301$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.499E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.907E-05
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.147E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.697E-04



6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.306E-04
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.109E-04
8	58	-6.101	0.157E-10	0.157E-10	0.336E-08	-0.336E-08	0.195E+09	0.944E-06
9	59	-6.079	0.171E-10	0.171E-10	0.349E-08	-0.349E-08	0.195E+09	0.405E-05
10	58	-6.090	0.164E-10	0.164E-10	0.343E-08	-0.343E-08	0.195E+09	0.155E-05
11	58	-6.096	0.160E-10	0.160E-10	0.340E-08	-0.340E-08	0.195E+09	0.304E-06
12	58	-6.098	0.159E-10	0.159E-10	0.338E-08	-0.338E-08	0.195E+09	0.320E-06
13	58	-6.097	0.160E-10	0.160E-10	0.339E-08	-0.339E-08	0.195E+09	0.830E-08

Factor of Safety [FS]:

Since Fic [58.62] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 58.623/2.0

= 29.311 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 * Fha(t / Do)

= 2 * 29309.625(15.0/7238.0)

= 121.482 KPa.

Determine the value of [n]:

$$= \sqrt[3]{(2 \cdot Do^{1.5} / (3 \cdot LB \cdot (t)))}$$
$$= \sqrt[3]{(2 \cdot 7238.01.5 / (3 \cdot 25700.5 \cdot (16.0)))}$$

= 2.000 Note: Rounded to Nearest Integer Value

Because n < 2, set n = 2

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

= 1.5 * Fhe * Ls * Rc² * t / (E * (n² - 1)) [eq 6-1] or [4.4.30]= 1.5 * 59 * 2500.0 * 3744.0² * 16.0 / (0.19629E+09 * (2.0² - 1))**= 7898.487 cm⁴ Failed, Inertia less than required !**

Factor of Safety [FS]:

Since Fic [115.85] < 0.55 * 234.4, FS = 2

Engineering Yield to Tensile Stress ratio [R]:

= σ_{ys} / σ_{uts}

= 234.4/482.6

= 0.486

Ratio [AE]:

= Fhe / E

= 150.6/1.9629E+09

= 0.000767

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / (\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

= 234.4 (1 + 0.002) / ln(1 + 0.002) 0.157301

= 624.374 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= m2f(m2f² - R)

= 0.6(1.0 - 0.486)

= 0.309

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

= (482.6 * exp(0.309)) / (0.309 0.309)

= 944.566 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= 1.5 * R1.5 - 0.5 * R2.5 - R3.5

= 1.5 * 0.4861.5 - 0.5 * 0.4862.5 - 0.4863.5

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.188E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.328E-04
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.457E-03
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.276E-03
5	105	-5.002	0.373E-09	0.373E-09	0.127E-07	-0.127E-07	0.171E+09	0.150E-03
6	113	-4.827	0.540E-09	0.540E-09	0.148E-07	-0.148E-07	0.162E+09	0.680E-04
7	116	-4.739	0.644E-09	0.643E-09	0.159E-07	-0.159E-07	0.157E+09	0.203E-04
8	118	-4.695	0.701E-09	0.701E-09	0.165E-07	-0.165E-07	0.154E+09	0.552E-05
9	117	-4.717	0.672E-09	0.672E-09	0.162E-07	-0.162E-07	0.155E+09	0.757E-05
10	118	-4.706	0.686E-09	0.686E-09	0.163E-07	-0.163E-07	0.154E+09	0.107E-05
11	118	-4.700	0.694E-09	0.693E-09	0.164E-07	-0.164E-07	0.154E+09	0.222E-05
12	118	-4.703	0.690E-09	0.690E-09	0.164E-07	-0.164E-07	0.154E+09	0.574E-06
13	118	-4.704	0.688E-09	0.688E-09	0.164E-07	-0.163E-07	0.154E+09	0.246E-06
14	118	-4.704	0.689E-09	0.689E-09	0.164E-07	-0.164E-07	0.154E+09	0.164E-06
15	118	-4.704	0.689E-09	0.688E-09	0.164E-07	-0.163E-07	0.154E+09	0.414E-07
16	118	-4.704	0.689E-09	0.689E-09	0.164E-07	-0.164E-07	0.154E+09	0.611E-07
17	118	-4.704	0.689E-09	0.689E-09	0.164E-07	-0.163E-07	0.154E+09	0.983E-08

Factor of Safety [FS]:

Since Fic [118.38] < 0.55 * 234.4, FS = 2

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

= 0.184 * Cv * Ms0.8 * t³ * Ls [eq 6-5] or [eq 4.4.43]

= 0.184 * 0.4626 * 10.730.8 * 15.0³ * 2500.0

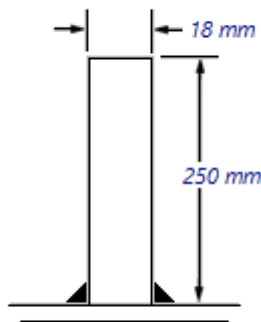
= 479.424 cm⁴

MDMT Calculations for: Ring R9, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R9, SA-516 70 , Bar Ring: 250.0 x 18.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$h/t \leq 0.375 * \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]

$250.0/18.0 \leq 0.375 * \sqrt{0.19629E+09/234}$

13.889 <= 10.852 Failed Geometry Check

Effective Length of Shell:

362 mm

Area (cm²) Distance (mm) Area*Dist

Shell:	54.367	7.5000	407.756
Ring :	45.000	139.9999	6299.993
Total:	99.367		6707.749

Centroid of Ring plus Shell: 68 mm

	Inertia	Distance	A*Dist ²
Shell:	10.194	60.0045	195752.234
Ring :	2343.744	-72.4954	236501.188
Total:	2353.938		432253.438

Available Moment of Inertia, Ring + Shell: 0.668E+04 cm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2550.26 / \sqrt{(3619.0 \cdot 15.0)}$$

$$= 10.946$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.946 - 0.579)$$

$$= 0.0887$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.089 \cdot 0.19629E+09 (15.0 / 7238.0)$$

$$= 57.765 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 57.8 / 234.4$$

$$= 0.246$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 57.8 / 0.19629E+09$$

$$= 0.000294$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.192E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.505E-03
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.153E-04
4	30	-6.760	0.454E-12	0.454E-12	0.765E-09	-0.765E-09	0.196E+09	0.141E-03
5	45	-6.409	0.398E-11	0.398E-11	0.190E-08	-0.190E-08	0.196E+09	0.634E-04
6	52	-6.233	0.910E-11	0.910E-11	0.268E-08	-0.268E-08	0.196E+09	0.244E-04
7	56	-6.145	0.132E-10	0.132E-10	0.313E-08	-0.313E-08	0.195E+09	0.464E-05
8	58	-6.101	0.157E-10	0.157E-10	0.336E-08	-0.336E-08	0.195E+09	0.531E-05
9	57	-6.123	0.144E-10	0.144E-10	0.324E-08	-0.324E-08	0.195E+09	0.326E-06
10	57	-6.134	0.138E-10	0.138E-10	0.319E-08	-0.318E-08	0.195E+09	0.216E-05
11	57	-6.128	0.141E-10	0.141E-10	0.321E-08	-0.321E-08	0.195E+09	0.917E-06
12	57	-6.126	0.142E-10	0.142E-10	0.323E-08	-0.323E-08	0.195E+09	0.296E-06
13	57	-6.124	0.143E-10	0.143E-10	0.324E-08	-0.324E-08	0.195E+09	0.153E-07
14	57	-6.125	0.143E-10	0.143E-10	0.323E-08	-0.323E-08	0.195E+09	0.140E-06
15	57	-6.125	0.143E-10	0.143E-10	0.323E-08	-0.323E-08	0.195E+09	0.624E-07
16	57	-6.125	0.143E-10	0.143E-10	0.324E-08	-0.324E-08	0.195E+09	0.236E-07
17	57	-6.124	0.143E-10	0.143E-10	0.324E-08	-0.324E-08	0.195E+09	0.414E-08

Factor of Safety [FS]:

Since Fic [57.44] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 57.441/2.0

= 28.720 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 * Fha(t / Do)

= 2 * 28718.645(15.0/7238.0)

= 119.033 KPa.

Determine the value of [n]:

= $\sqrt{(2 \cdot Do^{1.5} / (3 \cdot LB \cdot (t)))}$

= $\sqrt{(2 \cdot 7238.015 / (3 \cdot 25700.5 \cdot (16.0)))}$

= 2.000 Note: Rounded to Nearest Integer Value

Because n < 2, set n = 2

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

= 1.5 • Fhe • Ls • Rc² • t / (E • (n²-1)) [eq 6-1] or [4.4.30]

= 1.5 • 58 • 2550.26 • 3744.0² • 16.0 / (0.19629E+09 • (2.0²-1))

= 7889.619 cm⁴ Failed, Inertia less than required !

Factor of Safety [FS]:

Since Fic [115.68] < 0.55 * 234.4, FS = 2

Engineering Yield to Tensile Stress ratio [R]:

= σ_{ys} / σ_{uts}

= 234.4/482.6

= 0.486

Ratio [AE]:

= Fhe / E

= 148.9/19629E+09

= 0.000759

Factor [m1]:

= (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))

= (ln(0.486) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= σ_{ys}(1 + epsys) / ln(1 + epsys) m1

$$= 234.4(1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.566 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-07	0.271E-07	0.807E-07	-0.632E-07	0.124E+08	0.188E-01
2	120	-4.651	0.763E-09	0.763E-09	0.171E-07	-0.171E-07	0.151E+09	0.411E-04
3	60	-6.057	0.186E-10	0.186E-10	0.361E-08	-0.361E-08	0.195E+09	0.449E-03
4	90	-5.354	0.163E-09	0.163E-09	0.896E-08	-0.896E-08	0.184E+09	0.268E-03
5	105	-5.002	0.373E-09	0.373E-09	0.127E-07	-0.127E-07	0.171E+09	0.142E-03
6	113	-4.827	0.540E-09	0.540E-09	0.148E-07	-0.148E-07	0.162E+09	0.597E-04
7	116	-4.739	0.644E-09	0.643E-09	0.159E-07	-0.159E-07	0.157E+09	0.120E-04
8	118	-4.695	0.701E-09	0.701E-09	0.165E-07	-0.165E-07	0.154E+09	0.138E-04
9	117	-4.717	0.672E-09	0.672E-09	0.162E-07	-0.162E-07	0.155E+09	0.751E-06
10	117	-4.728	0.658E-09	0.657E-09	0.160E-07	-0.160E-07	0.156E+09	0.566E-05
11	117	-4.722	0.665E-09	0.664E-09	0.161E-07	-0.161E-07	0.156E+09	0.247E-05
12	117	-4.720	0.668E-09	0.668E-09	0.162E-07	-0.161E-07	0.155E+09	0.861E-06
13	117	-4.718	0.670E-09	0.670E-09	0.162E-07	-0.162E-07	0.155E+09	0.556E-07
14	117	-4.718	0.671E-09	0.671E-09	0.162E-07	-0.162E-07	0.155E+09	0.347E-06
15	117	-4.718	0.670E-09	0.670E-09	0.162E-07	-0.162E-07	0.155E+09	0.146E-06
16	117	-4.718	0.670E-09	0.670E-09	0.162E-07	-0.162E-07	0.155E+09	0.452E-07
17	117	-4.718	0.670E-09	0.670E-09	0.162E-07	-0.162E-07	0.155E+09	0.521E-08

Factor of Safety [FS]:
 Since Fic [117.78] < 0.55 * 234.4, FS = 2

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot C_v \cdot M s 0.8 \cdot t^3 \cdot L_s \quad [\text{eq 4.4.3}]$$

$$= 0.184 \cdot 0.4576 \cdot 10.9460.8 \cdot 15.0^3 \cdot 2550.26$$

$$= 491.524 \text{ cm}^4$$

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	16000	...	...	No Calc	No Calc
20	30	16000	...	...	No Calc	No Calc
30	40	16000	...	...	No Calc	No Calc
40	50	No Calc	6918	53	No Calc	No Calc
50	60	19667.7	7004	96	No Calc	No Calc
60	70	19667.7	7004	96	No Calc	No Calc
70	80	19667.7	7004	96	No Calc	No Calc
80	90	19667.7	6998	95	No Calc	No Calc
90	100	19667.7	6994	93	No Calc	No Calc
100	110	19667.7	6994	93	No Calc	No Calc
110	120	19667.7	6994	93	No Calc	No Calc
120	130	No Calc	6907.2	49.6	No Calc	No Calc
130	140	16000	...	...	No Calc	No Calc
140	150	16000	...	...	No Calc	No Calc
150	160	16000	...	...	No Calc	No Calc



160	170	16000	...	...	No Calc	No Calc
170	180	No Calc	7260	24	No Calc	No Calc
180	Ring	2850	7248	18	No Calc	No Calc
Ring	190	2750	7248	18	No Calc	No Calc
190	Ring	2750	7248	18	No Calc	No Calc
Ring	200	2500.01	7248	18	No Calc	No Calc
200	Ring	2500.01	7244	16	No Calc	No Calc
Ring	210	2499.96	7244	16	No Calc	No Calc
210	Ring	2499.96	7244	16	No Calc	No Calc
Ring	220	2500	7244	16	No Calc	No Calc
220	Ring	2500	7240	14	No Calc	No Calc
Ring	230	2500	7240	14	No Calc	No Calc
230	Ring	2500	7242	17	No Calc	No Calc
Ring	Ring	2500	7242	17	No Calc	No Calc
Ring	240	2500	7242	17	No Calc	No Calc
240	Ring	2500	7238	15	No Calc	No Calc
Ring	250	2500	7238	15	No Calc	No Calc
250	Ring	2500	7238	15	No Calc	No Calc
Ring	260	2600.52	7238	15	No Calc	No Calc
260	270	2600.52	7238	15	No Calc	No Calc
270	280	No Calc	7238	15	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	59	No Calc	...	2507.53
50	60	102	No Calc	...	1561.1
60	70	102	No Calc	...	1561.1
70	80	102	No Calc	...	1561.1
80	90	96	No Calc	...	1535.41
90	100	94	No Calc	...	1466.49
100	110	94	No Calc	...	1466.49
110	120	94	No Calc	...	1466.49
120	130	50.6	No Calc	...	2288.54
130	140	...	No Calc	...	No Calc
140	150	...	No Calc	...	No Calc
150	160	...	No Calc	...	No Calc
160	170	...	No Calc	...	No Calc
170	180	30	18.1564	51.73	200.817
180	Ring	24	17.2717	51.73	166.308
Ring	190	24	17.1046	51.73	172.252
190	Ring	24	17.1046	51.73	172.252
Ring	200	24	16.6821	51.73	188.739
200	Ring	22	16.6785	51.73	142.267
Ring	210	22	16.6785	51.73	142.267
210	Ring	22	16.6785	51.73	142.267
Ring	220	22	16.6785	51.73	142.267
220	Ring	20	16.6752	51.73	102.273
Ring	230	20	16.6752	51.73	102.273
230	Ring	18	11.6769	51.73	164.917
Ring	Ring	18	11.6769	51.73	164.909
Ring	240	18	11.6769	51.73	164.917
240	Ring	16	11.6733	51.73	121.482
Ring	250	16	11.6733	51.73	121.482
250	Ring	16	11.6733	51.73	121.482
Ring	260	16	11.8395	51.73	116.675
260	270	16	11.8395	51.73	116.675
270	280	16	13.1654	51.73	78.6246

Minimum

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External Pressure Calculations:



From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	16000	No Calc	No Calc	No Calc
20	30	16000	No Calc	No Calc	No Calc
30	40	16000	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	19667.7	No Calc	No Calc	No Calc
60	70	19667.7	No Calc	No Calc	No Calc
70	80	19667.7	No Calc	No Calc	No Calc
80	90	19667.7	No Calc	No Calc	No Calc
90	100	19667.7	No Calc	No Calc	No Calc
100	110	19667.7	No Calc	No Calc	No Calc
110	120	19667.7	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc
130	140	16000	No Calc	No Calc	No Calc
140	150	16000	No Calc	No Calc	No Calc
150	160	16000	No Calc	No Calc	No Calc
160	170	16000	No Calc	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc
180	Ring	2850	8727.11	No Calc	No Calc
Ring	190	2750	8727.03	12635.8 >>>	11847.8
190	Ring	2750	8727.03	No Calc	No Calc
Ring	200	2500.01	8726.64	12522.2 >>>	7478.23
200	Ring	2500.01	6590.59	No Calc	No Calc
Ring	210	2499.96	6590.47	9310.12 >>>	6902.5
210	Ring	2499.96	6590.47	No Calc	No Calc
Ring	220	2500	6590.57	9310.12 >>>	6902.5
220	Ring	2500	4792.6	No Calc	No Calc
Ring	230	2500	4792.6	6637 >>>	6440.16
230	Ring	2500	7621.01	No Calc	No Calc
Ring	Ring	2500	7620.75	10849.1 >>>	7116.84
Ring	240	2500	7621.02	10849.1 >>>	7116.84
240	Ring	2500	5654.68	No Calc	No Calc
Ring	250	2500	5654.68	7898.49 >>>	6676.47
250	Ring	2500	5654.68	No Calc	No Calc
Ring	260	2600.52	5654.6	7889.62 >>>	6676.47
260	270	2600.52	5654.6	No Calc	No Calc
270	280	No Calc	No Calc	No Calc	No Calc

One or more elements or rings failed code requirements for external pressure with the given thickness!



Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	24726.1	...	23922.5	...	...
20	30	18171.2	...	17367.6	...	...
30	40	9587.02	...	9185.51	...	...
40	50	39049.2	82332864	35665.7	82769520	...
50	60	51422.5	108969968	48439.7	109354904	...
60	70	51422.5	108969968	48439.7	109354904	...
70	80	32567.6	69014312	30678.5	69258104	...
80	90	48397.7	109162336	47900.5	109226504	...
90	100	47375.8	109162336	46878.6	109226504	...
100	110	31583.9	72774888	31252.4	72817672	...
110	120	23687.9	54581168	23439.3	54613252	...
120	130	33267.5	82550984	32703.4	82623784	...
130	140	5040.25	...	4705.33	...	...
140	150	12244.9	...	11839.7	...	...
150	160	12532.5	...	12117.6	...	...
160	170	5279.54	...	4928.7	...	...
170	180	16803.2	50902920	13838	51154460	...
180	190	12663.9	122167016	9505.87	122574576	...
190	200	12663.9	122167016	9505.87	122574576	...
200	210	11605.2	122167016	8447.12	122574576	...
210	220	11605.2	122167016	8447.12	122574576	...
220	230	9844.14	114022544	6896.62	114402936	...
230	240	9497.99	122370720	8971.64	122438648	...
240	250	8440.39	122370720	7914.03	122438648	...
250	260	7314.97	106054632	6858.84	106113496	...
260	270	2813.43	40790244	2637.98	40812884	...
270	280	8871.47	51028592	8378.61	51070524	...
Total		558479	0.189E+10	520866	0.190E+10	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Wght	800	...	2200	...	HP_manwy wt
20	Plat	1752.92	-2766.48	300	2766.48	SKRT PLAT-1
30	Plat	7014.89	0.38462E-04	1300	...	PLAT-1
40	Pack	13680.8	...	-1700	...	HP_Pack: 1_1
40	Liqd	87219.3	...	-1700	...	Liquid: 10
40	Insl	2456.44	...	-1700	...	Ins: 70
40	Nozl	13356.6	...	-3550.5	...	D
40	Wght	170	...	-3400	...	HP_VB
40	Wght	2020	...	...	...	HP_btm SF dish
50	Pack	6035.65	...	500	...	HP_Pack: 1_2
50	Liqd	115437	...	1500	...	Liquid: 20
50	Insl	1882.31	...	1500	...	Ins: 70
50	Wght	3155	-5319	0.0040691	...	HP_D_PP_OVH
60	Plat	3500.23	-2743.17	799.975	...	PLAT-2
60	Liqd	115437	...	1500	...	Liquid: 30
60	Insl	1882.31	...	1500	...	Ins: 70
60	Nozl	1656.48	-3221.93	1400	1502.41	MH-6
70	Plat	3500.23	2743.17	1799.97	...	PLAT-3
70	Pack	6035.77	...	1399.98	...	HP_Pack: 2_1
70	Liqd	23087.5	...	300	...	Liquid: 40
70	Insl	1192.13	...	950	...	Ins: 70
70	Wght	3100	...	899.974	...	HP_B#1_btm grd
70	Wght	300	...	899.974	...	HP_B#1_btm grd
80	Pack	6035.51	...	499.993	...	HP_Pack: 2_2
80	Insl	1880.71	...	1500	...	Ins: 70
80	Lini	1540.66	...	1500	...	SS304L_clad
80	Wght	960	...	999.974	...	HP_B#1_top HDG



80	Wght	5900	...	1599.97	...	HP_B#1_dis. dea
80	Wght	300	...	1599.97	...	HP_B#1_dis. TA
80	Wght	6000	...	2799.97	...	HP_Shroud
90	Plat	3496.22	-2745.06	899.975	...	PLAT-4
90	Insl	1879.65	...	1500	...	Ins: 70
90	Lini	1540.66	...	1500	...	SS304L_clad
90	Nozl	1648.77	-2515.89	1500	2515.89	MH-5
90	Wght	1400	...	1499.98	...	HP_Nozzle wt
100	Plat	3496.22	2745.06	1899.97	...	PLAT-5
100	Insl	1253.1	...	1000	...	Ins: 70
100	Lini	1027.11	...	1000	...	SS304L_clad
100	Nozl	1648.77	-2515.89	1000	2515.89	MH-4
110	Insl	939.827	...	750	...	Ins: 70
110	Lini	770.329	...	750	...	SS304L_clad
110	Nozl	2360.53	-3583	750	...	A
110	Wght	6800	...	1099.97	...	HP_Gallery
110	Wght	21960	-5319	749.975	...	HP_A_PP_OVH
110	Wght	2296	-4964	749.975	...	HP_E_PP_OVH
120	Plat	3461.44	2745.06	2399.97	...	PLAT-6
120	Insl	2437.23	...	1701.5	...	Ins: 70
120	Lini	1746.79	...	1701.5	...	SS304L_clad
120	Nozl	104.131	...	3507.7	...	E
120	Wght	1900	...	...	...	HP_top SF dish
120	Forc	...	...	3403	...	HP_E_FxFz
130	Insl	775.438	...	625	...	Ins: 70
140	Wght	10000	...	1649.95	...	margin
140	Wght	17400	...	1649.95	...	Cleat
140	Wght	20200	...	1649.95	...	Lifting wt
160	Plat	3618.85	-2870.5	899.95	...	PLAT-7
160	Insl	649.401	...	500	...	Ins: 70
170	Liqd	53924	...	-900	...	Liquid: 10
170	Insl	2068.53	...	-875	...	Ins: 70
170	Nozl	4427.32	...	-1990	...	C
170	Wght	250	...	-1800	...	LP_VB
170	Wght	7132	-5237	49.9504	...	LP_C_PP_OVH
180	Plat	3598.01	-2869.24	1849.95	...	PLAT-8
180	Liqd	129418	...	1500	...	Liquid: 20
180	Insl	1947.14	...	1500	...	Ins: 70
180	Ring	992.391	...	2200	...	Ring R8
190	Plat	3598.01	2869.24	2849.95	...	PLAT-9
190	Liqd	129418	...	1500	...	Liquid: 30
190	Insl	1947.14	...	1500	...	Ins: 70
190	Ring	828.307	...	1950	...	Ring R7
190	Nozl	505.793	-2655.19	600.001	2655.19	MH-3
200	Liqd	4313.92	...	50	...	Liquid: 40
200	Insl	1946.08	...	1500	...	Ins: 70
200	Ring	821.077	...	1450.01	...	Ring R1
200	Nozl	696.671	-3755	2300	...	B
200	Nozl	42.6542	3705	2500	...	K
200	Wght	7993	-4987	2500.05	...	LP_K_PP_OVH
200	Wght	30921	-5241	2299.95	...	LP_B_PP_OVH
210	Plat	3596.41	-2869.24	849.955	...	PLAT-10
210	Pack	8119.59	...	2400.03	...	LP_Pack_1
210	Insl	1946.08	...	1500	...	Ins: 70
210	Ring	821.077	...	949.975	...	Ring R5
210	Wght	4000	...	1800.05	...	LP_B#1_btm grid
210	Wght	350	...	1800.05	...	LP_B#1_TA
220	Plat	3594.81	2869.24	1849.95	...	PLAT-11
220	Pack	18946.5	...	1400	...	LP_Pack_2
220	Insl	1815.35	...	1400	...	Ins: 70
220	Ring	820.639	...	449.977	...	Ring R2
230	Pack	6766.61	...	500	...	LP_Pack_3
230	Insl	1945.55	...	1500	...	Ins: 70
230	Lini	1631.24	...	1500	...	Clad SS304L
230	Ring	821.076	...	149.976	...	Ring R4
230	Ring	821.076	...	2649.98	...	Ring R3
230	Nozl	504.406	-2657.31	2100	2657.31	MH-2
230	Wght	900	...	999.953	...	LP_B#1_HDG
230	Wght	5750	...	1599.95	...	LP_B#1_dist. de
230	Wght	350	...	1599.95	...	LP_B#1_dist.TA



230	Wght	9800	...	2800.05	...	LP_Shroud
230	Wght	850	...	1100.05	...	LP_Nozzle wt
240	Plat	3594.01	-2871.13	49.9551	...	PLAT-12
240	Insl	1944.49	...	1500	...	Ins: 70
240	Lini	1631.24	...	1500	...	Clad SS304L
240	Ring	820.638	...	2149.98	...	Ring R6
240	Nozl	505.793	-2657.31	1400	2657.31	MH-1
240	Wght	80	...	2749.95	...	LP_dwncm PP dea
250	Plat	5621.61	2871.13	1500.05	...	PLAT-13
250	Insl	1685.22	...	1300	...	Ins: 70
250	Lini	1413.74	...	1300	...	Clad SS304L
250	Ring	820.638	...	1649.98	...	Ring R9
250	Nozl	782.245	-3758	800	...	H1
250	Nozl	782.245	3758	800	...	H2
250	Tray	8831.58	...	1550.05	...	Tray#1-3
250	Wght	3100	...	650.053	...	LP_Gallery
250	Wght	150	...	2450.05	...	LP_FP_G_dead
250	Wght	20	...	2450.05	...	LP_FP_G_TA
250	Wght	29835	-5241	800.053	...	LP_H1_PP_OVH
250	Wght	29835	-5241	800.053	...	LP_H2_PP_OVH
260	Insl	648.163	...	500	...	Ins: 70
260	Lini	543.747	...	500	...	Clad SS304L
260	Wght	900	3600	0.046509	...	LP_Davit
260	Wght	35704	-5317	500.049	...	LP_F_PP_OVH
270	Plat	9733.92	...	2300.05	...	TOP PLT
270	Insl	2043.83	...	925.75	...	Ins: 70
270	Lini	1519.25	...	925.75	...	Clad SS304L
270	Nozl	837.881	...	2093.5	...	F
270	Wght	960	...	...	...	LP_Demister
270	Wght	7500	-3750	1851.5	...	PSV
270	Wght	150	...	...	...	LP_Demister_TA
270	Forc	...	...	1800	...	F_FxFz
70	Pliq	9620.00	...	1399.985	...	
80	Pliq	9636.56	...	499.993	...	
210	Pliq	12941.26	...	2400.025	...	
220	Pliq	30197.54	...	1400.000	...	
230	Pliq	10802.82	...	500.000	...	

Note: The individual tray liquid weights are listed below, but these weights are included in the tray weights above.

From	Tliq	Weight	X Offset	Y Offset
250	Tliq	0.00	0.000	1550.053

Total Weight of Each Detail Type:

Platforms	63177.8
Packing	65620.5
Packing Liquid	73198.2
Liquid	658254.2
Insulation	37166.1
Lining	13364.8
Stiffeners	7566.9
Nozzles	29860.3
Trays	8201.6
Tray Bolting Weight	630.0
Weights	281191.0

Sum of the Detail Weights 1238231.4 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	558479.9	558479.9	558479.9	558479.9	558479.9	558479.9
Stif. Rings	7566.9	7566.9	7566.9	7566.9	7566.9	7566.9
Nozzles	29860.3	29860.3	29860.3	29860.3	29860.3	29860.3
Tray Bar	630.0	630.0	630.0	630.0	630.0	630.0



Wld Weights	83140.0	83140.0	83140.0	83140.0	83140.0	83140.0
Platforms	...	...	63177.8	63177.8	63177.8	63177.8
Packing	...	...	65620.5	65620.5	65620.5	65620.5
Insulation	...	...	37166.1	37166.1	37166.1	37166.1
Lining	13364.8	13364.8	13364.8	13364.8	13364.8	13364.8
Trays	...	...	8201.6	8201.6	8201.6	8201.6
Empty Weights	...	...	...	198051.0	198051.0	...
Test Weights	...	198051.0	...	...	...	...
Ope Weights	...	...	...	...	...	198051.0
Ope. Liquid	...	...	...	...	...	658254.2
Packing Liq	...	...	...	...	...	73198.2
Test Liquid	...	1890903.5	...	...	...	...
Totals	693041.9	2781996.5	867207.8	1065258.9	1065258.9	1796711.1

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	693041.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	2781996.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	867207.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1065258.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1065258.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1796711.1 kg.
Field Test Wt.	- Empty weight + test medium (full)	2890541.8 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	428895.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	1177884.5 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	346307.0 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	654645	647859
20	30	654645	647859
30	40	327323	323741
40	50	755680	726336
50	60	660112	640885
60	70	660112	640885
70	80	418071	405894
80	90	659546	641450
90	100	659169	641450
100	110	439446	427634
110	120	329585	320725
120	130	752633	727619
130	140	271905	270020
140	150	668022	663498
150	160	683485	678961
160	170	284785	282900
170	180	584019	573248
180	190	683108	678584
190	200	683108	678584
200	210	682731	678584
210	220	682731	678584
220	230	636864	633345
230	240	682542	679150
240	250	682166	679150
250	260	591210	588596
260	270	227389	226383
270	280	579889	574194

Total	15594919.0	15356118.0 cm ²
Total	16786.2	16529.2 ft ²



Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	25526.1	25526.1	25526.1	...	25526.1
20	30	19924.1	19924.1	19924.1	4849.52	19924.1
30	40	16601.9	16601.9	16601.9	0.00026981	16601.9
40	50	70733.1	157952	139335	...	70733.1
50	60	62495.5	177933	165363	16781.8	62495.5
60	70	58461.5	173899	167365	14939.1	58461.5
70	80	46695.7	79403.1	109632	9601.92	46695.7
80	90	71014.6	80651.1	174075	...	71014.6
90	100	57341.1	57341.1	166437	13745.7	57341.1
100	110	39009.1	39009.1	111740	5449.32	39009.1
110	120	58814.6	58814.6	113362	136663	58814.6
120	130	42917	42917	125418	9502.04	42917
130	140	5815.69	5815.69	5815.69	...	5815.69
140	150	59844.9	59844.9	59844.9	...	59844.9
150	160	12532.5	12532.5	12532.5	...	12532.5
160	170	9547.8	9547.8	9547.8	10388.1	9547.8
170	180	30681.1	84605.1	81552.9	37351	30681.1
180	190	19201.5	148619	141294	10323.8	19201.5
190	200	19543.2	148961	141636	8980.77	19543.2
200	210	54025.7	58339.6	176118	204380	54025.7
210	220	30438.4	43379.6	144411	10319.2	30438.4
220	230	35021.4	65219	130028	10314.6	35021.4
230	240	39638	50440.8	155167	1340.39	39638
240	250	17016.6	17016.6	139313	11663.2	17016.6
250	260	90192.3	90192.3	196182	296596	90192.3
260	270	40609.3	40609.3	81374.7	186602	40609.3
270	280	31616.3	31616.3	82613.8	28125.6	31616.3

Empty Support Force + the Sum of the Y forces 1002909.56 Kg
 Operating Support Force + the Sum of the Y forces 1807560.00 Kg
 Hydro Support Force + the Sum of the Y forces 2903058.75 Kg

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	1065258	1796711	2892209
20	30	1039732	1771185	2866683
30	40	1019808	1751261	2846759
40	50	1003206	1734659	2830157
50	60	932474	1576706	2690822
60	70	869978	1398774	2525459
70	80	811517	1224875	2358094
80	90	764821	1145472	2248462
90	100	693806	1064821	2074387
100	110	636465	1007479	1907950
110	120	597456	968471	1796211
120	130	538642	909656	1682848
130	140	495725	866739	1557431
140	150	489909	860924	1551615
150	160	430064	801079	1491770
160	170	417532	788546	1479238
170	180	407984	778998	1469690
180	190	377303	694393	1388137
190	200	358101	545774	1246843
200	210	338558	396813	1105207
210	220	284532	338474	929090
220	230	254094	295094	784678
230	240	219072	229875	654651
240	250	179434	179434	499483
250	260	162418	162418	360171



260	270	72225.7	72225.7	163988
270	280	31616.3	31616.3	82613.8

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	1027916	1027916	1027916
20	30	1027916	1027916	1027916
30	40	1023067	1023067	1023067
40	50	1023067	1023067	1023067
50	60	1023067	1023067	1023067
60	70	1006285	1006285	1006285
70	80	991347	991347	991347
80	90	981745	981745	981745
90	100	981745	981745	981745
100	110	967999	967999	967999
110	120	962550	962550	962550
120	130	825887	825887	825887
130	140	816384	816384	816384
140	150	816384	816384	816384
150	160	816384	816384	816384
160	170	816384	816384	816384
170	180	805996	805996	805996
180	190	768645	768645	768645
190	200	758322	758322	758322
200	210	749341	749341	749341
210	220	544961	544961	544961
220	230	534642	534642	534642
230	240	524327	524327	524327
240	250	522987	522987	522987
250	260	511323	511323	511323
260	270	214727	214727	214727
270	280	28125.6	28125.6	28125.6



Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
D	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-6	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-5	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-4	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
A	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
E	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
C	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
MH-3	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
B	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
K	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
MH-2	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
MH-1	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
H1	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
H2	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
F	1656.0	1960.0	130	150	GR 1.1	...	...	...	...
Min Rating	1656.0	1960.0, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.25231 Hz.

The Natural Frequency for the Vessel (Ope...) is 1.07523 Hz.

The Natural Frequency for the Vessel (Filled) is 0.72058 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	2061.54	8284.27
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	...	...
230	240	...	...
240	250	...	...
250	260	...	...
260	270	...	...
270	280	8325.76	39464.6

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	2061.54	8284.27
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	...	...
230	240	...	...
240	250	...	...
250	260	...	...
260	270	...	...
270	280	8325.76	39464.6



User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
120	3403.00	1458.	-2018.	1458.	897.		897.
270	1800.00	5887.	-8830.	5887.	17308.		17308.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.163 \cdot 0.872) / (1 + 1.7 \cdot 3.4 \cdot 0.163)))$
 $= \min(0.85, 0.868)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	1.075 Hz
Natural Frequency of Vessel (Empty)	1.252 Hz
Natural Frequency of Vessel (Test)	0.721 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.607
Structure Height to Diameter ratio	8.329
Height to top of Structure	59797.496 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1500.0 + 0.0 = 1500.0$ mm
 $= 4.921$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (4.921 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.607 \cdot 269.158$$

$$= 2223.4 \text{ lbs. } [1008.5] \text{ Kg}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
HPmain skirt-3	1500.0	0.000	0.000	0.000	0.849	1.000	78.120
HPmain skirt-2	4500.0	0.000	0.000	0.000	0.849	1.000	78.120
HPmain skirt-1	6750.0	0.000	0.000	0.000	0.921	1.000	78.120
HPBTM dish	7500.0	0.000	0.000	0.000	0.942	1.000	78.120
HPshell-1	9000.0	0.000	0.000	0.000	0.979	1.000	78.120
HPshell-2	12000.0	0.000	0.000	0.000	1.040	1.000	78.120
HPshell-3	14450.0	0.000	0.000	0.000	1.082	1.000	78.120
HPshell-4	16900.0	0.000	0.000	0.000	1.118	1.000	78.120
HPshell-5	19900.0	0.000	0.000	0.000	1.157	1.000	78.120
HPshell-6	22400.0	0.000	0.000	0.000	1.186	1.000	78.120
HPshell-7	24150.0	0.000	0.000	0.000	1.205	1.000	78.120
HPTOP dish	24900.0	0.000	0.000	0.000	1.213	1.000	78.120
HPintskirt-1	25525.0	0.000	0.000	0.000	1.219	1.000	78.120
HPintskirt-2	27655.9	0.000	0.000	0.000	1.240	1.000	78.120
LPintskirt-2	30655.8	0.000	0.000	0.000	1.267	1.000	78.120
LPintskirt-1	32775.0	0.000	0.000	0.000	1.285	1.000	78.120
LPBTM dish	33425.0	0.000	0.000	0.000	1.290	1.000	78.120
LP-shell-1	34950.0	0.000	0.000	0.000	1.303	1.000	78.120
LP-shell-2	37950.0	0.000	0.000	0.000	1.325	1.000	78.120
LP-shell-3	40950.0	0.000	0.000	0.000	1.347	1.000	78.120
LP-shell-4	43950.0	0.000	0.000	0.000	1.367	1.000	78.120
LP-shell-5	46850.0	0.000	0.000	0.000	1.386	1.000	78.120
LP-shell-6	49750.0	0.000	0.000	0.000	1.403	1.000	78.120
LP-shell-7	52750.0	0.000	0.000	0.000	1.421	1.000	78.120
LP-shell-8	55550.0	0.000	0.000	0.000	1.436	1.000	78.120
LP-shell-9	57350.0	0.000	0.000	0.000	1.446	1.000	78.120
LP-TOP dish	58678.4	0.000	0.000	0.000	1.453	1.000	78.120

Note:

The height/(avg. diameter) ratio [8.0], was less than or equal to 15, therefore no wind vibration calculations have been performed.

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf	Cf
SKRT PLAT-1	15000.00	3300.00	78.12	117.18	1.00
PLAT-1	30000.00	7300.00	78.12	234.36	1.00
PLAT-2	30000.00	11299.97	78.12	234.36	1.00
PLAT-3	30000.00	15299.97	78.12	234.36	1.00
PLAT-4	30000.00	19299.98	78.12	234.36	1.00
PLAT-5	30000.00	23299.97	78.12	234.36	1.00
PLAT-6	30000.00	27299.94	78.12	234.36	1.00
PLAT-7	30000.00	33049.95	78.12	234.36	1.00
PLAT-8	30000.00	35299.95	78.12	234.36	1.00
PLAT-9	30000.00	39299.95	78.12	234.36	1.00
PLAT-10	30000.00	43299.96	78.12	234.36	1.00
PLAT-11	30000.00	47299.95	78.12	234.36	1.00



PLAT-12	30000.00	51299.95	78.12	234.36	1.00
PLAT-13	30000.00	55750.05	78.12	234.36	1.00
TOP PLT	75000.00	60150.05	78.12	585.90	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
HPmanwy wt	0.00	2200.00	78.12	0.00
HPVB	0.00	4099.97	78.12	0.00
HPbtm SF dis	0.00	7499.97	78.12	0.00
HPDPPQVH	0.00	7500.00	78.12	0.00
HPB#1btm gr	0.00	14399.98	78.12	0.00
HPB#1btm gr	0.00	14399.98	78.12	0.00
HPB#1top HD	0.00	16399.97	78.12	0.00
HPB#1dis. d	0.00	16999.98	78.12	0.00
HPB#1dis. T	0.00	16999.98	78.12	0.00
HPShroud	0.00	18199.97	78.12	0.00
HPNozzle wt	0.00	19899.98	78.12	0.00
HPGallary	0.00	24499.98	78.12	0.00
HPAPPOVH	0.00	24149.98	78.12	0.00
HPEPPOVH	0.00	24149.98	78.12	0.00
HPtop SF dis	0.00	24899.97	78.12	0.00
margin	0.00	27799.95	78.12	0.00
Cleat	0.00	27799.95	78.12	0.00
Lifting wt	0.00	27799.95	78.12	0.00
LPVB	0.00	31600.00	78.12	0.00
LPCPPOVH	0.00	33449.95	78.12	0.00
LPKPPPOVH	0.00	41950.05	78.12	0.00
LPBPPOVH	0.00	41749.95	78.12	0.00
LPB#1btm gr	0.00	44250.05	78.12	0.00
LPB#1TA	0.00	44250.05	78.12	0.00
LPB#1HDG	0.00	49249.95	78.12	0.00
LPB#1dist.	0.00	49849.96	78.12	0.00
LPB#1dist.T	0.00	49849.96	78.12	0.00
LPShroud	0.00	51050.05	78.12	0.00
LPNozzle wt	0.00	49350.05	78.12	0.00
LPdwnc PP d	0.00	53999.95	78.12	0.00
LPGallary	0.00	54900.05	78.12	0.00
LPFPGdead	0.00	56700.05	78.12	0.00
LPFPGTA	0.00	56700.05	78.12	0.00
LPH1PPOVH	0.00	55050.05	78.12	0.00
LPH2PPOVH	0.00	55050.05	78.12	0.00
LPDavit	0.00	56850.04	78.12	0.00
LPFPPOVH	0.00	57350.05	78.12	0.00
LPDemister	0.00	57850.00	78.12	0.00
PSV	0.00	59701.50	78.12	0.00
LPDemisterT	0.00	57850.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 1.07523 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1500	8335.2	250056	78.12	1109.37
20	30	4500	8335.2	250056	78.12	1203.9
30	40	6750	8335.2	125028	78.12	833.487
40	50	7499.98	8493.6	2.58885	78.12	0.011485
50	60	9000	8596.8	257904	78.12	1144.19
60	70	12000	8596.8	257904	78.12	1386.95
70	80	14450	8596.8	163339	78.12	997.613
80	90	16900	8589.6	257688	78.12	1143.23
90	100	19900	8584.8	257544	78.12	1392.61
100	110	22400	8584.8	171696	78.12	1029.88
110	120	24150	8584.8	128772	78.12	571.295



120	130	24900	8480.64	2.5849	78.12	282.655
130	140	25525	8500.8	106260	78.12	471.421
140	150	27655.9	8407.16	252215	78.12	1118.95
150	160	30655.8	8603.96	258119	78.12	1145.14
160	170	32775	8856	110700	78.12	751.078
170	180	33425	8904	4452	78.12	19.7512
180	190	34950	8889.6	266688	78.12	1443.54
190	200	37950	8889.6	266688	78.12	1450.12
200	210	40950	8884.8	266544	78.12	1182.52
210	220	43950	8884.8	266544	78.12	1436.5
220	230	46850	8880	248640	78.12	1363.36
230	240	49750	8882.4	266472	78.12	1182.2
240	250	52750	8877.6	266328	78.12	1432.27
250	260	55550	8877.6	230818	78.12	1282.74
260	270	57350	8877.6	88776	78.12	393.853
270	280	58678.4	8877.6	136741	78.12	1267.31

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(195.87110.75)$
 = 1.047 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/1.075)
 = 0.930

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 1.047$, 0.93) per 12.8.2
 = 0.930

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.157 / (1.25 / 1.25)$
 = 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.104 / (0.93(1.25 / 1.25))$
 = 0.112

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.1568 \cdot 1.25), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.1118 \cdot 1796711$
 = 200914.922 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 1796711$

= 56344.9 Kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$) and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.215

The Natural Frequency for the Vessel (Ope...) is 1.07523 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1500	25526.1	81.7613	108.278	54326.5
20	30	4500	19924.1	242.467	298.727	53526
30	40	6750	16601.9	330.663	396.668	52901.2
40	50	7499.98	157952	3575.6	1907.56	52380.5
50	60	9000	177933	5026.74	2078.29	47427.1
60	70	12000	173899	6968.34	2705.92	41847.2
70	80	14450	79403.1	3987.55	2675.78	36393.7
80	90	16900	80651.1	4899.18	4871.84	33903.6
90	100	19900	57341.1	4248.19	4746.47	31374.4
100	110	22400	39009.1	3336.95	3699.44	29576.2
110	120	24150	58814.6	5512.68	6081.37	28352.8
120	130	24900	42917	4174.89	4596.32	26508.4
130	140	25525	5815.69	583.039	640.849	27180.9
140	150	27650	59844.9	6611.82	7229.28	26998.6
150	160	30650	12532.5	1569.22	1704.18	25121.8
160	170	32775	9547.8	1296.95	1402.3	24728.8
170	180	33425	84605.1	11770.1	4609.04	24429.4
180	190	34950	148619	21827.2	3036.28	21776.2
190	200	37950	148961	24179.7	3397.08	17115.5
200	210	40950	58339.6	10387	10249.1	12444.1
210	220	43950	43379.6	8416.25	6263.16	10614.5
220	230	46850	65219	13674.9	7755.31	9254.16
230	240	49750	50440.8	11376.9	9404.88	7208.89
240	250	52750	17016.6	4121.08	4318.56	5627.07
250	260	55550	90192.3	23259.5	24291.3	5093.43
260	270	57350	40609.3	10886.4	11345.5	2265
270	280	57875	31616.3	8569.94	8926.02	991.489



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas Kgf	Cumulative Shr Eqk Cas Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1500	10387.3	10387.3	580737	580737
20	30	4500	10387.3	10387.3	549574	549574
30	40	6750	10387.3	10387.3	518412	518412
40	50	7499.98	10387.3	10387.3	502831	502831
50	60	9000	10387.3	10387.3	502831	502831
60	70	12000	10387.3	10387.3	471668	471668
70	80	14450	10387.3	10387.3	440505	440505
80	90	16900	10387.3	10387.3	420769	420769
90	100	19900	10387.3	10387.3	389607	389607
100	110	22400	10387.3	10387.3	358444	358444
110	120	24150	10387.3	10387.3	337669	337669
120	130	24900	10387.3	10387.3	322088	322088
130	140	25525	8325.76	8325.76	313804	313804
140	150	27650	8325.76	8325.76	303396	303396
150	160	30650	8325.76	8325.76	278418	278418
160	170	32775	8325.76	8325.76	253441	253441
170	180	33425	8325.76	8325.76	243033	243033
180	190	34950	8325.76	8325.76	242617	242617
190	200	37950	8325.76	8325.76	217639	217639
200	210	40950	8325.76	8325.76	192662	192662
210	220	43950	8325.76	8325.76	167684	167684
220	230	46850	8325.76	8325.76	142706	142706
230	240	49750	8325.76	8325.76	119393	119393
240	250	52750	8325.76	8325.76	94415.6	94415.6
250	260	55550	8325.76	8325.76	69437.9	69437.9
260	270	57350	8325.76	8325.76	47790.5	47790.5
270	280	57875	8325.76	8325.76	39464.6	39464.6

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1500	27035.9	200915	831855	7696828
20	30	4500	25926.6	200833	752410	7094194
30	40	6750	24722.7	200591	676435	6492046
40	50	7499.98	23889.2	200260	639976	6191408
50	60	9000	23889.2	196684	639975	6191402
60	70	12000	22745	191658	570022	5608878
70	80	14450	21358	184689	503867	5044346
80	90	16900	20360.4	180702	464233	4697218
90	100	19900	19217.2	175803	404866	4162450
100	110	22400	17824.6	171554	349302	3641403
110	120	24150	16794.7	168217	314682	3301625
120	130	24900	16223.4	162705	289918	3053433
130	140	25525	15940.8	158530	289918	3053429
140	150	27650	15469.3	157947	270286	2855627
150	160	30650	14350.4	151335	225556	2391695
160	170	32775	13205.2	149766	184221	1940034
170	180	33425	12454.2	148469	168184	1753634
180	190	34950	12434.4	136699	167562	1746504
190	200	37950	10990.9	114872	132423	1369141
200	210	40950	9540.75	90691.9	101625	1060790
210	220	43950	8358.23	80304.9	74776.1	804290
220	230	46850	6921.73	71888.7	51855.7	575996
230	240	49750	5558.37	58213.8	34383.2	393848
240	250	52750	4376.17	46836.9	19481.1	236270
250	260	55550	2943.9	42715.8	8500.78	101938
260	270	57350	1661.16	19456.3	2514.08	21112.6
270	280	57875	1267.31	8569.94	1049.82	7099.25

Note:
 The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	10387.3	0.0068979	0.027343	0.18063E-04
20	30	10387.3	0.060961	0.10738	0.35129E-04
30	40	10387.3	0.13519	0.1659	0.42857E-04
40	50	10387.3	0.1659	0.1659	0.42857E-04
50	60	10387.3	0.23239	0.3032	0.48612E-04
60	70	10387.3	0.37818	0.45721	0.54000E-04
70	80	10387.3	0.50928	0.56289	0.57221E-04
80	90	10387.3	0.65057	0.74186	0.62035E-04
90	100	10387.3	0.83667	0.93488	0.66578E-04
100	110	10387.3	1.00217	1.07088	0.69396E-04
110	120	10387.3	1.12331	1.17649	0.71400E-04
120	130	10387.3	1.17649	1.17649	0.71400E-04
130	140	8325.76	1.22211	1.26968	0.77662E-04
140	150	8325.76	1.39145	1.52347	0.91337E-04
150	160	8325.76	1.66499	1.81525	0.000103
160	170	8325.76	1.88031	1.94675	0.00010738
170	180	8325.76	1.94944	1.95212	0.00010754
180	190	8325.76	2.11832	2.29394	0.00012011
190	200	8325.76	2.47847	2.6714	0.0001313
200	210	8325.76	2.8727	3.0823	0.00014238
210	220	8325.76	3.29963	3.52412	0.00015191
220	230	8325.76	3.73999	3.96188	0.00016051
230	240	8325.76	4.20515	4.45306	0.0001667
240	250	8325.76	4.70534	4.96166	0.00017209
250	260	8325.76	5.18659	5.41369	0.00017542
260	270	8325.76	5.50153	5.5896	0.00017638
270	280	8325.76	5.59401	5.59842	0.00017638

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 289.500 Actual deflection : 5.598 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	27035.9	0.0099334	0.039087	0.25630E-04
20	30	25926.6	0.086508	0.15126	0.48743E-04
30	40	24722.7	0.18977	0.23212	0.58980E-04
40	50	23889.2	0.23213	0.23213	0.58980E-04
50	60	23889.2	0.32346	0.42029	0.66323E-04
60	70	22745	0.52232	0.62924	0.72838E-04
70	80	21358	0.69934	0.77121	0.76560E-04
80	90	20360.4	0.88813	1.00903	0.81862E-04
90	100	19217.2	1.13367	1.26184	0.86566E-04
100	110	17824.6	1.34912	1.43778	0.89327E-04
110	120	16794.7	1.50514	1.57321	0.91214E-04
120	130	16223.4	1.57321	1.57321	0.91214E-04
130	140	15940.8	1.63115	1.69092	0.97055E-04
140	150	15469.3	1.84126	2.00057	0.00010902
150	160	14350.4	2.16779	2.34193	0.00011824
160	170	13205.2	2.41634	2.49176	0.00012144
170	180	12454.2	2.49479	2.49783	0.00012155
180	190	12434.4	2.68356	2.87558	0.00012995
190	200	10990.9	3.07317	3.27567	0.00013649



200	210	9540.75	3.4827	3.6939	0.00014204
210	220	8358.23	3.90864	4.12636	0.00014602
220	230	6921.73	4.33194	4.53954	0.00014891
230	240	5558.37	4.7636	4.98886	0.00015049
240	250	4376.17	5.21504	5.44191	0.00015141
250	260	2943.9	5.63888	5.83606	0.00015172
260	270	1661.16	5.91192	5.9878	0.00015176
270	280	1267.31	5.99159	5.99539	0.00015176

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	160.886	1005.54
20	30	160.886	1005.54
30	40	160.886	1005.54
40	50	163.943	1024.65
50	60	165.935	1037.09
60	70	165.935	1037.09
70	80	165.935	1037.09
80	90	165.796	1036.23
90	100	165.704	1035.65
100	110	165.704	1035.65
110	120	165.704	1035.65
120	130	163.693	1023.08
130	140	164.082	1025.51
140	150	162.275	1014.22
150	160	166.073	1037.96
160	170	170.938	1068.36
170	180	171.865	1074.15
180	190	171.587	1072.42
190	200	171.587	1072.42
200	210	171.494	1071.84
210	220	171.494	1071.84
220	230	171.401	1071.26
230	240	171.448	1071.55
240	250	171.355	1070.97
250	260	171.355	1070.97
260	270	171.355	1070.97
270	280	171.355	1070.97

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 289.500 Actual deflection : 5.995 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 289.500 Actual deflection : 11.594 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	7815.05	7469.46	1343088000	1284288640
20	30	7815.05	7469.46	1343088000	1284288640
30	40	8246.78	7901.39	1416471808	1357771904
40	50	12713.4	11430.5	2.162E+09	1946844416
50	60	22116.9	20834	3.762E+09	3.549E+09
60	70	22116.9	20834	3.762E+09	3.549E+09
70	80	22116.9	20834	3.762E+09	3.549E+09
80	90	20815.9	20602.1	3.543E+09	3.508E+09
90	100	20376.4	20162.6	3.468E+09	3.433E+09
100	110	20376.4	20162.6	3.468E+09	3.433E+09
110	120	20376.4	20162.6	3.468E+09	3.433E+09
120	130	10899.6	10685.7	1854768000	1818904448
130	140	5202.5	4856.79	894333952	835290368
140	150	5202.5	4856.83	894331008	835293504
150	160	5326.09	4972.19	937402496	875512384
160	170	5449.8	5087.62	981531328	916705280
170	180	6814.15	5455.85	1226589568	983711936
180	190	5446.79	4088.5	980444928	737165312
190	200	5446.79	4088.5	980444928	737165312
200	210	4991.43	3633.13	898473664	655059584
210	220	4991.43	3633.13	898473664	655059584
220	230	4536.42	3178.12	816567680	573019136
230	240	4085.09	3858.7	735938560	695345344
240	250	3630.2	3403.81	653986688	613371072
250	260	3630.2	3403.81	653986688	613371072
260	270	3630.2	3403.81	653986688	613371072
270	280	3630.2	3403.81	653986688	613371072



Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	113.099	246.996	-91.5959	-91.5959
20	30	113.099	246.996	-91.5959	-91.5959
30	40	86.02	247	-93.171	-93.171
40	50	156.4	247	-102.771	-102.771
50	60	156.4	247	-114.153	-114.153
60	70	156.4	247	-114.153	-114.153
70	80	156.4	247	-114.153	-114.153
80	90	156.4	247	-113.997	-113.997
90	100	156.4	247	-113.659	-113.659
100	110	156.4	247	-113.659	-113.659
110	120	156.4	247	-113.659	-113.659
120	130	156.4	247	-101.256	-101.256
130	140	156.399	246.996	-78.5074	-78.5074
140	150	156.399	246.996	-78.5074	-78.5074
150	160	156.399	246.996	-77.7257	-77.7257
160	170	156.4	247	-76.9549	-76.9549
170	180	156.4	247	-79.2234	-79.2234
180	190	156.4	247	-62.0721	-62.0721
190	200	156.4	247	-62.0721	-62.0721
200	210	156.4	247	-55.2057	-55.2057
210	220	156.4	247	-55.2057	-55.2057
220	230	156.4	247	-48.3317	-48.3317
230	240	156.4	247	-58.6722	-58.6722
240	250	156.4	247	-51.7982	-51.7982
250	260	138	234	-51.7982	-51.7982
260	270	156.4	247	-51.7982	-51.7982
270	280	156.4	247	-51.7982	-51.7982

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	130.86	...	19.2574
50	60	71.7957	...	10.5655
60	70	71.7957	...	10.5655
70	80	71.7957	...	10.5655
80	90	72.5187	...	10.6719
90	100	74.0996	...	10.9045
100	110	74.0996	...	10.9045
110	120	74.0996	...	10.9045
120	130	139.816	...	20.5753
130	140	...	...	...
140	150	...	...	...
150	160	...	...	...
160	170	...	...	...
170	180	28.7538	-3.92526	45.2232
180	190	38.3702	-5.22072	60.3476
190	200	38.3702	-5.22072	60.3476
200	210	43.1794	-5.86858	67.9114
210	220	43.1794	-5.86858	67.9114
220	230	49.3613	-6.70137	77.6341
230	240	40.6102	-5.52248	63.8706
240	250	46.0374	-6.2536	72.4064
250	260	46.0374	-6.2536	72.4064
260	270	46.0374	-6.2536	72.4064
270	280	46.0374	-6.2536	72.4064

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-13.986	-23.5894	-37.9725	7.84902	7.84902
20	30	-13.6509	-23.2543	-37.6374	7.84902	7.84902
30	40	-12.6574	-21.7358	-35.3326	7.3892	7.3892
40	50	-8.60705	-11.9554	-8.60705	5.15339	5.15339
50	60	-4.38927	-6.22632	-4.38927	2.82663	2.82663
60	70	-4.09509	-5.93215	-4.09509	2.78026	2.78026
70	80	-3.81991	-5.65696	-3.81991	2.73899	2.73899
80	90	-3.64063	-5.45258	-3.64063	2.74463	2.74463
90	100	-3.37459	-5.17916	-3.37459	2.8045	2.8045
100	110	-3.09569	-4.90026	-3.09569	2.76524	2.76524
110	120	-2.90596	-4.71053	-2.90596	2.74967	2.74967
120	130	-4.94338	-8.34836	-4.94338	4.45277	4.45277
130	140	-10.0097	-17.5012	-31.4476	9.58467	9.58467
140	150	-9.89216	-17.3836	-31.33	9.58464	9.58464
150	160	-8.48232	-15.8	-29.4228	9.14434	9.14434
160	170	-8.04828	-15.2	-28.5136	8.73343	8.73343
170	180	-7.33346	-8.30305	-7.33346	8.03499	8.03499
180	190	-9.05013	-10.344	-9.05013	10.2254	10.2254
190	200	-8.58956	-9.88342	-8.58956	10.0881	10.0881
200	210	-9.13862	-10.5947	-9.13862	11.2181	11.2181



210	220	-7.68031	-9.13635	-7.68031	8.15839	8.15839
220	230	-7.84065	-9.10581	-7.84065	9.14984	9.14984
230	240	-5.5677	-5.84225	-5.5677	7.39472	7.39472
240	250	-5.16975	-5.16975	-5.16975	8.36156	8.36156
250	260	-4.67948	-4.67948	-4.67948	8.17509	8.17509
260	270	-2.08092	-2.08092	-2.08092	3.43308	3.43308
270	280	-0.91091	-0.91091	-0.91091	0.44967	0.44967

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	7.84902	6.35192	58.7718	3.17596	...
20	30	7.84902	5.74529	54.1702	2.87265	...
30	40	7.3892	4.88561	46.8894	2.44281	...
40	50	5.15339	3.22368	31.1873	1.61184	...
50	60	2.82663	1.76819	17.1062	0.88409	...
60	70	2.78026	1.57491	15.4968	0.78746	...
70	80	2.73899	1.39213	13.937	0.69607	...
80	90	2.74463	1.29784	13.1318	0.64892	...
90	100	2.8045	1.15656	11.8907	0.57828	...
100	110	2.76524	0.99783	10.4022	0.49892	...
110	120	2.74967	0.89894	9.43159	0.44947	...
120	130	4.45277	1.5631	16.4626	0.78155	...
130	140	9.58467	3.40375	35.8485	1.70187	...
140	150	9.58464	3.17325	33.5261	1.58663	...
150	160	9.14434	2.52646	26.7894	1.26323	...
160	170	8.73343	1.97075	20.7539	0.98537	...
170	180	8.03499	1.67663	17.482	0.83831	...
180	190	10.2254	2.2291	23.2341	1.11455	...
190	200	10.0881	1.76165	18.2139	0.88082	...
200	210	11.2181	1.52139	15.8807	0.76069	...
210	220	8.15839	1.11944	12.0407	0.55972	...
220	230	9.14984	0.88746	9.85757	0.44373	...
230	240	7.39472	0.48492	5.55455	0.24246	...
240	250	8.36156	0.31147	3.7775	0.15573	...
250	260	8.17509	0.13591	1.62979	0.067956	...
260	270	3.43308	0.040195	0.33755	0.020098	...
270	280	0.44967	0.016785	0.1135	0.0083923	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	42.8429
20	30	...	...	...	39.6659
30	40	...	...	...	34.5185
40	50	...	...	...	23.0302
50	60	...	...	...	12.6321
60	70	...	...	...	11.5132
70	80	...	...	...	10.4141
80	90	...	...	...	9.84753
90	100	...	...	...	8.98139
100	110	...	...	...	7.94164
110	120	...	...	...	7.27259
120	130	...	...	...	12.8185
130	140	...	...	...	27.9131
140	150	...	...	...	26.383
150	160	...	...	...	21.8
160	170	...	...	...	17.7442
170	180	...	...	...	15.3605
180	190	...	...	...	20.4371
190	200	...	...	...	16.9462
200	210	...	...	...	15.2862



210	220	...	...	...	11.8086
220	230	...	...	...	9.94776
230	240	...	...	...	5.74289
240	250	...	...	...	3.94024
250	260	...	...	...	1.69921
260	270	...	...	...	0.35163
270	280	...	...	...	0.11822

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	-0.0265	0.71326
20	30	-0.0265	0.70276
30	40	-0.025051	0.65658
40	50	-0.017317	0.4494
50	60	-0.0095009	0.22325
60	70	-0.0095009	0.19698
70	80	-0.0095009	0.17131
80	90	-0.0096078	0.16139
90	100	-0.0098173	0.1526
100	110	-0.0098173	0.14385
110	120	-0.0098173	0.1379
120	130	-0.018524	0.24328
130	140	...	0.54884
140	150	...	0.54515
150	160	...	0.49549
160	170	...	0.47667
170	180	...	0.43912
180	190	...	0.52233
190	200	...	0.41054
200	210	...	0.3359
210	220	...	0.28652
220	230	...	0.28556
230	240	...	0.18321
240	250	...	0.16212
250	260	...	0.14675
260	270	...	0.065258
270	280	...	0.028566

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	4.43442	4.43442	4.24028	4.24028
20	30	4.19647	4.19647	4.01275	4.01275
30	40	3.74428	3.74428	3.58911	3.58911
40	50	2.53286	2.53286	2.28122	2.28122
50	60	1.38927	1.38927	1.31093	1.31093
60	70	1.30317	1.30317	1.22968	1.22968
70	80	1.21707	1.21707	1.14844	1.14844
80	90	1.17633	1.17633	1.16458	1.16458
90	100	1.11297	1.11297	1.1016	1.1016
100	110	1.02395	1.02395	1.01349	1.01349
110	120	0.9646	0.9646	0.95475	0.95475
120	130	1.73654	1.73654	1.70296	1.70296
130	140	3.68418	3.68418	3.44095	3.44095
140	150	3.56198	3.56198	3.32684	3.32684
150	160	3.11857	3.11857	2.91268	2.91268
160	170	2.71123	2.71123	2.53217	2.53217
170	180	2.4228	2.4228	1.94306	1.94306
180	190	3.22758	3.22758	2.42671	2.42671
190	200	2.89529	2.89529	2.17688	2.17688
200	210	2.88426	2.88426	2.10286	2.10286



210	220	2.51033	2.51033	1.83023	1.83023
220	230	2.44227	2.44227	1.71384	1.71384
230	240	1.68384	1.68384	1.59096	1.59096
240	250	1.50953	1.50953	1.41578	1.41578
250	260	1.11018	1.11018	1.04123	1.04123
260	270	0.76408	0.76408	0.71663	0.71663
270	280	0.63096	0.63096	0.59178	0.59178

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...
220	230	...
230	240	...
240	250	...
250	260	...
260	270	...
270	280	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	32.65	113.10	-32.65	102.67	0.2887	0.3180	0.1702
20	31.47	113.10	-31.47	102.67	0.2782	0.3065	0.1625
30	28.70	86.02	-28.70	100.64	0.3337	0.2852	0.1497
40	19.53	156.40	-19.53	135.00	0.1249	0.1447	0.1960
50	10.38	156.40	-10.38	158.43	0.0664	0.0655	0.0377
60	9.76	156.40	-9.76	158.43	0.0624	0.0616	0.0357
70	9.18	156.40	-9.18	158.43	0.0587	0.0579	0.0337
80	8.87	156.40	-8.87	158.23	0.0567	0.0561	0.0329
90	8.46	156.40	-8.46	157.25	0.0541	0.0538	0.0322
100	7.89	156.40	-7.89	157.25	0.0505	0.0502	0.0304
110	7.53	156.40	-7.53	157.25	0.0481	0.0479	0.0293
120	12.71	156.40	-12.71	132.71	0.0813	0.0958	0.0696
130	26.68	156.40	-26.68	83.56	0.1706	0.3193	0.1764
140	26.21	156.40	-26.21	82.69	0.1676	0.3170	0.1738
150	23.27	156.40	-23.27	81.83	0.1488	0.2844	0.1584
160	21.46	156.40	-21.46	81.83	0.1372	0.2623	0.1437
170	19.47	156.40	-19.47	68.22	0.1245	0.2853	0.3044



180	24.73	156.40	-24.73	76.73	0.1581	0.3223	0.1787
190	23.33	156.40	-23.33	77.13	0.1492	0.3025	0.1680
200	24.76	156.40	-24.76	72.75	0.1583	0.3404	0.1843
210	19.47	156.40	-19.47	72.75	0.1245	0.2676	0.1390
220	20.32	156.40	-20.32	66.14	0.1299	0.3072	0.1534
230	15.13	156.40	-15.13	75.60	0.0967	0.2001	0.1108
240	15.35	156.40	-15.35	69.65	0.0982	0.2204	0.1225
250	14.10	138.00	-14.10	69.65	0.1022	0.2025	0.1133
260	6.32	156.40	-6.32	69.18	0.0404	0.0913	0.0510
270	2.01	156.40	-2.01	17.01	0.0128	0.1181	...

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	68.40	113.10	-68.40	102.67	0.6048	0.6662	0.5036
20	64.66	113.10	-64.66	102.67	0.5717	0.6298	0.4724
30	57.65	86.02	-57.65	100.64	0.6702	0.5729	0.4267
40	38.87	156.40	-38.87	135.73	0.2486	0.2864	0.4083
50	21.01	156.40	-21.01	158.43	0.1344	0.1326	0.1062
60	19.49	156.40	-19.49	158.43	0.1246	0.1231	0.0983
70	18.02	156.40	-18.02	158.43	0.1152	0.1137	0.0906
80	17.25	156.40	-17.25	158.23	0.1103	0.1090	0.0869
90	16.12	156.40	-16.12	157.25	0.1031	0.1025	0.0819
100	14.68	156.40	-14.68	157.25	0.0939	0.0934	0.0745
110	13.75	156.40	-13.75	157.25	0.0879	0.0875	0.0698
120	23.71	156.40	-23.71	132.89	0.1516	0.1784	0.1163
130	50.64	156.40	-50.64	83.56	0.3238	0.6061	0.4357
140	48.88	156.40	-48.88	82.69	0.3125	0.5911	0.4209
150	42.05	156.40	-42.05	81.83	0.2689	0.5139	0.3649
160	36.76	156.40	-36.76	81.83	0.2350	0.4492	0.3127
170	32.71	156.40	-32.71	69.94	0.2092	0.4677	0.5156
180	42.42	156.40	-42.42	76.73	0.2712	0.5528	0.3861
190	38.11	156.40	-38.11	77.13	0.2437	0.4941	0.3410
200	38.19	156.40	-38.19	72.75	0.2442	0.5250	0.3466
210	29.87	156.40	-29.87	72.75	0.1910	0.4106	0.2651
220	29.09	156.40	-29.09	66.14	0.1860	0.4399	0.2648
230	20.21	156.40	-20.21	75.60	0.1292	0.2673	0.1717
240	18.82	156.40	-18.82	69.65	0.1203	0.2702	0.1661
250	15.52	138.00	-15.52	69.65	0.1124	0.2228	0.1848
260	6.56	156.40	-6.56	69.18	0.0420	0.0949	0.1973
270	2.08	156.40	-2.08	17.01	0.0133	0.1224	...

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	42.25	113.10	-42.25	102.67	0.3736	0.4115	0.2507
20	41.07	113.10	-41.07	102.67	0.3631	0.4000	0.2471
30	37.78	86.02	-37.78	100.64	0.4392	0.3754	0.2355
40	22.88	156.40	-22.88	70.27	0.1463	0.3256	0.5058
50	12.22	156.40	-12.22	158.43	0.0781	0.0771	0.0443
60	11.60	156.40	-11.60	158.43	0.0742	0.0732	0.0422
70	11.01	156.40	-11.01	158.43	0.0704	0.0695	0.0402
80	10.68	156.40	-10.68	158.23	0.0683	0.0675	0.0388
90	10.26	156.40	-10.26	157.25	0.0656	0.0653	0.0371
100	9.70	156.40	-9.70	157.25	0.0620	0.0617	0.0351
110	9.33	156.40	-9.33	157.25	0.0597	0.0594	0.0337
120	16.12	156.40	-16.12	132.66	0.1031	0.1215	0.0832
130	34.17	156.40	-34.17	83.56	0.2185	0.4090	0.2259
140	33.70	156.40	-33.70	82.69	0.2155	0.4076	0.2260
150	30.59	156.40	-30.59	81.83	0.1956	0.3738	0.2069
160	28.62	156.40	-28.62	81.83	0.1830	0.3497	0.1990
170	20.44	156.40	-20.44	27.13	0.1307	0.7533	...
180	26.03	156.40	-26.03	76.73	0.1664	0.3392	0.1787
190	24.63	156.40	-24.63	77.13	0.1575	0.3193	0.1680
200	26.22	156.40	-26.22	72.75	0.1676	0.3604	0.1843
210	20.92	156.40	-20.92	72.75	0.1338	0.2876	0.1390
220	21.59	156.40	-21.59	66.14	0.1380	0.3263	0.1534
230	15.41	156.40	-15.41	75.60	0.0985	0.2038	0.1108



240	15.35	156.40	-15.35	69.65	0.0982	0.2204	0.1225
250	14.10	138.00	-14.10	69.65	0.1022	0.2025	0.1133
260	6.32	156.40	-6.32	69.18	0.0404	0.0913	0.0510
270	2.01	156.40	-2.01	17.01	0.0128	0.1181	...

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	93.93	113.10	-93.93	102.67	0.8305	0.9149	0.6491
20	88.77	113.10	-88.77	102.67	0.7849	0.8646	0.6049
30	79.10	86.02	-79.10	100.64	0.9196	0.7860	0.5424
40	50.38	156.40	-50.38	135.80	0.3221	0.3710	0.5660
50	27.33	156.40	-27.33	158.43	0.1747	0.1725	0.1344
60	25.32	156.40	-25.32	158.43	0.1619	0.1598	0.1235
70	23.38	156.40	-23.38	158.43	0.1495	0.1476	0.1128
80	22.34	156.40	-22.34	158.23	0.1429	0.1412	0.1076
90	20.83	156.40	-20.83	157.25	0.1332	0.1325	0.1004
100	18.95	156.40	-18.95	157.25	0.1211	0.1205	0.0901
110	17.72	156.40	-17.72	157.25	0.1133	0.1127	0.0835
120	30.76	156.40	-30.76	132.90	0.1967	0.2314	0.1461
130	66.07	156.40	-66.07	83.56	0.4224	0.7907	0.5197
140	63.51	156.40	-63.51	82.69	0.4061	0.7681	0.4969
150	54.36	156.40	-54.36	81.83	0.3476	0.6642	0.4184
160	46.92	156.40	-46.92	81.83	0.3000	0.5734	0.3450
170	35.80	156.40	-35.80	69.52	0.2289	0.5150	0.6649
180	46.51	156.40	-46.51	76.73	0.2974	0.6061	0.4179
190	40.67	156.40	-40.67	77.13	0.2600	0.5273	0.3554
200	40.24	156.40	-40.24	72.75	0.2573	0.5532	0.3536
210	31.56	156.40	-31.56	72.75	0.2018	0.4338	0.2679
220	30.27	156.40	-30.27	66.14	0.1935	0.4576	0.2636
230	20.29	156.40	-20.29	75.60	0.1297	0.2684	0.1695
240	18.66	156.40	-18.66	69.65	0.1193	0.2679	0.1641
250	15.45	138.00	-15.45	69.65	0.1119	0.2218	0.1848
260	6.55	156.40	-6.55	69.18	0.0419	0.0947	0.1973
270	2.08	156.40	-2.08	17.01	0.0133	0.1221	...

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	49.00	247.00	-49.00	91.60	0.1984	0.5349	0.4024
20	48.36	247.00	-48.36	91.60	0.1958	0.5280	0.3988
30	45.16	247.00	-45.16	93.17	0.1829	0.4848	0.3810
40	15.37	247.00	-15.37	102.77	0.0622	0.1496	0.6069
50	8.10	247.00	-8.10	114.15	0.0328	0.0710	0.0309
60	7.66	247.00	-7.66	114.15	0.0310	0.0671	0.0288
70	7.25	247.00	-7.25	114.15	0.0294	0.0636	0.0269
80	7.03	247.00	-7.03	114.00	0.0285	0.0617	0.0257
90	6.76	247.00	-6.76	113.66	0.0274	0.0595	0.0239
100	6.36	247.00	-6.36	113.66	0.0257	0.0560	0.0219
110	6.11	247.00	-6.11	113.66	0.0247	0.0537	0.0206
120	10.18	247.00	-10.18	101.26	0.0412	0.1005	0.0585
130	42.73	247.00	-42.73	78.51	0.1730	0.5443	0.4039
140	42.50	247.00	-42.50	78.51	0.1721	0.5414	0.4052
150	39.83	247.00	-39.83	77.73	0.1613	0.5124	0.3832
160	38.23	247.00	-38.23	76.95	0.1548	0.4968	0.3713
170	16.21	247.00	-16.21	79.22	0.0656	0.2046	0.5159
180	20.39	247.00	-20.39	62.07	0.0826	0.3285	0.1288
190	19.56	247.00	-19.56	62.07	0.0792	0.3151	0.1246
200	21.12	247.00	-21.12	55.21	0.0855	0.3825	0.1408
210	16.40	247.00	-16.40	55.21	0.0664	0.2970	0.1117
220	17.43	247.00	-17.43	48.33	0.0706	0.3607	0.1246
230	13.20	247.00	-13.20	58.67	0.0535	0.2251	0.0882
240	13.69	247.00	-13.69	51.80	0.0554	0.2642	0.1021
250	12.92	234.00	-12.92	51.80	0.0552	0.2495	0.0988
260	5.53	247.00	-5.53	51.80	0.0224	0.1068	0.0414
270	1.37	247.00	-1.37	51.80	0.0055	0.0264	0.0892

Analysis of Load Case 6 : NP+HW+HE



From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	45.82	247.00	-45.82	91.60	0.1855	0.5003	0.4024
20	45.49	247.00	-45.49	91.60	0.1842	0.4966	0.3988
30	42.72	247.00	-42.72	93.17	0.1730	0.4585	0.3810
40	13.76	247.00	-13.76	102.77	0.0557	0.1339	0.5724
50	7.22	247.00	-7.22	114.15	0.0292	0.0632	0.0309
60	6.88	247.00	-6.88	114.15	0.0278	0.0602	0.0288
70	6.56	247.00	-6.56	114.15	0.0266	0.0575	0.0269
80	6.39	247.00	-6.39	114.00	0.0259	0.0560	0.0257
90	6.18	247.00	-6.18	113.66	0.0250	0.0544	0.0239
100	5.86	247.00	-5.86	113.66	0.0237	0.0516	0.0219
110	5.66	247.00	-5.66	113.66	0.0229	0.0498	0.0206
120	9.40	247.00	-9.40	101.26	0.0380	0.0928	0.0551
130	41.03	247.00	-41.03	78.51	0.1661	0.5227	0.4039
140	40.91	247.00	-40.91	78.51	0.1656	0.5212	0.4052
150	38.57	247.00	-38.57	77.73	0.1561	0.4962	0.3832
160	37.25	247.00	-37.25	76.95	0.1508	0.4840	0.3713
170	15.37	247.00	-15.37	79.22	0.0622	0.1940	0.6255
180	19.28	247.00	-19.28	62.07	0.0780	0.3105	0.1252
190	18.68	247.00	-18.68	62.07	0.0756	0.3009	0.1183
200	20.36	247.00	-20.36	55.21	0.0824	0.3687	0.1329
210	15.84	247.00	-15.84	55.21	0.0641	0.2869	0.1117
220	16.99	247.00	-16.99	48.33	0.0688	0.3515	0.1246
230	12.96	247.00	-12.96	58.67	0.0525	0.2209	0.0854
240	13.53	247.00	-13.53	51.80	0.0548	0.2612	0.1002
250	12.85	234.00	-12.85	51.80	0.0549	0.2482	0.0980
260	5.51	247.00	-5.51	51.80	0.0223	0.1065	0.0411
270	1.36	247.00	-1.36	51.80	0.0055	0.0263	0.0916

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	42.25	113.10	-42.25	102.67	0.3736	0.4115	0.2507
20	41.07	113.10	-41.07	102.67	0.3631	0.4000	0.2471
30	37.78	86.02	-37.78	100.64	0.4392	0.3754	0.2355
40	132.04	156.40		70.27	0.8442		0.5058
50	129.02	156.40		158.43	0.8250		0.0443
60	127.95	156.40		158.43	0.8181		0.0422
70	126.88	156.40		158.43	0.8113		0.0402
80	127.89	156.40		158.23	0.8177		0.0388
90	130.59	156.40		157.25	0.8349		0.0371
100	130.54	156.40		157.25	0.8346		0.0351
110	130.51	156.40		157.25	0.8345		0.0337
120	139.51	156.40		101.26	0.8920		...
130	34.17	156.40	-34.17	83.56	0.2185	0.4090	0.2259
140	33.70	156.40	-33.70	82.69	0.2155	0.4076	0.2260
150	30.59	156.40	-30.59	81.83	0.1956	0.3738	0.2069
160	28.62	156.40	-28.62	81.83	0.1830	0.3497	0.1990
170	59.45	156.40		27.13	0.3801		...
180	84.14	156.40		76.73	0.5380		0.1787
190	77.43	156.40		77.13	0.4951		0.1680
200	79.67	156.40		72.75	0.5094		0.1843
210	77.85	156.40		72.75	0.4977		0.1390
220	88.36	156.40		66.14	0.5650		0.1534
230	85.99	156.40		75.60	0.5498		0.1108
240	96.92	156.40		69.65	0.6197		0.1225
250	96.62	138.00		69.65	0.7002		0.1133
260	95.13	156.40		69.18	0.6082		0.0510
270	84.87	156.40		51.80	0.5427		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	93.93	113.10	-93.93	102.67	0.8305	0.9149	0.6491
20	88.77	113.10	-88.77	102.67	0.7849	0.8646	0.6049
30	79.10	86.02	-79.10	100.64	0.9196	0.7860	0.5424



40	147.68	156.40		135.80	0.9442		0.5660
50	131.58	156.40		158.43	0.8413		0.1344
60	130.09	156.40		158.43	0.8318		0.1235
70	128.65	156.40		158.43	0.8226		0.1128
80	129.48	156.40		158.23	0.8279		0.1076
90	131.92	156.40		157.25	0.8435		0.1004
100	131.62	156.40		157.25	0.8416		0.0901
110	131.44	156.40		157.25	0.8404		0.0835
120	147.63	156.40		101.26	0.9439		...
130	66.07	156.40	-66.07	83.56	0.4224	0.7907	0.5197
140	63.51	156.40	-63.51	82.69	0.4061	0.7681	0.4969
150	54.36	156.40	-54.36	81.83	0.3476	0.6642	0.4184
160	46.92	156.40	-46.92	81.83	0.3000	0.5734	0.3450
170	66.97	156.40	-7.05	69.52	0.4282	0.1014	0.6649
180	93.96	156.40	-8.14	76.73	0.6008	0.1061	0.4179
190	84.56	156.40	-2.30	77.13	0.5407	0.0298	0.3554
200	85.35	156.40		72.75	0.5457		0.3536
210	81.36	156.40		72.75	0.5202		0.2679
220	90.88	156.40		66.14	0.5811		0.2636
230	87.41	156.40		75.60	0.5589		0.1695
240	97.78	156.40		69.65	0.6252		0.1641
250	96.94	138.00		69.65	0.7025		0.1848
260	95.16	156.40		69.18	0.6085		0.1973
270	84.88	156.40		51.80	0.5427		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	42.25	113.10	-42.25	102.84	0.3736	0.4109	0.2499
20	41.07	113.10	-41.07	102.84	0.3631	0.3994	0.2463
30	37.78	86.02	-37.78	100.89	0.4392	0.3745	0.2343
40	22.88	156.40	-22.88	70.92	0.1463	0.3226	0.5012
50	12.22	156.40	-12.22	159.82	0.0781	0.0765	0.0438
60	11.60	156.40	-11.60	159.82	0.0742	0.0726	0.0417
70	11.01	156.40	-11.01	159.82	0.0704	0.0689	0.0398
80	10.68	156.40	-10.68	159.61	0.0683	0.0669	0.0384
90	10.26	156.40	-10.26	158.66	0.0656	0.0647	0.0366
100	9.70	156.40	-9.70	158.66	0.0620	0.0611	0.0347
110	9.33	156.40	-9.33	158.66	0.0597	0.0588	0.0333
120	16.12	156.40	-16.12	133.80	0.1031	0.1205	0.0825
130	34.17	156.40	-34.17	83.56	0.2185	0.4090	0.2259
140	33.70	156.40	-33.70	82.69	0.2155	0.4076	0.2260
150	30.59	156.40	-30.59	81.83	0.1956	0.3738	0.2069
160	28.62	156.40	-28.62	81.83	0.1830	0.3497	0.1990
170	24.36	156.40	-24.36	27.13	0.1558	0.8980	...
180	31.25	156.40	-31.25	76.73	0.1998	0.4072	0.5892
190	29.85	156.40	-29.85	77.13	0.1909	0.3870	0.5626
200	32.09	156.40	-32.09	72.75	0.2052	0.4411	0.6697
210	26.79	156.40	-26.79	72.75	0.1713	0.3683	0.6341
220	28.29	156.40	-28.29	66.14	0.1809	0.4277	0.8439
230	20.93	156.40	-20.93	75.60	0.1338	0.2768	0.4332
240	21.61	156.40	-21.61	69.65	0.1381	0.3102	0.6295
250	20.35	138.00	-20.35	69.65	0.1475	0.2923	0.6138
260	12.57	156.40	-12.57	69.18	0.0804	0.1817	0.5151
270	8.26	156.40	-8.26	52.33	0.0528	0.1579	0.2138

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	93.93	113.10	-93.93	102.84	0.8305	0.9134	0.6483
20	88.77	113.10	-88.77	102.84	0.7849	0.8632	0.6041
30	79.10	86.02	-79.10	100.89	0.9196	0.7841	0.5413
40	50.38	156.40	-50.38	136.85	0.3221	0.3681	0.5617
50	27.33	156.40	-27.33	159.82	0.1747	0.1710	0.1333
60	25.32	156.40	-25.32	159.82	0.1619	0.1584	0.1224
70	23.38	156.40	-23.38	159.82	0.1495	0.1463	0.1118
80	22.34	156.40	-22.34	159.61	0.1429	0.1400	0.1067
90	20.83	156.40	-20.83	158.66	0.1332	0.1313	0.0995



100	18.95	156.40	-18.95	158.66	0.1211	0.1194	0.0893
110	17.72	156.40	-17.72	158.66	0.1133	0.1117	0.0827
120	30.76	156.40	-30.76	134.03	0.1967	0.2295	0.1449
130	66.07	156.40	-66.07	83.56	0.4224	0.7907	0.5197
140	63.51	156.40	-63.51	82.69	0.4061	0.7681	0.4969
150	54.36	156.40	-54.36	81.83	0.3476	0.6642	0.4184
160	46.92	156.40	-46.92	81.83	0.3000	0.5734	0.3450
170	39.73	156.40	-39.73	69.52	0.2540	0.5715	0.6649
180	51.73	156.40	-51.73	76.73	0.3308	0.6742	0.7335
190	45.89	156.40	-45.89	77.13	0.2934	0.5950	0.6663
200	46.11	156.40	-46.11	72.75	0.2948	0.6338	0.7495
210	37.43	156.40	-37.43	72.75	0.2393	0.5145	0.6780
220	36.97	156.40	-36.97	66.14	0.2364	0.5589	0.8835
230	25.81	156.40	-25.81	75.60	0.1651	0.3415	0.4544
240	24.91	156.40	-24.91	69.65	0.1593	0.3577	0.6486
250	21.70	138.00	-21.70	69.65	0.1573	0.3116	0.6330
260	12.80	156.40	-12.80	69.18	0.0819	0.1851	0.5341
270	8.33	156.40	-8.33	52.33	0.0533	0.1592	0.2138

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	49.00	247.00	-49.00	91.60	0.1984	0.5349	0.4024
20	48.36	247.00	-48.36	91.60	0.1958	0.5280	0.3988
30	45.16	247.00	-45.16	93.17	0.1829	0.4848	0.3810
40	18.54	247.00		102.77	0.0751		0.6069
50	21.56	247.00		114.15	0.0873		0.0309
60	21.39	247.00		114.15	0.0866		0.0288
70	21.23	247.00		114.15	0.0859		0.0269
80	22.51	247.00		114.00	0.0911		0.0257
90	22.83	247.00		113.66	0.0924		0.0239
100	22.69	247.00		113.66	0.0918		0.0219
110	22.60	247.00		113.66	0.0915		0.0206
120	21.85	247.00		101.26	0.0885		...
130	42.73	247.00	-42.73	78.51	0.1730	0.5443	0.4039
140	42.50	247.00	-42.50	78.51	0.1721	0.5414	0.4052
150	39.83	247.00	-39.83	77.73	0.1613	0.5124	0.3832
160	38.23	247.00	-38.23	76.95	0.1548	0.4968	0.3713
170	64.06	247.00		79.22	0.2594		0.5159
180	88.72	247.00		62.07	0.3592		0.1288
190	88.73	247.00		62.07	0.3592		0.1246
200	97.11	247.00		55.21	0.3932		0.1408
210	96.85	247.00		55.21	0.3921		0.1117
220	107.06	247.00		48.33	0.4334		0.1246
230	140.93	247.00		58.67	0.5706		0.0882
240	158.16	247.00		51.80	0.6403		0.1021
250	158.01	234.00		51.80	0.6753		0.0988
260	156.77	247.00		51.80	0.6347		0.0414
270	139.80	247.00		51.80	0.5660		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	45.82	247.00	-45.82	91.60	0.1855	0.5003	0.4024
20	45.49	247.00	-45.49	91.60	0.1842	0.4966	0.3988
30	42.72	247.00	-42.72	93.17	0.1730	0.4585	0.3810
40	17.93	247.00		102.77	0.0726		0.5724
50	21.21	247.00		114.15	0.0859		0.0309
60	21.09	247.00		114.15	0.0854		0.0288
70	20.97	247.00		114.15	0.0849		0.0269
80	22.28	247.00		114.00	0.0902		0.0257
90	22.62	247.00		113.66	0.0916		0.0239
100	22.52	247.00		113.66	0.0912		0.0219
110	22.45	247.00		113.66	0.0909		0.0206
120	21.52	247.00		101.26	0.0871		...
130	41.03	247.00	-41.03	78.51	0.1661	0.5227	0.4039
140	40.91	247.00	-40.91	78.51	0.1656	0.5212	0.4052
150	38.57	247.00	-38.57	77.73	0.1561	0.4962	0.3832



160	37.25	247.00	-37.25	76.95	0.1508	0.4840	0.3713
170	63.93	247.00		79.22	0.2588		0.6255
180	88.58	247.00		62.07	0.3586		0.1252
190	88.62	247.00		62.07	0.3588		0.1183
200	97.00	247.00		55.21	0.3927		0.1329
210	96.77	247.00		55.21	0.3918		0.1117
220	106.98	247.00		48.33	0.4331		0.1246
230	140.88	247.00		58.67	0.5704		0.0854
240	158.13	247.00		51.80	0.6402		0.1002
250	158.00	234.00		51.80	0.6752		0.0980
260	156.77	247.00		51.80	0.6347		0.0411
270	139.80	247.00		51.80	0.5660		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	21.84	113.10	-21.84	102.67	0.1931	0.2127	0.1485
20	21.50	113.10	-21.50	102.67	0.1901	0.2094	0.1449
30	20.05	86.02	-20.05	100.64	0.2330	0.1992	0.1370
40	130.92	156.40		70.27	0.8371		0.2535
50	128.56	156.40		158.43	0.8220		0.0312
60	127.54	156.40		158.43	0.8155		0.0291
70	126.53	156.40		158.43	0.8090		0.0271
80	127.57	156.40		158.23	0.8157		0.0259
90	130.30	156.40		157.25	0.8331		0.0241
100	130.28	156.40		157.25	0.8330		0.0221
110	130.27	156.40		157.25	0.8329		0.0208
120	139.57	156.40		101.26	0.8924		...
130	19.59	156.40	-19.59	83.56	0.1253	0.2345	0.1292
140	19.48	156.40	-19.48	82.69	0.1245	0.2355	0.1286
150	17.63	156.40	-17.63	81.83	0.1127	0.2154	0.1111
160	16.78	156.40	-16.78	81.83	0.1073	0.2051	0.1054
170	57.65	156.40		27.13	0.3686		...
180	81.77	156.40		76.73	0.5228		0.1260
190	75.49	156.40		77.13	0.4827		0.1190
200	77.92	156.40		72.75	0.4982		0.1339
210	76.63	156.40		72.75	0.4900		0.1125
220	87.35	156.40		66.14	0.5585		0.1258
230	85.38	156.40		75.60	0.5459		0.0856
240	96.49	156.40		69.65	0.6170		0.1006
250	96.34	138.00		69.65	0.6981		0.0983
260	95.01	156.40		69.18	0.6075		0.0413
270	84.83	156.40		51.80	0.5424		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	22.55	113.10	-22.55	102.67	0.1994	0.2196	0.2393
20	22.23	113.10	-22.23	102.67	0.1965	0.2165	0.2359
30	20.83	86.02	-20.83	100.64	0.2421	0.2069	0.2254
40	127.72	156.40		70.27	0.8166		0.2964
50	128.47	156.40		158.43	0.8214		0.0416
60	127.46	156.40		158.43	0.8150		0.0397
70	126.46	156.40		158.43	0.8086		0.0378
80	127.52	156.40		158.23	0.8153		0.0383
90	130.24	156.40		157.25	0.8328		0.0366
100	130.23	156.40		157.25	0.8327		0.0346
110	130.22	156.40		157.25	0.8326		0.0333
120	135.91	156.40		101.26	0.8690		...
130	16.34	156.40	-16.34	83.56	0.1045	0.1955	0.2109
140	16.23	156.40	-16.23	82.69	0.1038	0.1963	0.2110
150	14.75	156.40	-14.75	81.83	0.0943	0.1802	0.1931
160	14.19	156.40	-14.19	81.83	0.0907	0.1734	0.1858
170	55.52	156.40		27.13	0.3550		...
180	78.91	156.40		76.73	0.5046		0.1081
190	73.01	156.40		77.13	0.4668		0.1028
200	75.57	156.40		72.75	0.4832		0.1130
210	75.26	156.40		72.75	0.4812		0.0974



220	85.91	156.40	66.14	0.5493	0.1023
230	83.93	156.40	75.60	0.5366	0.0779
240	94.91	156.40	69.65	0.6069	0.0740
250	94.85	138.00	69.65	0.6873	0.0670
260	94.56	156.40	69.18	0.6046	0.0300
270	84.80	156.40	51.80	0.5422	...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	31.44	113.10	-31.44	102.67	0.2780	0.3062	0.2504
20	31.10	113.10	-31.10	102.67	0.2750	0.3030	0.2468
30	29.13	86.02	-29.13	100.64	0.3386	0.2894	0.2352
40	129.41	156.40		70.27	0.8274		0.3826
50	128.71	156.40		158.43	0.8229		0.0442
60	127.68	156.40		158.43	0.8164		0.0421
70	126.66	156.40		158.43	0.8098		0.0402
80	127.69	156.40		158.23	0.8164		0.0387
90	130.41	156.40		157.25	0.8338		0.0370
100	130.39	156.40		157.25	0.8337		0.0350
110	130.37	156.40		157.25	0.8336		0.0337
120	137.91	156.40		101.26	0.8818		...
130	27.09	156.40	-27.09	83.56	0.1732	0.3241	0.2259
140	26.97	156.40	-26.97	82.69	0.1724	0.3261	0.2260
150	24.94	156.40	-24.94	81.83	0.1595	0.3048	0.2069
160	23.93	156.40	-23.93	81.83	0.1530	0.2925	0.1990
170	57.97	156.40		27.13	0.3706		...
180	82.19	156.40		76.73	0.5255		0.1440
190	75.88	156.40		77.13	0.4852		0.1369
200	78.32	156.40		72.75	0.5008		0.1552
210	76.95	156.40		72.75	0.4920		0.1338
220	87.61	156.40		66.14	0.5601		0.1461
230	85.45	156.40		75.60	0.5463		0.0856
240	96.49	156.40		69.65	0.6170		0.1006
250	96.34	138.00		69.65	0.6981		0.0983
260	95.01	156.40		69.18	0.6075		0.0413
270	84.83	156.40		51.80	0.5424		...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	21.84	113.10	-21.84	102.67	0.1931	0.2127	0.1485
20	21.50	113.10	-21.50	102.67	0.1901	0.2094	0.1449
30	20.05	86.02	-20.05	100.64	0.2330	0.1992	0.1370
40	130.92	156.40		70.27	0.8371		0.2535
50	128.56	156.40		158.43	0.8220		0.0312
60	127.54	156.40		158.43	0.8155		0.0291
70	126.53	156.40		158.43	0.8090		0.0271
80	127.57	156.40		158.23	0.8157		0.0259
90	130.30	156.40		157.25	0.8331		0.0241
100	130.28	156.40		157.25	0.8330		0.0221
110	130.27	156.40		157.25	0.8329		0.0208
120	139.57	156.40		101.26	0.8924		...
130	19.59	156.40	-19.59	83.56	0.1253	0.2345	0.1292
140	19.48	156.40	-19.48	82.69	0.1245	0.2355	0.1286
150	17.63	156.40	-17.63	81.83	0.1127	0.2154	0.1111
160	16.78	156.40	-16.78	81.83	0.1073	0.2051	0.1054
170	57.65	156.40		27.13	0.3686		...
180	81.77	156.40		76.73	0.5228		0.1260
190	75.49	156.40		77.13	0.4827		0.1190
200	77.92	156.40		72.75	0.4982		0.1339
210	76.63	156.40		72.75	0.4900		0.1125
220	87.35	156.40		66.14	0.5585		0.1258
230	85.38	156.40		75.60	0.5459		0.0856
240	96.49	156.40		69.65	0.6170		0.1006
250	96.34	138.00		69.65	0.6981		0.0983
260	95.01	156.40		69.18	0.6075		0.0413
270	84.83	156.40		51.80	0.5424		...



Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	31.44	113.10	-31.44	102.67	0.2780	0.3062	0.2504
20	31.10	113.10	-31.10	102.67	0.2750	0.3030	0.2468
30	29.13	86.02	-29.13	100.64	0.3386	0.2894	0.2352
40	17.11	156.40	-17.11	70.27	0.1094	0.2435	0.3826
50	9.05	156.40	-9.05	158.43	0.0579	0.0571	0.0442
60	8.71	156.40	-8.71	158.43	0.0557	0.0550	0.0421
70	8.40	156.40	-8.40	158.43	0.0537	0.0530	0.0402
80	8.20	156.40	-8.20	158.23	0.0524	0.0518	0.0387
90	7.98	156.40	-7.98	157.25	0.0510	0.0508	0.0370
100	7.67	156.40	-7.67	157.25	0.0490	0.0487	0.0350
110	7.46	156.40	-7.46	157.25	0.0477	0.0474	0.0337
120	12.80	156.40	-12.80	68.45	0.0818	0.1870	0.1339
130	27.09	156.40	-27.09	83.56	0.1732	0.3241	0.2259
140	26.97	156.40	-26.97	82.69	0.1724	0.3261	0.2260
150	24.94	156.40	-24.94	81.83	0.1595	0.3048	0.2069
160	23.93	156.40	-23.93	81.83	0.1530	0.2925	0.1990
170	16.34	156.40	-16.34	27.13	0.1045	0.6022	...
180	20.57	156.40	-20.57	76.73	0.1315	0.2681	0.1440
190	19.97	156.40	-19.97	77.13	0.1277	0.2589	0.1369
200	21.81	156.40	-21.81	72.75	0.1395	0.2998	0.1552
210	17.29	156.40	-17.29	72.75	0.1106	0.2377	0.1338
220	18.26	156.40	-18.26	66.14	0.1167	0.2760	0.1461
230	13.24	156.40	-13.24	75.60	0.0846	0.1751	0.0856
240	13.53	156.40	-13.53	69.65	0.0865	0.1943	0.1006
250	12.85	138.00	-12.85	69.65	0.0931	0.1846	0.0983
260	5.51	156.40	-5.51	69.18	0.0353	0.0797	0.0413
270	1.36	156.40	-1.36	17.01	0.0087	0.0800	0.0941

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	35.16	113.10	-35.16	102.67	0.3109	0.3425	0.2428
20	34.60	113.10	-34.60	102.67	0.3059	0.3370	0.2394
30	32.21	86.02	-32.21	100.64	0.3745	0.3201	0.2281
40	130.75	156.40		70.27	0.8360		0.4266
50	128.82	156.40		158.43	0.8236		0.0426
60	127.77	156.40		158.43	0.8170		0.0407
70	126.74	156.40		158.43	0.8104		0.0389
80	127.77	156.40		158.23	0.8169		0.0376
90	130.48	156.40		157.25	0.8343		0.0359
100	130.45	156.40		157.25	0.8341		0.0340
110	130.43	156.40		157.25	0.8339		0.0327
120	138.87	156.40		101.26	0.8879		...
130	30.22	156.40	-30.22	83.56	0.1932	0.3617	0.2188
140	29.99	156.40	-29.99	82.69	0.1917	0.3626	0.2189
150	27.57	156.40	-27.57	81.83	0.1763	0.3369	0.2004
160	26.17	156.40	-26.17	81.83	0.1673	0.3198	0.1928
170	58.65	156.40		27.13	0.3750		...
180	83.12	156.40		76.73	0.5314		0.1367
190	76.68	156.40		77.13	0.4903		0.1312
200	79.07	156.40		72.75	0.5056		0.1503
210	77.48	156.40		72.75	0.4954		0.1296
220	88.08	156.40		66.14	0.5632		0.1415
230	85.82	156.40		75.60	0.5487		0.0799
240	96.81	156.40		69.65	0.6190		0.0765
250	96.56	138.00		69.65	0.6997		0.0692
260	95.11	156.40		69.18	0.6081		0.0310
270	84.87	156.40		51.80	0.5427		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	35.16	113.10	-35.16	102.84	0.3109	0.3419	0.2420



20	34.60	113.10	-34.60	102.84	0.3059	0.3364	0.2386
30	32.21	86.02	-32.21	100.89	0.3745	0.3193	0.2270
40	19.19	156.40	-19.19	70.92	0.1227	0.2706	0.4227
50	10.22	156.40	-10.22	159.82	0.0653	0.0639	0.0421
60	9.82	156.40	-9.82	159.82	0.0628	0.0614	0.0402
70	9.44	156.40	-9.44	159.82	0.0604	0.0591	0.0385
80	9.21	156.40	-9.21	159.61	0.0589	0.0577	0.0372
90	8.94	156.40	-8.94	158.66	0.0572	0.0564	0.0355
100	8.55	156.40	-8.55	158.66	0.0546	0.0539	0.0336
110	8.29	156.40	-8.29	158.66	0.0530	0.0522	0.0323
120	14.29	156.40	-14.29	133.78	0.0914	0.1069	0.0747
130	30.22	156.40	-30.22	83.56	0.1932	0.3617	0.2188
140	29.99	156.40	-29.99	82.69	0.1917	0.3626	0.2189
150	27.57	156.40	-27.57	81.83	0.1763	0.3369	0.2004
160	26.17	156.40	-26.17	81.83	0.1673	0.3198	0.1928
170	22.25	156.40	-22.25	27.13	0.1422	0.8200	...
180	28.50	156.40	-28.50	76.73	0.1822	0.3714	0.5536
190	27.68	156.40	-27.68	77.13	0.1770	0.3588	0.5306
200	30.23	156.40	-30.23	72.75	0.1933	0.4155	0.6380
210	25.39	156.40	-25.39	72.75	0.1623	0.3490	0.6131
220	27.11	156.40	-27.11	66.14	0.1734	0.4099	0.8219
230	20.26	156.40	-20.26	75.60	0.1295	0.2680	0.4171
240	21.13	156.40	-21.13	69.65	0.1351	0.3034	0.6123
250	20.07	138.00	-20.07	69.65	0.1454	0.2882	0.5977
260	12.47	156.40	-12.47	69.18	0.0797	0.1802	0.5090
270	8.22	156.40	-8.22	52.33	0.0525	0.1570	0.2138

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	31.44	113.10	-31.44	102.67	0.2780	0.3062	0.2504
20	31.10	113.10	-31.10	102.67	0.2750	0.3030	0.2468
30	29.13	86.02	-29.13	100.64	0.3386	0.2894	0.2352
40	129.41	156.40		70.27	0.8274		0.3826
50	128.71	156.40		158.43	0.8229		0.0442
60	127.68	156.40		158.43	0.8164		0.0421
70	126.66	156.40		158.43	0.8098		0.0402
80	127.69	156.40		158.23	0.8164		0.0387
90	130.41	156.40		157.25	0.8338		0.0370
100	130.39	156.40		157.25	0.8337		0.0350
110	130.37	156.40		157.25	0.8336		0.0337
120	137.91	156.40		101.26	0.8818		...
130	27.09	156.40	-27.09	83.56	0.1732	0.3241	0.2259
140	26.97	156.40	-26.97	82.69	0.1724	0.3261	0.2260
150	24.94	156.40	-24.94	81.83	0.1595	0.3048	0.2069
160	23.93	156.40	-23.93	81.83	0.1530	0.2925	0.1990
170	57.97	156.40		27.13	0.3706		...
180	82.19	156.40		76.73	0.5255		0.1440
190	75.88	156.40		77.13	0.4852		0.1369
200	78.32	156.40		72.75	0.5008		0.1552
210	76.95	156.40		72.75	0.4920		0.1338
220	87.61	156.40		66.14	0.5601		0.1461
230	85.45	156.40		75.60	0.5463		0.0856
240	96.49	156.40		69.65	0.6170		0.1006
250	96.34	138.00		69.65	0.6981		0.0983
260	95.01	156.40		69.18	0.6075		0.0413
270	84.83	156.40		51.80	0.5424		...

Absolute Maximum of the all of the Stress Ratio's

0.9442

Governing Element: HPBTM dish

Governing Load Case 8 : IP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
HP_main skirt-3	...	-22.9	7.8	63.2	48.2	+ Tens
HP_main skirt-3	...	-22.9	-7.8	-63.2	-93.9	- Comp



HP_main skirt-2	...	-22.6	7.8	58.4	43.7	+ Tens
HP_main skirt-2	...	-22.6	-7.8	-58.4	-88.8	- Comp
HP_main skirt-1	...	-21.1	7.4	50.6	36.9	+ Tens
HP_main skirt-1	...	-21.1	-7.4	-50.6	-79.1	- Comp
HP_BTМ dish	130.9	-11.5	5.2	33.7	158.2	+ Tens
HP_BTМ dish	130.9	-11.5	-5.2	-33.7	...	- Comp
HP_shell-1	71.8	-6.0	2.8	18.5	87.1	+ Tens
HP_shell-1	71.8	-6.0	-2.8	-18.5	...	- Comp
HP_shell-2	71.8	-5.7	2.8	16.8	85.6	+ Tens
HP_shell-2	71.8	-5.7	-2.8	-16.8	...	- Comp
HP_shell-3	71.8	-5.5	2.7	15.2	84.2	+ Tens
HP_shell-3	71.8	-5.5	-2.7	-15.2	...	- Comp
HP_shell-4	72.5	-5.3	2.7	14.3	84.3	+ Tens
HP_shell-4	72.5	-5.3	-2.7	-14.3	...	- Comp
HP_shell-5	74.1	-5.0	2.8	13.0	84.9	+ Tens
HP_shell-5	74.1	-5.0	-2.8	-13.0	...	- Comp
HP_shell-6	74.1	-4.8	2.8	11.4	83.5	+ Tens
HP_shell-6	74.1	-4.8	-2.8	-11.4	...	- Comp
HP_shell-7	74.1	-4.6	2.7	10.4	82.7	+ Tens
HP_shell-7	74.1	-4.6	-2.7	-10.4	...	- Comp
HP_TOP dish	139.8	-8.1	4.5	18.2	154.4	+ Tens
HP_TOP dish	139.8	-8.1	-4.5	-18.2	...	- Comp
HP_int_skirt-1	...	-17.0	9.6	39.5	32.2	+ Tens
HP_int_skirt-1	...	-17.0	-9.6	-39.5	-66.1	- Comp
HP_int_skirt-2	...	-16.8	9.6	37.1	29.8	+ Tens
HP_int_skirt-2	...	-16.8	-9.6	-37.1	-63.5	- Comp
LP_int_skirt-2	...	-15.3	9.1	29.9	23.7	+ Tens
LP_int_skirt-2	...	-15.3	-9.1	-29.9	-54.4	- Comp
LP_int_skirt-1	...	-14.7	8.7	23.5	17.5	+ Tens
LP_int_skirt-1	...	-14.7	-8.7	-23.5	-46.9	- Comp
LP_BTМ dish	28.8	-7.9	8.0	19.9	48.8	+ Tens
LP_BTМ dish	28.8	-7.9	-8.0	-19.9	-7.0	- Comp
LP-shell-1	38.4	-9.8	10.2	26.5	65.2	+ Tens
LP-shell-1	38.4	-9.8	-10.2	-26.5	-8.1	- Comp
LP-shell-2	38.4	-9.5	10.1	21.1	60.1	+ Tens
LP-shell-2	38.4	-9.5	-10.1	-21.1	-2.3	- Comp
LP-shell-3	43.2	-10.3	11.2	18.8	62.9	+ Tens
LP-shell-3	43.2	-10.3	-11.2	-18.8	...	- Comp
LP-shell-4	43.2	-8.8	8.2	14.6	57.0	+ Tens
LP-shell-4	43.2	-8.8	-8.2	-14.6	...	- Comp
LP-shell-5	49.4	-8.8	9.1	12.3	62.0	+ Tens
LP-shell-5	49.4	-8.8	-9.1	-12.3	...	- Comp
LP-shell-6	40.6	-5.7	7.4	7.2	49.6	+ Tens
LP-shell-6	40.6	-5.7	-7.4	-7.2	...	- Comp
LP-shell-7	46.0	-5.0	8.4	5.3	54.7	+ Tens
LP-shell-7	46.0	-5.0	-8.4	-5.3	...	- Comp



LP-shell-8	46.0	-4.5	8.2	2.7	52.4	+ Tens
LP-shell-8	46.0	-4.5	-8.2	-2.7	...	- Comp
LP-shell-9	46.0	-2.0	3.4	1.1	48.6	+ Tens
LP-shell-9	46.0	-2.0	-3.4	-1.1	...	- Comp
LP-TOP dish	46.0	-0.9	0.4	0.7	46.3	+ Tens
LP-TOP dish	46.0	-0.9	-0.4	-0.7	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm
 Fxa - Allowable comp mem stress due to compression with lamc ≤ 0.15 , N./mm²
 Fba - Allowable all comp mem stress due to bending moment, N./mm²
 Fva - Allowable shear stress when only shear is present, N./mm²
 Fha - Allowable hoop comp stress under ext pressure alone, N./mm²
 Fxha - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , N./mm²
 Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , N./mm²
 Faha - Allowable axial comp w/hoop compression and lamc > 0.15 , N./mm²
 Fbha - Allowable axial comp from bending w/hoop compression, N./mm²
 Fhba - Allowable hoop comp w/long compression due to moment, N./mm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						134	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120						132	6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						68	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						135	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120						132	6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895



150	0.428	16000	81	93	45	6895	81	6895	6895	6895
160	0.428	16000	81	93	45	6895	81	6895	6895	6895
170						69	6895	6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895
270						17	6895	6895	6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120						132	6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						135	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120						132	6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						69	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895



250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.484	16000	102	109	65		6895	102	6895	6895	6895
20	0.484	16000	102	109	65		6895	102	6895	6895	6895
30	0.478	16000	100	107	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
60	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
70	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
80	0.580	19667	159	159	93	51	6895	159	6895	6895	6895
90	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
100	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
110	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
120						133	6895		6895	6895	6895
130	0.446	16000	83	94	47		6895	83	6895	6895	6895
140	0.434	16000	82	94	46		6895	82	6895	6895	6895
150	0.422	16000	82	93	46		6895	82	6895	6895	6895
160	0.422	16000	82	93	46		6895	82	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.412	2849	77	88	63	34	6895	77	6895	6895	6895
190	0.413	2750	77	88	64	35	6895	77	6895	6895	6895
200	0.403	2500	73	85	61	33	6895	73	6895	6895	6895
210	0.403	2499	73	85	61	33	6895	73	6895	6895	6895
220	0.388	2500	66	81	57	27	6895	66	6895	6895	6895
230	0.410	2499	75	86	63	36	6895	75	6895	6895	6895
240	0.397	2500	70	83	59	30	6895	70	6895	6895	6895
250	0.397	2500	70	83	59	30	6895	70	6895	6895	6895
260	0.396	2600	69	83	59	28	6895	69	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.484	16000	102	109	65		6895	102	6895	6895	6895
20	0.484	16000	102	109	65		6895	102	6895	6895	6895
30	0.478	16000	100	107	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
60	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
70	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
80	0.580	19667	159	159	93	51	6895	159	6895	6895	6895
90	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
100	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
110	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
120						133	6895		6895	6895	6895
130	0.446	16000	83	94	47		6895	83	6895	6895	6895
140	0.434	16000	82	94	46		6895	82	6895	6895	6895
150	0.422	16000	82	93	46		6895	82	6895	6895	6895
160	0.422	16000	82	93	46		6895	82	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.412	2849	77	88	63	34	6895	77	6895	6895	6895
190	0.413	2750	77	88	64	35	6895	77	6895	6895	6895
200	0.403	2500	73	85	61	33	6895	73	6895	6895	6895
210	0.403	2499	73	85	61	33	6895	73	6895	6895	6895
220	0.388	2500	66	81	57	27	6895	66	6895	6895	6895
230	0.410	2499	75	86	63	36	6895	75	6895	6895	6895
240	0.397	2500	70	83	59	30	6895	70	6895	6895	6895
250	0.397	2500	70	83	59	30	6895	70	6895	6895	6895
260	0.396	2600	69	83	59	28	6895	69	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895



50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						135	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						69	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.481	16000	102	109	65		6895	102	6895	6895	6895
20	0.481	16000	102	109	65		6895	102	6895	6895	6895
30	0.476	16000	100	107	65		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
60	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
70	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
80	0.577	19667	159	159	93	51	6895	159	6895	6895	6895
90	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
100	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
110	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
120						133	6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895



150	0.428	16000	81	93	45	6895	81	6895	6895	6895
160	0.428	16000	81	93	45	6895	81	6895	6895	6895
170						27	6895	6895	6895	6895
180	0.417	2849	76	87	63	33	41	27	37	45
190	0.418	2750	77	87	63	34	41	28	37	44
200	0.408	2500	72	84	61	32	37	26	34	39
210	0.408	2499	72	84	61	32	34	27	31	30
220	0.392	2500	66	81	56	26	27	23	24	23
230	0.414	2499	75	86	63	35	32	31	29	29
240	0.401	2500	69	83	59	29	24	26	22	23
250	0.401	2500	69	83	59	29	23	26	21	21
260	0.400	2600	69	83	58	28	17	26	16	9
270						52	6895	6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.481	16000	102	109	65		6895	102	6895	6895	6895
20	0.481	16000	102	109	65		6895	102	6895	6895	6895
30	0.476	16000	100	107	65		6895	100	6895	6895	6895
40						136	6895		6895	6895	6895
50	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
60	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
70	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
80	0.577	19667	159	159	93	51	6895	159	6895	6895	6895
90	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
100	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
110	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
120						134	6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						69	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	40	27	36	72	20
190	0.418	2750	77	87	63	34	40	28	37	69	23
200	0.408	2500	72	84	61	32	36	26	33	62	24
210	0.408	2499	72	84	61	32	34	27	31	53	27
220	0.392	2500	66	81	56	26	26	23	24	39	24
230	0.414	2499	75	86	63	35	31	31	28	42	31
240	0.401	2500	69	83	59	29	23	26	21	30	27
250	0.401	2500	69	83	59	29	23	26	21	24	28
260	0.400	2600	69	83	58	28	17	26	16	10	27
270						52	6895		6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.484	16000	102	109	65		6895	102	6895	6895	6895
20	0.484	16000	102	109	65		6895	102	6895	6895	6895
30	0.478	16000	100	107	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
60	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
70	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
80	0.580	19667	159	159	93	51	6895	159	6895	6895	6895
90	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
100	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
110	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
120							6895		6895	6895	6895
130	0.446	16000	83	94	47		6895	83	6895	6895	6895
140	0.434	16000	82	94	46		6895	82	6895	6895	6895
150	0.422	16000	82	93	46		6895	82	6895	6895	6895
160	0.422	16000	82	93	46		6895	82	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.412	2849	77	88	63	34	6895	77	6895	6895	6895
190	0.413	2750	77	88	64	35	6895	77	6895	6895	6895
200	0.403	2500	73	85	61	33	6895	73	6895	6895	6895
210	0.403	2499	73	85	61	33	6895	73	6895	6895	6895
220	0.388	2500	66	81	57	27	6895	66	6895	6895	6895
230	0.410	2499	75	86	63	36	6895	75	6895	6895	6895
240	0.397	2500	70	83	59	30	6895	70	6895	6895	6895



250	0.397	2500	70	83	59	30	6895	70	6895	6895	6895
260	0.396	2600	69	83	59	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.484	16000	102	109	65		6895	102	6895	6895	6895
20	0.484	16000	102	109	65		6895	102	6895	6895	6895
30	0.478	16000	100	107	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
60	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
70	0.582	19667	159	159	93	51	6895	159	6895	6895	6895
80	0.580	19667	159	159	93	51	6895	159	6895	6895	6895
90	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
100	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
110	0.579	19667	158	158	92	50	6895	158	6895	6895	6895
120							6895		6895	6895	6895
130	0.446	16000	83	94	47		6895	83	6895	6895	6895
140	0.434	16000	82	94	46		6895	82	6895	6895	6895
150	0.422	16000	82	93	46		6895	82	6895	6895	6895
160	0.422	16000	82	93	46		6895	82	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.412	2849	77	88	63	34	6895	77	6895	6895	6895
190	0.413	2750	77	88	64	35	6895	77	6895	6895	6895
200	0.403	2500	73	85	61	33	6895	73	6895	6895	6895
210	0.403	2499	73	85	61	33	6895	73	6895	6895	6895
220	0.388	2500	66	81	57	27	6895	66	6895	6895	6895
230	0.410	2499	75	86	63	36	6895	75	6895	6895	6895
240	0.397	2500	70	83	59	30	6895	70	6895	6895	6895
250	0.397	2500	70	83	59	30	6895	70	6895	6895	6895
260	0.396	2600	69	83	59	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895



50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895



150	0.428	16000	81	93	45	6895	81	6895	6895	6895
160	0.428	16000	81	93	45	6895	81	6895	6895	6895
170						27	6895	6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895
270						6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120						68	6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895



250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.481	16000	102	109	65		6895	102	6895	6895	6895
20	0.481	16000	102	109	65		6895	102	6895	6895	6895
30	0.476	16000	100	107	65		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
60	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
70	0.579	19667	159	159	93	52	6895	159	6895	6895	6895
80	0.577	19667	159	159	93	51	6895	159	6895	6895	6895
90	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
100	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
110	0.576	19667	158	158	92	50	6895	158	6895	6895	6895
120						133	6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	40	27	36	10	33
190	0.418	2750	77	87	63	34	40	28	37	9	34
200	0.408	2500	72	84	61	32	36	26	33	7	31
210	0.408	2499	72	84	61	32	34	27	31	6	31
220	0.392	2500	66	81	56	26	26	23	24	4	26
230	0.414	2499	75	86	63	35	31	31	28	5	34
240	0.401	2500	69	83	59	29	23	26	21	3	29
250	0.401	2500	69	83	59	29	23	26	21	2	29
260	0.400	2600	69	83	58	28	17	26	16	1	28
270						52	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.487	16000	102	109	65		6895	102	6895	6895	6895
20	0.487	16000	102	109	65		6895	102	6895	6895	6895
30	0.485	16000	100	106	64		6895	100	6895	6895	6895
40						70	6895		6895	6895	6895
50	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
60	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
70	0.588	19667	158	158	93	51	6895	158	6895	6895	6895
80	0.586	19667	158	158	93	50	6895	158	6895	6895	6895
90	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
100	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
110	0.585	19667	157	157	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.452	16000	83	94	46		6895	83	6895	6895	6895
140	0.439	16000	82	93	45		6895	82	6895	6895	6895
150	0.428	16000	81	93	45		6895	81	6895	6895	6895
160	0.428	16000	81	93	45		6895	81	6895	6895	6895
170						27	6895		6895	6895	6895
180	0.417	2849	76	87	63	33	6895	76	6895	6895	6895
190	0.418	2750	77	87	63	34	6895	77	6895	6895	6895
200	0.408	2500	72	84	61	32	6895	72	6895	6895	6895
210	0.408	2499	72	84	61	32	6895	72	6895	6895	6895
220	0.392	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.414	2499	75	86	63	35	6895	75	6895	6895	6895
240	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
250	0.401	2500	69	83	59	29	6895	69	6895	6895	6895
260	0.400	2600	69	83	58	28	6895	69	6895	6895	6895
270							6895		6895	6895	6895

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90,



4.4.91 and 4.4.92.

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: LPintskirt-1

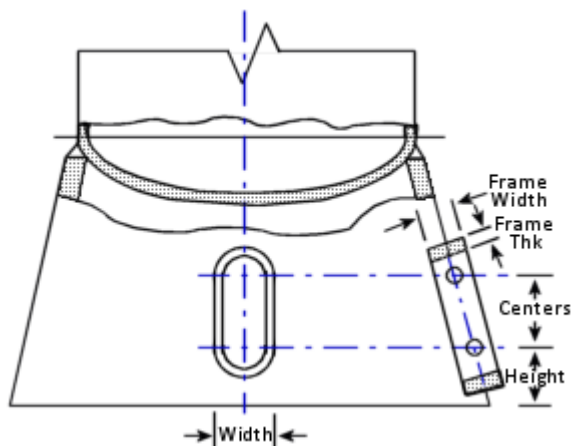
Total vessel (empty) weight	Fg	1050875.250 Kgf
Head Weight below section 2-2	DeltaFg	30681.086 Kgf
Weight of vessel contents (liquid)	Ff	658254.250 Kgf
Vessel Gross Weight	Fg + Ff	1709129.625 Kgf
Skirt Mean Diameter	Dz	7229.600 mm
Skirt Analysis Thickness (Corroded)	ez	22.400 mm
Skirt Height	H	1250.000 mm
Additional Axial Force at section 1-1	F1	-10848.891 Kgf
Moment at section 1-1	M1	1996667.375 Kg-m.
Applied Moment - Bottom of Skirt	Mb	2193475.250 Kg-m.
Weight at Bottom of Skirt	W	788546.188 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	7236.000 mm
Head Corroded Thickness	eb	24.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 1806.0 \text{ mm} \\
 K &= D_i / (2 * h_i) = 7212.0 / (2 * 1806.0) = 1.9967 \\
 R &= D_i (0.44 * K + 0.02) = 7212.0 * (0.44 * 1.997 + 0.02) = 6480.2573 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 7212.0 * ((0.5 / 1.997) - 0.08) = 1229.04 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	332391776.000 cm ⁴
Moment of Inertia	Iyy =	332391776.000 cm ⁴
Section Modulus	W =	919531008.000 mm ³
Section Metal Area	A =	5087.591 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(0.33239E+09, 0.33239E+09) / 5087.591)} \\
 &= 2556.05 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned}
 &= r_g / L \\
 &= 2556.05/1250.0 \\
 &= 2.045
 \end{aligned}$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 1996667 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:
 Downwind Side:
 $= -M / Z - W / A$
 $= -1996667/.36781E+10 - 788546/5087.59082$
 $= -20.524 \text{ N./mm}^2$

Upwind Side:
 $= +(M) / Z - W / A$
 $= +1996667/.36781E+10 - 788546/5087.59082$
 $= -9.876 \text{ N./mm}^2$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= -10848.891 - 1050875 - 658254 + 4 \cdot 1996667/7229.6$
 $= -615282.1 \text{ Kgf}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -10848.891 - 1050875 - 658254 - 4 \cdot 1996667/7229.6$
 $= -2824674.8 \text{ Kgf}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-0.6E+06 + 30681.086 + 658254) / (\pi \cdot 7236.0 \cdot 24.0) + 380.0 \cdot 7236.0 / (4 \cdot 24.0)$
 $= 29.97 \text{ N./mm}^2$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-0.3E+07 + 30681.086 + 658254) / (\pi \cdot 7236.0 \cdot 24.0) + 380.0 \cdot 7236.0 / (4 \cdot 24.0)$
 $= -9.75 \text{ N./mm}^2$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (658254 + 30681.086) / (\pi \cdot 7236.0 \cdot 24.0) + 380.0 \cdot 7236.0 / (4 \cdot 24.0)$
 $= 41.03 \text{ N./mm}^2$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= -615282.062 / (\pi \cdot 7229.6 \cdot 22.4)$
 $= -11.86 \text{ N./mm}^2$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -2824674.750 / (\pi \cdot 7229.6 \cdot 22.4)$
 $= -54.45 \text{ N./mm}^2$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:
 $= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5(7236.0 + 24.0 - 7229.6 + 22.4) / (1229.04 + 24.0)$
 $= 0.9789$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.979) = 11.782 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(24.0^2 + 22.4^2 + 2 \cdot 24.0 \cdot 22.4 \cdot 0.979)}$$

$$= 23.078 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 23.07 \cdot -615282.1 = -14199.53 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 23.07 \cdot -2824674.8 = -65188.07 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{24.0}{22.4} \right)^2$$

$$= 0.565$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb}(o)$ & $\sigma_{2pb}(o)$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.565 \cdot 6 \cdot -14199.529 / (\pi \cdot 7236.0 \cdot 24.0^2)$$

$$= -36.02 \text{ N./mm}^2$$

Stress [$\sigma_{1qb}(o)$ & $\sigma_{2qb}(o)$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.565 \cdot 6 \cdot -65188.066 / (\pi \cdot 7236.0 \cdot 24.0^2)$$

$$= -165.38 \text{ N./mm}^2$$

Stress [$\sigma_{3pb}(o)$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.565 \cdot 6 \cdot -14199.529 / (\pi \cdot 7229.6 \cdot 22.4^2)$$

$$= -41.39 \text{ N./mm}^2$$

Stress [$\sigma_{3qb}(o)$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.565 \cdot 6 \cdot -65188.066 / (\pi \cdot 7229.6 \cdot 22.4^2)$$

$$= -190.02 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.945/284.882 = 0.0033$$

$$y = 125 \cdot eb/D_b = 125 \cdot 24.0/7236.0 = 0.4146$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.415 + 3.994 = 4.2565$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b}(p)$ & $\sigma_{2b}(p)$]:

$$= (P + Ph) \cdot D_b \left(\left(\frac{y}{y_A} \right) \cdot \alpha - 1 \right) / (4 \cdot eb)$$

$$= (380.0 + 0.0) \cdot 7236.0 \left(\left(\frac{11.78}{40.0} \right) \cdot 4.257 - 1 \right) / (4 \cdot 24.0)$$

$$= 7.27 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb}(o) + \sigma_{1b}(p)$$

$$= 29.968 - -36.024 + 7.269$$

$$= 73.26 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb}(o) - \sigma_{1b}(p)$$

$$= 29.968 + -36.024 - 7.269$$

$$= -13.32 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb}(o) + \sigma_{1b}(p)$$



$$= -9.746 - -165.38 + 7.269$$

$$= 162.90 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= -9.746 + -165.38 - 7.269$$

$$= -182.39 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 41.028 + -36.024 + 7.269$$

$$= 12.27 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 41.028 - -36.024 - 7.269$$

$$= 69.78 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 41.028 + -165.38 + 7.269$$

$$= -117.08 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 41.028 - -165.38 - 7.269$$

$$= 199.14 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= -11.86 - -41.39$$

$$= 29.53 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= -11.86 + -41.39$$

$$= -53.25 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -54.448 - -190.018$$

$$= 135.57 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -54.448 + -190.018$$

$$= -244.47 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 156.4(3 - (1 / 1.5)) \cdot (29.968 / 156.4)^2$$

$$= 465.37 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 156.4(3 - (1 / 1.5)) \cdot (-9.746 / 156.4)^2$$

$$= 468.80 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 156.4(3 - (1 / 1.5)) \cdot (41.028 / 156.4)^2$$

$$= 462.02 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 156.4(3 - (1 / 1.5)) \cdot (41.028 / 156.4)^2$$

$$= 462.02 \text{ N./mm}^2$$

For $\sigma_{3pi\text{tot}}$ & $\sigma_{3po\text{tot}}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 156.4(3 - (1/1.5) \cdot (-11.86/156.4)^2)$$

$$= 468.60 \text{ N./mm}^2$$

For $\sigma_{3qi\text{tot}}$ & $\sigma_{3qo\text{tot}}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 156.4(3 - (1/1.5) \cdot (-54.448/156.4)^2)$$

$$= 456.56 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	73.26	465.37	Pass	46
Stot1po	-13.32	465.37	Pass	47
Stot1qi	162.90	468.80	Pass	48
Stot1qo	-182.39	468.80	Pass	49
Stot2pi	12.27	462.02	Pass	50
Stot2po	69.78	462.02	Pass	51
Stot2qi	-117.08	462.02	Pass	52
Stot2qo	199.14	462.02	Pass	53
Stot3pi	29.53	468.60	Pass	54
Stot3po	-53.25	468.60	Pass	55
Stot3qi	135.57	456.56	Pass	56
Stot3qo	-244.47	456.56	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
 Packing is installed in the Shop.
 Trays are installed in the Shop.
 Insulation is installed in the Shop.
 Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	34673.285 mm
Center of Gravity of the Packing	28881.410 mm
Center of Gravity of the Liquid	22189.188 mm
Center of Gravity of the Insulation	32463.994 mm
Center of Gravity of the Lining	38234.020 mm
Center of Gravity of the Stiffening Rings	45632.621 mm
Center of Gravity of the Nozzles	19397.875 mm
Center of Gravity of the Trays	55800.051 mm
Center of Gravity of the Added Weights (Operating)	40922.414 mm
Center of Gravity of the Added Weights (Empty)	40922.414 mm
Center of Gravity of Bare Shell New and Cold	21187.020 mm
Center of Gravity of Bare Shell Corroded	20684.367 mm
Vessel CG in the Operating Condition	26162.648 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	28688.059 mm
Vessel CG in the Test Condition	31609.285 mm

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	57900.00 mm
Total vessel weight (No Liquid) Twt	1065258.88 Kgf
Impact weight multiplication factor Imp	1.20
Design lifting weight, DWT = Imp * Twt	1278310.75 Kgf
Elevation of the Tailing Lug (bottom)	0.00 mm
Elevation of the Lifting Lug (top)	47000.00 mm
Design Reaction force at the tailing lug	498050.06 Kgf
Design Reaction force at the lifting lug	780260.69 Kgf
CG distance during field lifting operation	28688.06 mm
CG distance from tailing lug	28688.06 mm
CG distance from the nearer lifting lug/trunnion	18311.94 mm
Compressive Allowable Increase Factor	1.20

Critical Values: (N./mm², mm)

	Max Stress	Elevation	Allowables
Bending	57.92	24900.01	96.93
Shear	10.90	46570.02	122.50 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	341666.1	4.3
11100.00	4261701.5	11.1	164030.9	0.7
24000.01	5301228.0	15.0	177195.8	0.9
32900.02	3391774.8	33.9	328831.7	5.9
44850.02	1057866.8	11.5	480096.1	9.4
57850.02	34477.0	0.5	6323.3	0.2



Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.5976 at elevation 24900.0078 mm - Must be ≤ 1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	6946.0000	mm
Skirt Thickness	STHK	36.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.6000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	60.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	175.00	N./mm ²
Basing Yield Stress		262.00	N./mm ²
Inside Diameter of Basing	DI, D4	6778.0000	mm
Outside Diameter of Basing	DOU, D5	7378.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Root Area of a Single Bolt	Area	21.5100	cm ²
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	284.00	N./mm ²
Number of Bolts	RN, nb	48	
Diameter of Bolt Circle	DC, DBC	7162.0000	mm
Bolt Allowable Shear Stress		170.400	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	26.0000	mm
Width of Gussets at Top Plate	TWDT	206.0000	mm
Width of Gussets at Base Plate	BWDT	206.0000	mm
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm ²
Height of Gussets	HG, h1	356.0000	mm
Distance between Gussets	RG, b6	120.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	60.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	54.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	216.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	90.0000	mm
External Corrosion Allowance	CA	1.6000	mm
Dead Weight of Vessel	DW	1065258.9	Kgf
Operating Weight of Vessel	ROW	1807560.0	Kgf
Earthquake Moment on Basing	EQMOM	8608552.0	Kg-m.
Wind Moment on Basing	WIMOM	1743579.5	Kg-m.
User Moment at the Base [Earthquake Case]		580736.8	Kg-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		580736.8	Kg-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 9189289/7162.0) - 1751215)/48$$

$$= 70436.1 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 2324316/7162.0) - 1065259)/48$$

$$= 4851.1 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/7162.0) - 1807560)/48$$

$$= -37657.5 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 21.5100 cm²
 Area Available in all the Bolts Abss * RN : 1032.4800 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.211	0.317	1.253	2.636	0.457	0.776	622713.2	2373928.5	9275.2
4	0.370	0.396	1.690	2.290	0.422	0.784	693389.4	2444604.5	11885.9
6	0.383	0.376	1.723	2.261	0.419	0.784	699646.2	2450861.5	12146.5
8	0.380	0.381	1.714	2.269	0.420	0.784	698076.5	2449291.5	12080.6
10	0.380	0.380	1.716	2.267	0.420	0.784	698468.6	2449683.8	12097.0

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 698469 / (4.589 \cdot 7162.0 \cdot 2.267)$$

$$= 183.882 \text{ N./mm}^2, \text{ Should be } \leq 284.0 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 48 \cdot 21.51 / (\pi \cdot 7162.0)$$

$$= 4.589 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 11.477[(2 \cdot 0.3803 \cdot 7162.0 + 300.0) / (2 \cdot 0.3803 \cdot 7162.0)]$$

$$= 12.109 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 216.0, b = 234.376, l/b = 0.921596

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(5932.521, 8319.039) = 8319.039 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 8319.039 / 172.9)} + 1.6$$

$$= 54.806 \text{ mm}$$

Nomenclature:

a = (Dc-Ds)/2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 216.0, b = 120.0, b/l = 0.555556
 As b/l (0.556) is less than 1, inverting b/l = 1.8.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1)] - [(0.7 - g2)P / (4\pi)]$$

$$= 62291.68 / (4 \cdot \pi) [1.3(\ln((2 \cdot 216.0 \cdot \sin(\pi \cdot 108.0/216.0)) / (\pi \cdot 53.975))) + 1] - [(0.7 - 0.04) \cdot 62291.68 / (4 \cdot \pi)]$$

$$= 7710.2393 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb})} + \text{Ca}$$

$$= \sqrt{(6 \cdot \text{Abs}(7710.24) / 174.66)} + 1.6$$

$$= 52.5651 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 62291.68 / (124.11 \cdot 206.0) + 1.6$$

$$= 25.494 \text{ (not less than } 9.525 + 1.6 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 7162.0 / 48 = 468.752 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (62292 \cdot 108.0 / (468.75 \cdot 416.0 \cdot 242))^{2/3} \cdot 3473.01^{1/3} + \text{Ca}$$

$$= 33.292 + 1.6 = 34.892 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 211302.2 / (21.51 \cdot 48)$$

$$= 20.1 \text{ N./mm}^2. \text{ Must be less than } 170.4 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	54.8057	mm
Actual Basing Thickness as entered by user	60.0000	mm
Required Thickness of Chair Cap	52.5651	mm
Actual Top Ring Thickness as entered by user	54.0000	mm
Required Gusset thickness, + CA	25.4938	mm
Actual Gusset Thickness as entered by user	26.0000	mm
Required Thickness of Skirt for Local Stress	34.8923	mm
Given Thickness of Skirt	36.0000	mm
Required Gusset Height to meet local stress	336.0703	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 93.931 \cdot (36.0 - 1.6)$$

$$= 329.49 \text{ Kg/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 329 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$

$$= 23.549 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 62291.7 / (137.0 + 2 \cdot (356.0 + 54.0))$$

$$= 65.091 \text{ Kg/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 62291.7 \cdot 108.0 / (137.0 \cdot (410.0) + 0.6667 \cdot (410.0)^2)$$

$$= 39.988 \text{ Kg/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{65.09^2 + 39.99^2}$$

$$= 76.393 \text{ Kg/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 76.39 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$



= 5.46 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= 62291.7 / (2 • 206.0)
= 151.193 Kgf/mm

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= 62291.7 • 108.0 / (2 • 410.0 • 206.0)
= 39.827 Kgf/mm

Resultant Weld Load [Wr]:
= $\sqrt{Wv^2 + Wh^2}$
= $\sqrt{151.19^2 + 39.83^2}$
= 156.351 Kgf/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 156.35 / [(0.4 • 243) • 2 • 0.707]
= 11.175 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	23.5491	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	11.1746	mm
Required Gusset to Top Plate Double Fillet Weld Size	11.1746	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D

From: 40

Pressure for Reinforcement Calculations	P	4211.901	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	6800.00	mm
Head Finished (Minimum) Thickness	t	59.0000	mm
Head Internal Corrosion Allowance	c	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

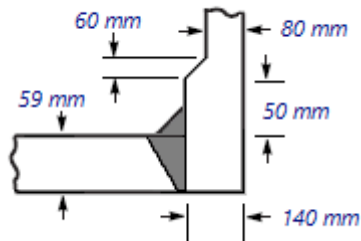
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	80.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		140.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	360.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		B	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		4200.0	mm
Centerline Distance to Tangent		5400.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: D

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	3.150	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 926.4 \cdot (\exp(4211.901/(147.598 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 13.4092 + 6.0 + 0.0 = 19.4092 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.598 \cdot 1.0 \cdot \ln((2 \cdot 74.0 + 926.4) / 926.4) - 0.0$
 $= 21874.459 - 0.0 = 21874.459 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c :	52.176 mm
Nozzle neck minimum required thickness :	14.340 mm

Minimum nozzle neck thickness 80.010 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	16.8,	Allowable :	147.6 N./mm ²	Passed
Expansion :	0.0,	Allowable :	374.0 N./mm ²	Passed
Occasional :	12.2,	Allowable :	196.3 N./mm ²	Passed
Shear :	2.6,	Allowable :	103.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 140.0 - 6.0$
 $= 134.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 80.0 - 6.0$
 $= 74.000 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 6800.0/2 + 6.0$

$$= 3406.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(3406.0 \cdot 53.0)}, 2 \cdot 463.2)$$

$$= 424.874 \text{ mm}$$

Intermediate value [Cn]:

$$= \min((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min((53.0 + 0.0)/134.0)^{0.35}, 1)$$

$$= \min(0.723, 1)$$

$$= 0.723$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(53.0 + 0.0 + 0.723 \cdot \sqrt{463.2 \cdot 134.0}, 360.0 + 53.0)$$

$$= 233.072 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 53.0 + ((0.0 \cdot 1.0) / 424.874)$$

$$= 53.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 53.0 \cdot 424.874$$

$$= 225.183 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$

$$= 138.02 + 106.863$$

$$= 244.883 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 225.183 + 0.944(244.883 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 456.784 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 134.0 / \ln[1 + 134.0/463.2]$$

$$= 527.366 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/\text{Reff}]$$

$$= 53.0 / \ln[1 + 53.0/3406.0]$$

$$= 3432.438 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N/mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 4211.901 \cdot 527.366 \cdot 233.072$$

$$= 52793.1 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$



$$= 4211.901 \cdot 3432.438 \cdot (424.874 + 134.0) / 2$$

$$= 411966.0 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (4211.901 \cdot 3432.438 \cdot 463.2) / 2$$

$$= 341441.2 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (411966.0 + 341441.2 + 52793.1) / 4211.901$$

$$= 18770.2 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 234.6 / (2 \cdot 18770.174 / 456.784 - 3432.438 / (2 \cdot 53.0))$$

$$= 4710.3 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 156.4[53.0/3432.438]$$

$$= 4829.6 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4710.337, 4829.639)$$

$$= 4710.337 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (52793.078 + 411966.031 + 341441.25) / 456.784$$

$$= 173.085 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 4211.901 \cdot 3432.438 / (2 \cdot 53.0)$$

$$= 136.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 173.085 - 136.396, 136.396)$$

$$= 209.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	209.78	N./mm ²
Maximum Allowable Working Pressure	Pmax	4710.34	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (463.2 + 134.0) / 463.2$$

$$= 1.289 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (463.2 + 134.0)$$

$$= 938.080 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 134.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(341441 \cdot 1.29, 1.5 \cdot 147.6(244.883 + 0.0), \pi / 4 \cdot 4211.9 \cdot 463.2^2 \cdot 1.29^2)$$

$$= 120310.680 \text{ Kgf}$$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 120310.68 / (938.08(0.49 \cdot 7.071 + 0.6 \cdot 53.0 + 0.49 \cdot 0.0))$
 $= 35.666 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.162

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (11.977 \cdot 1.0) / (80.0 - 6.0 - 0.0)$
 $= 0.162$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 52.8588 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 471.8587 mm

Input Echo, WRC107/537 Item	1,	Description: D	:
Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	6.0000	mm
Vessel Diameter	Dv	6800.000	mm
Vessel Thickness	Tv	59.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	914.400	mm
Nozzle Thickness	Tn	140.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.60	N./mm ²
Design Internal Pressure	Dp	4211.901	KPa.
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

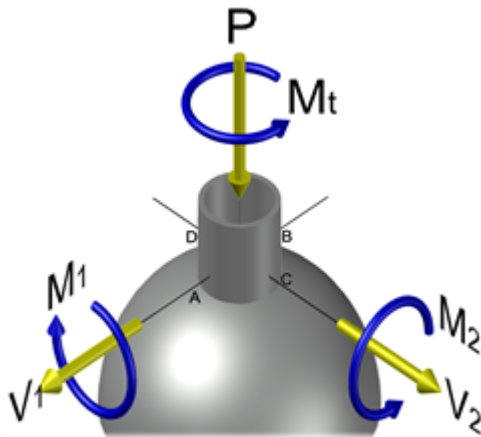
Radial Load (SUS)	P	8257.1	Kgf
Longitudinal Shear (SUS)	(V1) V1	5963.4	Kgf
Circumferential Shear (SUS)	(Vc) V2	5963.4	Kgf
Circumferential Moment (SUS)	(Mc) M1	16515.0	Kg-m.

Longitudinal Moment (SUS) (Ml) M2 16515.0 Kg-m.
 Torsional Moment (SUS) Mt 18992.2 Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 1194.4 + 2 \cdot 1.65 \cdot \sqrt{(3432.5 (59.0 - 6.0))}$$

$$= 2601.928 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	8257.1	Kgf
Circumferential Shear	(VC)	V2	5963.4	Kgf
Longitudinal Shear	(VL)	V1	5963.4	Kgf
Circumferential Moment	(MC)	M1	16515.0	Kg-m.
Longitudinal Moment	(ML)	M2	16515.0	Kg-m.
Torsional Moment		MT	18992.2	Kg-m.

Dimensionless Param: U = 1.40 TAU = 5.00 (3.96) RHO = 0.40

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02832	(A,B,C,D)
$M(x) / P$	SP 1	0.02742	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.02870	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.03895	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.02870	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.03895	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04179	(A,B,C,D)
$M(y) / P$	SP 1	0.00768	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03885	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01214	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03885	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01214	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8
Rad. Bend. P		-4.7	4.7	-4.7	4.7	-4.7	4.7	-4.7	4.7
Rad. Memb. MC		0.0	0.0	0.0	0.0	-3.9	-3.9	3.9	3.9
Rad. Bend. MC		0.0	0.0	0.0	0.0	-31.6	31.6	31.6	-31.6
Rad. Memb. ML		-3.9	-3.9	3.9	3.9	0.0	0.0	0.0	0.0
Rad. Bend. ML		-31.6	31.6	31.6	-31.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-41.0	31.6	29.9	-23.8	-41.0	31.6	29.9	-23.8
Tang. Memb. P		-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2
Tang. Bend. P		-1.3	1.3	-1.3	1.3	-1.3	1.3	-1.3	1.3
Tang. Memb. MC		0.0	0.0	0.0	0.0	-5.3	-5.3	5.3	5.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-9.8	9.8	9.8	-9.8
Tang. Memb. ML		-5.3	-5.3	5.3	5.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-9.8	9.8	9.8	-9.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-17.6	4.7	12.6	-4.5	-17.6	4.7	12.6	-4.5
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Tot. Shear		2.2	2.2	1.0	1.0	1.0	1.0	2.2	2.2
Str. Int.		41.2	31.8	30.0	23.8	41.1	31.7	30.2	24.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		136.4	136.4	136.4	136.4	136.4	136.4	136.4	136.4
Rad. Pl (SUS)		-4.7	-4.7	3.1	3.1	-4.7	-4.7	3.1	3.1
Rad. Q (SUS)		-36.3	36.3	26.8	-26.8	-36.3	36.3	26.8	-26.8
Long. Pm (SUS)		136.4	136.4	136.4	136.4	136.4	136.4	136.4	136.4
Long. Pl (SUS)		-6.5	-6.5	4.0	4.0	-6.5	-6.5	4.0	4.0
Long. Q (SUS)		-11.2	11.2	8.5	-8.5	-11.2	11.2	8.5	-8.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Pm (SUS)		136.4	136.4	136.4	136.4	136.4	136.4	136.4	136.4
Pm+Pl (SUS)		131.9	131.9	140.7	140.7	131.9	131.9	140.7	140.7
Pm+Pl+Q (Total)		119.0	168.2	166.4	132.0	118.8	168.1	166.6	132.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.40	156.40	Passed
Pm+Pl (SUS)	140.72	234.60	Passed
Pm+Pl+Q (TOTAL)	168.20	469.20	Passed



The $Pm+PI+Q$ allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The $Pm+PI+Q$ allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: MH-6

From: 60

Pressure for Reinforcement Calculations	P	4126.858	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6800.00	mm
Shell Finished (Minimum) Thickness	t	102.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4400.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

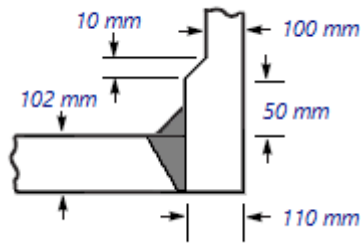
Material		SA-266 2 +IT	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		205.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	100.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	10.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	98.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-6

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.937	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 621.6 \cdot (\exp(4126.858/(147.598 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 8.8131 + 6.0 + 0.0 = 14.8131 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.598 \cdot 1.0 \cdot \ln((2 \cdot 94.0 + 621.6) / 621.6) - 0.0$
 $= 38999.547 - 0.0 = 38999.5469 \text{ KPa.}$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 110.0 - 6.0$
 $= 104.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 100.0 - 6.0$
 $= 94.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 6812.0/96.0$
 $= 70.958 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 6800.0/2 + 6.0$
 $= 3406.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{3406.0 \cdot 96.0}, 2 \cdot 310.8)$
 $= 571.818 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$
 $= \min[1.5 \cdot 96.0, 0.0] + \sqrt{310.8 \cdot 104.0}$
 $= 179.787 \text{ mm}$

Thickness Limit Candidate [LH2]:
 $= L_{pr1}$

$$= 310.0$$

$$= 310.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(96.0 + 0.0)$$

$$= 768.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[179.787, 310.0, 768.0] + 96.0$$

$$= 275.787 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 96.0 + ((0.0 \cdot 1.0) / 571.818)$$

$$= 96.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + t_n) / (\sqrt{ (Di + teff) \cdot teff }))$$

$$= \min(12, (2 \cdot 310.8 + 104.0) / (\sqrt{ (6812.0 + 96.0) \cdot 96.0 }))$$

$$= 0.891$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 96.0 \cdot 571.818 \cdot \max((0.891/5)^{0.85}, 1)$$

$$= 548.945 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 151.84 + 122.499$$

$$= 274.339 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 548.945 + 0.944(274.339 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 808.345 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/Rn]$$

$$= 104.0 / \ln[1 + 104.0/310.8]$$

$$= 360.302 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 96.0 / \ln[1 + 96.0/3406.0]$$

$$= 3453.771 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 4126.858 \cdot 360.302 \cdot 275.787$$

$$= 41817.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + t_n)$$

$$= 4126.858 \cdot 3453.771(571.818 + 104.0)$$

$$= 982289.5 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 4126.858 \cdot 3453.771 \cdot 310.8$$

$$= 451742.2 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (982290 + 451742.2 + 41817.3) / 4126.858$$

$$= 35069.2 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 35069.199 / 808.345 - 3453.771 / 96.0)$$

$$= 4618.7 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 156.4 [96.0 / 3453.771]$$

$$= 4347.0 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(4618.654, 4346.995)$$

$$= 4346.995 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (41817.266 + 982290 + 451742.156) / 808.345$$

$$= 179.050 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 4126.858 \cdot 3453.771 / 96.0$$

$$= 148.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 179.05 - 148.48, 148.48)$$

$$= 209.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	209.62	N./mm ²
Maximum Allowable Working Pressure	Pmax	4347.00	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (310.8 + 104.0) / 310.8$$

$$= 1.335 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (310.8 + 104.0)$$

$$= 651.566 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 104.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(451742 \cdot 1.33, 1.5 \cdot 147.6(274.339 + 0.0), \pi / 4 \cdot 4126.9 \cdot 310.8^2 \cdot 1.33^2)$$

$$= 56870.012 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

= $56870.012 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 96.0 + 0.49 \cdot 0.0))$
= 14.017 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.084

The computed thickness ratio [sr]:
= $(tr \cdot E^*) / (tnom - ci - cext)$
= $(7.872 \cdot 1.0) / (100.0 - 6.0 - 0.0)$
= 0.084

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -22 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : D
From Figure 3.8M- with PWHT, the MDMT is : -15 °C
Temperature reduction per Graph 3.13 : 8 °C
The computed stress ratio is : 0.847

The computed thickness ratio [sr]:
= $(tr \cdot E^*) / (tnom - ci - cext)$
= $(81.28 \cdot 1.0) / (102.0 - 6.0 - 0.0)$
= 0.847

Gov. MDMT of the nozzle to shell joint welded assembly : -22 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 25.3977 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 437.3976 mm

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Input, Nozzle Desc: MH-5

From: 90

Pressure for Reinforcement Calculations	P	4104.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6806.00	mm
Shell Finished (Minimum) Thickness	t	94.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12400.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

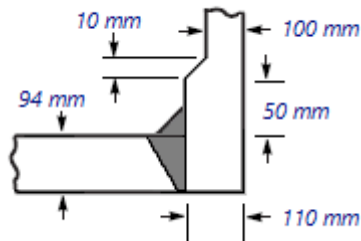
Material		SA-266 2 +IT	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	100.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	10.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	0.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	94.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-5

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.937	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 609.6 \cdot (\exp(4104.0/(147.598 \cdot 1.0)) - 1) + 0.0 + 0.0$$

$$= 8.5945 + 0.0 + 0.0 = 8.5945 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.598 \cdot 1.0 \cdot \ln((2 \cdot 100.0 + 609.6) / 609.6) - 0.0$$

$$= 41876.629 - 0.0 = 41876.6289 \text{ KPa.}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 110.0 - 0.0$$

$$= 110.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 100.0 - 0.0$$

$$= 100.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6808.0/93.0$$

$$= 73.204 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6806.0/2 + 1.0$$

$$= 3404.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3404.0 \cdot 93.0}, 2 \cdot 304.8)$$

$$= 562.647 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 93.0, 0.0] + \sqrt{304.8 \cdot 110.0}$$

$$= 183.107 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 310.0$$

$$= 310.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(93.0 + 0.0)$$

$$= 744.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[183.107, 310.0, 744.0] + 93.0$$

$$= 276.107 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 93.0 + ((0.0 \cdot 1.0) / 562.647)$$

$$= 93.000 \text{ mm}$$

Parameter [λ]:

$$= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right)$$

$$= \min\left(12, \left(2 \cdot 304.8 + 110.0\right) / \left(\sqrt{\left(6808.0 + 93.0\right) \cdot 93.0}\right)\right)$$

$$= 0.898$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right)$$

$$= 93.0 \cdot 562.647 \cdot \max\left(\left(0.898/5\right)^{0.85}, 1\right)$$

$$= 523.262 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 157.3 + 133.607$$

$$= 290.907 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 523.262 + 0.944(290.907 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 798.297 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 110.0 / \ln[1 + 110.0/304.8]$$

$$= 356.980 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 93.0 / \ln[1 + 93.0/3404.0]$$

$$= 3450.298 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 4104.0 \cdot 356.98 \cdot 276.107$$

$$= 41250.0 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$

$$= 4104.0 \cdot 3450.298(562.647 + 110.0)$$

$$= 971287.8 \text{ Kg}$$



Discontinuity Force from Internal Pressure [FY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 4104.0 \cdot 3450.298 \cdot 304.8$$

$$= 440124.4 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (971288 + 440124.4 + 41250.0) / 4104.0$$

$$= 34710.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 234.6 / (2 \cdot 34710.492 / 798.297 - 3450.298 / 93.0)$$

$$= 4704.8 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 156.4 [93.0 / 3450.298]$$

$$= 4215.4 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4704.768, 4215.391)$$

$$= 4215.391 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (41250.023 + 971288 + 440124.438) / 798.297$$

$$= 178.455 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 4104.0 \cdot 3450.298 / 93.0$$

$$= 152.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 178.455 - 152.267, 152.267)$$

$$= 204.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	204.64	N./mm ²
Maximum Allowable Working Pressure	Pmax	4215.39	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (304.8 + 110.0) / 304.8$$

$$= 1.361 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (304.8 + 110.0)$$

$$= 651.566 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 110.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(440124 \cdot 1.36, 1.5 \cdot 147.6(290.907 + 0.0), \pi / 4 \cdot 4104.0 \cdot 304.8^2 \cdot 1.36^2)$$

$$= 56555.020 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

= 56555.02/(651.566(0.49*7.071 + 0.6*93.0 + 0.49*0.0))
 = 14.363 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.077

The computed thickness ratio [sr]:
 = (tr • E*)/(tnom - ci - cext)
 = (7.677 • 1.0)/(100.0 - 0.0 - 0.0)
 = 0.077

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -15 °C
 Temperature reduction per Graph 3.13 : 7 °C
 The computed stress ratio is : 0.869

The computed thickness ratio [sr]:
 = (tr • E*)/(tnom - ci - cext)
 = (80.777 • 1.0)/(94.0 - 1.0 - 0.0)
 = 0.869

Gov. MDMT of the nozzle to shell joint welded assembly : -22 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 25.3752 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 429.3752 mm

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Input, Nozzle Desc: MH-4

From: 100

Pressure for Reinforcement Calculations	P	4104.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6806.00	mm
Shell Finished (Minimum) Thickness	t	94.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		14900.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

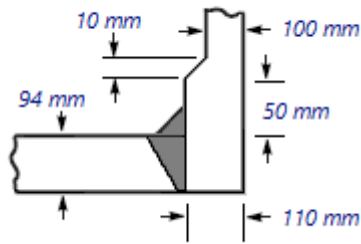
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	100.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	10.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	0.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	94.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.937	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 609.6 \cdot (\exp(4104.0/(147.598 \cdot 1.0)) - 1) + 0.0 + 0.0$$

$$= 8.5945 + 0.0 + 0.0 = 8.5945 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.598 \cdot 1.0 \cdot \ln((2 \cdot 110.0 + 609.6) / 609.6) - 0.0$$

$$= 41876.629 - 0.0 = 41876.6289 \text{ KPa.}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 110.0 - 0.0$$

$$= 110.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 100.0 - 0.0$$

$$= 100.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6808.0/93.0$$

$$= 73.204 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6806.0/2 + 1.0$$

$$= 3404.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3404.0 \cdot 93.0}, 2 \cdot 304.8)$$

$$= 562.647 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 93.0, 0.0] + \sqrt{304.8 \cdot 110.0}$$

$$= 183.107 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 310.0$$

$$= 310.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(93.0 + 0.0)$$

$$= 744.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[183.107, 310.0, 744.0] + 93.0$$

$$= 276.107 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 93.0 + ((0.0 \cdot 1.0) / 562.647)$$

$$= 93.000 \text{ mm}$$

Parameter [λ]:

$$= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right)$$

$$= \min\left(12, \left(2 \cdot 304.8 + 110.0\right) / \left(\sqrt{\left(6808.0 + 93.0\right) \cdot 93.0}\right)\right)$$

$$= 0.898$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right)$$

$$= 93.0 \cdot 562.647 \cdot \max\left(\left(0.898/5\right)^{0.85}, 1\right)$$

$$= 523.262 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 157.3 + 133.607$$

$$= 290.907 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 523.262 + 0.944(290.907 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 798.297 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 110.0 / \ln[1 + 110.0/304.8]$$

$$= 356.980 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 93.0 / \ln[1 + 93.0/3404.0]$$

$$= 3450.298 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 4104.0 \cdot 356.98 \cdot 276.107$$

$$= 41250.0 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$

$$= 4104.0 \cdot 3450.298(562.647 + 110.0)$$

$$= 971287.8 \text{ Kg}$$

Discontinuity Force from Internal Pressure [FY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 4104.0 \cdot 3450.298 \cdot 304.8$$

$$= 440124.4 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (971288 + 440124.4 + 41250.0) / 4104.0$$

$$= 34710.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 234.6 / (2 \cdot 34710.492 / 798.297 - 3450.298 / 93.0)$$

$$= 4704.8 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 156.4 [93.0 / 3450.298]$$

$$= 4215.4 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4704.768, 4215.391)$$

$$= 4215.391 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (41250.023 + 971288 + 440124.438) / 798.297$$

$$= 178.455 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 4104.0 \cdot 3450.298 / 93.0$$

$$= 152.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 178.455 - 152.267, 152.267)$$

$$= 204.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	204.64	N./mm ²
Maximum Allowable Working Pressure	Pmax	4215.39	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (304.8 + 110.0) / 304.8$$

$$= 1.361 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (304.8 + 110.0)$$

$$= 651.566 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 110.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(440124 \cdot 1.36, 1.5 \cdot 147.6(290.907 + 0.0), \pi / 4 \cdot 4104.0 \cdot 304.8^2 \cdot 1.36^2)$$

$$= 56555.020 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



= $56555.02 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 93.0 + 0.49 \cdot 0.0))$
= 14.363 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.077

The computed thickness ratio [sr]:
= $(tr \cdot E^*) / (tnom - ci - cext)$
= $(7.677 \cdot 1.0) / (100.0 - 0.0 - 0.0)$
= 0.077

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 25.3752 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 429.3752 mm

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Input, Nozzle Desc: A

From: 110

Pressure for Reinforcement Calculations	P	4104.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	6806.00	mm
Shell Finished (Minimum) Thickness	t	94.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		16650.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

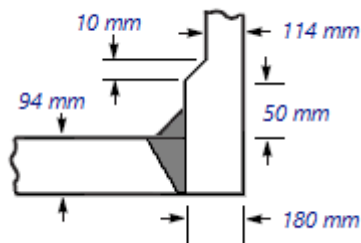
Material		SA-266 2 +IT	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	114.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	10.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	360.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	94.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	4.488	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 926.4 \cdot (\exp(4104.0/(147.598 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 13.0609 + 6.0 + 0.0 = 19.0609 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.598 \cdot 1.0 \cdot \ln((2 \cdot 108.0 + 926.4) / 926.4) - 0.0$
 $= 30931.865 - 0.0 = 30931.8652 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 91.515 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 113.995 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	9.3,	Allowable	:	147.6 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	381.5 N./mm ²	Passed
Occasional	:	7.9,	Allowable	:	196.3 N./mm ²	Passed
Shear	:	0.9,	Allowable	:	103.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 180.0 - 6.0$
 $= 174.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 114.0 - 6.0$
 $= 108.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D/t$
 $= 6808.0/93.0$



= 73.204 must be less than 400.

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6806.0/2 + 1.0$$

$$= 3404.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3404.0 \cdot 93.0}, 2 \cdot 463.2)$$

$$= 562.647 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 93.0, 0.0] + \sqrt{463.2 \cdot 174.0}$$

$$= 283.896 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 360.0$$

$$= 360.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(93.0 + 0.0)$$

$$= 744.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[283.896, 360.0, 744.0] + 93.0$$

$$= 376.896 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 93.0 + ((0.0 \cdot 1.0) / 562.647)$$

$$= 93.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 463.2 + 174.0) / (\sqrt{(6808.0 + 93.0) \cdot 93.0}))$$

$$= 1.374$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 93.0 \cdot 562.647 \cdot \max((1.374/5)^{0.85}, 1)$$

$$= 523.262 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 248.82 + 188.415$$

$$= 437.235 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 523.262 + 0.944(437.235 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 936.390 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 174.0 / \ln[1 + 174.0/463.2]$$

$$= 545.583 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 93.0 / \ln[1 + 93.0/3404.0] \\
 &= 3450.298 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 156.4 \cdot 1.0 \\
 &= 234.6 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 4104.0 \cdot 545.583 \cdot 376.896 \\
 &= 86057.0 \text{ Kg}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 4104.0 \cdot 3450.298 (562.647 + 174.0) \\
 &= 1063702.4 \text{ Kg}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 4104.0 \cdot 3450.298 \cdot 463.2 \\
 &= 668850.6 \text{ Kg}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (1063702 + 668851 + 86057.0) / 4104.0 \\
 &= 43454.6 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\
 &= 234.6 / (2 \cdot 43454.594 / 936.39 - 3450.298 / 93.0) \\
 &= 4210.6 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S / t / R_{xs} \\
 &= 156.4 [93.0 / 3450.298] \\
 &= 4215.4 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(4210.616, 4215.391) \\
 &= 4210.616 \text{ KPa.}
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (86057.023 + 1063702 + 668851) / 936.39 \\
 &= 190.463 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot R_{xs} / t_{eff} \\
 &= 4104.0 \cdot 3450.298 / 93.0 \\
 &= 152.3 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 190.463 - 152.267, 152.267) \\
 &= 228.7 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	228.66	N./mm ²
Maximum Allowable Working Pressure	Pmax	4210.62	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (R_{nc} + t_n) / R_{nc} \\
 &= (463.2 + 174.0) / 463.2 \\
 &= 1.376 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (463.2 + 174.0)$$

$$= 1000.911 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 174.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(668851 \cdot 1.38, 1.5 \cdot 147.6(437.235 + 0.0), \pi/4 \cdot 4104.0 \cdot 463.2^2 \cdot 1.38^2)$$

$$= 133458.219 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_w1 + 0.49 \cdot L43T))$$

$$= 133458.219 / (1000.911(0.49 \cdot 7.071 + 0.6 \cdot 93.0 + 0.49 \cdot 0.0))$$

$$= 22.064 < \text{or } = \text{ to } 156.400 \text{ Weld Stress Passed}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 451.371 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -22 °C
User entered Minimum Design Temperature	: -12 °C
The Curve for this Material is	: D
From Figure 3.8M- with PWHT, the MDMT is	: -15 °C
Temperature reduction per Graph 3.13	: 7 °C
The computed stress ratio is	: 0.869

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (80.777 \cdot 1.0) / (94.0 - 1.0 - 0.0)$$

$$= 0.869$$

Gov. MDMT of the nozzle to shell joint welded assembly : -22 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 60.1890 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 514.1890 mm

Input Echo, WRC297 Item 1, Description: A :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6806.000	mm
Shell Thickness	94.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	



Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		6.0000	mm
Nozzle Diameter		914.400	mm
Nozzle Thickness		180.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.60	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	4104.00	KPa.
Radial Load	P	5504.71	Kgf
Circumferential Shear	Vc	3975.63	Kgf
Longitudinal Shear	Vl	3975.63	Kgf
Circumferential Moment	Mc	7340.00	Kg-m.
Longitudinal Moment	Ml	7340.00	Kg-m.
Torsional Moment	Mt	8441.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (7.080) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1274.4 + 2 \cdot 1.65 \cdot \sqrt{(3450.5(94.0 - 1.0))}$$

$$= 3143.775 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6901.000	mm
Nozzle Outside Diameter	(d) =	1274.400	mm
Vessel Thickness used	(T) =	93.000	mm
Nozzle Thickness used	(t) =	174.000	mm
T / t	=	0.534	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.591$$

Nr/P	=	0.080
Mr/P	=	0.164
M0/P	=	0.048
N0/P	=	0.109
MrD/Mc	=	0.292
NrDL/Mc	=	0.102
M0d/Mc	=	0.092
N0DL/Mc	=	0.060
MrD/Ml	=	0.137
NrDL/Ml	=	0.075
M0D/Ml	=	0.045
N0DL/Ml	=	0.131

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-1	1	1	-1
Normal Pressure Stress	148	152	148	152

Outplane Stress Summary	143	154	148	152
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-6	6	-6	6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	74	74	74	74

Inplane Stress Summary	61	84	73	74
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	143	154	148	152
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	74	74	74	74

Outplane Stress Summary	67	78	75	72
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-6	6	-6	6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-11	11	11	-11
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	148	152	148	152



Inplane Stress Summary 129 168 153 147

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary 0 0 0 0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress 129 168 153 147

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	15	11	15

Hoop Stress Summary 9 14 11 16

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-1	1	-1	1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	5	5	5	5

Axial Stress Summary 2 7 5 6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 0 0 0 0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress 9 14 11 16

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses in the

	Cu Left	Cl Left	Du Right	Dl Right
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hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	15	11	15
Hoop Stress Summary	9	14	10	15

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-1	1	-1	1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-2	2	2	-2
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	0	8	7	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	9	14	10	15

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	148	152	148	152	148	152	148	152
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-2	2	0	0	-17	17	5	-5
Long. Pm (SUS)	74	74	74	74	74	74	74	74
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-11	11	-1	1	-4	4	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	148.0	152.0	148.0	152.0	148.0	152.0	148.0	152.0
Pm+Pl (SUS)	148.0	152.0	148.0	152.0	148.0	152.0	148.0	152.0
Pm+Pl+Q (Total)	146.0	154.0	148.0	152.0	131.0	169.0	153.0	147.0

Vessel Stress Summation Comparison (N./mm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	152.00	156.40	Passed
Pm+Pl (SUS)	152.00	234.60	Passed
Pm+Pl+Q (TOTAL)	169.00	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	15	11	15	11	15	11	15
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-2	2	0	0	-3	3	1	-1
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	11.0	15.0	11.0	15.0	11.0	15.0	11.0	15.0
Pm+Pl (SUS)	11.0	15.0	11.0	15.0	11.0	15.0	11.0	15.0
Pm+Pl+Q (Total)	11.0	15.0	11.0	15.0	11.0	15.0	11.0	15.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	147.60	Passed
Pm+Pl (SUS)	15.00	221.40	Passed
Pm+Pl+Q (TOTAL)	15.00	442.79	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: E

From: 120

Pressure for Reinforcement Calculations	P	4100.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	6806.00	mm
Head Finished (Minimum) Thickness	t	50.6000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

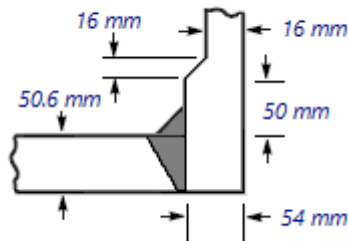
Material		SA-266 2 +IT	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.7402	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	16.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		54.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	260.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: E

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	6.740	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 177.2 \cdot (\exp(4100.0/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 2.672 + 3.0 + 0.0 = 5.672 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.0 + 177.2) / 177.2) - 0.0$$

$$= 18892.713 - 0.0 = 18892.7129 \text{ KPa.}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 10.160 mm
 Min. required shell thickness, internal p + c : 45.915 mm

Nozzle neck minimum required thickness : 10.160 mm

Minimum nozzle neck thickness 16.002 mm, is greater than required thickness 10.160 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47.9,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 297.1 N./mm ²	Passed
Occasional	: 13.0,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 19.4,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 54.0 - 3.0$$

$$= 51.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 3.0$$

$$= 13.000 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6806.0/2 + 1.0$$

$$= 3404.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{(3404.0 \cdot 49.6)}, 2 \cdot 88.6)$$

$$= 177.200 \text{ mm}$$

Intermediate value [Cn]:

$$= \min((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min((49.6 + 0.0)/51.0)^{0.35}, 1)$$

$$= \min(0.99, 1)$$

$$= 0.990$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(49.6 + 0.0 + 0.99 \cdot \sqrt{(88.6 \cdot 51.0)}, 260.0 + 49.6)$$

$$= 116.169 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 49.6 + ((0.0 \cdot 1.0) / 177.2)$$

$$= 49.600 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 49.6 \cdot 177.2$$

$$= 87.891 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$

$$= 50.796 + 3.441$$

$$= 54.237 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 20.0^2$$

$$= 2.000 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 87.891 + 0.882(54.237 + 0.0) + 2.0 + 0.0 + 0.0$$

$$= 137.748 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 51.0 / \ln[1 + 51.0/88.6]$$

$$= 112.174 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 49.6 / \ln[1 + 49.6/3404.0]$$

$$= 3428.745 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 4100.0 \cdot 112.174 \cdot 116.169$$

$$= 5448.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 4100.0 \cdot 3428.745 \cdot (177.2 + 51.0)/2$$

$$= 163569.1 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (4100.0 \cdot 3428.745 \cdot 88.6) / 2$$

$$= 63506.7 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (163569.1 + 63506.7 + 5448.3) / 4100.0$$

$$= 5561.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 234.6 / (2 \cdot 5561.445 / 137.748 - 3428.745 / (2 \cdot 49.6))$$

$$= 5079.4 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 156.4[49.6/3428.745]$$

$$= 4524.7 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(5079.351, 4524.681)$$

$$= 4524.681 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (5448.333 + 163569.109 + 63506.672) / 137.748$$

$$= 165.544 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 4100.0 \cdot 3428.745 / (2 \cdot 49.6)$$

$$= 141.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 165.544 - 141.72, 141.72)$$

$$= 189.4 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	189.37	N./mm ²
Maximum Allowable Working Pressure	Pmax	4524.68	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (88.6 + 51.0) / 88.6$$

$$= 1.576 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (88.6 + 51.0)$$

$$= 219.283 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

tc, L41T = 14.142, must be $\geq \min(0.7 \cdot 51.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(63507 \cdot 1.58, 1.5 \cdot 138.0(54.237 + 0.0), \pi / 4 \cdot 4100.0 \cdot 88.6^2 \cdot 1.58^2)$$

$$= 6399.429 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 6399.429 / (219.283(0.49 \cdot 14.142 + 0.6 \cdot 14.5 + 0.49 \cdot 0.0))$$

= 18.311 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.206

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (2.672 \cdot 1.0) / (16.0 - 3.0 - 0.0)$
 $= 0.206$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -37 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -26 °C
 Temperature reduction per Graph 3.13 : 10 °C
 The computed stress ratio is : 0.809

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (40.112 \cdot 1.0) / (50.6 - 1.0 - 0.0)$
 $= 0.809$

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 2.8646 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 313.4646 mm

Input Echo, WRC107/537 Item 1, Description: E :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	6806.000	mm
Vessel Thickness	Tv	50.600	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	171.200	mm
Nozzle Thickness	Tn	54.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	4100.000	KPa.

Include Pressure Thrust No

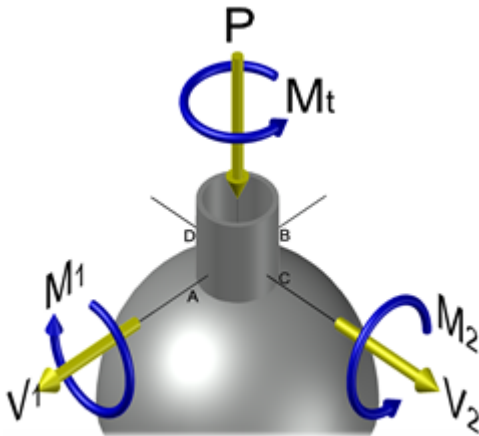
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	1834.9	Kgf
Longitudinal Shear	(SUS)	(V1) V1	1325.2	Kgf
Circumferential Shear	(SUS)	(Vc) V2	1325.2	Kgf
Circumferential Moment	(SUS)	(Mc) M1	815.6	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	815.6	Kg-m.
Torsional Moment	(SUS)	Mt	937.9	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 279.2 + 2 \cdot 1.65 \cdot \sqrt{(3428.8 (50.6 - 1.0))}$$

$$= 1640.099 \text{ mm}$$

WRC 107 Stress Calculation for SUSstained loads:

Radial Load		P	1834.9	Kgf
Circumferential Shear	(VC)	V2	1325.2	Kgf
Longitudinal Shear	(VL)	V1	1325.2	Kgf
Circumferential Moment	(MC)	M1	815.6	Kg-m.
Longitudinal Moment	(ML)	M2	815.6	Kg-m.
Torsional Moment		MT	937.9	Kg-m.

Dimensionless Param: U = 0.34 TAU = 5.00 (2.24) RHO = 0.97

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.09509	(A,B,C,D)
$M(x) / P$	SP 1	0.12415	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.20115	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.38306	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.20115	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.38306	(A,B,C,D)
$N(y) * T / P$	SP 1	0.16478	(A,B,C,D)



M(y) / P SP 1 0.04789 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / MC SM 1 0.04375 (A,B,C,D)
 M(y) * SQRT(Rm * T) / MC SM 1 0.16967 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / ML SM 1 0.04375 (A,B,C,D)
 M(y) * SQRT(Rm * T) / ML SM 1 0.16967 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Rad. Bend. P		-5.4	5.4	-5.4	5.4	-5.4	5.4	-5.4	5.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-18.1	18.1	18.1	-18.1
Rad. Memb. ML		-1.6	-1.6	1.6	1.6	0.0	0.0	0.0	0.0
Rad. Bend. ML		-18.1	18.1	18.1	-18.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-25.8	21.3	13.6	-11.8	-25.8	21.3	13.6	-11.8
Tang. Memb. P		-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2
Tang. Bend. P		-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-8.0	8.0	8.0	-8.0
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-8.0	8.0	8.0	-8.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-11.7	8.6	5.1	-6.8	-11.7	8.6	5.1	-6.8
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear		2.1	2.1	0.9	0.9	0.9	0.9	2.1	2.1
Str. Int.		26.2	21.6	13.7	11.9	25.9	21.4	14.1	12.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		141.7	141.7	141.7	141.7	141.7	141.7	141.7	141.7
Rad. Pl (SUS)		-2.3	-2.3	0.9	0.9	-2.3	-2.3	0.9	0.9
Rad. Q (SUS)		-23.6	23.6	12.7	-12.7	-23.6	23.6	12.7	-12.7
Long. Pm (SUS)		141.7	141.7	141.7	141.7	141.7	141.7	141.7	141.7
Long. Pl (SUS)		-1.6	-1.6	-0.9	-0.9	-1.6	-1.6	-0.9	-0.9
Long. Q (SUS)		-10.1	10.1	5.9	-5.9	-10.1	10.1	5.9	-5.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		141.7	141.7	141.7	141.7	141.7	141.7	141.7	141.7
Pm+Pl (SUS)		140.5	140.5	142.8	142.8	140.5	140.5	142.8	142.8
Pm+Pl+Q (Total)		130.4	163.3	155.4	135.1	130.1	163.1	155.8	135.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	141.72	156.40	Passed
Pm+Pl (SUS)	142.80	234.60	Passed
Pm+Pl+Q (TOTAL)	163.35	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: C
From: 170

Pressure for Reinforcement Calculations	P	470.601	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Elliptical Head	D	7200.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	c	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

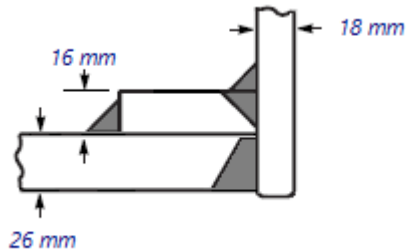
Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	N./mm ²
Allowable Stress At Ambient	Sna	175.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		46.5827	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	2439.2000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		609.9999	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.00	
Horizontal Run Length		4200.0	mm
Centerline Distance to Tangent		3800.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: C

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 46.583 in.
 Actual Thickness Used in Calculation 0.709 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1195.2 \cdot (\exp(470.601/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.8009 + 6.0 + 0.0 = 7.8009 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 12.0 + 1195.2) / 1195.2) - 0.0$
 $= 3109.267 - 0.0 = 3109.2666 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 21.336 mm
 Min. required shell thickness, external p + c : 24.156 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 18.009 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 34.5	Allowable	: 156.4 N./mm ²	Passed
Expansion	: 0.0	Allowable	: 379.7 N./mm ²	Passed
Occasional	: 11.6	Allowable	: 208.0 N./mm ²	Passed
Shear	: 14.5	Allowable	: 109.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 18.0 - 6.0$
 $= 12.000 \text{ mm}$

Effective Pressure Radius [Reff]:

$$= 0.9/6 \cdot Di [2 + (Di/(2h))^2]$$

$$= 0.9/6 \cdot 7212.0 [2 + (7212.0/(2 \cdot 1806.0))^2]$$

$$= 6476.436 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t) + W}$$

$$= \sqrt{(6476.436 \cdot 20.0) + 610.0}$$

$$= 969.901 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((Reff + t) \cdot (t + te))}$$

$$= \sqrt{((6476.436 + 20.0) \cdot (20.0 + 16.0))}$$

$$= 483.603 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot Rn$$

$$= 2 \cdot 597.6$$

$$= 1195.200 \text{ mm}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(969.901, 483.603, 1195.2)$$

$$= 483.603 \text{ mm}$$

Intermediate value [Cn]:

$$= \min(((t + te)/tn)^{0.35}, 1)$$

$$= \min(((20.0 + 16.0)/12.0)^{0.35}, 1)$$

$$= \min(1.469, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= \exp((0.35 \cdot di - Xo)/(16t))$$

$$= \exp((0.35 \cdot 7212.0 - 609.6)/(16 \cdot 20.0))$$

$$= 396.678$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + te + Fp \cdot \sqrt{Rn \cdot tn}, Lpr1 + t)$$

$$= \min(20.0 + 16.0 + 1.0 \cdot \sqrt{(597.6 \cdot 12.0)}, 310.0 + 20.0)$$

$$= 120.683 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 20.0 + ((77.376 \cdot 1.0) / 483.603)$$

$$= 36.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 20.0 \cdot 483.603$$

$$= 96.721 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 12.0 \cdot 120.683$$

$$= 14.482 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 16.0^2$$

$$= 1.280 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot te, LR \cdot te)$$

$$= \min(610.0 \cdot 16.0, 483.603 \cdot 16.0)$$

$$= 77.376 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:
 $= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$
 $= 96.721 + 1.0(14.482 + 0.0) + 1.28 + 0.0 + 0.0 + 1.0(77.376)$
 $= 189.859 \text{ cm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= t_n / \ln[1 + t_n/R_n]$
 $= 12.0 / \ln[1 + 12.0/597.6]$
 $= 603.580 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:
 $= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 36.0 / \ln[1 + 36.0/6476.436]$
 $= 6494.422 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 156.4 \cdot 1.0$
 $= 234.6 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot R_{xn} \cdot L_H$
 $= 470.601 \cdot 603.58 \cdot 120.683$
 $= 3495.7 \text{ Kgf}$

Force acting on the Shell [fS]:
 $= P \cdot R_{xs} \cdot (L_R + t_n)/2$
 $= 470.601 \cdot 6494.422 \cdot (483.603 + 12.0)/2$
 $= 77231.4 \text{ Kgf}$

Discontinuity Force from Internal Pressure [fY]:
 $= (P \cdot R_{xs} \cdot R_{nc})/2$
 $= (470.601 \cdot 6494.422 \cdot 597.6)/2$
 $= 93126.0 \text{ Kgf}$

Area Resisting Internal Pressure [Ap]:
 $= (f_S + f_Y + f_N)/P$
 $= (77231.4 + 93126.0 + 3495.7)/470.601$
 $= 36227.0 \text{ cm}^2$

Maximum Allowable Working Pressure Candidate [Pmax1]:
 $= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$
 $= 234.6 / (2 \cdot 36227.023 / 189.859 - 6494.422 / (2 \cdot 36.0))$
 $= 805.0 \text{ KPa.}$

Maximum Allowable Working Pressure Candidate [Pmax2]:
 $= 2 \cdot S[t/R_{xs}]$
 $= 2 \cdot 156.4[20.0/6494.422]$
 $= 963.2 \text{ KPa.}$

Maximum Allowable Working Pressure [Pmax]:
 $= \min(P_{max1}, P_{max2})$
 $= \min(804.977, 963.232)$
 $= 804.977 \text{ KPa.}$

Average Primary Membrane Stress [σ_{Avg}]:
 $= (f_N + f_S + f_Y) / AT$
 $= (3495.67 + 77231.43 + 93125.992) / 189.859$
 $= 89.801 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot R_{xs} / (2 \cdot t_{eff})$
 $= 470.601 \cdot 6494.422 / (2 \cdot 36.0)$
 $= 42.5 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 89.801 - 42.451, 42.451)$

= 137.2 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	137.15	N./mm ²
Maximum Allowable Working Pressure	Pmax	804.98	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (597.6 + 12.0) / 597.6$$

$$= 1.020 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (597.6 + 12.0)$$

$$= 957.557 \text{ mm}$$

Weld Length of Pad to Shell Weld [L_{TP}]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (597.6 + 12.0 + 610.0)$$

$$= 1915.743 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 11.314, \text{ must be } \geq \min(0.7 \cdot 12.000, 6.000) \text{ Passed}$$

$$t_{f1}, L_{42T} = 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 20.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(93126 \cdot 1.02, 1.5 \cdot 156.4(14.482 + 0.0), \pi/4 \cdot 470.6 \cdot 597.6^2 \cdot 1.02^2)$$

$$= 14006.499 \text{ Kgf}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 14006.5 \cdot 20.0 \cdot 156 / (20.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 7781.388 \text{ Kgf}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 14006.5 \cdot 16.0 \cdot 156 / (20.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 6225.110 \text{ Kgf}$$

Shear Stress [τ₁]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 7781.388 / (957.557 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0))$$

$$= 8.301 \text{ N./mm}^2$$

Shear Stress [τ₂]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 6225.11 / (957.557 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314))$$

$$= 4.210 \text{ N./mm}^2$$

Shear Stress [τ₃]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 6225.11 / (1915.743 \cdot (0.49 \cdot 11.314))$$

$$= 5.748 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(8.301, 4.21, 5.748)$$

$$= 8.3 \text{ must be less than or equal to } 156.4 \text{ N./mm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 12.0 / \ln[1 + 12.0/597.6]$$

$$= 603.580 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 36.0 / \ln[1 + 36.0/6476.436]$$

$$= 6494.422 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 156.4$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 51.73 \cdot 603.58 \cdot 120.683$$

$$= 384.3 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n)/2$$

$$= 51.73 \cdot 6494.422 \cdot (483.603 + 12.0)/2$$

$$= 8489.5 \text{ Kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (51.73 \cdot 6494.422 \cdot 597.6)/2$$

$$= 10236.7 \text{ Kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (8489.5 + 10236.7 + 384.3)/51.73$$

$$= 36227.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 234.6 / (2 \cdot 36227.023 / 189.859 - 6494.422 / (2 \cdot 36.0))$$

$$= 805.0 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / (t / R_{xs})$$

$$= 2 \cdot 156.4 / (20.0 / 6494.422)$$

$$= 963.2 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(804.977, 963.232)$$

$$= 804.977 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (384.255 + 8489.523 + 10236.704) / 189.859$$

$$= 9.871 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 51.73 \cdot 6494.422 / (2 \cdot 36.0)$$

$$= 4.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 9.871 - 4.666, 4.666)$$

$$= 15.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	15.08	N./mm ²
Maximum Allowable External Pressure	Pmax	804.98	KPa.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.134

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (1.609 \cdot 1.0) / (18.0 - 6.0 - 0.0)$$

$$= 0.134$$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -30 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (10.819 \cdot 1.0) / (16.0 - 6.0 - 0.0)$$

$$= 1.000$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.134

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (1.609 \cdot 1.0) / (18.0 - 6.0 - 0.0)$$

$$= 0.134$$

Gov. MDMT of the Nozzle : -12 °C
 Gov. MDMT of the Reinforcement Pad : -12 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -30 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -104 °C

The Drop for this Nozzle is : 28.6452 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 364.6451 mm

Input Echo, WRC107/537 Item 1, Description: C :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	6.0000 mm
Vessel Diameter	Dv	12959.999 mm

Vessel Thickness	Tv	26.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	1183.200	mm
Nozzle Thickness	Tn	18.000	mm
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	175.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	156.40	N./mm ²
Thickness of Reinforcing Pad	Tpad	16.000	mm
Diameter of Reinforcing Pad	Dpad	2439.200	mm
Design Internal Pressure	Dp	470.601	KPa.
Include Pressure Thrust		No	

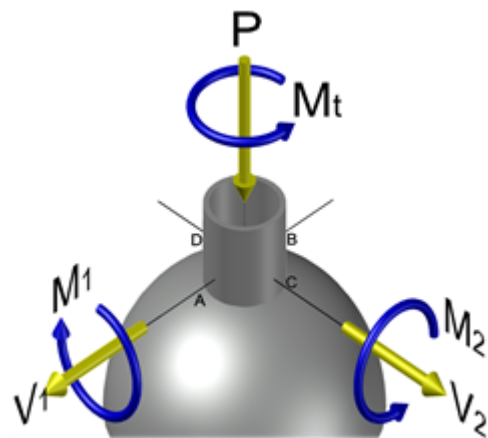
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	9174.6	Kgf
Longitudinal Shear (SUS)	(V1) V1	6116.3	Kgf
Circumferential Shear (SUS)	(Vc) V2	6116.3	Kgf
Circumferential Moment (SUS)	(Mc) M1	20551.9	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	20551.9	Kg-m.
Torsional Moment (SUS)	Mt	29360.1	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1219.2 + 2 \cdot 1.65 \cdot \sqrt{(6496.0 (26.0 - 6.0))}$$

= 2408.666 mm

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	9174.6	Kgf
Circumferential Shear	(VC) V2	6116.3	Kgf
Longitudinal Shear	(VL) V1	6116.3	Kgf
Circumferential Moment	(MC) M1	20551.9	Kg-m.
Longitudinal Moment	(ML) M2	20551.9	Kg-m.
Torsional Moment	MT	29360.1	Kg-m.

Unitless Prm: U = 1.26 TAU = 50.00 (50.30) RHO = 4.00 (3.00)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 9	0.01053	(A,B,C,D)
M(x) / P	SP 9	0.00883	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 9	0.01591	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 9	0.01410	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 9	0.01591	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 9	0.01410	(A,B,C,D)
N(y) * T / P	SP 9	0.19924	(A,B,C,D)
M(y) / P	SP 9	0.01835	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 9	0.21481	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 9	0.03478	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 9	0.21481	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 9	0.03478	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Rad.	Bend. P	-3.7	3.7	-3.7	3.7	-3.7	3.7	-3.7	3.7
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-5.1	-5.1	5.1	5.1
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-27.2	27.2	27.2	-27.2
Rad.	Memb. ML	-5.1	-5.1	5.1	5.1	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-27.2	27.2	27.2	-27.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-36.7	25.0	27.9	-19.1	-36.7	25.0	27.9	-19.1
Tang.	Memb. P	-13.8	-13.8	-13.8	-13.8	-13.8	-13.8	-13.8	-13.8
Tang.	Bend. P	-7.6	7.6	-7.6	7.6	-7.6	7.6	-7.6	7.6
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-69.0	-69.0	69.0	69.0
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-67.1	67.1	67.1	-67.1
Tang.	Memb. ML	-69.0	-69.0	69.0	69.0	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-67.1	67.1	67.1	-67.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-157.6	-8.2	114.6	-4.2	-157.6	-8.2	114.6	-4.2
Shear VC		0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		3.4	3.4	3.4	3.4	3.4	3.4	3.4	3.4
Tot. Shear		4.3	4.3	2.6	2.6	2.6	2.6	4.3	4.3
Str. Int.		157.7	34.3	114.7	19.6	157.6	33.6	114.8	20.3

Unitless Prm: U = 3.38 TAU = 0.00 (101.13) RHO = 0.00 (1.67)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
----------------------	--------	-------	----------

N(y) * T / P	SR 2	0.00650	(A,B,C,D)
M(y) / P	SR 2	0.00300	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00317	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00317	(A,B,C,D)

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-4.9	-4.9	-4.9	-4.9	-4.9	-4.9	-4.9	-4.9
Rad.	Bend. P	-13.4	13.4	-13.4	13.4	-13.4	13.4	-13.4	13.4
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-23.6	-23.6	23.6	23.6
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-88.3	88.3	88.3	-88.3
Rad.	Memb. ML	-23.6	-23.6	23.6	23.6	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-88.3	88.3	88.3	-88.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-130.2	73.1	93.7	-56.2	-130.2	73.1	93.7	-56.2
Tang. Memb. P		-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
Tang. Bend. P		-4.0	4.0	-4.0	4.0	-4.0	4.0	-4.0	4.0
Tang. Memb. MC		0.0	0.0	0.0	0.0	-7.1	-7.1	7.1	7.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-26.6	26.6	26.6	-26.6
Tang. Memb. ML		-7.1	-7.1	7.1	7.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-26.6	26.6	26.6	-26.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-39.2	22.1	28.2	-17.0	-39.2	22.1	28.2	-17.0
Shear VC		0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear		2.3	2.3	0.8	0.8	0.8	0.8	2.3	2.3
Str. Int.		130.3	73.2	93.7	56.2	130.2	73.1	93.8	56.3

Vessel Stress Summation at Attachment Junction (N./mm²)

[illegible]



Pm+Pl (SUS)	77.0	77.0	97.7	97.7	77.0	77.0	97.7	97.7
Pm+Pl+Q (Total)	121.2	68.1	157.2	38.7	121.0	67.7	157.3	39.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.51	156.40	Passed
Pm+Pl (SUS)	97.73	234.60	Passed
Pm+Pl+Q (TOTAL)	157.34	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	42.5	42.5	42.5	42.5	42.5	42.5	42.5	42.5
Rad.	Pl (SUS)	-28.5	-28.5	18.8	18.8	-28.5	-28.5	18.8	18.8
Rad.	Q (SUS)	-101.7	101.7	74.9	-74.9	-101.7	101.7	74.9	-74.9
Long.	Pm (SUS)	42.5	42.5	42.5	42.5	42.5	42.5	42.5	42.5
Long.	Pl (SUS)	-8.5	-8.5	5.6	5.6	-8.5	-8.5	5.6	5.6
Long.	Q (SUS)	-30.7	30.7	22.6	-22.6	-30.7	30.7	22.6	-22.6
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear	Q (SUS)	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		42.5	42.5	42.5	42.5	42.5	42.5	42.5	42.5
Pm+Pl (SUS)		34.0	34.0	61.3	61.3	34.0	34.0	61.3	61.3
Pm+Pl+Q (Total)		91.1	115.7	136.2	39.3	91.0	115.6	136.3	39.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.51	156.40	Passed
Pm+Pl (SUS)	61.31	234.60	Passed
Pm+Pl+Q (TOTAL)	136.28	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



Input, Nozzle Desc: MH-3

From: 190

Pressure for Reinforcement Calculations	P	409.975	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7200.00	mm
Design Length of Section	L	2750.0002	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		29550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

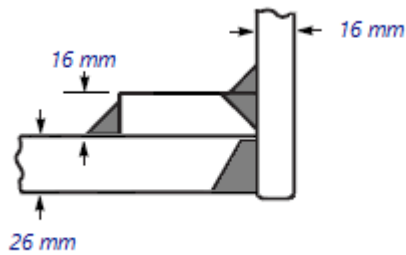
Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	N./mm ²
Allowable Stress At Ambient	Sna	175.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		22.7402	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1209.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		300.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.740	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 589.6 \cdot (\exp(409.975/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 0.7738 + 6.0 + 0.0 = 6.7738 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 589.6) / 589.6) - 0.0$
 $= 5216.982 - 0.0 = 5216.9824 \text{ KPa.}$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.0 - 6.0$
 $= 10.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7212.0/20.0$
 $= 360.600 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 7200.0/2 + 6.0$
 $= 3606.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:
 $= \sqrt{(Reff \cdot t) + W}$
 $= \sqrt{(3606.0 \cdot 20.0) + 300.0}$
 $= 568.552 \text{ mm}$

Limit [LR2]:
 $= \sqrt{((Reff + t) \cdot (t + t_e))}$
 $= \sqrt{((3606.0 + 20.0) \cdot (20.0 + 16.0))}$
 $= 361.298 \text{ mm}$

Limit [LR3]:

$$\begin{aligned}
 &= 2 \cdot R_n \\
 &= 2 \cdot 294.8 \\
 &= 589.600 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(568.552, 361.298, 589.6) \\
 &= 361.298 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 20.0, 16.0] + \sqrt{(294.8 \cdot 10.0)} \\
 &= 70.295 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 310.0 \\
 &= 310.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(20.0 + 16.0) \\
 &= 288.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[70.295, 310.0, 288.0] + 20.0 \\
 &= 90.295 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 20.0 + ((48.0 \cdot 1.0) / 361.298) \\
 &= 33.285 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff} })) \\
 &= \min(12, (2 \cdot 294.8 + 10.0) / (\sqrt{(7212.0 + 33.285) \cdot 33.285 })) \\
 &= 1.221
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 20.0 \cdot 361.298 \cdot \max((1.221/5)^{0.85}, 1) \\
 &= 72.260 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 10.0 \cdot 90.295 \\
 &= 9.030 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg42}^2 \\
 &= 0.5 \cdot 16.0^2 \\
 &= 1.280 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg41}^2 \\
 &= 0.5 \cdot 16.0^2 \\
 &= 1.280 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned}
 &= \min(W \cdot t_e, LR \cdot t_e) \\
 &= \min(300.0 \cdot 16.0, 361.298 \cdot 16.0) \\
 &= 48.000 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned}
 &= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5) \\
 &= 72.26 + 1.0(9.03 + 0.0) + 1.28 + 1.28 + 0.0 + 1.0(48.0) \\
 &= 131.849 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 10.0/\ln[1 + 10.0/294.8]$$

$$= 299.773 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 33.285/\ln[1 + 33.285/3606.0]$$

$$= 3622.601 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 409.975 \cdot 299.773 \cdot 90.295$$

$$= 1131.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 409.975 \cdot 3622.601 (361.298 + 10.0)$$

$$= 56233.7 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 409.975 \cdot 3622.601 \cdot 294.8$$

$$= 44648.0 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (56233.7 + 44648.0 + 1131.7)/409.975$$

$$= 24400.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff})$$

$$= 234.6 / (2 \cdot 24400.742/131.849 - 3622.601/33.285)$$

$$= 897.8 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}]$$

$$= 156.4[20.0/3622.601]$$

$$= 863.4 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(897.777, 863.418)$$

$$= 863.418 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (1131.651 + 56233.73 + 44648.016) / 131.849$$

$$= 75.877 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 409.975 \cdot 3622.601 / 33.285$$

$$= 44.6 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 75.877 - 44.622, 44.622)$$

$$= 107.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	107.13	N./mm ²
Maximum Allowable Working Pressure	Pmax	863.42	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (294.8 + 10.0) / 294.8$$

$$= 1.034 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (294.8 + 10.0)$$

$$= 478.779 \text{ mm}$$

Weld Length of Pad to Shell Weld [L_{TP}]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (294.8 + 10.0 + 300.0)$$

$$= 950.018 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 11.314, \text{ must be } \geq \min(0.7 \cdot 10.000, 6.000) \text{ Passed}$$

$$t_{f1}, L_{42T} = 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 20.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(44648 \cdot 1.03, 1.5 \cdot 156.4(9.03 + 0.0), \pi/4 \cdot 410.0 \cdot 294.8^2 \cdot 1.03^2)$$

$$= 3050.521 \text{ Kg}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 3050.5 \cdot 20.0 \cdot 156 / (20.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 1694.734 \text{ Kg}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 3050.5 \cdot 16.0 \cdot 156 / (20.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 1355.787 \text{ Kg}$$

Shear Stress [τ₁]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 1694.734 / (478.779 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0))$$

$$= 3.616 \text{ N./mm}^2$$

Shear Stress [τ₂]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 1355.787 / (478.779 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314))$$

$$= 1.834 \text{ N./mm}^2$$

Shear Stress [τ₃]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 1355.787 / (950.018 \cdot (0.49 \cdot 11.314))$$

$$= 2.525 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(3.616, 1.834, 2.525)$$

$$= 3.6 \text{ must be less than or equal to } 156.4 \text{ N./mm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 10.0 / \ln[1 + 10.0/294.8]$$

$$= 299.773 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 33.285 / \ln[1 + 33.285/3606.0]$$

= 3622.601 mm

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 156.4$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 51.73 \cdot 299.773 \cdot 90.295$$

$$= 142.8 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 51.73 \cdot 3622.601 (361.298 + 10.0)$$

$$= 7095.5 \text{ Kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 51.73 \cdot 3622.601 \cdot 294.8$$

$$= 5633.6 \text{ Kgf}$$

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (7095.5 + 5633.6 + 142.8) / 51.73$$

$$= 24400.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 24400.74 / 131.849 - 3622.601 / 33.285)$$

$$= 897.8 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 156.4 [20.0 / 3622.601]$$

$$= 863.4 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(897.777, 863.418)$$

$$= 863.418 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (142.79 + 7095.479 + 5633.613) / 131.849$$

$$= 9.574 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 51.73 \cdot 3622.601 / 33.285$$

$$= 5.6 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 9.574 - 5.63, 5.63)$$

$$= 13.5 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	13.52	N./mm ²
Maximum Allowable External Pressure	Pmax	863.42	KPa.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-12 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.069

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (0.691 \cdot 1.0) / (16.0 - 6.0 - 0.0)$$

$$= 0.069$$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -37 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 7 °C
 The computed stress ratio is : 0.846

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (8.458 \cdot 1.0) / (16.0 - 6.0 - 0.0)$$

$$= 0.846$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.069

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (0.691 \cdot 1.0) / (16.0 - 6.0 - 0.0)$$

$$= 0.069$$

Gov. MDMT of the Nozzle : -12 °C
 Gov. MDMT of the Reinforcement Pad : -12 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -104 °C

The Drop for this Nozzle is : 12.9263 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.9263 mm

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Input, Nozzle Desc: B
From: 200

Pressure for Reinforcement Calculations	P	384.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7200.00	mm
Design Length of Section	L	2499.9648	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		34250.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

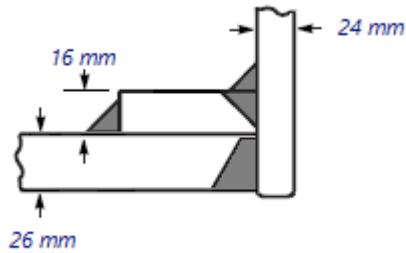
Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	N./mm ²
Allowable Stress At Ambient	Sna	175.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		34.1102	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1824.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		455.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: B

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	34.110	in.
Actual Thickness Used in Calculation	0.945	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 878.4 \cdot (\exp(384.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.0798 + 6.0 + 0.0 = 7.0798 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 878.4) / 878.4) - 0.0$
 $= 6281.614 - 0.0 = 6281.6143 \text{ KPa.}$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 441.87 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 818.50 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 14.865 mm
 Min. required shell thickness, external p + c : 22.679 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 24.003 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.7,	Allowable	: 156.4 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 396.6 N./mm ²	Passed
Occasional	: 4.6,	Allowable	: 208.0 N./mm ²	Passed
Shear	: 10.0,	Allowable	: 109.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 24.0 - 6.0$
 $= 18.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned}
 &= D_i/t \\
 &= 7212.0/20.0 \\
 &= 360.600 \text{ must be less than } 400.
 \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= D_i/2 + \text{corrosion allowance} \\
 &= 7200.0/2 + 6.0 \\
 &= 3606.000 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned}
 &= \sqrt{(Reff \cdot t) + W} \\
 &= \sqrt{(3606.0 \cdot 20.0) + 455.0} \\
 &= 723.552 \text{ mm}
 \end{aligned}$$

Limit [LR2]:

$$\begin{aligned}
 &= \sqrt{((Reff + t) \cdot (t + t_e))} \\
 &= \sqrt{((3606.0 + 20.0) \cdot (20.0 + 16.0))} \\
 &= 361.298 \text{ mm}
 \end{aligned}$$

Limit [LR3]:

$$\begin{aligned}
 &= 2 \cdot R_n \\
 &= 2 \cdot 439.2 \\
 &= 878.400 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(723.552, 361.298, 878.4) \\
 &= 361.298 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 20.0, 16.0] + \sqrt{(439.2 \cdot 18.0)} \\
 &= 104.913 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 310.0 \\
 &= 310.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(20.0 + 16.0) \\
 &= 288.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\
 &= \min[104.913, 310.0, 288.0] + 20.0 \\
 &= 124.913 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 20.0 + ((57.808 \cdot 1.0) / 361.298) \\
 &= 36.000 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, ((2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))) \\
 &= \min(12, (2 \cdot 439.2 + 18.0) / (\sqrt{(7212.0 + 36.0) \cdot 36.0})) \\
 &= 1.755
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 20.0 \cdot 361.298 \cdot \max((1.755/5)^{0.85}, 1) \\
 &= 72.260 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot L_H$$

$$= 18.0 \cdot 124.913$$

$$= 22.484 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 16.0^2$$

$$= 1.280 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(455.0 \cdot 16.0, 361.298 \cdot 16.0)$$

$$= 57.808 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 72.26 + 1.0(22.484 + 0.0) + 1.28 + 0.0 + 0.0 + 1.0(57.808)$$

$$= 153.832 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 18.0 / \ln[1 + 18.0/439.2]$$

$$= 448.140 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 36.0 / \ln[1 + 36.0/3606.0]$$

$$= 3623.991 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 384.0 \cdot 448.14 \cdot 124.913$$

$$= 2192.1 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 384.0 \cdot 3623.991 (361.298 + 18.0)$$

$$= 53826.4 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 384.0 \cdot 3623.991 \cdot 439.2$$

$$= 62327.2 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (53826.4 + 62327.2 + 2192.1) / 384.0$$

$$= 30222.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 30222.076 / 153.832 - 3623.991 / 36.0)$$

$$= 802.7 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 156.4 / 20.0 / 3623.991$$

$$= 863.1 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(802.669, 863.086)$$

$$= 802.669 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:
 $= (fN + fS + fY) / AT$
 $= (2192.052 + 53826.371 + 62327.156) / 153.832$
 $= 75.446 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot Rxs / teff$
 $= 384.0 \cdot 3623.991 / 36.0$
 $= 38.7 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 75.446 - 38.658, 38.658)$
 $= 112.2 \text{ N./mm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	112.23	N./mm ²
Maximum Allowable Working Pressure	Pmax	802.67	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 $= (Rnc + tn) / Rnc$
 $= (439.2 + 18.0) / 439.2$
 $= 1.041 \text{ For set-in Nozzles}$

Weld Length of Nozzle to Shell Weld [LT]:
 $= \pi/2 \cdot (Rn + tn)$
 $= \pi/2 \cdot (439.2 + 18.0)$
 $= 718.168 \text{ mm}$

Weld Length of Pad to Shell Weld [LTP]:
 $= \pi/2 \cdot (Rn + tn + W)$
 $= \pi/2 \cdot (439.2 + 18.0 + 455.0)$
 $= 1432.881 \text{ mm}$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:
 $= 11.314, 11.314, 0.000, \text{ mm}$

Required Weld Throat Dimensions :
 tc, L41T = 11.314, must be $\geq \min(0.7 \cdot 18.000, 6.000)$ Passed
 tf1, L42T = 11.314, must be $\geq \min(0.6 \cdot 16.000, 0.6 \cdot 20.000)$ Passed

Weld Load Value [fwelds]:
 $= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$
 $= \min(62327 \cdot 1.04, 1.5 \cdot 156.4(22.484 + 0.0), \pi/4 \cdot 384.0 \cdot 439.2^2 \cdot 1.04^2)$
 $= 6428.805 \text{ Kgf}$

Discontinuity Force [fws]:
 $= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp)$
 $= 6428.8 \cdot 20.0 \cdot 156 / (20.0 \cdot 156 + 16.0 \cdot 156)$
 $= 3571.558 \text{ Kgf}$

Discontinuity Force [fwp]:
 $= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp)$
 $= 6428.8 \cdot 16.0 \cdot 156 / (20.0 \cdot 156 + 16.0 \cdot 156)$
 $= 2857.247 \text{ Kgf}$

Shear Stress [τ_1]:
 $= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T))$
 $= 3571.558 / (718.168 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0))$
 $= 5.080 \text{ N./mm}^2$

Shear Stress [τ_2]:
 $= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T))$
 $= 2857.247 / (718.168 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314))$
 $= 2.576 \text{ N./mm}^2$

Shear Stress [τ_3]:

$$\begin{aligned}
 &= fwp / (L_t P \cdot (0.49 \cdot L_{42T})) \\
 &= 2857.247 / (1432.881 \cdot (0.49 \cdot 11.314)) \\
 &= 3.528 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned}
 &= \max(\tau_1, \tau_2, \tau_3) \\
 &= \max(5.08, 2.576, 3.528) \\
 &= 5.1 \text{ must be less than or equal to } 156.4 \text{ N./mm}^2
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n / R_n] \\
 &= 18.0 / \ln[1 + 18.0 / 439.2] \\
 &= 448.140 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\
 &= 36.0 / \ln[1 + 36.0 / 3606.0] \\
 &= 3623.991 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 156.4 \\
 &= 234.6 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 51.73 \cdot 448.14 \cdot 124.913 \\
 &= 295.3 \text{ Kg}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 51.73 \cdot 3623.991 (361.298 + 18.0) \\
 &= 7251.1 \text{ Kg}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 51.73 \cdot 3623.991 \cdot 439.2 \\
 &= 8396.3 \text{ Kg}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (7251.1 + 8396.3 + 295.3) / 51.73 \\
 &= 30222.1 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\
 &= 234.6 / (2 \cdot 30222.072 / 153.832 - 3623.991 / 36.0) \\
 &= 802.7 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S / [t / R_{xs}] \\
 &= 156.4 [20.0 / 3623.991] \\
 &= 863.1 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(802.669, 863.086) \\
 &= 802.669 \text{ KPa.}
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (295.299 + 7251.141 + 8396.312) / 153.832 \\
 &= 10.164 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot R_{xs} / t_{eff} \\
 &= 51.73 \cdot 3623.991 / 36.0 \\
 &= 5.2 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 10.164 - 5.208, 5.208)$$

$$= 15.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	15.12	N./mm ²
Maximum Allowable External Pressure	Pmax	802.67	KPa.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: -12 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.054

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (0.965 \cdot 1.0) / (24.0 - 6.0 - 0.0)$$

$$= 0.054$$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	: -12 °C
Gov. MDMT of the Reinforcement Pad	: -12 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	-104 °C
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The Drop for this Nozzle is : 29.1502 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 365.1501 mm

Input Echo, WRC297 Item 1, Description: B :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7200.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40 N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	866.400	mm



Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	175.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	156.40 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1824.400	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	384.00	KPa.
Radial Load	P	-6883.20	Kgf
Circumferential Shear	Vc	4588.80	Kgf
Longitudinal Shear	Vl	4588.80	Kgf
Circumferential Moment	Mc	11564.40	Kg-m.
Longitudinal Moment	Ml	11564.40	Kg-m.
Torsional Moment	Mt	16520.55	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(3624.0(26.0 - 6.0))}$
 $= 1802.829 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7248.000	mm
Nozzle Outside Diameter	(d) =	914.400	mm
Vessel Thickness used	(T) =	36.000	mm
Nozzle Thickness used	(t) =	18.000	mm
T / t	=	2.000	
d / t	=	50.800	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.790$$

Nr/P	=	0.050
Mr/P	=	0.097
M0/P	=	0.054
N0/P	=	0.158
MrD/Mc	=	0.194
NrDL/Mc	=	0.093
M0d/Mc	=	0.123
N0DL/Mc	=	0.199
MrD/Ml	=	0.113
NrDL/Ml	=	0.060
M0D/Ml	=	0.062
N0DL/Ml	=	0.298

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-28	-28	28	28



Outplane Bending (ML)	-35	35	35	-35
Normal Pressure Stress	38	38	38	38

Outplane Stress Summary	0	36	127	23
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	19	19	19	19

Inplane Stress Summary	-19	50	122	-67
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	20	52	128	90
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-19	-19	19	19
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19

Outplane Stress Summary	-45	62	134	-41
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-111	111	111	-111
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	38	38	38

Inplane Stress Summary	-49	113	191	-91
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	50	113	191	91

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-28	-28	28	28
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	11	11	11
Hoop Stress Summary	-9	-8	47	48

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	89	-89	89	-89
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-191	191	191	-191
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-105	98	297	-263

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	5	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	106	108	297	311

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-19	-19	19	19



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	11	11	11

Hoop Stress Summary	0	0	38	38
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	89	-89	89	-89
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-339	339	339	-339
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-253	246	445	-411
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary	5	5	8	8
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	254	247	446	450
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(3616.0(26.0 - 6.0))}$$

$$= 1801.849 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7232.000 mm
Nozzle Outside Diameter	(d) =	1824.400 mm
Vessel Thickness used	(T) =	20.000 mm
Nozzle Thickness used	(t) =	473.000 mm
T / t	=	0.042
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.797$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.048
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.021
MrD/Mc	=	0.241
NrDL/Mc	=	0.105
M0d/Mc	=	0.072
N0DL/Mc	=	0.044
MrD/M1	=	0.049
NrDL/M1	=	0.054
M0D/M1	=	0.015

NØDL/M1 = 0.023

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	69	69	69	69
Outplane Stress Summary	74	63	109	43

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	72	-72	72	-72
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-45	45	45	-45
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	60	7	168	-66

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	75	63	168	109

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-66	66	66	-66
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	-16	78	131	-41

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
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Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	72	-72	72	-72
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-224	224	224	-224
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	69	69	69	69

Inplane Stress Summary	-91	213	390	-202
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	91	213	390	202

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	38	38	38	38	38	38	38
Circ. Pl (SUS)	-20	-20	36	36	-6	-6	10	10
Circ. Q (SUS)	-19	19	51	-51	-81	81	141	-141
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-3	-3	7	7	-11	-11	27	27
Long. Q (SUS)	-35	35	95	-95	-54	54	86	-86
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	38.0	38.0	38.0	38.0	38.0	38.0	38.0	38.0
Pm+Pl (SUS)	18.0	18.0	74.0	74.0	32.0	32.0	48.0	48.0
Pm+Pl+Q (Total)	19.5	51.6	126.6	92.2	50.9	113.2	189.2	93.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	156.40	Passed
Pm+Pl (SUS)	74.00	234.60	Passed
Pm+Pl+Q (TOTAL)	189.16	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	69	69	69	69	69	69	69	69
Circ. Pl (SUS)	0	0	6	6	-8	-8	24	24
Circ. Q (SUS)	6	-6	32	-32	-152	152	296	-296
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	0	0	16	16	-3	-3	9	9
Long. Q (SUS)	27	-27	117	-117	-47	47	85	-85
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	69.0	69.0	69.0	69.0	69.0	69.0	69.0	69.0
Pm+Pl (SUS)	69.0	69.0	75.0	75.0	61.0	61.0	93.0	93.0
Pm+Pl+Q (Total)	75.1	63.0	167.0	110.0	91.0	213.0	389.0	203.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	69.00	156.40	Passed
Pm+Pl (SUS)	93.00	234.60	Passed
Pm+Pl+Q (TOTAL)	389.00	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	11	11	11	11	11	11	11
Circ. Pl (SUS)	-20	-20	36	36	-11	-11	27	27
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-102	102	280	-280	-250	250	428	-428
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0
Pm+Pl (SUS)	9.2	9.2	47.0	47.0	4.5	4.5	38.0	38.0
Pm+Pl+Q (Total)	106.5	107.9	296.1	311.2	254.2	246.2	444.1	450.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	156.40	Passed
Pm+Pl (SUS)	47.03	234.60	Passed
Pm+Pl+Q (TOTAL)	450.22	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K
From: 200

Pressure for Reinforcement Calculations	P	384.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7200.00	mm
Design Length of Section	L	2499.9648	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		34450.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

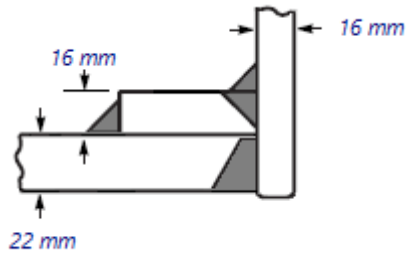
Material		SA-106 B	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	143.60	N./mm ²
Allowable Stress At Ambient	Sna	161.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	210.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	322.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		77.0625	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	5.368	in.
Actual Thickness Used in Calculation	0.628	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 142.36 \cdot (\exp(384.0/(143.6 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 0.1906 + 3.0 + 0.0 = 3.1906 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 143.6 \cdot 1.0 \cdot \ln((2 \cdot 12.958 + 142.36) / 142.36) - 0.0$
 $= 24014.438 - 0.0 = 24014.4375 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.220 mm
 Min. required shell thickness, internal p + c : 14.865 mm
 Min. required shell thickness, external p + c : 22.679 mm

Nozzle neck minimum required thickness : 9.220 mm

Minimum nozzle neck thickness 15.951 mm , is greater than required thickness 9.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 22.3,	Allowable	: 143.6 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 358.5 N./mm ²	Passed
Occasional	: 1.0,	Allowable	: 191.0 N./mm ²	Passed
Shear	: 14.6,	Allowable	: 100.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[143.6/156.4, 1]$
 $= 0.918$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 15.958 - 3.0$
 $= 12.958 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7212.0/16.0$
 $= 450.750 \text{ must be less than 400; Calculation not possible.}$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.013

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (0.17 \cdot 1.0) / (15.958 - 3.0 - 0.0)$$

$$= 0.013$$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -12 °C
 Gov. MDMT of the Reinforcement Pad : -12 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -104 °C

The Drop for this Nozzle is : 0.9833 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.9831 mm

Input Echo, WRC297 Item 1, Description: K :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7200.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 156.40	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	136.360	mm
Nozzle Thickness	15.9576	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 161.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 143.60	N./mm ²
Diameter of Reinforcing Pad	322.400	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	384.00	KPa.
Radial Load	P	1146.82	Kgf
Circumferential Shear	Vc	764.54	Kgf
Longitudinal Shear	Vl	764.54	Kgf
Circumferential Moment	Mc	321.13	Kg-m.
Longitudinal Moment	Ml	321.13	Kg-m.
Torsional Moment	Mt	458.75	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(3622.0(22.0 - 6.0))}$
 $= 962.691 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7244.000 mm
Nozzle Outside Diameter	(d) =	168.275 mm
Vessel Thickness used	(T) =	32.000 mm
Nozzle Thickness used	(t) =	12.958 mm
T / t	=	2.470
d / t	=	12.987

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.350$$

Nr/P	=	0.210
Mr/P	=	0.211
M0/P	=	0.164
N0/P	=	0.207
MrD/Mc	=	0.212
NrDL/Mc	=	0.182
M0d/Mc	=	0.198
N0DL/Mc	=	-0.125
MrD/Ml	=	0.205
NrDL/Ml	=	0.180
M0D/Ml	=	0.189
N0DL/Ml	=	-0.059

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-10	10	-10	10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	1	1	-1	-1
Outplane Bending (Ml)	-20	20	20	-20
Normal Pressure Stress	43	43	43	43
Outplane Stress Summary	10	73	49	30

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-13	13	-13	13
Inplane Membrane (Mc)	0	0	0	0



Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	21	21	21	21

Inplane Stress Summary	-20	52	31	13
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	31	74	49	30
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-10	10	-10	10
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-21	21	21	-21
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21

Outplane Stress Summary	-10	54	27	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-13	13	-13	13
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-23	23	23	-23
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	43	43	43	43

Inplane Stress Summary	0	75	53	35
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 12 75 54 35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	1	1	-1	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-42	42	-42	42
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-76	76	76	-76
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-133	105	47	-20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7
Shear Stress Summary	9	9	5	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	136	106	49	23

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	2	2	-2	-1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1



Axial Bending (P)	-42	42	-42	42
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-80	80	80	-80
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-138	109	51	-25
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	140	109	57	30

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(3614.0(22.0 - 6.0))}$$

$$= 961.813 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7228.000 mm
Nozzle Outside Diameter	(d) =	322.400 mm
Vessel Thickness used	(T) =	16.000 mm
Nozzle Thickness used	(t) =	90.020 mm
T / t	=	0.178
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.948$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.116
Mr/P	=	0.236
M0/P	=	0.068
N0/P	=	0.112
MrD/Mc	=	0.314
NrDL/Mc	=	0.071
M0d/Mc	=	0.095
N0DL/Mc	=	0.042
MrD/Ml	=	0.201
NrDL/Ml	=	0.099
M0D/Ml	=	0.063
N0DL/Ml	=	0.097

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-17	17	-17	17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-14	14	14	-14



Normal Pressure Stress	86	86	86	86
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Outplane Stress Summary	45	110	81	89
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-62	62	-62	62
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-46	46	46	-46
Inplane Pressure Stress	43	43	43	43

Inplane Stress Summary	-73	142	25	57
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	119	142	81	89
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-17	17	-17	17
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-21	21	21	-21
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43

Outplane Stress Summary	-3	76	43	36
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-62	62	-62	62
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-71	71	71	-71
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	86	86	86	86

Inplane Stress Summary	-55	212	93	74
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right

circumferential plane)		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1
Shear Stress Summary		0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	55	212	93	74

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	43	43	43	43	43	43	43
Circ. Pl (SUS)	-1	-1	-3	-3	-5	-5	1	1
Circ. Q (SUS)	-30	30	10	-10	-36	36	10	-10
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-5	-5	1	1	0	0	-4	-4
Long. Q (SUS)	-35	35	9	-9	-31	31	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	43.0	43.0	43.0	43.0	43.0	43.0	43.0	43.0
Pm+Pl (SUS)	42.0	42.0	40.0	40.0	38.0	38.0	44.0	44.0
Pm+Pl+Q (Total)	31.6	72.4	50.5	30.5	13.4	74.4	54.3	34.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43.00	156.40	Passed
Pm+Pl (SUS)	44.00	234.60	Passed
Pm+Pl+Q (TOTAL)	74.40	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	86	86	86	86	86	86	86	86
Circ. Pl (SUS)	-7	-7	-1	-1	-7	-7	-3	-3
Circ. Q (SUS)	-31	31	-3	3	-133	133	9	-9
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-8	-8	-2	-2	-5	-5	-3	-3
Long. Q (SUS)	-108	108	-16	16	-38	38	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0



Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0
Pm+P1 (SUS)	79.0	79.0	85.0	85.0	79.0	79.0	83.0	83.0
Pm+P1+Q (Total)	121.0	143.0	82.0	88.0	54.0	212.0	92.0	74.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	156.40	Passed
Pm+P1 (SUS)	85.00	234.60	Passed
Pm+P1+Q (TOTAL)	212.01	469.20	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. P1 (SUS)	-1	-1	-3	-3	0	0	-4	-4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. P1 (SUS)	-14	-14	12	12	-14	-14	12	12
Long. Q (SUS)	-118	118	34	-34	-122	122	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+P1 (SUS)	14.6	14.6	14.6	14.6	15.5	15.5	15.5	15.5
Pm+P1+Q (Total)	133.2	105.8	49.0	22.4	137.4	109.2	56.0	29.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	143.60	Passed
Pm+P1 (SUS)	15.52	215.40	Passed
Pm+P1+Q (TOTAL)	137.36	430.80	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-2
From: 230

Pressure for Reinforcement Calculations	P	384.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7206.00	mm
Design Length of Section	L	2500.0039	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		42850.01	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

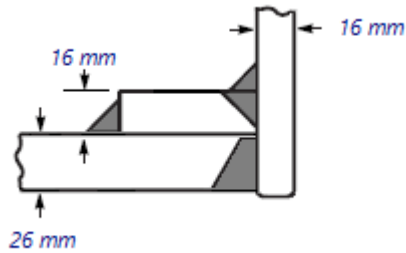
Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	N./mm ²
Allowable Stress At Ambient	Sna	175.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		22.7402	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1209.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		300.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.740	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 589.6 \cdot (\exp(384.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 0.7248 + 6.0 + 0.0 = 6.7248 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 589.6) / 589.6) - 0.0$
 $= 5216.982 - 0.0 = 5216.9824 \text{ KPa.}$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 690.53 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.0 - 6.0$
 $= 10.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7208.0/25.0$
 $= 288.320 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 7206.0/2 + 1.0$
 $= 3604.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:
 $= \sqrt{(Reff \cdot t) + W}$
 $= \sqrt{(3604.0 \cdot 25.0) + 300.0}$
 $= 600.167 \text{ mm}$

Limit [LR2]:
 $= \sqrt{((Reff + t) \cdot (t + t_e))}$



$$= \sqrt{(3604.0 + 25.0) \cdot (25.0 + 16.0)}$$

$$= 385.732 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 294.8$$

$$= 589.600 \text{ mm}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(600.167, 385.732, 589.6)$$

$$= 385.732 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 25.0, 16.0] + \sqrt{294.8 \cdot 10.0}$$

$$= 70.295 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 310.0$$

$$= 310.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(25.0 + 16.0)$$

$$= 328.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[70.295, 310.0, 328.0] + 25.0$$

$$= 95.295 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot f_{rp}) / LR)$$

$$= 25.0 + ((48.0 \cdot 1.0) / 385.732)$$

$$= 37.444 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 294.8 + 10.0) / (\sqrt{(7208.0 + 37.444) \cdot 37.444}))$$

$$= 1.151$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 25.0 \cdot 385.732 \cdot \max((1.151/5)^{0.85}, 1)$$

$$= 96.433 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 10.0 \cdot 95.295$$

$$= 9.530 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}42^2$$

$$= 0.5 \cdot 16.0^2$$

$$= 1.280 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}41^2$$

$$= 0.5 \cdot 16.0^2$$

$$= 1.280 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(300.0 \cdot 16.0, 385.732 \cdot 16.0)$$

$$= 48.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:
 $= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$
 $= 96.433 + 1.0(9.53 + 0.0) + 1.28 + 1.28 + 0.0 + 1.0(48.0)$
 $= 156.523 \text{ cm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= t_n / \ln[1 + t_n/R_n]$
 $= 10.0 / \ln[1 + 10.0/294.8]$
 $= 299.773 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:
 $= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 37.444 / \ln[1 + 37.444/3604.0]$
 $= 3622.678 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 156.4 \cdot 1.0$
 $= 234.6 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot R_{xn} \cdot L_H$
 $= 384.0 \cdot 299.773 \cdot 95.295$
 $= 1118.6 \text{ Kg}$

Force acting on the Shell [fS]:
 $= P \cdot R_{xs} (L_R + t_n)$
 $= 384.0 \cdot 3622.678 (385.732 + 10.0)$
 $= 56138.2 \text{ Kg}$

Discontinuity Force from Internal Pressure [fY]:
 $= P \cdot R_{xs} \cdot R_{nc}$
 $= 384.0 \cdot 3622.678 \cdot 294.8$
 $= 41820.1 \text{ Kg}$

Area Resisting Internal Pressure [Ap]:
 $= (fS + fY + fN) / P$
 $= (56138.2 + 41820.1 + 1118.6) / 384.0$
 $= 25301.4 \text{ cm}^2$

Maximum Allowable Working Pressure Candidate [Pmax1]:
 $= \text{Sallow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$
 $= 234.6 / (2 \cdot 25301.414 / 156.523 - 3622.678 / 37.444)$
 $= 1035.5 \text{ KPa.}$

Maximum Allowable Working Pressure Candidate [Pmax2]:
 $= S[t/R_{xs}]$
 $= 156.4[25.0/3622.678]$
 $= 1079.2 \text{ KPa.}$

Maximum Allowable Working Pressure [Pmax]:
 $= \min(P_{max1}, P_{max2})$
 $= \min(1035.497, 1079.249)$
 $= 1035.497 \text{ KPa.}$

Average Primary Membrane Stress [σ_{Avg}]:
 $= (fN + fS + fY) / AT$
 $= (1118.645 + 56138.199 + 41820.098) / 156.523$
 $= 62.076 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot R_{xs} / t_{eff}$
 $= 384.0 \cdot 3622.678 / 37.444$
 $= 37.2 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 62.076 - 37.154, 37.154)$
 $= 87.0 \text{ N./mm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	87.00	N./mm ²
Maximum Allowable Working Pressure	Pmax	1035.50	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (294.8 + 10.0) / 294.8$$

$$= 1.034 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (294.8 + 10.0)$$

$$= 478.779 \text{ mm}$$

Weld Length of Pad to Shell Weld [LTp]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (294.8 + 10.0 + 300.0)$$

$$= 950.018 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 11.314, \text{ must be } \geq \min(0.7 \cdot 10.000, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 25.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(41820 \cdot 1.03, 1.5 \cdot 156.4(9.53 + 0.0), \pi/4 \cdot 384.0 \cdot 294.8^2 \cdot 1.03^2)$$

$$= 2857.245 \text{ Kgf}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 2857.2 \cdot 25.0 \cdot 156 / (25.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 1742.223 \text{ Kgf}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 2857.2 \cdot 16.0 \cdot 156 / (25.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 1115.023 \text{ Kgf}$$

Shear Stress [τ1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 1742.223 / (478.779 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0))$$

$$= 3.717 \text{ N./mm}^2$$

Shear Stress [τ2]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L41T))$$

$$= 1115.023 / (478.779 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314))$$

$$= 1.508 \text{ N./mm}^2$$

Shear Stress [τ3]:

$$= fwp / (L_{Tp} \cdot (0.49 \cdot L42T))$$

$$= 1115.023 / (950.018 \cdot (0.49 \cdot 11.314))$$

$$= 2.076 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(3.717, 1.508, 2.076)$$

$$= 3.7 \text{ must be less than or equal to } 156.4 \text{ N./mm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 10.0 / \ln[1 + 10.0/294.8]$$

$$= 299.773 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 37.444 / \ln[1 + 37.444/3604.0]$$

$$= 3622.678 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 156.4$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 51.73 \cdot 299.773 \cdot 95.295$$

$$= 150.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 51.73 \cdot 3622.678 (385.732 + 10.0)$$

$$= 7562.6 \text{ Kg}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 51.73 \cdot 3622.678 \cdot 294.8$$

$$= 5633.7 \text{ Kg}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (7562.6 + 5633.7 + 150.7) / 51.73$$

$$= 25301.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 25301.418 / 156.523 - 3622.678 / 37.444)$$

$$= 1035.5 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 156.4 [25.0 / 3622.678]$$

$$= 1079.2 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(1035.497, 1079.249)$$

$$= 1035.497 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (150.697 + 7562.576 + 5633.733) / 156.523$$

$$= 8.362 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 51.73 \cdot 3622.678 / 37.444$$

$$= 5.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 8.362 - 5.005, 5.005)$$

$$= 11.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	11.72	N./mm ²
Maximum Allowable External Pressure	Pmax	1035.50	KPa.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.065

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.648 \cdot 1.0) / (16.0 - 6.0 - 0.0)$
 $= 0.065$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -49 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 19 °C
 The computed stress ratio is : 0.528

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (7.917 \cdot 1.0) / (16.0 - 1.0 - 0.0)$
 $= 0.528$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.065

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.648 \cdot 1.0) / (16.0 - 6.0 - 0.0)$
 $= 0.065$

Gov. MDMT of the Nozzle : -12 °C
 Gov. MDMT of the Reinforcement Pad : -12 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -49 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -104 °C

The Drop for this Nozzle is : 12.9155 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.9154 mm

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Input, Nozzle Desc: MH-1

From: 240

Pressure for Reinforcement Calculations	P	384.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7206.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		45150.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

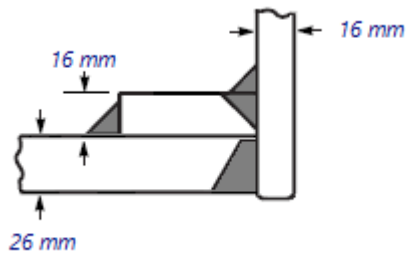
Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	N./mm ²
Allowable Stress At Ambient	Sna	175.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		22.7402	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1209.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		300.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.740	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 589.6 \cdot (\exp(384.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 0.7248 + 6.0 + 0.0 = 6.7248 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 589.6) / 589.6) - 0.0$
 $= 5216.982 - 0.0 = 5216.9824 \text{ KPa.}$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.0 - 6.0$
 $= 10.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7208.0/25.0$
 $= 288.320 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 7206.0/2 + 1.0$
 $= 3604.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:
 $= \sqrt{(Reff \cdot t) + W}$
 $= \sqrt{(3604.0 \cdot 25.0) + 300.0}$
 $= 600.167 \text{ mm}$

Limit [LR2]:
 $= \sqrt{((Reff + t) \cdot (t + t_e))}$
 $= \sqrt{((3604.0 + 25.0) \cdot (25.0 + 16.0))}$
 $= 385.732 \text{ mm}$

Limit [LR3]:



$$\begin{aligned}
 &= 2 \cdot R_n \\
 &= 2 \cdot 294.8 \\
 &= 589.600 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(600.167, 385.732, 589.6) \\
 &= 385.732 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 25.0, 16.0] + \sqrt{(294.8 \cdot 10.0)} \\
 &= 70.295 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 310.0 \\
 &= 310.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(25.0 + 16.0) \\
 &= 328.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[70.295, 310.0, 328.0] + 25.0 \\
 &= 95.295 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 25.0 + ((48.0 \cdot 1.0) / 385.732) \\
 &= 37.444 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff} })) \\
 &= \min(12, (2 \cdot 294.8 + 10.0) / (\sqrt{(7208.0 + 37.444) \cdot 37.444 })) \\
 &= 1.151
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 25.0 \cdot 385.732 \cdot \max((1.151/5)^{0.85}, 1) \\
 &= 96.433 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 10.0 \cdot 95.295 \\
 &= 9.530 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned}
 &= 0.5 \cdot Leg42^2 \\
 &= 0.5 \cdot 16.0^2 \\
 &= 1.280 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot Leg41^2 \\
 &= 0.5 \cdot 16.0^2 \\
 &= 1.280 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned}
 &= \min(W \cdot t_e, LR \cdot t_e) \\
 &= \min(300.0 \cdot 16.0, 385.732 \cdot 16.0) \\
 &= 48.000 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned}
 &= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5) \\
 &= 96.433 + 1.0(9.53 + 0.0) + 1.28 + 1.28 + 0.0 + 1.0(48.0) \\
 &= 156.523 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 10.0 / \ln[1 + 10.0/294.8]$$

$$= 299.773 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 37.444 / \ln[1 + 37.444/3604.0]$$

$$= 3622.678 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 384.0 \cdot 299.773 \cdot 95.295$$

$$= 1118.6 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 384.0 \cdot 3622.678 (385.732 + 10.0)$$

$$= 56138.2 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 384.0 \cdot 3622.678 \cdot 294.8$$

$$= 41820.1 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (56138.2 + 41820.1 + 1118.6) / 384.0$$

$$= 25301.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 25301.414 / 156.523 - 3622.678 / 37.444)$$

$$= 1035.5 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 156.4 [25.0 / 3622.678]$$

$$= 1079.2 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(1035.497, 1079.249)$$

$$= 1035.497 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1118.645 + 56138.199 + 41820.098) / 156.523$$

$$= 62.076 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 384.0 \cdot 3622.678 / 37.444$$

$$= 37.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 62.076 - 37.154, 37.154)$$

$$= 87.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	87.00	N./mm ²
Maximum Allowable Working Pressure	Pmax	1035.50	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (294.8 + 10.0) / 294.8$$

$$= 1.034 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (294.8 + 10.0)$$

$$= 478.779 \text{ mm}$$

Weld Length of Pad to Shell Weld [L_{TP}]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (294.8 + 10.0 + 300.0)$$

$$= 950.018 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 11.314, \text{ must be } \geq \min(0.7 \cdot 10.000, 6.000) \text{ Passed}$$

$$t_{f1}, L_{42T} = 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 25.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(41820 \cdot 1.03, 1.5 \cdot 156.4(9.53 + 0.0), \pi/4 \cdot 384.0 \cdot 294.8^2 \cdot 1.03^2)$$

$$= 2857.245 \text{ Kg}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 2857.2 \cdot 25.0 \cdot 156 / (25.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 1742.223 \text{ Kg}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 2857.2 \cdot 16.0 \cdot 156 / (25.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 1115.023 \text{ Kg}$$

Shear Stress [τ₁]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 1742.223 / (478.779 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0))$$

$$= 3.717 \text{ N./mm}^2$$

Shear Stress [τ₂]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 1115.023 / (478.779 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314))$$

$$= 1.508 \text{ N./mm}^2$$

Shear Stress [τ₃]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 1115.023 / (950.018 \cdot (0.49 \cdot 11.314))$$

$$= 2.076 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(3.717, 1.508, 2.076)$$

$$= 3.7 \text{ must be less than or equal to } 156.4 \text{ N./mm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 10.0 / \ln[1 + 10.0/294.8]$$

$$= 299.773 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 37.444 / \ln[1 + 37.444/3604.0]$$

= 3622.678 mm

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 156.4$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 51.73 \cdot 299.773 \cdot 95.295$$

$$= 150.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 51.73 \cdot 3622.678 (385.732 + 10.0)$$

$$= 7562.6 \text{ Kg}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 51.73 \cdot 3622.678 \cdot 294.8$$

$$= 5633.7 \text{ Kg}$$

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (7562.6 + 5633.7 + 150.7) / 51.73$$

$$= 25301.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 25301.418 / 156.523 - 3622.678 / 37.444)$$

$$= 1035.5 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 156.4 [25.0 / 3622.678]$$

$$= 1079.2 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(1035.497, 1079.249)$$

$$= 1035.497 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (150.697 + 7562.576 + 5633.733) / 156.523$$

$$= 8.362 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 51.73 \cdot 3622.678 / 37.444$$

$$= 5.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 8.362 - 5.005, 5.005)$$

$$= 11.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	11.72	N./mm ²
Maximum Allowable External Pressure	Pmax	1035.50	KPa.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-12 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.065

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.648 \cdot 1.0) / (16.0 - 6.0 - 0.0)$
 $= 0.065$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -49 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 19 °C
 The computed stress ratio is : 0.528

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (7.917 \cdot 1.0) / (16.0 - 1.0 - 0.0)$
 $= 0.528$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.065

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.648 \cdot 1.0) / (16.0 - 6.0 - 0.0)$
 $= 0.065$

Gov. MDMT of the Nozzle : -12 °C
 Gov. MDMT of the Reinforcement Pad : -12 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -49 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -104 °C

The Drop for this Nozzle is : 12.9155 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.9154 mm

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Input, Nozzle Desc: H1

From: 250

Pressure for Reinforcement Calculations	P	384.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7206.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		47550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

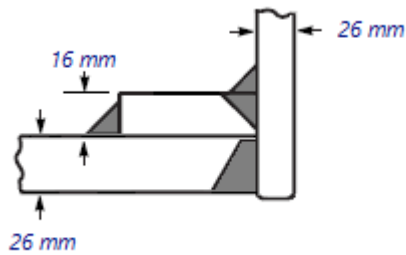
Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	N./mm ²
Allowable Stress At Ambient	Sna	175.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		33.9528	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1824.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		455.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	33.953	in.
Actual Thickness Used in Calculation	1.024	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 874.4 \cdot (\exp(384.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.0748 + 6.0 + 0.0 = 7.0748 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 20.0 + 874.4) / 874.4) - 0.0$
 $= 6995.386 - 0.0 = 6995.3862 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c :	11.043 mm
Min. required shell thickness, external p + c :	12.673 mm

Nozzle neck minimum required thickness	: 12.673 mm
--	-------------

Minimum nozzle neck thickness 26.010 mm , is greater than required thickness 12.673 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 18.3,	Allowable	: 156.4 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 395.9 N./mm ²	Passed
Occasional	: 4.1,	Allowable	: 208.0 N./mm ²	Passed
Shear	: 9.1,	Allowable	: 109.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/138.0, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/138.0, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 26.0 - 6.0$
 $= 20.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7208.0/25.0$
 $= 288.320 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= Di/2 + \text{corrosion allowance} \\
 &= 7206.0/2 + 1.0 \\
 &= 3604.000 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned}
 &= \sqrt{(Reff \cdot t) + W} \\
 &= \sqrt{(3604.0 \cdot 25.0) + 455.0} \\
 &= 755.167 \text{ mm}
 \end{aligned}$$

Limit [LR2]:

$$\begin{aligned}
 &= \sqrt{((Reff + t) \cdot (t + te))} \\
 &= \sqrt{((3604.0 + 25.0) \cdot (25.0 + 16.0))} \\
 &= 385.732 \text{ mm}
 \end{aligned}$$

Limit [LR3]:

$$\begin{aligned}
 &= 2 \cdot Rn \\
 &= 2 \cdot 437.2 \\
 &= 874.400 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(755.167, 385.732, 874.4) \\
 &= 385.732 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, te] + \sqrt{(Rn \cdot tn)} \\
 &= \min[1.5 \cdot 25.0, 16.0] + \sqrt{(437.2 \cdot 20.0)} \\
 &= 109.509 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= Lpr1 \\
 &= 310.0 \\
 &= 310.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + te) \\
 &= 8(25.0 + 16.0) \\
 &= 328.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[109.509, 310.0, 328.0] + 25.0 \\
 &= 134.509 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 25.0 + ((61.717 \cdot 1.0) / 385.732) \\
 &= 41.000 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot Rn + tn) / \sqrt{((Di + teff) \cdot teff)}) \\
 &= \min(12, (2 \cdot 437.2 + 20.0) / \sqrt{((7208.0 + 41.0) \cdot 41.0)}) \\
 &= 1.641
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 25.0 \cdot 385.732 \cdot \max((1.641/5)^{0.85}, 1) \\
 &= 96.433 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= tn \cdot LH \\
 &= 20.0 \cdot 134.509 \\
 &= 26.902 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 16.0^2$$

$$= 1.280 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(455.0 \cdot 16.0, 385.732 \cdot 16.0)$$

$$= 61.717 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 96.433 + 1.0(26.902 + 0.0) + 1.28 + 0.0 + 0.0 + 1.0(61.717)$$

$$= 186.332 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 20.0 / \ln[1 + 20.0/437.2]$$

$$= 447.126 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 41.0 / \ln[1 + 41.0/3604.0]$$

$$= 3624.457 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 138.0 \cdot 1.0$$

$$= 207.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 384.0 \cdot 447.126 \cdot 134.509$$

$$= 2355.1 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 384.0 \cdot 3624.457 (385.732 + 20.0)$$

$$= 57585.1 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 384.0 \cdot 3624.457 \cdot 437.2$$

$$= 62051.3 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (57585.1 + 62051.3 + 2355.1) / 384.0$$

$$= 31153.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 207.0 / (2 \cdot 31153.129 / 186.332 - 3624.457 / 41.0)$$

$$= 841.5 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 138.0 [25.0 / 3624.457]$$

$$= 951.8 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(841.477, 951.811)$$

$$= 841.477 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (2355.104 + 57585.055 + 62051.309) / 186.332$$

$$= 64.205 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 384.0 \cdot 3624.457 / 41.0$$

$$= 33.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 64.205 - 33.948, 33.948)$$

$$= 94.5 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	94.46	N./mm ²
Maximum Allowable Working Pressure	Pmax	841.48	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (437.2 + 20.0) / 437.2$$

$$= 1.046 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (437.2 + 20.0)$$

$$= 718.168 \text{ mm}$$

Weld Length of Pad to Shell Weld [LTp]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (437.2 + 20.0 + 455.0)$$

$$= 1432.880 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 11.314, \text{ must be } \geq \min(0.7 \cdot 20.000, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 25.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(62051 \cdot 1.05, 1.5 \cdot 156.4(26.902 + 0.0), \pi/4 \cdot 384.0 \cdot 437.2^2 \cdot 1.05^2)$$

$$= 6428.803 \text{ Kgf}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 6428.8 \cdot 25.0 \cdot 138 / (25.0 \cdot 138 + 16.0 \cdot 156)$$

$$= 3726.122 \text{ Kgf}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 6428.8 \cdot 16.0 \cdot 156 / (25.0 \cdot 138 + 16.0 \cdot 156)$$

$$= 2702.680 \text{ Kgf}$$

Shear Stress [τ_1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 3726.122 / (718.168 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0))$$

$$= 5.300 \text{ N./mm}^2$$

Shear Stress [τ_2]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 2702.68 / (718.168 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314))$$

$$= 2.437 \text{ N./mm}^2$$

Shear Stress [τ_3]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 2702.68 / (1432.88 \cdot (0.49 \cdot 11.314))$$

$$= 3.337 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(5.3, 2.437, 3.337) \\ &= 5.3 \text{ must be less than or equal to } 138.0 \text{ N./mm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 20.0 / \ln[1 + 20.0/437.2] \\ &= 447.126 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 41.0 / \ln[1 + 41.0/3604.0] \\ &= 3624.457 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 138.0 \\ &= 207.0 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 51.73 \cdot 447.126 \cdot 134.509 \\ &= 317.3 \text{ Kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 51.73 \cdot 3624.457 (385.732 + 20.0) \\ &= 7757.5 \text{ Kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 51.73 \cdot 3624.457 \cdot 437.2 \\ &= 8359.2 \text{ Kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (7757.5 + 8359.2 + 317.3) / 51.73 \\ &= 31153.1 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 207.0 / (2 \cdot 31153.133 / 186.332 - 3624.457 / 41.0) \\ &= 841.5 \text{ KPa.} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 138.0 [25.0 / 3624.457] \\ &= 951.8 \text{ KPa.} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(841.477, 951.811) \\ &= 841.477 \text{ KPa.} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (317.264 + 7757.487 + 8359.151) / 186.332 \\ &= 8.649 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 51.73 \cdot 3624.457 / 41.0 \\ &= 4.6 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 8.649 - 4.573, 4.573) \end{aligned}$$

= 12.7 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	12.73	N./mm ²
Maximum Allowable External Pressure	Pmax	841.48	KPa.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-12 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.048

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (0.96 \cdot 1.0) / (26.0 - 6.0 - 0.0)$$

$$= 0.048$$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-12 °C
Gov. MDMT of the Reinforcement Pad	:	-12 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	:	-104 °C
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The Drop for this Nozzle is : 29.1257 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 365.1257 mm

Input Echo, WRC297 Item 1, Description: H1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7206.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	862.400	mm
Nozzle Thickness	26.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	175.00 N./mm ²



Nozzle Hot S.I. Allowable SNmh 156.40 N./mm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 1824.400 mm
 Thickness of Reinforcing Pad 16.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	384.00	KPa.
Radial Load	P	6880.89	Kgf
Circumferential Shear	Vc	4587.26	Kgf
Longitudinal Shear	Vl	4587.26	Kgf
Circumferential Moment	Mc	11560.50	Kg-m.
Longitudinal Moment	Ml	11560.50	Kg-m.
Torsional Moment	Mt	16515.02	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(3624.5(26.0 - 1.0))}$$

$$= 1907.763 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7249.001 mm
Nozzle Outside Diameter	(d) =	914.400 mm
Vessel Thickness used	(T) =	41.000 mm
Nozzle Thickness used	(t) =	20.000 mm
T / t	=	2.050
d / t	=	45.720

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.677$$

Nr/P	=	0.060
Mr/P	=	0.099
M0/P	=	0.057
N0/P	=	0.163
MrD/Mc	=	0.194
NrDL/Mc	=	0.098
M0d/Mc	=	0.128
N0DL/Mc	=	0.191
MrD/Ml	=	0.118
NrDL/Ml	=	0.067
M0D/Ml	=	0.068
N0DL/Ml	=	0.298

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-21	-21	21	21
Outplane Bending (Ml)	-29	29	29	-29
Normal Pressure Stress	33	33	33	33
Outplane Stress Summary	-38	49	65	33



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-23	23	-23	23
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	-66	85	47	-9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	67	86	65	42

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-56	56	56	-56
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	-74	66	67	-18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-23	23	-23	23
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-86	86	86	-86
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	33	33	33
Inplane Stress Summary	-85	134	100	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0



Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	86	134	101	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10
Hoop Stress Summary	-18	-18	25	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-69	69	-69	69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-158	158	158	-158
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-233	222	101	-76

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	7	7	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	233	240	102	102

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10

Hoop Stress Summary	-10	-10	17	17
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-69	69	-69	69
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-270	270	270	-270
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-344	333	213	-187
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary	4	4	7	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	344	344	213	206

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{3616.5(26.0 - 1.0)}$
 $= 1906.666 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7233.000 mm
Nozzle Outside Diameter	(d) =	1824.400 mm
Vessel Thickness used	(T) =	25.000 mm
Nozzle Thickness used	(t) =	475.000 mm
T / t	=	0.053
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.290$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.079
M0/P	=	0.022
N0/P	=	0.025
MrD/Mc	=	0.248
NrDL/Mc	=	0.108
M0d/Mc	=	0.074
N0DL/Mc	=	0.048
MrD/Ml	=	0.055
NrDL/Ml	=	0.061
M0D/Ml	=	0.017
N0DL/Ml	=	0.028

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	55	55	55	55
Outplane Stress Summary	25	74	51	59

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-51	51	-51	51
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-32	32	32	-32
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	-67	100	9	46

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	93	100	51	59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27
Outplane Stress Summary	-38	78	59	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-51	51	-51	51
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-147	147	147	-147

Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	55	55	55	55
Inplane Stress Summary	-160	238	156	-35

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	160	238	156	35

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	33	33	33	33	33	33	33
Circ. Pl (SUS)	-27	-27	15	15	-9	-9	5	5
Circ. Q (SUS)	-42	42	16	-16	-109	109	63	-63
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-6	-6	2	2	-20	-20	8	8
Long. Q (SUS)	-75	75	29	-29	-69	69	43	-43
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	33.0	33.0	33.0	33.0	33.0	33.0	33.0	33.0
Pm+Pl (SUS)	10.0	10.0	48.0	48.0	28.0	28.0	38.0	38.0
Pm+Pl+Q (Total)	65.3	85.2	64.5	43.4	85.7	133.1	101.3	26.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	48.00	207.00	Passed
Pm+Pl+Q (TOTAL)	133.13	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	55	55	55	55	55	55	55	55
Circ. Pl (SUS)	-4	-4	0	0	-15	-15	5	5
Circ. Q (SUS)	-23	23	-5	5	-198	198	96	-96
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-11	-11	1	1	-6	-6	2	2
Long. Q (SUS)	-83	83	-19	19	-58	58	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0
Pm+Pl (SUS)	51.0	51.0	55.0	55.0	40.0	40.0	60.0	60.0
Pm+Pl+Q (Total)	95.0	99.0	50.0	60.1	158.0	238.0	156.0	36.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.00	138.00	Passed
Pm+Pl (SUS)	60.00	207.00	Passed
Pm+Pl+Q (TOTAL)	238.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. Pl (SUS)	-27	-27	15	15	-20	-20	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-10	-10	8	8	-10	-10	8	8
Long. Q (SUS)	-227	227	89	-89	-339	339	201	-201
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+Pl (SUS)	18.1	17.1	24.1	25.1	11.2	10.2	17.2	18.2
Pm+Pl+Q (Total)	233.2	238.4	101.3	102.5	345.1	343.1	213.2	207.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	156.40	Passed
Pm+Pl (SUS)	25.08	234.60	Passed
Pm+Pl+Q (TOTAL)	345.07	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: H2

From: 250

Pressure for Reinforcement Calculations	P	384.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7206.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		47550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

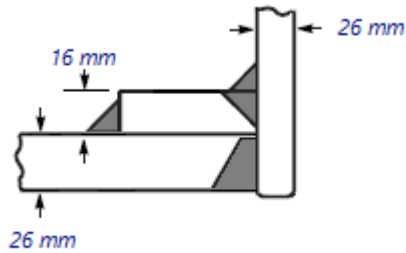
Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	N./mm ²
Allowable Stress At Ambient	Sna	175.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		33.9528	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1824.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		455.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	33.953	in.
Actual Thickness Used in Calculation	1.024	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 874.4 \cdot (\exp(384.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.0748 + 6.0 + 0.0 = 7.0748 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 20.0 + 874.4) / 874.4) - 0.0$
 $= 6995.386 - 0.0 = 6995.3862 \text{ KPa.}$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 220.81 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 842.93 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 11.043 mm
 Min. required shell thickness, external p + c : 12.673 mm

Nozzle neck minimum required thickness : 12.673 mm

Minimum nozzle neck thickness 26.010 mm , is greater than required thickness 12.673 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.3,	Allowable	:	156.4 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	395.9 N./mm ²	Passed
Occasional	:	4.1,	Allowable	:	208.0 N./mm ²	Passed
Shear	:	9.1,	Allowable	:	109.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/138.0, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/138.0, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 26.0 - 6.0$
 $= 20.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:



$$\begin{aligned}
 &= D_i/t \\
 &= 7208.0/25.0 \\
 &= 288.320 \text{ must be less than } 400.
 \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= D_i/2 + \text{corrosion allowance} \\
 &= 7206.0/2 + 1.0 \\
 &= 3604.000 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned}
 &= \sqrt{(R_{eff} \cdot t) + W} \\
 &= \sqrt{(3604.0 \cdot 25.0) + 455.0} \\
 &= 755.167 \text{ mm}
 \end{aligned}$$

Limit [LR2]:

$$\begin{aligned}
 &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\
 &= \sqrt{((3604.0 + 25.0) \cdot (25.0 + 16.0))} \\
 &= 385.732 \text{ mm}
 \end{aligned}$$

Limit [LR3]:

$$\begin{aligned}
 &= 2 \cdot R_n \\
 &= 2 \cdot 437.2 \\
 &= 874.400 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(755.167, 385.732, 874.4) \\
 &= 385.732 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 25.0, 16.0] + \sqrt{(437.2 \cdot 20.0)} \\
 &= 109.509 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 310.0 \\
 &= 310.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(25.0 + 16.0) \\
 &= 328.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[109.509, 310.0, 328.0] + 25.0 \\
 &= 134.509 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A_5 \cdot frp) / LR) \\
 &= 25.0 + ((61.717 \cdot 1.0) / 385.732) \\
 &= 41.000 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\
 &= \min(12, (2 \cdot 437.2 + 20.0) / (\sqrt{(7208.0 + 41.0) \cdot 41.0})) \\
 &= 1.641
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 25.0 \cdot 385.732 \cdot \max((1.641/5)^{0.85}, 1) \\
 &= 96.433 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot L_H$$

$$= 20.0 \cdot 134.509$$

$$= 26.902 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 16.0^2$$

$$= 1.280 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(455.0 \cdot 16.0, 385.732 \cdot 16.0)$$

$$= 61.717 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 96.433 + 1.0(26.902 + 0.0) + 1.28 + 0.0 + 0.0 + 1.0(61.717)$$

$$= 186.332 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 20.0 / \ln[1 + 20.0/437.2]$$

$$= 447.126 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 41.0 / \ln[1 + 41.0/3604.0]$$

$$= 3624.457 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 138.0 \cdot 1.0$$

$$= 207.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 384.0 \cdot 447.126 \cdot 134.509$$

$$= 2355.1 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 384.0 \cdot 3624.457 (385.732 + 20.0)$$

$$= 57585.1 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 384.0 \cdot 3624.457 \cdot 437.2$$

$$= 62051.3 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (57585.1 + 62051.3 + 2355.1) / 384.0$$

$$= 31153.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 207.0 / (2 \cdot 31153.129 / 186.332 - 3624.457 / 41.0)$$

$$= 841.5 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 138.0 [25.0 / 3624.457]$$

$$= 951.8 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(841.477, 951.811)$$

$$= 841.477 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:
 $= (f_N + f_S + f_Y) / AT$
 $= (2355.104 + 57585.055 + 62051.309) / 186.332$
 $= 64.205 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot R_{xs} / t_{eff}$
 $= 384.0 \cdot 3624.457 / 41.0$
 $= 33.9 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 64.205 - 33.948, 33.948)$
 $= 94.5 \text{ N./mm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	94.46	N./mm ²
Maximum Allowable Working Pressure	Pmax	841.48	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 $= (R_{nc} + t_n) / R_{nc}$
 $= (437.2 + 20.0) / 437.2$
 $= 1.046 \text{ For set-in Nozzles}$

Weld Length of Nozzle to Shell Weld [LT]:
 $= \pi/2 \cdot (R_n + t_n)$
 $= \pi/2 \cdot (437.2 + 20.0)$
 $= 718.168 \text{ mm}$

Weld Length of Pad to Shell Weld [LTp]:
 $= \pi/2 \cdot (R_n + t_n + W)$
 $= \pi/2 \cdot (437.2 + 20.0 + 455.0)$
 $= 1432.880 \text{ mm}$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:
 $= 11.314, 11.314, 0.000, \text{ mm}$

Required Weld Throat Dimensions :
 tc, L41T = 11.314, must be $\geq \min(0.7 \cdot 20.000, 6.000)$ Passed
 tf1, L42T = 11.314, must be $\geq \min(0.6 \cdot 16.000, 0.6 \cdot 25.000)$ Passed

Weld Load Value [fwelds]:
 $= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(62051 \cdot 1.05, 1.5 \cdot 156.4(26.902 + 0.0), \pi/4 \cdot 384.0 \cdot 437.2^2 \cdot 1.05^2)$
 $= 6428.803 \text{ Kgf}$

Discontinuity Force [fws]:
 $= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$
 $= 6428.8 \cdot 25.0 \cdot 138 / (25.0 \cdot 138 + 16.0 \cdot 156)$
 $= 3726.122 \text{ Kgf}$

Discontinuity Force [fwp]:
 $= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$
 $= 6428.8 \cdot 16.0 \cdot 156 / (25.0 \cdot 138 + 16.0 \cdot 156)$
 $= 2702.680 \text{ Kgf}$

Shear Stress [τ_1]:
 $= fws / (L_T \cdot (0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
 $= 3726.122 / (718.168 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0))$
 $= 5.300 \text{ N./mm}^2$

Shear Stress [τ_2]:
 $= fwp / (L_T \cdot (0.6 \cdot tw_2 + 0.49 \cdot L_{41T}))$
 $= 2702.68 / (718.168 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314))$
 $= 2.437 \text{ N./mm}^2$

Shear Stress [τ_3]:

$$\begin{aligned}
 &= fwp / (L_tP \cdot (0.49 \cdot L_{42T})) \\
 &= 2702.68 / (1432.88 \cdot (0.49 \cdot 11.314)) \\
 &= 3.337 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned}
 &= \max(\tau_1, \tau_2, \tau_3) \\
 &= \max(5.3, 2.437, 3.337) \\
 &= 5.3 \text{ must be less than or equal to } 138.0 \text{ N./mm}^2
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 20.0 / \ln[1 + 20.0/437.2] \\
 &= 447.126 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 41.0 / \ln[1 + 41.0/3604.0] \\
 &= 3624.457 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 138.0 \\
 &= 207.0 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 51.73 \cdot 447.126 \cdot 134.509 \\
 &= 317.3 \text{ Kg}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 51.73 \cdot 3624.457 (385.732 + 20.0) \\
 &= 7757.5 \text{ Kg}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 51.73 \cdot 3624.457 \cdot 437.2 \\
 &= 8359.2 \text{ Kg}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (7757.5 + 8359.2 + 317.3) / 51.73 \\
 &= 31153.1 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\
 &= 207.0 / (2 \cdot 31153.133 / 186.332 - 3624.457 / 41.0) \\
 &= 841.5 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S / [t / R_{xs}] \\
 &= 138.0 [25.0 / 3624.457] \\
 &= 951.8 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(841.477, 951.811) \\
 &= 841.477 \text{ KPa.}
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (317.264 + 7757.487 + 8359.151) / 186.332 \\
 &= 8.649 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot R_{xs} / t_{eff} \\
 &= 51.73 \cdot 3624.457 / 41.0 \\
 &= 4.6 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 8.649 - 4.573, 4.573)$
 $= 12.7 \text{ N./mm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	12.73	N./mm ²
Maximum Allowable External Pressure	Pmax	841.48	KPa.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.048

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.96 \cdot 1.0) / (26.0 - 6.0 - 0.0)$
 $= 0.048$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-12 °C
Gov. MDMT of the Reinforcement Pad	:	-12 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-104 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -104 °C

The Drop for this Nozzle is : 29.1257 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 365.1257 mm

Input Echo, WRC297 Item 1, Description: H2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7206.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	862.400	mm



Nozzle Thickness	26.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	175.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	156.40 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1824.400	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	384.00	KPa.
Radial Load	P	6880.89	Kgf
Circumferential Shear	Vc	4587.26	Kgf
Longitudinal Shear	Vl	4587.26	Kgf
Circumferential Moment	Mc	11560.50	Kg-m.
Longitudinal Moment	Ml	11560.50	Kg-m.
Torsional Moment	Mt	16515.02	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(3624.5(26.0 - 1.0))}$
 $= 1907.763 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7249.001	mm
Nozzle Outside Diameter	(d) =	914.400	mm
Vessel Thickness used	(T) =	41.000	mm
Nozzle Thickness used	(t) =	20.000	mm
T / t	=	2.050	
d / t	=	45.720	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.677$$

Nr/P	=	0.060
Mr/P	=	0.099
M0/P	=	0.057
N0/P	=	0.163
MrD/Mc	=	0.194
NrDL/Mc	=	0.098
M0d/Mc	=	0.128
N0DL/Mc	=	0.191
MrD/Ml	=	0.118
NrDL/Ml	=	0.067
M0D/Ml	=	0.068
N0DL/Ml	=	0.298

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-21	-21	21	21



Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	33	33	33	33

Outplane Stress Summary	-38	49	65	33
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-23	23	-23	23
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	16	16	16	16

Inplane Stress Summary	-66	85	47	-9
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	3	3	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	67	86	65	42
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-56	56	56	-56
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-74	66	67	-18
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-23	23	-23	23
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-86	86	86	-86
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	33	33	33

Inplane Stress Summary	-85	134	100	-23
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	86	134	101	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10
Hoop Stress Summary	-18	-18	25	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-69	69	-69	69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-158	158	158	-158
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-233	222	101	-76

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	7	7	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	233	240	102	102

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10

Hoop Stress Summary	-10	-10	17	17
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-69	69	-69	69
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-270	270	270	-270
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-344	333	213	-187
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary	4	4	7	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	344	344	213	206
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(3616.5(26.0 - 1.0))}$$

$$= 1906.666 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7233.000 mm
Nozzle Outside Diameter	(d) =	1824.400 mm
Vessel Thickness used	(T) =	25.000 mm
Nozzle Thickness used	(t) =	475.000 mm
T / t	=	0.053
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.290$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.079
M0/P	=	0.022
N0/P	=	0.025
MrD/Mc	=	0.248
NrDL/Mc	=	0.108
M0d/Mc	=	0.074
N0DL/Mc	=	0.048
MrD/M1	=	0.055
NrDL/M1	=	0.061
M0D/M1	=	0.017

NØDL/M1 = 0.028

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	55	55	55	55
Outplane Stress Summary	25	74	51	59

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-51	51	-51	51
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-32	32	32	-32
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	-67	100	9	46

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	93	100	51	59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27
Outplane Stress Summary	-38	78	59	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-51	51	-51	51
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-147	147	147	-147
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	55	55	55	55

Inplane Stress Summary	-160	238	156	-35
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	160	238	156	35

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	33	33	33	33	33	33	33
Circ. Pl (SUS)	-27	-27	15	15	-9	-9	5	5
Circ. Q (SUS)	-42	42	16	-16	-109	109	63	-63
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-6	-6	2	2	-20	-20	8	8
Long. Q (SUS)	-75	75	29	-29	-69	69	43	-43
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	33.0	33.0	33.0	33.0	33.0	33.0	33.0	33.0
Pm+Pl (SUS)	10.0	10.0	48.0	48.0	28.0	28.0	38.0	38.0
Pm+Pl+Q (Total)	65.3	85.2	64.5	43.4	85.7	133.1	101.3	26.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	48.00	207.00	Passed
Pm+Pl+Q (TOTAL)	133.13	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	55	55	55	55	55	55	55	55
Circ. Pl (SUS)	-4	-4	0	0	-15	-15	5	5
Circ. Q (SUS)	-23	23	-5	5	-198	198	96	-96
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-11	-11	1	1	-6	-6	2	2
Long. Q (SUS)	-83	83	-19	19	-58	58	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0
Pm+Pl (SUS)	51.0	51.0	55.0	55.0	40.0	40.0	60.0	60.0
Pm+Pl+Q (Total)	95.0	99.0	50.0	60.1	158.0	238.0	156.0	36.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.00	138.00	Passed
Pm+Pl (SUS)	60.00	207.00	Passed
Pm+Pl+Q (TOTAL)	238.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. Pl (SUS)	-27	-27	15	15	-20	-20	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-10	-10	8	8	-10	-10	8	8
Long. Q (SUS)	-227	227	89	-89	-339	339	201	-201
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+Pl (SUS)	18.1	17.1	24.1	25.1	11.2	10.2	17.2	18.2
Pm+Pl+Q (Total)	233.2	238.4	101.3	102.5	345.1	343.1	213.2	207.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	156.40	Passed
Pm+Pl (SUS)	25.08	234.60	Passed
Pm+Pl+Q (TOTAL)	345.07	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: F
From: 270

Pressure for Reinforcement Calculations	P	380.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.73	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Elliptical Head	D	7206.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

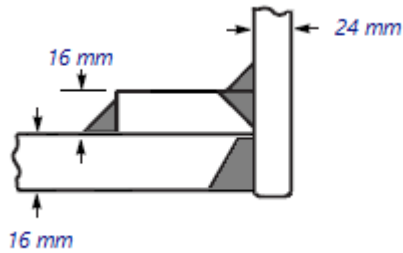
Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	N./mm ²
Allowable Stress At Ambient	Sna	175.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		40.1102	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	156.40	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	2126.8000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		530.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: F

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	40.110	in.
Actual Thickness Used in Calculation	0.945	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1030.8 \cdot (\exp(380.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.2538 + 6.0 + 0.0 = 7.2538 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 1030.8) / 1030.8) - 0.0$
 $= 5368.645 - 0.0 = 5368.6445 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c :	13.578 mm
Min. required shell thickness, external p + c :	14.165 mm

Nozzle neck minimum required thickness	: 14.165 mm
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Minimum nozzle neck thickness 24.003 mm , is greater than required thickness 14.165 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 20.9,	Allowable	: 156.4 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 393.3 N./mm ²	Passed
Occasional	: 5.3,	Allowable	: 208.0 N./mm ²	Passed
Shear	: 9.9,	Allowable	: 109.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 24.0 - 6.0$
 $= 18.000 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
 $= 0.9/6 \cdot 7208.0 [2 + (7208.0/(2 \cdot 1802.5))^2]$
 $= 6484.801 \text{ mm}$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned} \text{Limit [LR1]:} \\ &= \sqrt{(R_{eff} \cdot t) + W} \\ &= \sqrt{(6484.801 \cdot 15.0) + 530.0} \\ &= 841.885 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Limit [LR2]:} \\ &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\ &= \sqrt{((6484.801 + 15.0) \cdot (15.0 + 16.0))} \\ &= 448.881 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Limit [LR3]:} \\ &= 2 \cdot R_n \\ &= 2 \cdot 515.4 \\ &= 1030.800 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Final value of [LR]:} \\ &= \min(LR1, LR2, LR3) \\ &= \min(841.885, 448.881, 1030.8) \\ &= 448.881 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Intermediate value [Cn]:} \\ &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((15.0 + 16.0)/18.0)^{0.35}, 1) \\ &= \min(1.21, 1) \\ &= 1.000 \end{aligned}$$

$$\begin{aligned} \text{Intermediate value [Cp]:} \\ &= \exp((0.35 \cdot d_i - X_o)/(16t)) \\ &= \exp((0.35 \cdot 7208.0 - 533.4)/(16 \cdot 15.0)) \\ &= 3980.515 \end{aligned}$$

$$\begin{aligned} \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\ &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\ &= \min(15.0 + 16.0 + 1.0 \cdot \sqrt{(515.4 \cdot 18.0)}, 500.0 + 15.0) \\ &= 127.318 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Effective Vessel Thickness [teff]:} \\ &= t + ((A_5 \cdot f_{rp}) / LR) \\ &= 15.0 + ((71.821 \cdot 1.0) / 448.881) \\ &= 31.000 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned} \text{Area Contributed by the Vessel Wall [A1]:} \\ &= t \cdot LR \\ &= 15.0 \cdot 448.881 \\ &= 67.332 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\ &= t_n \cdot LH \\ &= 18.0 \cdot 127.318 \\ &= 22.917 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Outside Fillet Weld [A41]:} \\ &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 1.280 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Reinforcing Pad [A5]:} \\ &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(530.0 \cdot 16.0, 448.881 \cdot 16.0) \\ &= 71.821 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{The total area contributed by A1 through A5 [AT]:} \\ &= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 + f_{rp}(A5) \\ &= 67.332 + 1.0(22.917 + 0.0) + 1.28 + 0.0 + 0.0 + 1.0(71.821) \end{aligned}$$

$$= 163.350 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 18.0 / \ln[1 + 18.0/515.4]$$

$$= 524.349 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 31.0 / \ln[1 + 31.0/6484.801]$$

$$= 6500.285 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 380.0 \cdot 524.349 \cdot 127.318$$

$$= 2587.0 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n) / 2$$

$$= 380.0 \cdot 6500.285 \cdot (448.881 + 18.0) / 2$$

$$= 58801.5 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (380.0 \cdot 6500.285 \cdot 515.4) / 2$$

$$= 64912.3 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (58801.5 + 64912.3 + 2587.0) / 380.0$$

$$= 32593.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 234.6 / (2 \cdot 32593.115 / 163.35 - 6500.285 / (2 \cdot 31.0))$$

$$= 797.3 \text{ KPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 156.4 [15.0 / 6500.285]$$

$$= 721.8 \text{ KPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(797.331, 721.773)$$

$$= 721.773 \text{ KPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (2586.971 + 58801.504 + 64912.301) / 163.35$$

$$= 75.825 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 380.0 \cdot 6500.285 / (2 \cdot 31.0)$$

$$= 39.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 75.825 - 39.843, 39.843)$$

$$= 111.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	111.81	N./mm ²

Maximum Allowable Working Pressure Pmax 721.77 KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (515.4 + 18.0) / 515.4$$

$$= 1.035 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (515.4 + 18.0)$$

$$= 837.863 \text{ mm}$$

Weld Length of Pad to Shell Weld [LTP]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (515.4 + 18.0 + 530.0)$$

$$= 1670.385 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 11.314, \text{ must be } \geq \min(0.7 \cdot 18.000, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 15.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(64912 \cdot 1.03, 1.5 \cdot 156.4(22.917 + 0.0), \pi/4 \cdot 380.0 \cdot 515.4^2 \cdot 1.03^2)$$

$$= 8659.167 \text{ Kgf}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 8659.2 \cdot 15.0 \cdot 156 / (15.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 4189.919 \text{ Kgf}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 8659.2 \cdot 16.0 \cdot 156 / (15.0 \cdot 156 + 16.0 \cdot 156)$$

$$= 4469.247 \text{ Kgf}$$

Shear Stress [τ1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 4189.919 / (837.863 \cdot (0.6 \cdot 15.0 + 0.49 \cdot 0.0))$$

$$= 5.449 \text{ N./mm}^2$$

Shear Stress [τ2]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 4469.247 / (837.863 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314))$$

$$= 3.454 \text{ N./mm}^2$$

Shear Stress [τ3]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 4469.247 / (1670.385 \cdot (0.49 \cdot 11.314))$$

$$= 4.733 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(5.449, 3.454, 4.733)$$

$$= 5.4 \text{ must be less than or equal to } 156.4 \text{ N./mm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 18.0 / \ln[1 + 18.0/515.4]$$

$$= 524.349 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 31.0 / \ln[1 + 31.0 / 6484.801]$$

$$= 6500.285 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 156.4$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 51.73 \cdot 524.349 \cdot 127.318$$

$$= 352.2 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n) / 2$$

$$= 51.73 \cdot 6500.285 \cdot (448.881 + 18.0) / 2$$

$$= 8004.7 \text{ Kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (51.73 \cdot 6500.285 \cdot 515.4) / 2$$

$$= 8836.6 \text{ Kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (8004.7 + 8836.6 + 352.2) / 51.73$$

$$= 32593.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 234.6 / (2 \cdot 32593.115 / 163.35 - 6500.285 / (2 \cdot 31.0))$$

$$= 797.3 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / R_{xs}]$$

$$= 2 \cdot 156.4[15.0 / 6500.285]$$

$$= 721.8 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(797.331, 721.773)$$

$$= 721.773 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (352.168 + 8004.741 + 8836.613) / 163.35$$

$$= 10.322 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 51.73 \cdot 6500.285 / (2 \cdot 31.0)$$

$$= 5.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 10.322 - 5.424, 5.424)$$

$$= 15.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	15.22	N./mm ²
Maximum Allowable External Pressure	Pmax	721.77	KPa.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-12 °C



The Curve for this Material is : A
 As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.062

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (1.12 \cdot 1.0) / (24.0 - 6.0 - 0.0)$$

$$= 0.062$$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -45 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 14 °C
 The computed stress ratio is : 0.649

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (9.73 \cdot 1.0) / (16.0 - 1.0 - 0.0)$$

$$= 0.649$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: D

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -45 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : D
 From Figure 3.7M- Non PWHT, the MDMT is : -30 °C
 Temperature reduction per Graph 3.12 : 14 °C
 The computed stress ratio is : 0.649

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (9.73 \cdot 1.0) / (16.0 - 1.0 - 0.0)$$

$$= 0.649$$

Gov. MDMT of the Nozzle : -12 °C
 Gov. MDMT of the Reinforcement Pad : -12 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -45 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -104 °C

The Drop for this Nozzle is : 21.9289 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 537.9291 mm

Input Echo, WRC107/537 Item 1, Description: F :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	12970.801	mm
Vessel Thickness	Tv	16.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	

Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	1018.800	mm
Nozzle Thickness	Tn	24.000	mm
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	175.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	156.40	N./mm ²

Thickness of Reinforcing Pad	Tpad	16.000	mm
Diameter of Reinforcing Pad	Dpad	2126.800	mm

Design Internal Pressure	Dp	380.000	KPa.
Include Pressure Thrust		No	

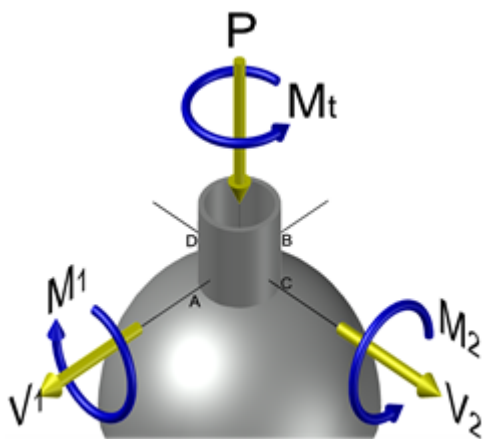
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	8027.7	Kgf
Longitudinal Shear	(SUS)	(V1) V1	5351.8	Kgf
Circumferential Shear	(SUS)	(Vc) V2	5351.8	Kgf
Circumferential Moment	(SUS)	(Mc) M1	15735.1	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	15735.1	Kg-m.
Torsional Moment	(SUS)	Mt	22478.7	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 1066.8 + 2 \cdot 1.65 \cdot \sqrt{(6493.9 (16.0 - 1.0))}$$

$$= 2096.741 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	8027.7	Kgf
Circumferential Shear	(VC) V2	5351.8	Kgf



Longitudinal Shear (VL) V1 5351.8 Kgf
 Circumferential Moment (MC) M1 15735.1 Kg-m.
 Longitudinal Moment (ML) M2 15735.1 Kg-m.
 Torsional Moment MT 22478.7 Kg-m.

Dimensionless Param: U = 1.19 TAU = 29.13 RHO = 4.00 (1.72)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 7	0.02046	(A,B,C,D)
M(x) / P	SP 7	0.00762	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 7	0.02940	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 7	0.01241	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 7	0.02940	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 7	0.01241	(A,B,C,D)
N(y) * T / P	SP 7	0.17824	(A,B,C,D)
M(y) / P	SP 7	0.01877	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.20933	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.03772	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.20933	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.03772	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7
Rad.	Bend. P	-3.7	3.7	-3.7	3.7	-3.7	3.7	-3.7	3.7
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-10.5	-10.5	10.5	10.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-26.6	26.6	26.6	-26.6
Rad.	Memb. ML	-10.5	-10.5	10.5	10.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-26.6	26.6	26.6	-26.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-42.6	18.2	31.7	-14.1	-42.6	18.2	31.7	-14.1
Tang.	Memb. P	-14.6	-14.6	-14.6	-14.6	-14.6	-14.6	-14.6	-14.6
Tang.	Bend. P	-9.2	9.2	-9.2	9.2	-9.2	9.2	-9.2	9.2
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-74.9	-74.9	74.9	74.9
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-80.9	80.9	80.9	-80.9
Tang.	Memb. ML	-74.9	-74.9	74.9	74.9	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-80.9	80.9	80.9	-80.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-179.6	0.7	132.0	-11.4	-179.6	0.7	132.0	-11.4
Shear VC		1.0	1.0	-1.0	-1.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Tot. Shear		5.0	5.0	3.0	3.0	3.0	3.0	5.0	5.0
Str. Int.		179.8	20.1	132.1	16.0	179.7	18.7	132.2	17.9

Unitless Prm: U = 3.41 TAU = 0.00 (58.58) RHO = 0.00 (0.83)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.02175	(A,B,C,D)
M(x) / P	SR 2	0.00990	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01053	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.01691	(A,B,C,D)



$M(x) * \text{SQRT}(R_m * T) / ML$ SR 3 0.01053 (A,B,C,D)
 $N(y) * T / P$ SR 2 0.00650 (A,B,C,D)
 $M(y) / P$ SR 2 0.00300 (A,B,C,D)
 $N(y) * T * \text{SQRT}(R_m * T) / MC$ SR 3 0.00507 (A,B,C,D)
 $M(y) * \text{SQRT}(R_m * T) / MC$ SR 3 0.00317 (A,B,C,D)
 $N(y) * T * \text{SQRT}(R_m * T) / ML$ SR 3 0.00507 (A,B,C,D)
 $M(y) * \text{SQRT}(R_m * T) / ML$ SR 3 0.00317 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-7.6	-7.6	-7.6	-7.6	-7.6	-7.6	-7.6	-7.6
Rad.	Bend. P	-20.8	20.8	-20.8	20.8	-20.8	20.8	-20.8	20.8
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-37.2	-37.2	37.2	37.2
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-138.8	138.8	138.8	-138.8
Rad.	Memb. ML	-37.2	-37.2	37.2	37.2	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-138.8	138.8	138.8	-138.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-204.4	114.8	147.6	-88.5	-204.4	114.8	147.6	-88.5
Tang.	Memb. P	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3
Tang.	Bend. P	-6.3	6.3	-6.3	6.3	-6.3	6.3	-6.3	6.3
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-11.1	-11.1	11.1	11.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-41.9	41.9	41.9	-41.9
Tang.	Memb. ML	-11.1	-11.1	11.1	11.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-41.9	41.9	41.9	-41.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-61.6	34.7	44.4	-26.7	-61.6	34.7	44.4	-26.7
Shear VC		1.0	1.0	-1.0	-1.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Tot. Shear		3.1	3.1	1.0	1.0	1.0	1.0	3.1	3.1
Str. Int.		204.4	114.9	147.6	88.5	204.4	114.8	147.7	88.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	39.9	39.9	39.9	39.9	39.9	39.9	39.9	39.9
Rad.	Pl (SUS)	-12.2	-12.2	8.8	8.8	-12.2	-12.2	8.8	8.8
Rad.	Q (SUS)	-30.4	30.4	22.9	-22.9	-30.4	30.4	22.9	-22.9
Long.	Pm (SUS)	39.9	39.9	39.9	39.9	39.9	39.9	39.9	39.9
Long.	Pl (SUS)	-89.5	-89.5	60.3	60.3	-89.5	-89.5	60.3	60.3
Long.	Q (SUS)	-90.2	90.2	71.7	-71.7	-90.2	90.2	71.7	-71.7
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.0	1.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0
Shear	Q (SUS)	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pm (SUS)		39.9	39.9	39.9	39.9	39.9	39.9	39.9	39.9
Pm+Pl (SUS)		77.3	77.3	100.1	100.1	77.3	77.3	100.1	100.1
Pm+Pl+Q (Total)		140.0	59.4	171.9	30.3	139.8	58.5	172.1	32.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.85	156.40	Passed
Pm+Pl (SUS)	100.14	234.60	Passed
Pm+Pl+Q (TOTAL)	172.08	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	39.9	39.9	39.9	39.9	39.9	39.9	39.9	39.9
Rad.	Pl (SUS)	-44.8	-44.8	29.6	29.6	-44.8	-44.8	29.6	29.6
Rad.	Q (SUS)	-159.6	159.6	118.0	-118.0	-159.6	159.6	118.0	-118.0
Long.	Pm (SUS)	39.9	39.9	39.9	39.9	39.9	39.9	39.9	39.9
Long.	Pl (SUS)	-13.4	-13.4	8.9	8.9	-13.4	-13.4	8.9	8.9
Long.	Q (SUS)	-48.2	48.2	35.6	-35.6	-48.2	48.2	35.6	-35.6
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.0	1.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0
Shear	Q (SUS)	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Pm (SUS)		39.9	39.9	39.9	39.9	39.9	39.9	39.9	39.9
Pm+Pl (SUS)		31.4	31.4	69.5	69.5	31.4	31.4	69.5	69.5
Pm+Pl+Q (Total)		164.6	154.8	187.4	61.8	164.5	154.7	187.5	62.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.85	156.40	Passed
Pm+Pl (SUS)	69.46	234.60	Passed
Pm+Pl+Q (TOTAL)	187.51	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

- ** Warning: Insufficient Element Thickness due to External Pressure!**
- ** Warning: Inadequate Stiffener Ring Specified, Inertia Inadequate!**
- ** Warning: The computed vessel MAWP seems to be less than the design pressure!**

Please review all reports carefully!

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